

ets_marl - Default RQ Analysis - STRATEGY #2

May 13, 2026

1 ETS MARL — Default-Config Research-Question Analysis

This notebook focuses on what the **default configuration** alone can tell us about the thesis research questions. It deliberately does **not** attempt cross-config (variant) comparisons — those will be addressed in a separate experiments notebook.

Research questions (full text in docs/research_questions.md and the top-level README.md):

- **Main RQ** — How can MARL be used to simulate the EU ETS, and what does the simulation reveal about its regulatory mechanisms and the strategic behaviour of power-sector participants?
- **Sub-RQ 1** — What environment / algorithm design choices are required to build a *stable, behaviourally credible* simulation?
- **Sub-RQ 2** — How sensitive are market outcomes (price stability, compliance, decarbonisation pace) to the regulatory mechanisms of the EU ETS? (*Default-only descriptive baseline; cross-variant claims deferred.*)
- **Sub-RQ 3** — How do financial vs environmental objectives shape company strategies, and how do those aggregate into market-level outcomes?

Structure

1. Setup — locate project root.
2. Imports.
3. Multi-sweep loader (allows pulling in **several** default-config sweeps at once — extends the single-spec pattern from ets_marl - Sweep Analysis.ipynb).
4. Discovery & loading.
5. Sanity overview (counts, episode coverage, default-only filter).
6. **§RQ1** — convergence, cross-seed reproducibility, behavioural credibility, pathology.
7. **§RQ2** — within-default regulatory mechanism behaviour (cap → price, MSR / TNAC, penalty incidence, decarbonisation pace).
8. **§RQ3** — strategic behaviour by archetype and reward weighting, aggregated to system-level outcomes.

The notebook is intentionally compact: each section ships the headline plot or table that supports the corresponding RQ claim, and stops there.

```
[2]: # Install notebook dependencies if needed.
# Set INSTALL_NOTEBOOK_DEPS = True (or run the cell with required packages
↳missing)
# to pip-install everything in notebooks/requirements-notebooks.txt using the
```

```

# CURRENT kernel's interpreter. This is more reliable than `!pip install -r
↳ {path}`
# (IPython magics don't expand `{var}` at runtime) and works on Azure / Colab /
↳ local.
import subprocess, sys, importlib, re
from pathlib import Path

INSTALL_NOTEBOOK_DEPS = False # flip to True to force-reinstall

# Map PyPI distribution name -> import module name when they differ.
_DIST_TO_IMPORT = {
    "pyyaml": "yaml",
    "scikit-learn": "sklearn",
    "imbalanced-learn": "imblearn",
    "opencv-python": "cv2",
    "pillow": "PIL",
    "beautifulsoup4": "bs4",
    "ipython": "IPython",
}

def _find_requirements():
    req_name = "requirements-notebooks.txt"
    here = Path.cwd()
    candidates = [here / req_name, here / "notebooks" / req_name]
    candidates.extend(p / "notebooks" / req_name for p in here.parents)
    return next((p for p in candidates if p.exists()), None)

def _parse_requirements(req_path):
    """Extract distribution names from a requirements file (skip comments,
    ↳ options, env markers)."""
    dists = []
    for raw in req_path.read_text(encoding="utf-8").splitlines():
        line = raw.strip()
        if not line or line.startswith("#") or line.startswith("-"):
            continue
        # Strip env markers and extras: "pkg[extra]>=1.0; python_version>='3.9'"
        line = line.split(";", 1)[0].strip()
        m = re.match(r"^([A-Za-z0-9_\.\\-]+)", line)
        if m:
            dists.append(m.group(1).lower())
    return dists

def _missing_required(dists):
    missing = []
    for dist in dists:
        mod = _DIST_TO_IMPORT.get(dist, dist.replace("-", "_"))
        try:

```

```

        importlib.import_module(mod)
    except ImportError:
        missing.append(dist)
    return missing

req_path = _find_requirements()
if req_path is None:
    raise FileNotFoundError("Could not find notebooks/requirements-notebooks.
↳txt")

dists = _parse_requirements(req_path)
missing = _missing_required(dists)

if INSTALL_NOTEBOOK_DEPS or missing:
    if missing:
        print(f"Missing packages: {missing}")
        print(f"Installing notebook dependencies from: {req_path}")
        subprocess.check_call([sys.executable, "-m", "pip", "install", "-r",
↳str(req_path)])
    else:
        print(f"All {len(dists)} notebook dependencies present; skipping install.")
        print("(Set INSTALL_NOTEBOOK_DEPS = True to force a reinstall.)")

```

All 14 notebook dependencies present; skipping install.
(Set INSTALL_NOTEBOOK_DEPS = True to force a reinstall.)

1.1 1. Setup — Locate Project Root

```

[3]: import os, sys
from pathlib import Path

def _is_project_root(path: str) -> bool:
    return os.path.isdir(os.path.join(path, "src")) and os.path.isdir(os.path.
↳join(path, "configs"))

_candidates = ["..", ".", os.path.join("..", ".."),
               os.path.join("Thesis-Energy-Auction", "ets_marl_happo_current"),
↳"Thesis-Energy-Auction"]
_found = _is_project_root(".")
if not _found:
    for _c in _candidates:
        if _is_project_root(_c):
            os.chdir(_c); _found = True; break
if not _found:
    raise FileNotFoundError(f"Cannot find project root (needs src/ + configs/).
↳\nCWD: {os.getcwd()} | Contents: {os.listdir('.')}")

```

```
PROJECT_ROOT = Path.cwd().resolve()
sys.path.insert(0, str(PROJECT_ROOT))
print(f"Working directory: {PROJECT_ROOT}")
```

Working directory: C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-Energy-Auction

1.2 2. Imports

```
[4]: import re
import yaml
import warnings
from pathlib import Path

import numpy as np
import pandas as pd
from src.utils.run_data import load_run_csv, glob_run_logs # parquet-cached
↳ log loader
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style('whitegrid')
pd.set_option('display.max_columns', 200)
pd.set_option('display.width', 200)

from src.utils.sweep import load_sweep_spec, expand_dotted_overrides, deep_merge
```

1.3 3. Sweep Specs to Load

List one or more sweep specs (paths relative to the project root). All of them are loaded and merged into a single dict keyed by (`sweep_name`, `variant_name`, `seed`). The default below loads only `default_seeds.yaml`, but additional default-config sweeps can be appended freely — every run that survives the §5 filter is treated as another seed of the *same* default-config population.

If a sweep contains variants whose `overrides` deviate from the base config, those variants are flagged in §5 and excluded from the analysis by default (this notebook is for the default config only). To include them anyway, set `STRICT_DEFAULT_ONLY = False` in §5.

```
[5]: # Add additional default-config sweeps here as the project produces them.
SWEEP_SPECS = [
    PROJECT_ROOT / 'configs' / 'sweeps' / 'default_seeds.yaml',
    # PROJECT_ROOT / 'configs' / 'sweeps' / 'default_seeds_old.yaml',
    # PROJECT_ROOT / 'configs' / 'sweeps' / 'default_seeds_expanded.yaml'
]

# Optional: override the output_dir from the spec (useful if results were
# moved). One entry per spec, or None to use the spec's own output_dir.
```

```

OUTPUT_DIR_OVERRIDES = [
    PROJECT_ROOT / 'results' / 'sweeps' / 'default',
    # PROJECT_ROOT / 'results' / 'sweeps',
    # PROJECT_ROOT / 'results' / 'sweeps' / 'default'
]
assert len(OUTPUT_DIR_OVERRIDES) == len(SWEEP_SPECS), "one override (or None)␣
↳per spec"

for sp in SWEEP_SPECS:
    if not Path(sp).exists():
        warnings.warn(f"Spec not found: {sp}")

print(f"Loading {len(SWEEP_SPECS)} sweep spec(s):")
for sp, od in zip(SWEEP_SPECS, OUTPUT_DIR_OVERRIDES):
    print(f"  - {sp} (override={od})")

```

Loading 1 sweep spec(s):

```

- C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-Energy-
Auction\configs\sweeps\default_seeds.yaml
(override=C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-Energy-
Auction\results\sweeps\default)

```

1.4 4. Multi-Sweep Discovery & Loading

```

[6]: from src.utils.run_data import log_exists, default_ep_columns,␣
↳default_yr_columns
from tqdm.auto import tqdm

def _resolve_run_files(variant_name, seed, variant_dir, output_dir):
    """Return (training_log, year_log) paths or (None, None). Tagged then␣
↳untagged."""
    search_dirs = []

    # Existing locations
    for d in (variant_dir, output_dir):
        if d is not None:
            d = Path(d)
            if d not in search_dirs:
                search_dirs.append(d)

    # Extra fallback: results/sweep/<tag>[/<variant>]
    # (tag inferred from output_dir name, e.g. default_seeds)
    if output_dir is not None:
        tag = Path(output_dir).name
        alt_base = PROJECT_ROOT / "results" / "sweep" / tag
        for d in (alt_base / variant_name, alt_base):

```

```

        if d not in search_dirs:
            search_dirs.append(d)

    training_names = [f"training_log_{variant_name}_s{seed}.csv",
↪f"training_log_s{seed}.csv"]
    year_names = [f"year_log_{variant_name}_s{seed}.csv", f"year_log_s{seed}.
↪csv"]

    tlog = ylog = None
    for base in search_dirs:
        if tlog is None:
            for n in training_names:
                cand = base / n
                if log_exists(cand):
                    tlog = cand
                    break
        if ylog is None:
            for n in year_names:
                cand = base / n
                if log_exists(cand):
                    ylog = cand
                    break
        if tlog is not None and ylog is not None:
            break

    return tlog, ylog

# Project both logs onto the analysis-superset columns at load time. The
# year_log carries 270+ diagnostic columns; pulling only the ~80 actually
# used by this notebook cuts in-RAM size by ~5-10x. Unknown columns are
# silently dropped by load_run_csv.
EP_COLUMNS = default_ep_columns()
YR_COLUMNS = list(dict.fromkeys(
    default_yr_columns()
    + [f"sec_price_mult_A{i}" for i in range(1, 17)]
))

def _load_csv_or_none(path, columns=None):
    if path is None: return None
    return load_run_csv(path, columns=columns)

def _empty_ep(yr=None):
    if yr is not None and 'episode' in yr.columns:
        return pd.DataFrame({'episode': np.sort(yr['episode'].dropna().
↪unique())})
    return pd.DataFrame({'episode': pd.Series(dtype='int64')})

```

```

def _empty_yr(ep=None):
    if ep is not None and 'episode' in ep.columns and len(ep) > 0:
        return pd.DataFrame({'episode': ep['episode'].astype('int64').values,
                              'year': np.zeros(len(ep), dtype='int64')})
    return pd.DataFrame({'episode': pd.Series(dtype='int64'), 'year': pd.
        ↳Series(dtype='int64')})

specs = []
for spec_path, override in zip(SWEEP_SPECS, OUTPUT_DIR_OVERRIDES):
    if not Path(spec_path).exists():
        continue
    spec = load_sweep_spec(str(spec_path))
    sweep_name = Path(spec_path).stem
    with open(spec['base_config'], encoding='utf-8') as f:
        base_cfg = yaml.safe_load(f)
    output_dir = Path(override) if override is not None else (PROJECT_ROOT /
        ↳spec['output_dir']).resolve()
    specs.append({'name': sweep_name, 'spec': spec, 'base_cfg': base_cfg,
        ↳'output_dir': output_dir})

# Pre-enumerate every (sweep, variant, seed) we'll attempt so tqdm has a real
↳total.
load_jobs = []
for s in specs:
    for v in s['spec']['variants']:
        seeds = v['seeds'] if v['seeds'] is not None else s['spec']['seeds']
        variant_dir = s['output_dir'] / v['name']
        merged_cfg = deep_merge(s['base_cfg'], expand_dotted_overrides(v.
        ↳get('overrides') or {}))
        flat_overrides = expand_dotted_overrides(v.get('overrides') or {})
        for seed in seeds:
            load_jobs.append((s, v, seed, variant_dir, merged_cfg,
        ↳flat_overrides))

# Build (sweep, variant, seed) -> run dict.
runs = {}
discovery = []
for s, v, seed, variant_dir, merged_cfg, flat_overrides in tqdm(
    load_jobs, desc='Loading sweep CSVs', unit='run', leave=True
):
    tlog, ylog = _resolve_run_files(v['name'], seed, variant_dir,
        ↳s['output_dir'])
    row = {
        'sweep': s['name'], 'variant': v['name'], 'seed': seed,
        'has_overrides': bool(flat_overrides),
        'training_log': str(tlog) if tlog else None,
    }

```

```

        'year_log': str(ylog) if ylog else None,
        'has_training_log': tlog is not None,
        'has_year_log': ylog is not None,
        'found': (tlog is not None) or (ylog is not None),
    }
    discovery.append(row)
    if not row['found']:
        continue
    ep_df = _load_csv_or_none(tlog, columns=EP_COLUMNS)
    yr_df = _load_csv_or_none(ylog, columns=YR_COLUMNS)
    if ep_df is None and yr_df is None:
        continue
    if ep_df is None: ep_df = _empty_ep(yr_df)
    if yr_df is None: yr_df = _empty_yr(ep_df)
    runs[(s['name'], v['name'], seed)] = {
        'ep_df': ep_df, 'yr_df': yr_df, 'config': merged_cfg,
        'has_overrides': bool(flat_overrides),
        'overrides': flat_overrides,
        'has_training_log': bool(tlog), 'has_year_log': bool(ylog),
    }

discovery_df = pd.DataFrame(discovery)
n_found = int(discovery_df['found'].sum()) if not discovery_df.empty else 0
print(f"Discovered {len(discovery_df)} (sweep, variant, seed) triples; loaded_{n_found}")
if not runs:
    raise RuntimeError("No runs loaded. Check SWEEP_SPECS /_{OUTPUT_DIR_OVERRIDES}")
discovery_df

# Free chunked CSV/parquet readers held by pandas/pyarrow before continuing.
import gc; gc.collect()

```

Loading sweep CSVs: 0%| | 0/4 [00:00<?, ?run/s]

Discovered 4 (sweep, variant, seed) triples; loaded 4.

[6]: 25

1.5 5. Default-Only Filter & Sanity Overview

```

[7]: STRICT_DEFAULT_ONLY = True # if True, drop runs whose variant has any override.

if STRICT_DEFAULT_ONLY:
    n_before = len(runs)
    runs = {k: v for k, v in runs.items() if not v['has_overrides']}
    n_after = len(runs)

```



```

    if n_after < n_before:
        print(f"Filtered out {n_before - n_after} run(s) whose variant_
↳ overrides the base config.")
    if n_after == 0:
        raise RuntimeError("STRICT_DEFAULT_ONLY filtered out every run. Set_
↳ STRICT_DEFAULT_ONLY = False to inspect non-default variants.")

# Treat every surviving run as a default-config seed. Build a flat list.
ALL_RUN_KEYS = sorted(runs.keys())
print(f"\nUsable default-config runs: {len(ALL_RUN_KEYS)}")
for (sw, var, seed), run in sorted(runs.items()):
    n_eps = len(run['ep_df'])
    n_yrs = len(run['yr_df'])
    print(f" {sw}/{var}/seed={seed:>4} ({n_eps:>6} episodes, {n_yrs:>7}_
↳ year-rows)")

# Pick a reference config. All defaults agree on schema; just take the first.
REFERENCE_CONFIG = runs[ALL_RUN_KEYS[0]]['config']
N_AGENTS = int(REFERENCE_CONFIG.get('companies', {}).get('n_agents', 8))
N_BOTS = int(REFERENCE_CONFIG.get('companies', {}).get('n_bot_agents', 0))
N_TOTAL = N_AGENTS + N_BOTS
N_YEARS = int(REFERENCE_CONFIG.get('environment', {}).get('n_years', 12))
YEAR_DISPLAY_START = 1
YEAR_DISPLAY_END = N_YEARS
print(f"\nReference config: n_agents={N_AGENTS}, n_bot_agents={N_BOTS},_
↳ n_years={N_YEARS}")

def _year_log_to_display_years(yr_df):
    if yr_df is None or yr_df.empty or 'year' not in yr_df.columns:
        return yr_df, False
    year = pd.to_numeric(yr_df['year'], errors='coerce')
    finite_year = year.dropna()
    if finite_year.empty:
        return yr_df, False
    year_min = finite_year.min()
    year_max = finite_year.max()
    if year_min >= YEAR_DISPLAY_START:
        return yr_df, False
    if year_min == 0 and year_max <= max(N_YEARS - 1, 0):
        shifted_year = year + YEAR_DISPLAY_START
        if shifted_year.notna().all():
            rounded_year = shifted_year.round()
            if shifted_year.eq(rounded_year).all():
                shifted_year = rounded_year.astype('int16', copy=False)
        yr_df['year'] = shifted_year
    return yr_df, True

```

```

        raise ValueError(f"Unexpected year range in year_log: {year_min}..
        ↳{year_max}")

n_shifted_year_frames = 0
for run in runs.values():
    run['yr_df'], shifted = _year_log_to_display_years(run.get('yr_df'))
    n_shifted_year_frames += int(shifted)

if n_shifted_year_frames:
    print(f"Converted {n_shifted_year_frames} year_log frame(s) from years_
    ↳0-{N_YEARS - 1} to years {YEAR_DISPLAY_START}-{YEAR_DISPLAY_END}.")
else:
    print(f"Year_log frames already use display years_
    ↳{YEAR_DISPLAY_START}-{YEAR_DISPLAY_END}.")

# Archetype/reward labelling (default config: 4 archetypes x {financial, ESG}).
ARCHETYPES = ['Coal-heavy', 'Gas-dominant', 'Transitioner', 'Green-leader']
def _agent_archetype(idx0):
    return ARCHETYPES[idx0 // 2] if idx0 // 2 < len(ARCHETYPES) else_
    ↳f'Custom-{idx0//2}'
def _agent_weighting(idx0):
    # Even indices (A1, A3, ...) are pure financial; odd are balanced ESG. (Per_
    ↳README.)
    return 'Financial' if idx0 % 2 == 0 else 'Balanced ESG'

AGENT_LABELS = [f'A{i+1} ({_agent_archetype(i)}, {_agent_weighting(i)})' for i_
    ↳in range(N_AGENTS)]
print("Agent roster (1-based):")
for lbl in AGENT_LABELS:
    print(f"  {lbl}")

```

Usable default-config runs: 4

default_seeds/default/seed=1729	(120000 episodes, 1440000 year-rows)
default_seeds/default/seed=5041	(120000 episodes, 1440000 year-rows)
default_seeds/default/seed=6561	(120000 episodes, 1440000 year-rows)
default_seeds/default/seed=8191	(120000 episodes, 1440000 year-rows)

Reference config: n_agents=8, n_bot_agents=0, n_years=12

Converted 4 year_log frame(s) from years 0-11 to years 1-12.

Agent roster (1-based):

```

A1 (Coal-heavy, Financial)
A2 (Coal-heavy, Balanced ESG)
A3 (Gas-dominant, Financial)
A4 (Gas-dominant, Balanced ESG)
A5 (Transitioner, Financial)
A6 (Transitioner, Balanced ESG)

```

A7 (Green-leader, Financial)
A8 (Green-leader, Balanced ESG)

1.5.1 5a. Helpers

A handful of helpers reused across §RQ1–§RQ3. We intentionally treat every loaded run as one seed of the same default-config population (no variant axis) — the helpers therefore aggregate over ALL_RUN_KEYS without any variant grouping.

```
[8]: ROLL = 50                # episode-axis smoothing window
TAIL_EPISODES = 1000        # converged-window definition for steady-state metrics

def agent_cols(df, prefix, max_idx=N_AGENTS):
    """Return columns 'prefix_A1' ... 'prefix_A{max_idx}' that exist in df,
    sorted by agent index."""
    rx = re.compile(rf'^{re.escape(prefix)}_A(\d+)\$')
    out = []
    for c in df.columns:
        m = rx.match(c)
        if m and int(m.group(1)) <= max_idx:
            out.append((int(m.group(1)), c))
    return [c for _, c in sorted(out)]

def per_run_series(metric_fn, source='ep'):
    """Return dict {run_key: pd.Series indexed by episode} from each run."""
    out = {}
    for k, run in runs.items():
        df = run['yr_df'] if source == 'yr' else run['ep_df']
        if source == 'run':
            s = metric_fn(run)
        else:
            s = metric_fn(df)
        if s is None or len(s) == 0:
            continue
        s = s.copy()
        if source == 'ep' and 'episode' in run['ep_df'].columns:
            s.index = run['ep_df']['episode']
        out[k] = s
    return out

def stack_runs(metric_fn, source='ep'):
    """Return wide DataFrame of episode-indexed series, one column per run."""
    series = per_run_series(metric_fn, source=source)
    if not series:
        return pd.DataFrame()
    return pd.concat(series, axis=1).sort_index()

def fan_chart(df, ax, color='steelblue', label='runs', smooth=ROLL):
```

```

"""Plot mean  $\pm 1$  std fan + thin per-run lines for a wide DataFrame."""
if df.empty:
    ax.text(0.5, 0.5, 'no data', ha='center', va='center', transform=ax.
↳transAxes)
    return
sm = df.rolling(smooth, min_periods=1).mean()
mean = sm.mean(axis=1)
std = sm.std(axis=1).fillna(0.0)
for col in sm.columns:
    ax.plot(sm.index, sm[col].values, color=color, lw=0.6, alpha=0.30)
    ax.fill_between(sm.index, (mean - std).values, (mean + std).values,
                    color=color, alpha=0.20, label=f'{label}  $\pm 1$  std (n={sm.
↳shape[1]})')
    ax.plot(sm.index, mean.values, color=color, lw=2.2, label=f'{label} mean')
    ax.grid(True, alpha=0.3); ax.legend(fontsize=8, loc='best')

def tail_episodes(ep_df, n=TAIL_EPISODES):
    if ep_df.empty: return ep_df
    return ep_df.tail(min(n, len(ep_df)))

def tail_year_rows(run, n=TAIL_EPISODES):
    ep = tail_episodes(run['ep_df'], n)
    if ep.empty or 'episode' not in run['yr_df'].columns:
        return run['yr_df'].head(0)
    return run['yr_df'][run['yr_df']['episode'].isin(ep['episode'].unique())]

def format_year_axis(ax, *, pad=0.35):
    start = globals().get('YEAR_DISPLAY_START', 1)
    end = globals().get('YEAR_DISPLAY_END', globals().get('N_YEARS', 12))
    ax.set_xlim(start - pad, end + pad)
    ax.set_xticks(np.arange(start, end + 1, dtype=int))

```

1.6 §RQ1 — Stable & Behaviourally Credible Simulation

Three sub-checks:

1. **Convergence** — system reward, composite quality score, and PPO actor/critic losses settle by training end.
2. **Cross-seed reproducibility** — fan charts on headline metrics across loaded seeds; the spread of converged-window means quantifies how reliably the same equilibrium is reached.
3. **Behavioural credibility** — final-window clearing price sits in the EU-ETS-plausible range (~€60–110 per the README's design notes), compliance rate plateaus high, and warning-flag counts in the converged window are not pathological.

1.6.1 RQ1.1 — Convergence dynamics (rewards, quality score, losses)

```
[9]: def _system_reward(df):
    cols = agent_cols(df, 'reward')
    return df[cols].sum(axis=1) if cols else pd.Series(dtype=float)

def _system_base_reward(df):
    cols = agent_cols(df, 'reward_base')
    return df[cols].sum(axis=1) if cols else pd.Series(dtype=float)

def _quality(df):
    return df['quality_score'] if 'quality_score' in df.columns else pd.
    ↳Series(dtype=float)

def _mean_actor_loss(df):
    cols = agent_cols(df, 'actor_loss')
    return df[cols].mean(axis=1) if cols else pd.Series(dtype=float)

def _mean_critic_loss(df):
    cols = agent_cols(df, 'critic_loss')
    return df[cols].mean(axis=1) if cols else pd.Series(dtype=float)

fig, axes = plt.subplots(2, 2, figsize=(16, 9))
fan_chart(stack_runs(_system_reward), axes[0, 0], color='steelblue',
    ↳label='system reward')
axes[0, 0].set(title='System total reward ( $\Sigma$  reward_A*)', xlabel='Episode',
    ↳ylabel='reward')

fan_chart(stack_runs(_quality), axes[0, 1], color='seagreen',
    ↳label='quality_score')
axes[0, 1].set(title='Composite quality score', xlabel='Episode', ylabel='Q
    ↳[-5, 5]')
axes[0, 1].set_ylim(0, 3)

fan_chart(stack_runs(_mean_actor_loss), axes[1, 0], color='firebrick',
    ↳label='actor loss')
axes[1, 0].set(title='Mean actor loss across agents', xlabel='Episode',
    ↳ylabel='loss')

fan_chart(stack_runs(_mean_critic_loss), axes[1, 1], color='darkorange',
    ↳label='critic loss')
axes[1, 1].set(title='Mean critic loss across agents', xlabel='Episode',
    ↳ylabel='loss')

plt.suptitle('RQ1.1 - Training-time convergence (mean  $\pm$ 1 std across loaded
    ↳seeds)',
    fontsize=13, y=1.02)
```

```
plt.tight_layout(); plt.show()
```

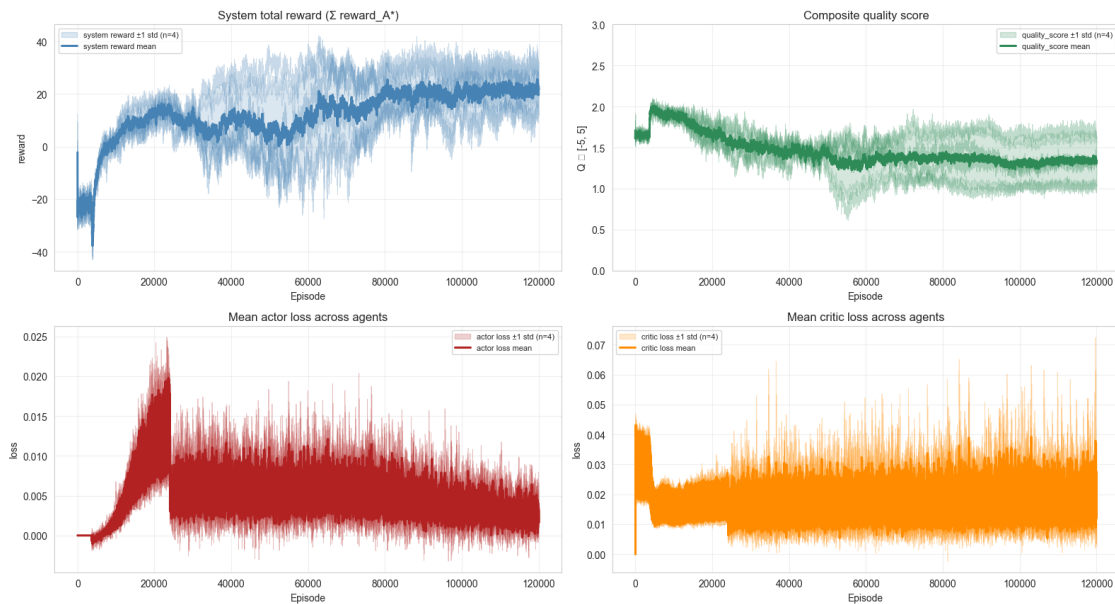
C:\Users\danie\AppData\Local\Temp\ipykernel_5056\61372440.py:36: UserWarning: Glyph 8712 (\N{ELEMENT OF}) missing from font(s) Arial.

```
plt.tight_layout(); plt.show()
```

C:\Users\danie\AppData\Roaming\Python\Python312\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 8712 (\N{ELEMENT OF}) missing from font(s) Arial.

```
fig.canvas.print_figure(bytes_io, **kw)
```

RQ1.1 — Training-time convergence (mean $\pm$ 1 std across loaded seeds)



1.6.2 RQ1.2 — Cross-seed reproducibility on headline metrics

```
[10]: # Per-seed converged-window means for a small set of headline metrics.
def _converged_metrics_for(run):
    ep = tail_episodes(run['ep_df'])
    yr = tail_year_rows(run)
    out = {}

    rew_cols = agent_cols(ep, 'reward')
    out['system_reward'] = float(ep[rew_cols].sum(axis=1).mean()) if
↪rew_cols else np.nan
    out['quality_score'] = float(ep['quality_score'].mean()) if
↪'quality_score' in ep.columns else np.nan

    if 'clearing_price_last' in ep.columns:
```

```

        out['mean_clearing_price'] = float(ep['clearing_price_last'].mean())
    elif 'clearing_price' in yr.columns:
        out['mean_clearing_price'] = float(yr['clearing_price'].mean())
    else:
        out['mean_clearing_price'] = np.nan

    sf_cols = agent_cols(yr, 'shortfall')
    out['compliance_rate'] = float((yr[sf_cols] <= 1e-6).mean(axis=1).mean())
    if sf_cols else np.nan

    pen_cols = agent_cols(ep, 'penalty')
    out['system_penalty'] = float(ep[pen_cols].sum(axis=1).mean()) if pen_cols
    else np.nan

    inv_cols = agent_cols(ep, 'invest_cost')
    out['system_invest'] = float(ep[inv_cols].sum(axis=1).mean()) if inv_cols
    else np.nan

    gf_cols = agent_cols(ep, 'green_frac')
    out['mean_final_green_frac'] = float(ep[gf_cols].mean(axis=1).mean()) if
    gf_cols else np.nan

    if 'secondary_volume' in ep.columns:
        out['secondary_volume'] = float(ep['secondary_volume'].mean())
    else:
        out['secondary_volume'] = np.nan

    return out

per_seed_rows = []
for k, run in runs.items():
    row = {'sweep': k[0], 'variant': k[1], 'seed': k[2]}
    row.update(_converged_metrics_for(run))
    per_seed_rows.append(row)
per_seed_df = pd.DataFrame(per_seed_rows)
print(f"Per-seed converged-window means (last {TAIL_EPISODES} episodes):")
display(per_seed_df.round(3))

agg_cols = ['system_reward', 'quality_score', 'mean_clearing_price',
    'compliance_rate',
    'system_penalty', 'system_invest', 'mean_final_green_frac',
    'secondary_volume']
summary = per_seed_df[agg_cols].agg(['mean', 'std', 'min', 'max']).round(3)
summary.loc['cv (%)'] = (summary.loc['std'] / summary.loc['mean'].abs() * 100).
    round(1)
print(f"\nCross-seed summary (n={len(per_seed_df)}):")

```

```
summary
```

Per-seed converged-window means (last 1000 episodes):

```
      sweep variant seed system_reward quality_score
↳mean_clearing_price compliance_rate system_penalty system_invest
↳mean_final_green_frac secondary_volume
0 default_seeds default 1729      20.079      1.712      82.
↳345      0.875      1920.884      5139.825      0.735
↳      43.932
1 default_seeds default 8191      31.476      1.331      58.
↳164      0.853      2529.066      6426.351      0.765
↳      40.209
2 default_seeds default 6561      12.398      1.345      81.
↳395      0.797      3793.327      6858.905      0.771
↳      27.110
3 default_seeds default 5041      24.875      1.015      48.
↳424      0.787      4301.157      5121.147      0.748
↳      30.116
```

Cross-seed summary (n=4):

```
[10]:      system_reward quality_score mean_clearing_price compliance_rate
system_penalty system_invest mean_final_green_frac secondary_volume
mean      22.207      1.351      67.582      0.828
3136.108      5886.557      0.755      35.342
std      8.037      0.285      16.975      0.043
1100.688      890.749      0.017      8.012
min      12.398      1.015      48.424      0.787
1920.884      5121.147      0.735      27.110
max      31.476      1.712      82.345      0.875
4301.157      6858.905      0.771      43.932
cv (%)      36.200      21.100      25.100      5.200
35.100      15.100      2.300      22.700
```

Reading the table. A coefficient of variation (cv (%)) below ~10% on the headline outcomes (mean clearing price, compliance rate, mean final green fraction) is the bar we use to call the simulation *reproducible* across seeds — anything significantly larger means the RQ2 / RQ3 claims need to be hedged seed-by-seed.

1.6.3 RQ1.3 — Behavioural credibility in the converged window

```
[11]: # (a) Final-window clearing-price distribution; (b) per-seed compliance plateau.
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

ax = axes[0]
boxes = []
```



```

labels = []
for k, run in runs.items():
    yr = tail_year_rows(run)
    if 'clearing_price' not in yr.columns or yr.empty:
        continue
    boxes.append(yr['clearing_price'].dropna().values)
    labels.append(f"{k[0][:8]}/s{k[2]}")
if boxes:
    ax.boxplot(boxes, tick_labels=labels, showfliers=False)
    ax.axhspan(60, 110, color='green', alpha=0.10, label='EU-ETS plausible band',
    ↪( €60-110/t)')
    ax.set(title='Converged-window clearing price by seed', ylabel='€/tCO2',
    ↪xlabel='Run')
    ax.legend(fontsize=8); ax.tick_params(axis='x', rotation=30)
else:
    ax.text(0.5, 0.5, 'No clearing_price column in any year_log', ha='center',
    ↪va='center', transform=ax.transAxes)

ax = axes[1]
for k, run in runs.items():
    sf_cols = agent_cols(run['yr_df'], 'shortfall')
    if not sf_cols or 'episode' not in run['yr_df'].columns:
        continue
    rate = (run['yr_df'][sf_cols] <= 1e-6).mean(axis=1).
    ↪groupby(run['yr_df']['episode']).mean()
    rate_smooth = rate.rolling(ROLL, min_periods=1).mean()
    ax.plot(rate_smooth.index, rate_smooth.values, lw=1.0, alpha=0.85,
    label=f"{k[0][:8]}/s{k[2]}")
ax.set(title='Compliance rate over training (mean across (agent, year) cells)',
    xlabel='Episode', ylabel='compliance rate [0,1]')
ax.set_ylim(0.0, 1.02); ax.grid(True, alpha=0.3); ax.legend(fontsize=8,
    ↪loc='best')

plt.suptitle('RQ1.3 - Behavioural credibility: realistic price band +
    ↪compliance plateau',
    fontsize=13, y=1.02)
plt.tight_layout(); plt.show()

```

C:\Users\danie\AppData\Local\Temp\ipykernel_5056\2922376889.py:36: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

plt.tight_layout(); plt.show()

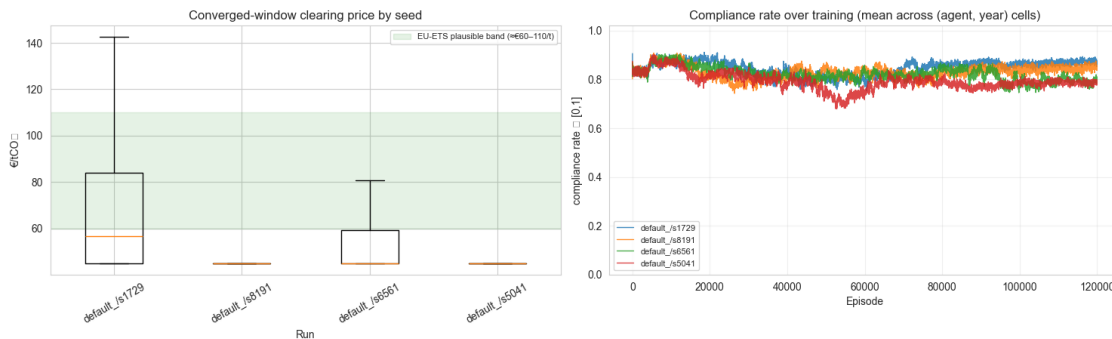
C:\Users\danie\AppData\Local\Temp\ipykernel_5056\2922376889.py:36: UserWarning: Glyph 8712 (\N{ELEMENT OF}) missing from font(s) Arial.

plt.tight_layout(); plt.show()

C:\Users\danie\AppData\Roaming\Python\Python312\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

```
fig.canvas.print_figure(bytes_io, **kw)
C:\Users\danie\AppData\Roaming\Python\Python312\site-
packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 8712 (\N{ELEMENT
OF}) missing from font(s) Arial.
fig.canvas.print_figure(bytes_io, **kw)
```

RQ1.3 — Behavioural credibility: realistic price band + compliance plateau



1.6.4 RQ1.4 — Pathology check: warnings in the converged window

```
[12]: # Sum the warn_* columns across the converged window for each run.
WARN_PREFIX = 'warn_'
warn_rows = []
warn_columns_seen = set()
for k, run in runs.items():
    ep = tail_episodes(run['ep_df'])
    warn_cols = [c for c in ep.columns if c.startswith(WARN_PREFIX)]
    warn_columns_seen.update(warn_cols)
    if not warn_cols:
        continue
    row = {'run': f"{k[0][:8]}/{s[k[2]]}"}
    # Per-episode mean warning count (interpretable as "how often per episode
    # does this warn fire").
    row.update({c: float(ep[c].mean()) for c in warn_cols})
    warn_rows.append(row)

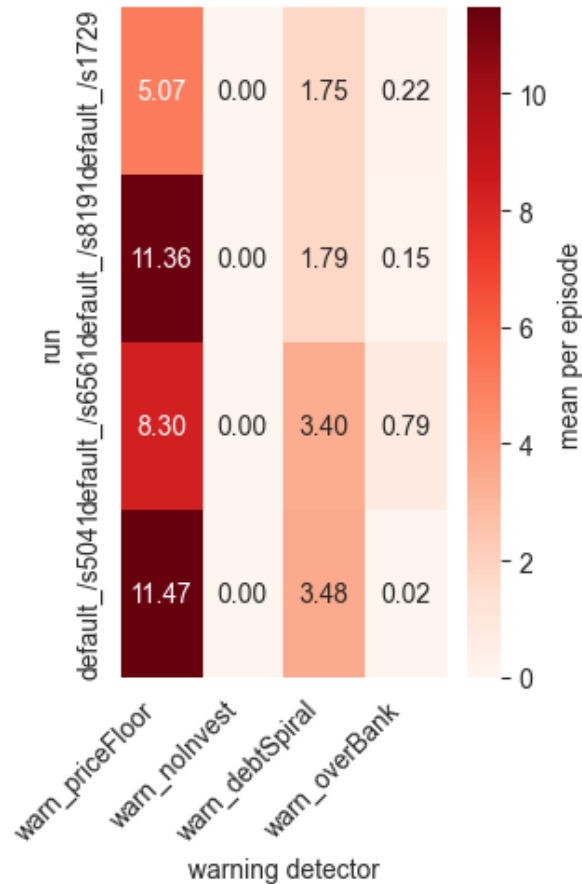
if warn_rows:
    warn_df = pd.DataFrame(warn_rows).set_index('run').fillna(0.0)
    # Drop columns that are exactly zero across all runs.
    warn_df = warn_df.loc[:, (warn_df > 0).any(axis=0)]
    fig, ax = plt.subplots(figsize=(min(14, 1.5 + 0.45 * len(warn_df.columns)),
    4 + 0.3 * len(warn_df)))
    if warn_df.empty:
        ax.text(0.5, 0.5, 'No non-zero warnings in the converged window - clean
run.',
```

```

        ha='center', va='center', transform=ax.transAxes, fontsize=12)
    ax.axis('off')
    else:
        sns.heatmap(warn_df.round(2), annot=True, cmap='Reds', fmt='.2f',
cbar_kws={'label': 'mean per episode'}, ax=ax)
        ax.set(title=f'Mean warning-flag counts per episode in the converged_
window (last {TAIL_EPISODES} eps)',
              xlabel='warning detector', ylabel='run')
        plt.xticks(rotation=45, ha='right')
        plt.tight_layout(); plt.show()
        print(f'Inspected {len(warn_columns_seen)} warning columns; showing only_
the {warn_df.shape[1]} with > 0 events.')
    else:
        print('No warn_* columns present in episode logs.')

```

Mean warning-flag counts per episode in the converged window (last 1000 eps)



Inspected 6 warning columns; showing only the 4 with > 0 events.

1.6.5 RQ1.5 — Whole-training overview (all loaded runs)

A single 2×2 panel summarising the entire training trajectory across *every* loaded run (i.e. before the §RQ2 default-seeds filter kicks in). Top-left: system reward; top-right: final-year clearing price (per-episode `clearing_price` at the maximum simulation year); bottom-left: compliance rate; bottom-right: mean green generation share. The shaded band on each panel is ± 1 **standard deviation around the mean** of the loaded runs at every smoothed episode, with thin lines showing the individual runs.

The clearing-price panel adds two horizontal references taken from `configs/default.yaml`: the auction reserve (floor) price and the final-year penalty rate (`penalty.rate × (1 + inflation_rate)^(N_YEARS - 1)`). The green-share panel marks the cross-agent mean of the *initial* (display year 1) green fractions read from `companies.initial_mix × technologies.is_green`. Compliance and green-share use a tight padded y-axis so changes during training are easy to read.

```
[13]: # 2x2 whole-training overview across all loaded runs.
# Episode-axis fan charts (mean  $\pm 1$  std band across runs, smoothed by ROLL).

def _final_year_clearing_by_ep(run):
    """Per-episode clearing price in the final simulation year."""
    yr = run['yr_df']
    needed = {'clearing_price', 'episode', 'year'}
    if not needed.issubset(yr.columns):
        return pd.Series(dtype=float)
    idx = yr.groupby('episode')['year'].idxmax()
    s = yr.loc[idx].set_index('episode')['clearing_price']
    return s.sort_index()

def _compliance_rate_by_ep(run):
    yr = run['yr_df']
    sf_cols = agent_cols(yr, 'shortfall')
    if not sf_cols or 'episode' not in yr.columns:
        return pd.Series(dtype=float)
    return (yr[sf_cols] <= 1e-6).mean(axis=1).groupby(yr['episode']).mean()

def _mean_green_frac_by_ep(df):
    cols = agent_cols(df, 'green_frac')
    return df[cols].mean(axis=1) if cols else pd.Series(dtype=float)

# Display year 1 (start-of-simulation) mean green fraction from the default_
↳ config.
_mix = REFERENCE_CONFIG.get('companies', {}).get('initial_mix', []) or []
_is_green = REFERENCE_CONFIG.get('technologies', {}).get('is_green', []) or []
_green_idx = [i for i, g in enumerate(_is_green) if g]
if _mix and _green_idx:
    _agent_start_green = [sum(row[i] for i in _green_idx) for row in _mix[:
↳ N_AGENTS]]
```

```

START_GREEN_FRAC = float(np.mean(_agent_start_green))
else:
    START_GREEN_FRAC = float('nan')

def _std_fan(df_wide, ax, *, color, label, smooth=ROLL):
    """Plot mean  $\pm 1$  std band (across runs) of `df_wide` on `ax`, plus thin
    per-run lines."""
    if df_wide.empty:
        ax.text(0.5, 0.5, 'no data', ha='center', va='center', transform=ax.
        transAxes)
        return
    sm = df_wide.rolling(smooth, min_periods=1).mean()
    mean = sm.mean(axis=1)
    std = sm.std(axis=1).fillna(0.0)
    n = sm.shape[1]
    for col in sm.columns:
        ax.plot(sm.index, sm[col].values, color=color, lw=0.6, alpha=0.30)
    ax.fill_between(sm.index, (mean - std).values, (mean + std).values,
                    color=color, alpha=0.20, label=f'{label}  $\pm 1$  std (n={n})')
    ax.plot(sm.index, mean.values, color=color, lw=2.2, label=f'{label} mean')
    ax.grid(True, alpha=0.3)
    ax.legend(fontsize=8, loc='best')

def _padded_ylim_std(df_wide, *, smooth=ROLL, pad_frac=0.10, include=None,
                    lo_clip=None, hi_clip=None):
    """Tight padded (ymin, ymax) around the smoothed mean  $\pm 1$  std band."""
    if df_wide.empty:
        return None
    sm = df_wide.rolling(smooth, min_periods=1).mean()
    mean = sm.mean(axis=1)
    std = sm.std(axis=1).fillna(0.0)
    lo = float((mean - std).min())
    hi = float((mean + std).max())
    if include is not None and not np.isnan(include):
        lo = min(lo, float(include))
        hi = max(hi, float(include))
    span = max(hi - lo, 1e-9)
    pad = pad_frac * span
    ylo, yhi = lo - pad, hi + pad
    if lo_clip is not None:
        ylo = max(ylo, lo_clip)
    if hi_clip is not None:
        yhi = min(yhi, hi_clip)
    return (ylo, yhi)

```

```

# Reserve (auction floor) price and final-year penalty rate from the
# default config - used as horizontal reference lines on the clearing-

```

```

# price panel, and reported in the companion table.
_ets_cfg = REFERENCE_CONFIG.get('ets', {}) or {}
RESERVE_PRICE = float(_ets_cfg.get('reserve_price', float('nan')))
_pen_cfg = REFERENCE_CONFIG.get('penalty', {}) or {}
_pen_base = float(_pen_cfg.get('rate', float('nan')))
_pen_infl = float(_pen_cfg.get('inflation_rate', 0.0) or 0.0)
# Final display year is N_YEARS; inflation still accrues over N_YEARS - 1
↳ intervals.
FINAL_YEAR_PENALTY_RATE = (
    _pen_base * (1.0 + _pen_infl) ** max(N_YEARS - 1, 0)
    if not np.isnan(_pen_base) else float('nan')
)

# Episode-indexed wide frames (one column per run).
overview_panels = {
    'System reward': (stack_runs(_system_reward,
↳ source='ep'), 'reward', 'steelblue'),
    'Final-year clearing price': (stack_runs(_final_year_clearing_by_ep,
↳ source='run'), '€/tCO2', 'firebrick'),
    'Compliance rate': (stack_runs(_compliance_rate_by_ep,
↳ source='run'), 'compliance in [0,1]', 'seagreen'),
    'Mean green generation share': (stack_runs(_mean_green_frac_by_ep,
↳ source='ep'), 'green frac in [0,1]', 'darkorange'),
}

fig, axes = plt.subplots(2, 2, figsize=(15, 8))
for ax, (title, (df_wide, ylabel, color)) in zip(axes.ravel(), overview_panels.
↳ items()):
    _std_fan(df_wide, ax, color=color, label=title)
    ax.set(title=title, xlabel='Episode', ylabel=ylabel)
    ax.margins(x=0)
    ax.set_xticks(range(0, 120001, 20000))

    if title == 'Final-year clearing price':
        _refs = [v for v in (RESERVE_PRICE, FINAL_YEAR_PENALTY_RATE) if not np.
↳ isnan(v)]
        ylim = _padded_ylim_std(df_wide, pad_frac=0.08,
                                include=None)

        if ylim is not None:
            lo, hi = ylim
            for r in _refs:
                lo = min(lo, r); hi = max(hi, r)
            ax.set_ylim(lo, hi+5)
        if not np.isnan(RESERVE_PRICE):
            ax.axhline(RESERVE_PRICE, color='black', lw=1.0, linestyle='--',
↳ alpha=0.7,

```

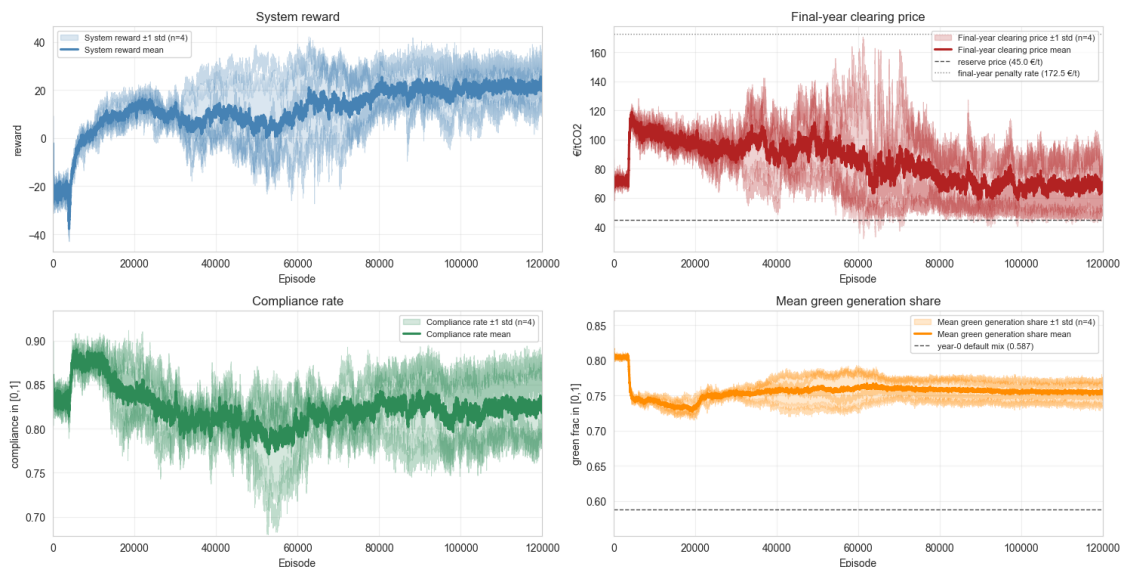
```

        label=f'reserve price ({RESERVE_PRICE:.1f} €/t)')
    if not np.isnan(FINAL_YEAR_PENALTY_RATE):
        ax.axhline(FINAL_YEAR_PENALTY_RATE, color='dimgray', lw=1.0,
        ↳linestyle=':', alpha=0.8,
        label=f'final-year penalty rate_
        ↳({FINAL_YEAR_PENALTY_RATE:.1f} €/t)')
    if _refs:
        ax.legend(fontsize=8, loc='best')
    elif title == 'Compliance rate':
        ylim = _padded_ylim_std(df_wide, pad_frac=0.15, lo_clip=0.0, hi_clip=1.
        ↳02)
    if ylim is not None:
        ax.set_ylim(*ylim)
    elif title == 'Mean green generation share':
        ylim = _padded_ylim_std(df_wide, pad_frac=0.15,
        ↳include=START_GREEN_FRAC,
        lo_clip=0.0, hi_clip=1.0)
    if ylim is not None:
        ax.set_ylim(*ylim)
    if not np.isnan(START_GREEN_FRAC):
        ax.axhline(START_GREEN_FRAC, color='black', lw=1.0, linestyle='--',
        ↳alpha=0.7,
        label=f'year-0 default mix ({START_GREEN_FRAC:.3f})')
    ax.legend(fontsize=8, loc='best')

plt.suptitle(f'RQ1.5 - Whole-training overview across {len(runs)} loaded run(s)
↳'
            f'(mean ±1 std across runs, ROLL={ROLL})', fontsize=13, y=1.02)
plt.tight_layout(); plt.show()

```

RQ1.5 — Whole-training overview across 4 loaded run(s) (mean ±1 std across runs, ROLL=50)



```
[14]: # Companion data table for the RQ1.5 overview figure: absolute values
# of the smoothed mean series (the line drawn in each panel) reduced
# to the headline scalars used in thesis prose.

_overview_rows = []
for title, (df_wide, ylabel, _color) in overview_panels.items():
    if df_wide.empty:
        _overview_rows.append({
            'panel': title, 'unit': ylabel, 'n_runs': 0, 'episodes': 0,
            'mean': np.nan, 'median': np.nan, 'std': np.nan,
            'min': np.nan, 'max': np.nan,
            'first': np.nan, 'last': np.nan,
        })
        continue
    sm = df_wide.rolling(ROLL, min_periods=1).mean()
    mean = sm.mean(axis=1)
    _overview_rows.append({
        'panel': title, 'unit': ylabel,
        'n_runs': int(sm.shape[1]), 'episodes': int(len(mean)),
        'mean': float(mean.mean()), 'median': float(mean.median()),
        'std': float(mean.std()), 'min': float(mean.min()), 'max': float(mean.
↳max()),
        'first': float(mean.iloc[0]), 'last': float(mean.iloc[-1]),
    })
overview_df = pd.DataFrame(_overview_rows)
print('Whole-training overview - summary of the smoothed mean series shown in_
↳each panel:')
display(overview_df.round(4))
print('Reference lines used in the figure:')
print(f' Year-1 default-mix mean green fraction: {START_GREEN_FRAC:.4f}')
print(f' Auction reserve (floor) price: {RESERVE_PRICE:.2f} €/t')
print(f' Final-year penalty rate (year {N_YEARS}, base {_pen_base:.2f} @_
↳{_pen_infl:.1%}/yr): '
      f'{FINAL_YEAR_PENALTY_RATE:.2f} €/t')
```

Whole-training overview - summary of the smoothed mean series shown in each panel:

		panel	unit	n_runs	episodes	mean	
↳median	std	min	max	first	last		
0		System reward	reward	4	120000	11.9544	↳
↳12.7906	9.7120	-37.7479	25.5983	-2.1723	22.8865		
1		Final-year clearing price	€/tCO2	4	120000	83.7786	↳
↳84.3478	13.7480	58.1888	119.0259	58.1888	67.0147		

2	Compliance rate	compliance in [0,1]	4	120000	0.8232	▢
	↪0.8215	0.0206	0.7709	0.8894	0.8516	0.8221
3	Mean green generation share	green frac in [0,1]	4	120000	0.7554	▢
	↪0.7560	0.0115	0.7271	0.8165	0.8158	0.7534

Reference lines used in the figure:

Year-1 default-mix mean green fraction: 0.5875

Auction reserve (floor) price: 45.00 €/t

Final-year penalty rate (year 12, base 138.75 @ 2.0%/yr): 172.52 €/t

RQ1 summary. The convergence panels (1.1) plus the cross-seed table (1.2) and credibility plots (1.3 / 1.4) together provide the evidence base for Sub-RQ 1: with the default config, the simulation converges, reproduces a plausible EU-ETS price band and compliance plateau across loaded seeds, and the converged window is not dominated by pathology warnings.

1.7 §RQ2 — Regulatory-Mechanism Behaviour Within the Default Config

Scope caveat. Real *sensitivity* analysis (cap LRF / MSR on/off / penalty level) requires sweep variants and is deferred to a separate experiments notebook. What we *can* show with default-config data is a descriptive characterisation of how the regulatory mechanisms operate in equilibrium: how the cap maps to the clearing price, how the MSR withholds and releases, when the penalty actually binds, and how quickly the system decarbonises. This is the baseline that the sensitivity analysis will perturb.

Four panels:

- **2.1** — Cap → price linkage by year (mean across episodes/seeds in the converged window).
- **2.2** — TNAC trajectory and MSR reserve / flow behaviour.
- **2.3** — Penalty incidence: how often does it bind, and how big is it when it does?
- **2.4** — Decarbonisation pace: green-share trajectory and MAC-driven reductions.

1.7.1 RQ2.1 — Cap and clearing-price trajectory by year

```
[15]: # Average across all seeds and converged-window episodes.
year_frames = []
for k, run in runs.items():
    yr = tail_year_rows(run)
    if yr.empty: continue
    keep = ['episode', 'year', 'cap', 'clearing_price', 'auction_volume',
    ↪'tnac',
            'msr_reserve', 'msr_withhold_this_year', 'msr_release_this_year',
            'msr_total_cancelled', 'secondary_price']
    keep = [c for c in keep if c in yr.columns]
    if 'year' not in keep: continue
    df = yr[keep].copy(); df['__seed__'] = k[2]
    year_frames.append(df)
if not year_frames:
```

```

        raise RuntimeError('No year_log data available - RQ2 requires year-level
        ↪logs.')
yr_pool = pd.concat(year_frames, ignore_index=True)

by_year = yr_pool.groupby('year').agg(['mean', 'std'])

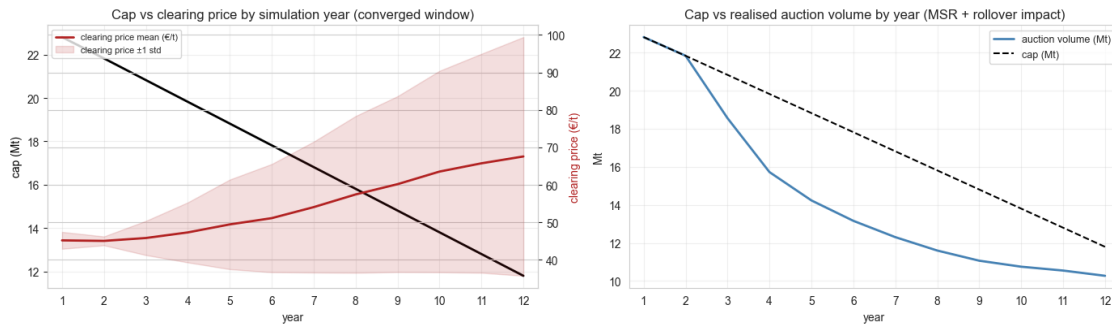
fig, axes = plt.subplots(1, 2, figsize=(14, 4.5))

ax = axes[0]
if ('cap', 'mean') in by_year.columns:
    ax.plot(by_year.index, by_year[('cap', 'mean')], lw=2, color='black',
    ↪label='cap (Mt, mean)')
    ax.set_ylabel('cap (Mt)', color='black')
    ax2 = ax.twinx()
    if ('clearing_price', 'mean') in by_year.columns:
        clearing_mean = by_year[('clearing_price', 'mean')]
        clearing_std = by_year[('clearing_price', 'std')].fillna(0.0)
        ax2.plot(by_year.index, clearing_mean, lw=2, color='firebrick',
        label='clearing price mean (€/t)')
        ax2.fill_between(by_year.index,
        (clearing_mean - clearing_std).values,
        (clearing_mean + clearing_std).values,
        color='firebrick', alpha=0.15,
        label='clearing price ±1 std')
        ax2.set_ylabel('clearing price (€/t)', color='firebrick')
        ax2.legend(fontsize=8, loc='best')
    ax.set(title='Cap vs clearing price by simulation year (converged window)',
    ↪xlabel='year')
    ax.grid(True, alpha=0.3)
    format_year_axis(ax)

ax = axes[1]
if ('auction_volume', 'mean') in by_year.columns:
    ax.plot(by_year.index, by_year[('auction_volume', 'mean')], lw=2,
    ↪color='steelblue', label='auction volume (Mt)')
if ('cap', 'mean') in by_year.columns:
    ax.plot(by_year.index, by_year[('cap', 'mean')], lw=1.5, color='black',
    ↪linestyle='--', label='cap (Mt)')
ax.set(title='Cap vs realised auction volume by year (MSR + rollover impact)',
    xlabel='year', ylabel='Mt')
ax.legend(fontsize=9); ax.grid(True, alpha=0.3)
format_year_axis(ax)

plt.suptitle('RQ2.1 - Regulatory cap schedule and its realised market impact',
    fontsize=13, y=1.02)
plt.tight_layout(); plt.show()

```



1.7.2 RQ2.2 — TNAC, MSR reserve, and MSR flow

```
[16]: fig, axes = plt.subplots(1, 3, figsize=(18, 4.5))

ax = axes[0]
if ('tnac', 'mean') in by_year.columns:
    tnac_mean = by_year[('tnac', 'mean')]
    tnac_std = by_year[('tnac', 'std')].fillna(0.0)
    ax.plot(by_year.index, tnac_mean, lw=2, color='darkblue', label='TNAC mean')
    ax.fill_between(by_year.index,
                    (tnac_mean - tnac_std).values,
                    (tnac_mean + tnac_std).values,
                    color='darkblue', alpha=0.15, label='TNAC ±1 std')
    ax.set(title='TNAC trajectory (Total Number of Allowances in Circulation)',
           xlabel='year', ylabel='Mt')
    ax.legend(fontsize=8); ax.grid(True, alpha=0.3)

ax = axes[1]
if ('msr_reserve', 'mean') in by_year.columns:
    ax.plot(by_year.index, by_year[('msr_reserve', 'mean')], lw=2,
            color='purple', label='MSR reserve')
if ('msr_total_cancelled', 'mean') in by_year.columns:
    ax.plot(by_year.index, by_year[('msr_total_cancelled', 'mean')], lw=2,
            color='red',
            linestyle='--', label='cumulative cancelled')
ax.set(title='MSR reserve & cancellations by year', xlabel='year', ylabel='Mt')
ax.legend(fontsize=9); ax.grid(True, alpha=0.3)

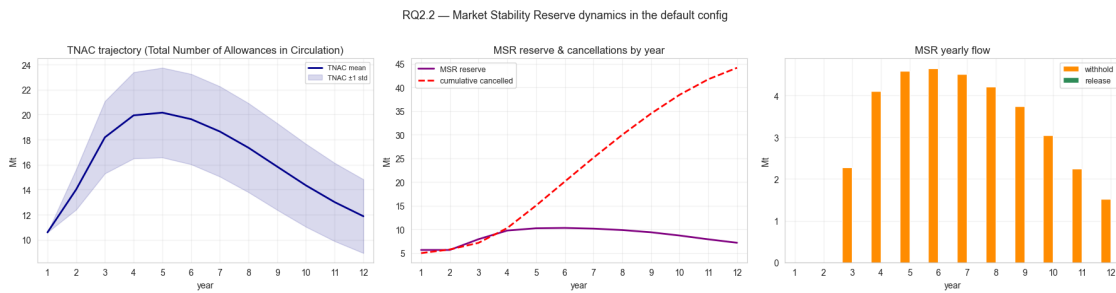
ax = axes[2]
years = by_year.index.values
if ('msr_withhold_this_year', 'mean') in by_year.columns:
    wh = by_year[('msr_withhold_this_year', 'mean')]
    ax.bar(years - 0.18, wh, width=0.35, color='darkorange', label='withhold')
```

```

if ('msr_release_this_year', 'mean') in by_year.columns:
    rel = by_year[('msr_release_this_year', 'mean')]
    ax.bar(years + 0.18, rel, width=0.35, color='seagreen', label='release')
ax.axhline(0, color='black', lw=0.8)
ax.set(title='MSR yearly flow', xlabel='year', ylabel='Mt')
ax.legend(fontsize=9); ax.grid(True, alpha=0.3)
for ax in axes:
    format_year_axis(ax)

plt.suptitle('RQ2.2 - Market Stability Reserve dynamics in the default config',
            fontsize=13, y=1.02)
plt.tight_layout(); plt.show()

```



1.7.3 RQ2.3 — Penalty incidence and magnitude

```

[17]: # (a) Share of (agent, year) cells with shortfall > 0; (b) mean per-cell
      ↪penalty when it bites.
share_rows = []
for k, run in runs.items():
    yr = tail_year_rows(run)
    sf_cols = agent_cols(yr, 'shortfall')
    pen_cols = agent_cols(yr, 'penalty')
    if not sf_cols: continue
    sf = yr[sf_cols].values.flatten()
    pen = yr[pen_cols].values.flatten() if pen_cols else None
    nc_mask = sf > 1e-6
    share_rows.append({
        'run': f'{k[0][:8]}/s{k[2]}',
        'pct_cells_short': 100.0 * nc_mask.mean() if sf.size else np.nan,
        'mean_shortfall_when_short_Mt': float(sf[nc_mask].mean()) if nc_mask.
    ↪any() else 0.0,
        'mean_penalty_when_short_Meur': float(pen[nc_mask].mean()) if (pen is
    ↪not None and nc_mask.any()) else 0.0,
    })
share_df = pd.DataFrame(share_rows)

```

```

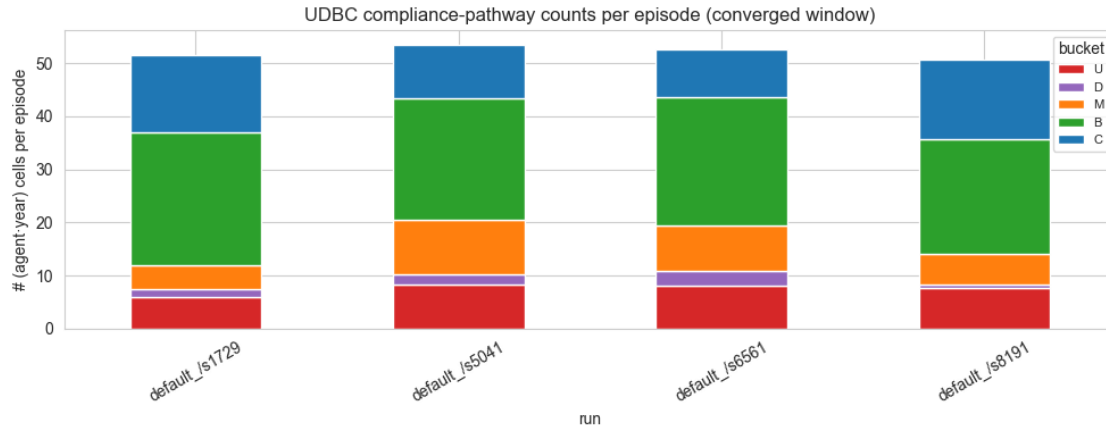
print(f'Penalty incidence in the converged window (last {TAIL_EPISODES}
↳ episodes):')
display(share_df.round(3))

# UDBC mix at the system level (averaged across seeds).
ud_rows = []
for k, run in runs.items():
    ep = tail_episodes(run['ep_df'])
    for b in ['U', 'D', 'M', 'B', 'C']:
        cols = agent_cols(ep, f'udbc_{b}_total')
        if cols:
            ud_rows.append({'run': f'{k[0][:8]}/s{k[2]}', 'bucket': b,
                            'mean_total_per_episode': float(ep[cols].
↳ sum(axis=1).mean())})
if ud_rows:
    ud_df = pd.DataFrame(ud_rows).pivot(index='run', columns='bucket',
↳ values='mean_total_per_episode').fillna(0.0)
    ud_df = ud_df[[c for c in ['U', 'D', 'M', 'B', 'C'] if c in ud_df.columns]]
    fig, ax = plt.subplots(figsize=(10, 4))
    ud_df.plot(kind='bar', stacked=True, ax=ax,
               color=['#d62728', '#9467bd', '#ff7f0e', '#2ca02c', '#1f77b4'])
    ax.set(title='UDBC compliance-pathway counts per episode (converged
↳ window)',
           ylabel='# (agent·year) cells per episode', xlabel='run')
    ax.legend(title='bucket', fontsize=8); ax.tick_params(axis='x', rotation=30)
    plt.tight_layout(); plt.show()
    print('U = pure under-bid; D = pure debt-cascade; M = mixed; B =
↳ bank-covered; C = secondary-covered.')

```

Penalty incidence in the converged window (last 1000 episodes):

	run	pct_cells_short	mean_shortfall_when_short_Mt	↳
				↳ mean_penalty_when_short_Meur
0	default_/s1729	12.494	1.002	↳
	↳ 160.154			
1	default_/s8191	14.730	1.123	↳
	↳ 178.846			
2	default_/s6561	20.312	1.229	↳
	↳ 194.530			
3	default_/s5041	21.330	1.333	↳
	↳ 210.048			



U = pure under-bid; D = pure debt-cascade; M = mixed; B = bank-covered; C = secondary-covered.

1.7.4 RQ2.4 — Decarbonisation pace

[18]: # (a) Mean green_frac by year across episodes/seeds; (b) MAC reduction by year.

```

yr_pool2 = []
for k, run in runs.items():
    yr = tail_year_rows(run)
    keep = ['year']
    gf_cols = agent_cols(yr, 'green_frac')
    mac_cols = agent_cols(yr, 'mac_reduction')
    em_cols = agent_cols(yr, 'emissions')
    if 'year' not in yr.columns: continue
    df = yr[['year']].copy()
    if gf_cols: df['green_frac_mean'] = yr[gf_cols].mean(axis=1)
    if mac_cols: df['mac_reduction_total'] = yr[mac_cols].sum(axis=1)
    if em_cols: df['emissions_total'] = yr[em_cols].sum(axis=1)
    yr_pool2.append(df)
if yr_pool2:
    pool = pd.concat(yr_pool2, ignore_index=True).groupby('year').agg(['mean',
    ↪ 'std'])

fig, axes = plt.subplots(1, 3, figsize=(18, 4.5))
ax = axes[0]
if ('green_frac_mean', 'mean') in pool.columns:
    green_mean = pool[('green_frac_mean', 'mean')]
    green_std = pool[('green_frac_mean', 'std')].fillna(0.0)
    ax.plot(pool.index, green_mean, lw=2, color='seagreen', label='mean')
    ax.fill_between(pool.index,
                    (green_mean - green_std).values,
                    (green_mean + green_std).values,

```

```

        color='seagreen', alpha=0.15, label='±1 std')
    ax.set(title='Mean green generation share by year', xlabel='year',
    ↪ylabel='green frac in [0,1]')
    ax.set_ylim(0, 1); ax.legend(fontsize=8); ax.grid(True, alpha=0.3)

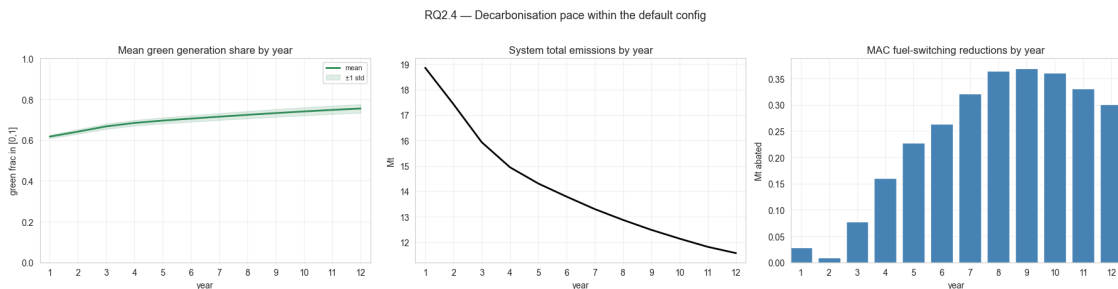
    ax = axes[1]
    if ('emissions_total', 'mean') in pool.columns:
        ax.plot(pool.index, pool[('emissions_total', 'mean')], lw=2,
    ↪color='black')
        ax.set(title='System total emissions by year', xlabel='year',
    ↪ylabel='Mt')
        ax.grid(True, alpha=0.3)

    ax = axes[2]
    if ('mac_reduction_total', 'mean') in pool.columns:
        ax.bar(pool.index, pool[('mac_reduction_total', 'mean')],
    ↪color='steelblue')
        ax.set(title='MAC fuel-switching reductions by year', xlabel='year',
    ↪ylabel='Mt abated')
        ax.grid(True, alpha=0.3)

    for ax in axes:
        format_year_axis(ax)

    plt.suptitle('RQ2.4 - Decarbonisation pace within the default config',
        fontsize=13, y=1.02)
    plt.tight_layout(); plt.show()

```



RQ2 summary. Within the default config, RQ2.1–2.4 give the *descriptive* baseline of how cap, MSR, penalty, and emission outcomes move together. The sensitivity question — *would these trajectories look qualitatively different under a tighter LRF, or with the MSR off?* — is what the cross-config experiments notebook will answer.

1.8 §RQ3 — Strategic Behaviour and Aggregation

The default config crosses 4 archetypes (coal-heavy, gas-dominant, transitioner, green-leader) with 2 reward weightings (pure financial, balanced ESG). That gives us the financial-vs-environmental contrast that RQ3 asks about, *inside one config*.

Five panels:

- **3.1** — Bidding behaviour by archetype $\times$ weighting.
- **3.2** — Investment behaviour by archetype $\times$ weighting.
- **3.3** — Secondary-market role by archetype.
- **3.4** — UDBC compliance pathway by archetype.
- **3.5** — Reward decomposition (financial vs ESG vs penalty) by weighting.

1.8.1 RQ3.1 — Bidding behaviour by archetype $\times$ reward-weighting

```
[19]: # Per-agent converged-window means, then group by (archetype, weighting).
def _per_agent_table(run, episode_cols, year_cols):
    """Return list of dicts (one per agent) with mean values from converged_
    window."""
    ep = tail_episodes(run['ep_df'])
    yr = tail_year_rows(run)
    rows = []
    for i in range(N_AGENTS):
        row = {'agent_idx': i, 'archetype': _agent_archetype(i), 'weighting': _
        agent_weighting(i)}
        for prefix in episode_cols:
            col = f'{prefix}_A{i+1}'
            row[prefix] = float(ep[col].mean()) if col in ep.columns else np.nan
        for prefix in year_cols:
            col = f'{prefix}_A{i+1}'
            row[prefix] = float(yr[col].mean()) if col in yr.columns else np.nan
        rows.append(row)
    return rows

agent_rows = []
for k, run in runs.items():
    for r in _per_agent_table(
        run,
        episode_cols=['bid_price', 'avg_bid_mult', 'avg_bid_coverage',
                     'sec_buy_vol', 'sec_sell_vol', 'sec_buy_avg_px',
                     'sec_sell_avg_px',
                     'sec_buy_years', 'sec_sell_years',
                     'invest_cost', 'green_frac', 'delta_green',
                     'mean_alloc', 'penalty', 'shortfall', 'reward',
                     'reward_base',
                     'inv_onshore_share', 'inv_offshore_share',
                     'inv_solar_share'],
```



```

        year_cols=['mac_reduction', 'mac_cost'],
    ):
        r.update({'sweep': k[0], 'seed': k[2]})
        agent_rows.append(r)
agent_df = pd.DataFrame(agent_rows)
print(f'Per-(run, agent) rows: {len(agent_df)}')

fig, axes = plt.subplots(1, 3, figsize=(18, 4.5))
order = [(arch, w) for arch in ARCHETYPES for w in ['Financial', 'Balanced',
↳ ESG']]
labels = [f'{a}\n{w}' for a, w in order]

def _bar(ax, col, title, ylabel):
    means = []; stds = []
    for arch, w in order:
        sub = agent_df[(agent_df['archetype'] == arch) & (agent_df['weighting'] == w)][col]
        means.append(sub.mean()); stds.append(sub.std())
    x = np.arange(len(order))
    colors = ['#d62728' if w == 'Financial' else '#2ca02c' for _, w in order]
    ax.bar(x, means, yerr=stds, color=colors, capsize=3, alpha=0.85)
    ax.set_xticks(x); ax.set_xticklabels(labels, fontsize=8, rotation=30, ha='right')
    ax.set(title=title, ylabel=ylabel); ax.grid(True, alpha=0.3, axis='y')

_bar(axes[0], 'bid_price', 'Mean Phase-1 bid price', '€/tCO2 ')
_bar(axes[1], 'avg_bid_mult', 'Mean qty multiplier on `estimate_need`, ↳
↳ 'multiplier')
_bar(axes[2], 'avg_bid_coverage', 'Mean coverage = alloc / need', 'ratio ↳
↳ (clipped to 1)')
plt.suptitle('RQ3.1 - Bidding behaviour: red = pure financial reward; green = ↳
↳ balanced ESG',
            fontsize=12, y=1.04)
plt.tight_layout(); plt.show()

```

Per-(run, agent) rows: 32

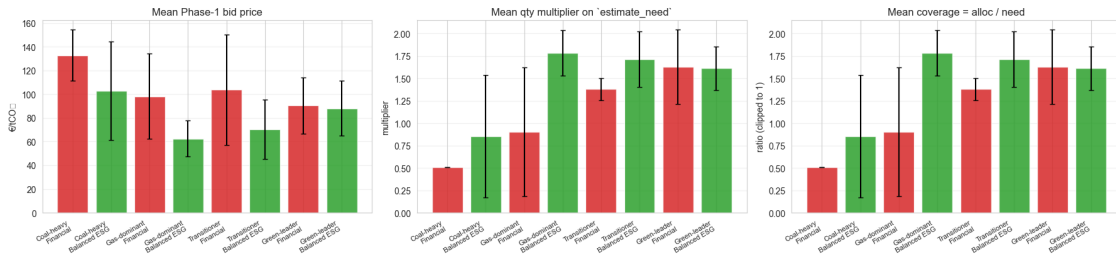
C:\Users\danie\AppData\Local\Temp\ipykernel_5056\309587121.py:55: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

```
plt.tight_layout(); plt.show()
```

C:\Users\danie\AppData\Roaming\Python\Python312\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

```
fig.canvas.print_figure(bytes_io, **kw)
```

RQ3.1 — Bidding behaviour: red = pure financial reward; green = balanced ESG



1.8.2 RQ3.2 — Investment behaviour

```
[20]: fig, axes = plt.subplots(1, 3, figsize=(18, 4.5))
_bar(axes[0], 'invest_cost', 'Mean episode invest cost', 'M€')
_bar(axes[1], 'green_frac', 'End-of-episode green fraction', 'green frac',
    ↳ [0,1])
_bar(axes[2], 'delta_green', 'Δ green fraction over the episode', 'Δ green',
    ↳ frac')
plt.suptitle('RQ3.2 - Investment outcomes: red = pure financial; green =',
    ↳ balanced ESG',
    fontsize=12, y=1.04)
plt.tight_layout(); plt.show()

# Tech mix by archetype × weighting (stacked bar).
tech_cols = ['inv_onshore_share', 'inv_offshore_share', 'inv_solar_share']
tech_labels = ['onshore wind', 'offshore wind', 'solar']
tech_colors = ['#1f77b4', '#17becf', '#ffbf00']
tech_label_colors = ['white', '#1f1f1f', '#1f1f1f']
tech_means = (
    agent_df.groupby(['archetype', 'weighting'])[tech_cols]
    .mean()
    .rename(columns=dict(zip(tech_cols, tech_labels)))
)

fig, ax = plt.subplots(figsize=(11, 5))
tech_means.plot(kind='bar', stacked=True, ax=ax, color=tech_colors)

for x_pos, (_, row) in enumerate(tech_means.iterrows()):
    bottom = 0.0
    for label, text_color in zip(tech_labels, tech_label_colors):
        value = row[label]
        if pd.notna(value) and value >= 0.025:
            ax.text(
                x_pos,
                bottom + value / 2,
```

```

        f'{value * 100:.1f}%',
        ha='center',
        va='center',
        fontsize=8,
        fontweight='bold',
        color=text_color,
    )
    bottom += 0.0 if pd.isna(value) else value

ax.set(title='Tech mix of episode investments by archetype × weighting',
       ylabel='share of invested MW', xlabel='')
ax.set_ylim(0, 1.05)
ax.legend(tech_labels, fontsize=9)
plt.xticks(rotation=30, ha='right')
plt.tight_layout(); plt.show()

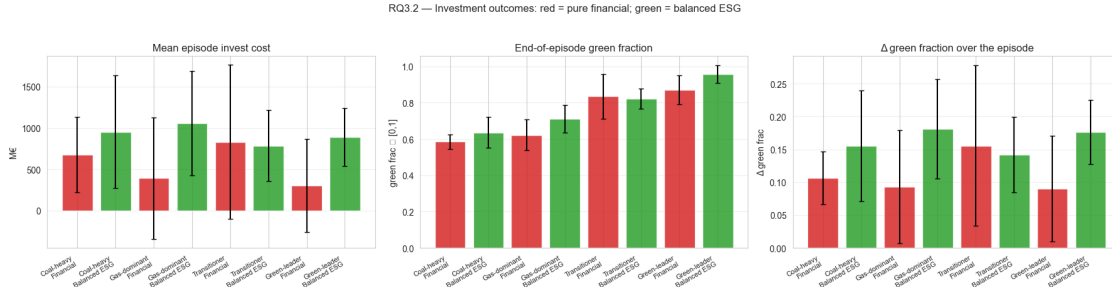
```

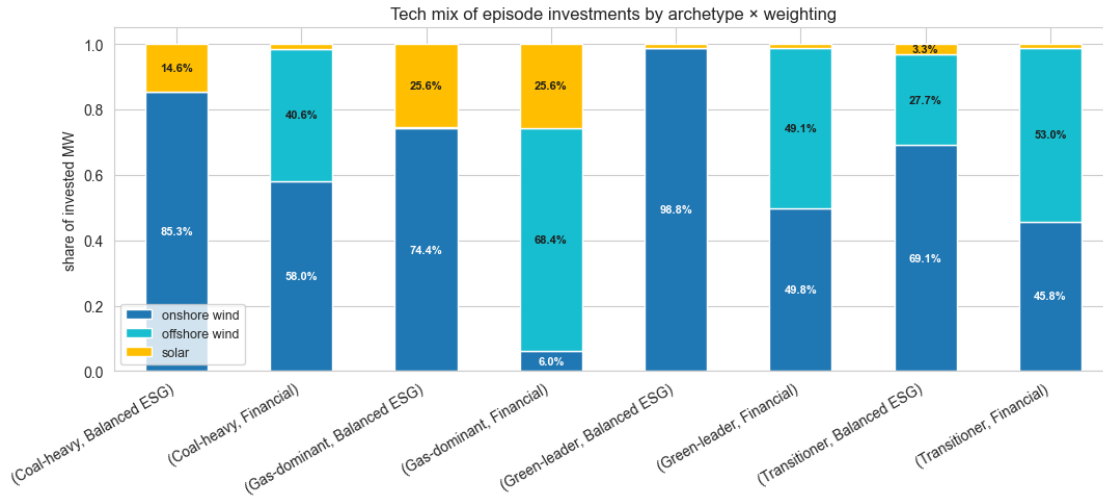
C:\Users\danie\AppData\Local\Temp\ipykernel_5056\3890071108.py:7: UserWarning: Glyph 8712 (\N{ELEMENT OF}) missing from font(s) Arial.

```
plt.tight_layout(); plt.show()
```

C:\Users\danie\AppData\Roaming\Python\Python312\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 8712 (\N{ELEMENT OF}) missing from font(s) Arial.

```
fig.canvas.print_figure(bytes_io, **kw)
```





```
[21]: # Tech mix by reward type (stacked bar) with thesis-ready table.
tech_cols = ['inv_onshore_share', 'inv_offshore_share', 'inv_solar_share']
tech_labels = ['onshore wind', 'offshore wind', 'solar']
tech_colors = ['#1f77b4', '#17becf', '#ffbf00']
tech_label_colors = ['white', '#1f1f1f', '#1f1f1f']
reward_type_order = [w for w in ['Financial', 'Balanced ESG'] if w in
    ↪ agent_df['weighting'].dropna().unique()]

tech_mix_by_reward = (
    agent_df.groupby('weighting')[tech_cols]
    .mean()
    .reindex(reward_type_order)
    .rename(columns=dict(zip(tech_cols, tech_labels)))
)

fig, ax = plt.subplots(figsize=(7.5, 4.5))
tech_mix_by_reward.plot(kind='bar', stacked=True, ax=ax, color=tech_colors,
    ↪ width=0.62)

for x_pos, (_, row) in enumerate(tech_mix_by_reward.iterrows()):
    bottom = 0.0
    for label, text_color in zip(tech_labels, tech_label_colors):
        value = row[label]
        if pd.notna(value) and value > 0:
            ax.text(
                x_pos,
                bottom + value / 2,
                f'{value * 100:.1f}%',
                ha='center',
                va='center',
                color=text_color
            )
            bottom = bottom + value
```

```

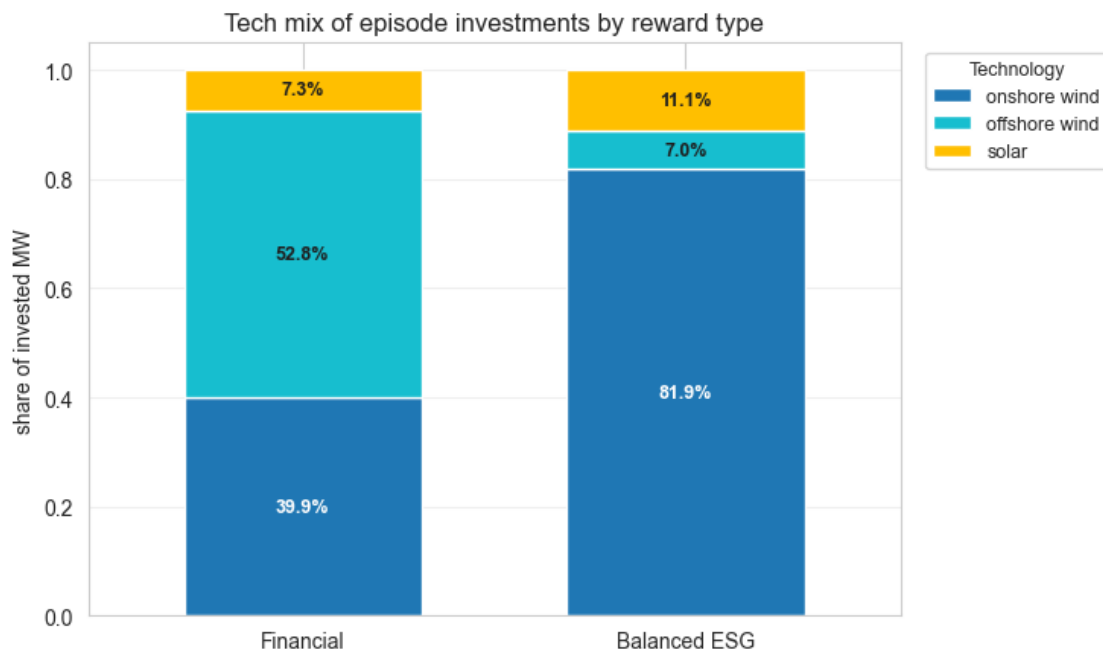
        fontsize=9,
        fontweight='bold',
        color=text_color,
    )
    bottom += 0.0 if pd.isna(value) else value

ax.set(title='Tech mix of episode investments by reward type',
       ylabel='share of invested MW', xlabel='')
ax.set_xticklabels(tech_mix_by_reward.index, rotation=0)
ax.set_ylim(0, 1.05)
ax.legend(title='Technology', fontsize=9, title_fontsize=9, loc='upper left',
         bbox_to_anchor=(1.02, 1))
ax.grid(True, alpha=0.3, axis='y')
plt.tight_layout(); plt.show()

tech_mix_by_reward_table = (tech_mix_by_reward * 100).round(1)
tech_mix_by_reward_table['total'] = tech_mix_by_reward_table.sum(axis=1).
    round(1)
tech_mix_by_reward_table.index.name = 'reward type'

styled_tech_mix_by_reward_table = (
    tech_mix_by_reward_table
    .style
    .format('{:.1f}%')
    .set_caption('Tech mix by reward type (% of invested MW)')
)
display(styled_tech_mix_by_reward_table)

```



<pandas.io.formats.style.Styler at 0x1648a6797c0>

1.8.3 RQ3.3 — Secondary-market role

```
[22]: fig, axes = plt.subplots(1, 3, figsize=(18, 4.5))
_bar(axes[0], 'sec_buy_vol', 'Mean Mt bought on secondary', 'Mt')
_bar(axes[1], 'sec_sell_vol', 'Mean Mt sold on secondary', 'Mt')
# Net buyer/seller years (out of 12).
def _net_buyer_years(row):
    by = row.get('sec_buy_years', np.nan); sy = row.get('sec_sell_years', np.
    ↪nan)
    if pd.isna(by) or pd.isna(sy): return np.nan
    return by - sy
agent_df['net_buyer_years'] = agent_df.apply(_net_buyer_years, axis=1)
_bar(axes[2], 'net_buyer_years', '(buy - sell) years per episode', 'years')
plt.suptitle('RQ3.3 - Who buys, who sells: secondary-market role by archetype ×
    ↪weighting',
            fontsize=12, y=1.04)
plt.tight_layout(); plt.show()
```



1.8.4 RQ3.4 — Compliance pathway (UDBC) by archetype × weighting

```
[23]: # Mean per-agent UDBC totals + grouped by archetype × weighting + stacked bar
    ↪of shares.
udbc_buckets = ['U', 'D', 'M', 'B', 'C']
udbc_rows = []
for k, run in runs.items():
    ep = tail_episodes(run['ep_df'])
    for i in range(N_AGENTS):
        row = {'archetype': _agent_archetype(i), 'weighting':
        ↪_agent_weighting(i)}
        for b in udbc_buckets:
            col = f'udbc_{b}_total_A{i+1}'
```

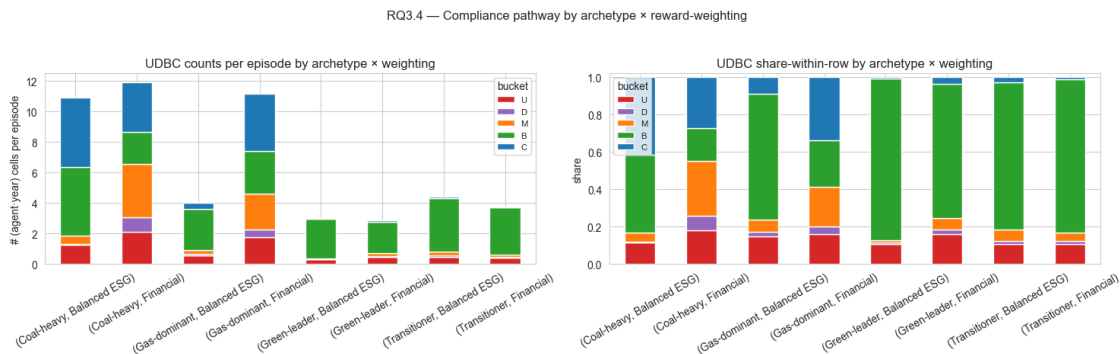
```

        row[b] = float(ep[col].mean()) if col in ep.columns else 0.0
    udbc_rows.append(row)
udbc_df = pd.DataFrame(udbc_rows)
udbc_grouped = udbc_df.groupby(['archetype', 'weighting'])[udbc_buckets].mean()
udbc_share = udbc_grouped.div(udbc_grouped.sum(axis=1).replace(0, np.nan),
    ↪axis=0).fillna(0.0)

fig, axes = plt.subplots(1, 2, figsize=(15, 4.5))
udbc_grouped.plot(kind='bar', stacked=True, ax=axes[0],
    color=['#d62728', '#9467bd', '#ff7f0e', '#2ca02c', '#1f77b4'])
axes[0].set(title='UDBC counts per episode by archetype × weighting',
    ylabel='# (agent·year) cells per episode', xlabel='')
axes[0].legend(title='bucket', fontsize=8)
axes[0].tick_params(axis='x', rotation=30)

udbc_share.plot(kind='bar', stacked=True, ax=axes[1],
    color=['#d62728', '#9467bd', '#ff7f0e', '#2ca02c', '#1f77b4'])
axes[1].set(title='UDBC share-within-row by archetype × weighting',
    ylabel='share', xlabel='')
axes[1].set_ylim(0, 1.02)
axes[1].legend(title='bucket', fontsize=8)
axes[1].tick_params(axis='x', rotation=30)
plt.suptitle('RQ3.4 - Compliance pathway by archetype × reward-weighting',
    fontsize=12, y=1.04)
plt.tight_layout(); plt.show()

```



1.8.5 RQ3.5 - A1-A8 reward balance and ESG-vs-financial twin contrast

[24]: *# RQ3.5 unified per-agent action + reward decomposition.*
Single source of truth: build per-(run, episode, agent_no) frame from ep_df
(action / strategy metrics, per-episode values) and merge it with reward
channels reconstructed from the year CSV. Two reward views are carried:
*# * REWARD_CHANNELS → mean across the 12 sim years*

```

# * <channel>_episode_sum      -> SUM across the 12 sim years (full_
↳episode,
#                               includes the final/terminal year)
# * reward_episode_total      -> authoritative full-episode reward taken
#                               straight from ep_df.reward_A{i} (sum
#                               over years, post-shaping, INCLUDES the
#                               terminal-year payoff).
# The decomposition bar plot is driven by the *episode-sum* view so the
# terminal-year payoffs are visible. The individual per-year-mean overviews
# remain available in the summary tables alongside the new sum columns.

from matplotlib.colors import TwoSlopeNorm

# Mass tolerance for treating a shortfall / coverage gap as zero (Mt CO2).
# Defined here so this cell does not depend on the RQ4.5 cell having been run.
EPS_MT = globals().get('EPS_MT', 1e-6)

OBJECTIVE_ORDER = ['Financial', 'ESG']
WEIGHTING_TO_OBJECTIVE = {'Financial': 'Financial', 'Balanced ESG': 'ESG'}
UDBC_BUCKETS = ['U', 'D', 'M', 'B', 'C']

# Action / strategy metrics from ep_df (per-episode), with display labels.
# (mac_reduction, mac_cost) live in yr_df and are aggregated to per-episode_
↳means.
STRATEGY_PREFIX_LABELS = [
    ('bid_price', 'bid_price'),
    ('avg_bid_mult', 'bid_multiplier'),
    ('avg_bid_coverage', 'auction_coverage'),
    ('mean_alloc', 'allocation_mt'),
    ('penalty', 'penalty_meur'),
    ('shortfall', 'shortfall_mt'),
    ('invest_cost', 'invest_cost_meur'),
    ('green_frac', 'green_fraction'),
    ('delta_green', 'delta_green'),
    ('mac_reduction', 'mac_reduction_mt'),
    ('mac_cost', 'mac_cost_meur'),
    ('sec_buy_vol', 'sec_buy_mt'),
    ('sec_sell_vol', 'sec_sell_mt'),
    ('sec_buy_years', 'sec_buy_years'),
    ('sec_sell_years', 'sec_sell_years'),
    ('inv_onshore_share', 'invest_onshore_share'),
    ('inv_offshore_share', 'invest_offshore_share'),
    ('inv_solar_share', 'invest_solar_share'),
    ('udbc_U_total', 'udbc_U_count'),
    ('udbc_D_total', 'udbc_D_count'),
    ('udbc_M_total', 'udbc_M_count'),
    ('udbc_B_total', 'udbc_B_count'),

```



```

        ('udbc_C_total', 'udbc_C_count'),
        # Year-level only (no per-episode column in ep_df) -- per-episode value is
        # the mean across the 12 sim years of that episode.
        ('invest_frac_post_clip', 'invest_frac_post_clip'),
        ('bank_end', 'bank_end'),
        ('trade_qty', 'trade_qty'),
    ]
    STRATEGY_LABELS = [label for _, label in STRATEGY_PREFIX_LABELS]
    YEAR_LEVEL_PREFIXES = {'mac_reduction', 'mac_cost', 'invest_frac_post_clip',
        ↪ 'bank_end', 'trade_qty'}

    # Derived per-year binary indicators -> per-episode share in [0, 1].
    DERIVED_YEAR_SHARE_LABELS = ['noncompliant_share', 'stressed_share']
    STRATEGY_LABELS = STRATEGY_LABELS + DERIVED_YEAR_SHARE_LABELS

    # Reward channel labels in display order (sign convention preserved: positive
    # values help reward, penalty_norm_proxy is reported as a positive cost).
    REWARD_CHANNELS = [
        'reward', 'reward_base', 'reward_shaping',
        'financial_unweighted_proxy', 'financial_reward_proxy',
        'weighted_esg_proxy', 'esg_stock_reward', 'esg_flow_reward',
        ↪ 'esg_speed_reward',
        'penalty_norm_proxy', 'penalty_realized_norm_proxy',
        ↪ 'penalty_remediation_norm_proxy',
        'banking_soft_residual',
    ]
    REWARD_CHANNEL_SUM_LABELS = [f'{c}_episode_sum' for c in REWARD_CHANNELS]

def _numeric_column(frame, column, default=0.0):
    if column in frame.columns:
        return pd.to_numeric(frame[column], errors='coerce').astype(float)
    return pd.Series(default, index=frame.index, dtype=float)

def _agent_output_twh(agent_idx):
    output_cfg = REFERENCE_CONFIG.get('companies', {}).get('output_twh', 10.0)
    if isinstance(output_cfg, (list, tuple)):
        return float(output_cfg[agent_idx]) if agent_idx < len(output_cfg) else
        ↪ float(output_cfg[-1])
    return float(output_cfg)

def _agent_reward_weights(agent_idx):
    weights = REFERENCE_CONFIG.get('companies', {}).get('reward_weights', [])
    ↪ or []
    if agent_idx < len(weights):

```

```

        return float(weights[agent_idx][0]), float(weights[agent_idx][1])
    return (1.0, 0.0) if agent_idx % 2 == 0 else (0.5, 0.5)

def _agent_objective_label(agent_no):
    return 'ESG' if agent_no % 2 == 0 else 'Financial'

def _initial_emission_factors():
    tech_efs = np.asarray(REFERENCE_CONFIG.get('technologies', {})).
    ↳get('emission_factors', []), dtype=float)
    initial_mix = REFERENCE_CONFIG.get('companies', {}).get('initial_mix', [])
    ↳or []
    factors = []
    for agent_idx in range(N_AGENTS):
        if agent_idx < len(initial_mix) and len(tech_efs) > 0:
            factors.append(float(np.dot(np.asarray(initial_mix[agent_idx],
    ↳dtype=float), tech_efs)))
        else:
            factors.append(np.nan)
    return factors

def _reward_context():
    esg_cfg = REFERENCE_CONFIG.get('esg', {}) or {}
    reward_cfg = REFERENCE_CONFIG.get('reward', {}) or {}
    banking_cfg = reward_cfg.get('banking_signal', {}) or {}
    return {
        'initial_efs': _initial_emission_factors(),
        'reward_weights': [_agent_reward_weights(agent_idx) for agent_idx in
    ↳range(N_AGENTS)],
        'base_penalty_rate': float(REFERENCE_CONFIG.get('penalty', {}).
    ↳get('rate', 100.0)),
        'annual_budgets': REFERENCE_CONFIG.get('budget', {}).
    ↳get('annual_budgets', []) or [1000.0 * N_AGENTS,
        'esg_scale': float(esg_cfg.get('scale', 0.5)),
        'esg_stock_weight': float(esg_cfg.get('stock_weight', 1.0)),
        'esg_flow_weight': float(esg_cfg.get('flow_weight', 1.5)),
        'esg_speed_early': float(esg_cfg.get('speed_coef', 0.3)),
        'esg_speed_late': float(esg_cfg.get('speed_coef_late', esg_cfg.
    ↳get('speed_coef', 0.3))),
        'gate_threshold': float(reward_cfg.
    ↳get('compliance_gate_blend_threshold', 0.90)),
        'gate_width': max(float(reward_cfg.get('compliance_gate_blend_width', 0.
    ↳30)), 1e-9),
        'imputed_bank_weight': float(banking_cfg.get('w_imputed', 1.0)),

```

```

        'imputed_cap_factor': float(banking_cfg.get('imputed_cap_factor', 5.0)),
    }

def _prepare_core_year_frame(run_data, include_final_year=False):
    """Slice the year frame and build the `common` context.

    By default the final/terminal year is dropped to keep the per-year-mean
    decomposition stable; pass include_final_year=True to compute the
    full-episode SUM view (which must include the terminal year so the
    end-of-episode payoffs land in the bars).
    """
    year_frame = tail_year_rows(run_data).copy()
    if year_frame.empty or 'year' not in year_frame.columns:
        return None, None

    year_frame = year_frame.sort_values(['episode', 'year']).copy()
    if {'episode', 'fundamental_anchor'}.issubset(year_frame.columns):
        year_frame['_anchor_next'] = year_frame.
        ↳groupby('episode')['fundamental_anchor'].shift(-1)
    else:
        year_frame['_anchor_next'] = np.nan
        year_frame['_anchor_next'] = year_frame['_anchor_next'].fillna(year_frame.
        ↳get('fundamental_anchor', np.nan))

    initial_year_label = YEAR_DISPLAY_START if 'YEAR_DISPLAY_START' in _
    ↳globals() else 1
    final_year_label = YEAR_DISPLAY_END if 'YEAR_DISPLAY_END' in globals() else _
    ↳N_YEARS
    if include_final_year:
        core_year_frame = year_frame.copy()
    else:
        core_year_frame = year_frame[year_frame['year'] < final_year_label].
        ↳copy()
    if core_year_frame.empty:
        core_year_frame = year_frame.copy()

    initial_year_rows = year_frame[year_frame['year'] == initial_year_label]
    anchor0_real = np.nan
    if {'fundamental_anchor', 'inflation_factor'}.issubset(initial_year_rows.
    ↳columns) and not initial_year_rows.empty:
        anchor0_real = float((initial_year_rows['fundamental_anchor'] / _
        ↳initial_year_rows['inflation_factor']).replace(0, np.nan)).median())
    if not np.isfinite(anchor0_real) or anchor0_real <= 0:
        anchor0_real = 1.0

```

```

    cap0 = float(initial_year_rows['cap'].median()) if 'cap' in_
↪initial_year_rows.columns and not initial_year_rows.empty else np.nan
    if not np.isfinite(cap0) or cap0 <= 0:
        cap0 = float(core_year_frame['cap'].max()) if 'cap' in core_year_frame.
↪columns else 1.0

    common = {
        'initial_year_label': initial_year_label,
        'anchor0_real': anchor0_real,
        'cap0': cap0,
        'inflation': _numeric_column(core_year_frame, 'inflation_factor', 1.0).
↪replace(0, np.nan).fillna(1.0),
        'anchor': _numeric_column(core_year_frame, 'fundamental_anchor', np.
↪nan).replace(0, np.nan),
        'anchor_next': _numeric_column(core_year_frame, '_anchor_next', np.nan).
↪replace(0, np.nan),
        'clearing_price': _numeric_column(core_year_frame, 'clearing_price', 0.
↪0),
        'cap': _numeric_column(core_year_frame, 'cap', cap0),
    }
    common['anchor_next'] = common['anchor_next'].fillna(common['anchor'])
    return core_year_frame, common

def _agent_reward_channel_frame(core_year_frame, common, context, run_key,
↪agent_idx):
    agent_no = agent_idx + 1
    reward_col = f'reward_A{agent_no}'
    base_col = f'reward_base_A{agent_no}'
    need_col = f'estimate_need_A{agent_no}'
    if base_col not in core_year_frame.columns or need_col not in_
↪core_year_frame.columns:
        return None

    inflation = common['inflation']
    anchor = common['anchor']
    anchor_next = common['anchor_next']
    cap = common['cap']
    cap0 = common['cap0']
    scarcity_amp = 1.0 + np.maximum(0.0, 1.0 - cap / max(cap0, 1e-9))
    year_offset = (_numeric_column(core_year_frame, 'year',
↪common['initial_year_label']) - common['initial_year_label']).clip(lower=0.0)
    year_frac = year_offset / max(N_YEARS - 1, 1)
    speed_coef = context['esg_speed_early'] + year_frac *
↪(context['esg_speed_late'] - context['esg_speed_early'])

```

```

weight_cost, weight_green = context['reward_weights'][agent_idx]
annual_budgets = context['annual_budgets']
annual_budget = float(annual_budgets[agent_idx]) if agent_idx <
↳ len(annual_budgets) else float(annual_budgets[-1])
budget_real = annual_budget / inflation
soft_denom = np.maximum(budget_real, 1.0)

need = _numeric_column(core_year_frame, need_col, np.nan).clip(lower=1e-6)
compliance_denom = np.maximum((anchor / inflation) * need, 1.0)
nominal_compliance_denom = np.maximum(inflation * compliance_denom, 1.0)

auction_cost = _numeric_column(core_year_frame,
↳ f'auction_cost_A{agent_no}', 0.0)
trade_cost = _numeric_column(core_year_frame, f'trade_cost_A{agent_no}', 0.
↳ 0)
mac_cost = _numeric_column(core_year_frame, f'mac_cost_A{agent_no}', 0.0)
investment_cost = _numeric_column(core_year_frame,
↳ f'invest_cost_A{agent_no}', 0.0)
compliance_norm_cash = ((auction_cost + trade_cost + mac_cost) / inflation)
↳ / compliance_denom

bank_start = _numeric_column(core_year_frame, f'bank_start_A{agent_no}', 0.
↳ 0).clip(lower=0.0)
allocation = _numeric_column(core_year_frame, f'alloc_A{agent_no}', 0.0).
↳ clip(lower=0.0)
trade_qty = _numeric_column(core_year_frame, f'trade_qty_A{agent_no}', 0.0)
emissions = _numeric_column(core_year_frame, f'emissions_A{agent_no}', 0.0).
↳ clip(lower=0.0)
carry_forward_start = _numeric_column(core_year_frame,
↳ f'carry_forward_start_A{agent_no}', 0.0).clip(lower=0.0)
carry_forward_end = _numeric_column(core_year_frame,
↳ f'carry_forward_end_A{agent_no}', 0.0).clip(lower=0.0)

total_new_allowances = allocation + trade_qty.clip(lower=0.0)
total_obligation = emissions + carry_forward_start
bank_drawdown = np.minimum(bank_start, (total_obligation -
↳ total_new_allowances).clip(lower=0.0)).clip(lower=0.0)
imputed_bank_real = bank_drawdown * common['clearing_price'] / inflation
imputed_bank_norm = np.minimum(imputed_bank_real,
↳ context['imputed_cap_factor'] * compliance_denom) / compliance_denom

compliance_norm = compliance_norm_cash + context['imputed_bank_weight'] *
↳ imputed_bank_norm
capital_norm = (investment_cost / inflation) / soft_denom
cost_norm_centered = compliance_norm + capital_norm - 1.0

```

```

    bid_coverage = _numeric_column(core_year_frame,
    ↪f'bid_coverage_A{agent_no}', np.nan)
    coverage_auction = bid_coverage.fillna(allocation / need).clip(lower=0.0,
    ↪upper=1.0)
    financial_unweighted_basis = -cost_norm_centered
    financial_unweighted_proxy = financial_unweighted_basis.where(
        cost_norm_centered >= 0.0,
        coverage_auction * financial_unweighted_basis,
    )
    savings_reward = weight_cost * financial_unweighted_basis
    financial_reward = savings_reward.where(cost_norm_centered >= 0.0,
    ↪coverage_auction * savings_reward)

    shortfall = _numeric_column(core_year_frame, f'shortfall_A{agent_no}', np.
    ↪nan)
    if shortfall.isna().all():
        effective_penalty_rate = _numeric_column(core_year_frame,
    ↪f'effective_penalty_rate_A{agent_no}', np.nan).replace(0, np.nan)
        shortfall = _numeric_column(core_year_frame, f'penalty_A{agent_no}', 0.
    ↪0) / effective_penalty_rate
        shortfall = shortfall.fillna(0.0).clip(lower=0.0)
        penalty_realized_norm = shortfall * context['base_penalty_rate'] /
    ↪nominal_compliance_denom
        penalty_remediation_norm = carry_forward_end * anchor_next * scarcity_amp /
    ↪nominal_compliance_denom
        penalty_norm = penalty_realized_norm + penalty_remediation_norm

    output_twh = max(_agent_output_twh(agent_idx), 1e-9)
    current_ef = emissions / output_twh
    initial_ef = context['initial_efs'][agent_idx]
    if np.isfinite(initial_ef) and initial_ef > 1e-9:
        ef_ratio = ((initial_ef - current_ef) / initial_ef).clip(lower=0.0)
    else:
        ef_ratio = pd.Series(0.0, index=core_year_frame.index)
    anchor_ratio = ((anchor / inflation) / max(common['anchor0_real'], 1.0)).
    ↪clip(lower=0.0)
    green_delta = _numeric_column(core_year_frame, f'delta_green_A{agent_no}',
    ↪0.0).clip(lower=0.0)
    pre_compliance_holdings = (bank_start + allocation + trade_qty).
    ↪clip(lower=0.0)
    coverage_pre_compliance = (pre_compliance_holdings / need).clip(lower=0.0,
    ↪upper=1.0)
    gate_blend = ((context['gate_threshold'] - coverage_pre_compliance) /
    ↪context['gate_width']).clip(lower=0.0, upper=1.0)
    compliance_gate = coverage_pre_compliance ** (1.0 + gate_blend)

```

```

    esg_stock_reward = weight_green * context['esg_scale'] *
    ↪context['esg_stock_weight'] * ef_ratio * compliance_gate
    esg_flow_reward = weight_green * context['esg_scale'] *
    ↪context['esg_flow_weight'] * ef_ratio * anchor_ratio * compliance_gate
    esg_speed_reward = weight_green * context['esg_scale'] * speed_coef *
    ↪green_delta * compliance_gate
    weighted_esg_reward = esg_stock_reward + esg_flow_reward + esg_speed_reward

    reward_base = _numeric_column(core_year_frame, base_col, np.nan)
    reward = _numeric_column(core_year_frame, reward_col, np.nan)
    reward_shaping = _numeric_column(core_year_frame,
    ↪f'reward_shaping_A{agent_no}', 0.0)
    residual = reward_base - financial_reward - weighted_esg_reward +
    ↪penalty_norm
    objective = _agent_objective_label(agent_no)
    archetype = _agent_archetype(agent_idx)

    return pd.DataFrame({
        'sweep': run_key[0],
        'variant': run_key[1],
        'seed': run_key[2],
        'episode': _numeric_column(core_year_frame, 'episode', np.nan),
        'year': _numeric_column(core_year_frame, 'year', np.nan),
        'agent_no': agent_no,
        'agent': f'A{agent_no}',
        'agent_label': f'A{agent_no} {objective}\n{archetype}',
        'archetype': archetype,
        'objective': objective,
        'weighting': _agent_weighting(agent_idx),
        'reward': reward,
        'reward_base': reward_base,
        'reward_shaping': reward_shaping,
        'financial_unweighted_proxy': financial_unweighted_proxy,
        'financial_reward_proxy': financial_reward,
        'weighted_esg_proxy': weighted_esg_reward,
        'esg_stock_reward': esg_stock_reward,
        'esg_flow_reward': esg_flow_reward,
        'esg_speed_reward': esg_speed_reward,
        'penalty_norm_proxy': penalty_norm,
        'penalty_realized_norm_proxy': penalty_realized_norm,
        'penalty_remediation_norm_proxy': penalty_remediation_norm,
        'banking_soft_residual': residual,
    })

def _build_reward_decomp_year(include_final_year=False):
    context = _reward_context()

```

```

frames = []
for run_key, run_data in runs.items():
    core_year_frame, common = _prepare_core_year_frame(
        run_data, include_final_year=include_final_year
    )
    if core_year_frame is None:
        continue
    for agent_idx in range(N_AGENTS):
        agent_frame = _agent_reward_channel_frame(core_year_frame, common,
↪context, run_key, agent_idx)
        if agent_frame is not None:
            frames.append(agent_frame)
    if not frames:
        return pd.DataFrame()
    decomp = pd.concat(frames, ignore_index=True)
    numeric_cols = decomp.select_dtypes(include=[np.number]).columns
    decomp[numeric_cols] = decomp[numeric_cols].replace([np.inf, -np.inf], np.
↪nan)
    return decomp.dropna(subset=['reward_base', 'financial_reward_proxy',
↪'penalty_norm_proxy'])

def _build_strategy_episode_frame():
    """Per-(run, episode, agent_no) frame of strategy/action metrics from ep_df.
↪"""
    rows = []
    for run_key, run_data in runs.items():
        ep = tail_episodes(run_data['ep_df']).copy()
        if ep.empty:
            continue
        if 'episode' in ep.columns:
            episodes = pd.to_numeric(ep['episode'], errors='coerce').
↪astype(float).values
        else:
            episodes = np.arange(len(ep), dtype=float)

        yr = tail_year_rows(run_data)
        yr_ep_means = {}
        yr_share_means = {}
        if yr is not None and not yr.empty and 'episode' in yr.columns:
            yr_ep = pd.to_numeric(yr['episode'], errors='coerce')
            for prefix in YEAR_LEVEL_PREFIXES:
                for agent_idx in range(N_AGENTS):
                    col = f'{prefix}_A{agent_idx + 1}'
                    if col in yr.columns:
                        yr_ep_means[(agent_idx + 1, prefix)] = (
                            pd.to_numeric(yr[col], errors='coerce')

```



```

        .groupby(yr_ep).mean()
    )
    for agent_idx in range(N_AGENTS):
        agent_no = agent_idx + 1
        sf_col = f'shortfall_A{agent_no}'
        em_col = f'emissions_A{agent_no}'
        al_col = f'alloc_A{agent_no}'
        cf_col = f'carry_forward_start_A{agent_no}'
        if sf_col not in yr.columns:
            continue
        sf = pd.to_numeric(yr[sf_col], errors='coerce').fillna(0.0)
        noncomp_flag = (sf > EPS_MT).astype(float)
        yr_share_means[(agent_no, 'noncompliant_share')] = (
            noncomp_flag.groupby(yr_ep).mean()
        )
        if all(c in yr.columns for c in (em_col, al_col, cf_col)):
            emiss = pd.to_numeric(yr[em_col], errors='coerce').fillna(0.
↪0).clip(lower=0.0)
            alloc = pd.to_numeric(yr[al_col], errors='coerce').fillna(0.
↪0)
            cf_start = pd.to_numeric(yr[cf_col], errors='coerce').
↪fillna(0.0).clip(lower=0.0)
            auction_gap = (emiss + cf_start - alloc).clip(lower=0.0)
            stressed_flag = ((auction_gap > EPS_MT) & (sf <= EPS_MT)).
↪astype(float)
            yr_share_means[(agent_no, 'stressed_share')] = (
                stressed_flag.groupby(yr_ep).mean()
            )

    for agent_idx in range(N_AGENTS):
        agent_no = agent_idx + 1
        frame = pd.DataFrame({
            'sweep': run_key[0],
            'variant': run_key[1],
            'seed': run_key[2],
            'episode': episodes,
            'agent_no': agent_no,
            'agent': f'A{agent_no}',
            'archetype': _agent_archetype(agent_idx),
            'weighting': _agent_weighting(agent_idx),
            'objective': _agent_objective_label(agent_no),
        })
    for prefix, label in STRATEGY_PREFIX_LABELS:
        if prefix in YEAR_LEVEL_PREFIXES:
            series = yr_ep_means.get((agent_no, prefix))
            if series is not None:

```

```

        frame[label] = pd.Series(episodes).map(series).
↳astype(float).values
        else:
            frame[label] = np.nan
    else:
        col = f'{prefix}_A{agent_no}'
        if col in ep.columns:
            frame[label] = pd.to_numeric(ep[col], errors='coerce').
↳astype(float).values
        else:
            frame[label] = np.nan
    for share_label in DERIVED_YEAR_SHARE_LABELS:
        series = yr_share_means.get((agent_no, share_label))
        if series is not None:
            frame[share_label] = pd.Series(episodes).map(series).
↳astype(float).values
        else:
            frame[share_label] = np.nan
    rows.append(frame)
if not rows:
    return pd.DataFrame()
out = pd.concat(rows, ignore_index=True)
numeric_cols = out.select_dtypes(include=[np.number]).columns
out[numeric_cols] = out[numeric_cols].replace([np.inf, -np.inf], np.nan)
return out

def _build_unified_agent_episode_frame(reward_decomp_year,
↳reward_decomp_year_full):
    """Per-(run, episode, agent) frame:
        * strategy/action metrics (per-episode)
        * REWARD_CHANNELS -> mean across the 12 sim years (final year excluded)
        * <channel>_episode_sum -> sum across all 12 sim years (terminal incl.)
        * reward_episode_total -> authoritative full-episode reward from
            ep_df.reward_A{i} (post-shaping, includes
            the terminal-year payoff).

    """
    strategy_df = _build_strategy_episode_frame()
    ep_keys = ['sweep', 'variant', 'seed', 'episode', 'agent_no']

    if not reward_decomp_year.empty:
        reward_ep_mean = (
            reward_decomp_year.groupby(ep_keys,
↳as_index=False)[REWARD_CHANNELS].mean()
        )
    else:
        reward_ep_mean = pd.DataFrame(columns=ep_keys)

```

```

    if not reward_decomp_year_full.empty:
        reward_ep_sum = (
            reward_decomp_year_full.groupby(ep_keys,
            ↪as_index=False)[REWARD_CHANNELS].sum()
            .rename(columns={c: f'{c}_episode_sum' for c in REWARD_CHANNELS})
        )
    else:
        reward_ep_sum = pd.DataFrame(columns=ep_keys)

    ep_total_rows = []
    for run_key, run_data in runs.items():
        ep = tail_episodes(run_data['ep_df']).copy()
        if ep.empty:
            continue
        if 'episode' in ep.columns:
            episodes = pd.to_numeric(ep['episode'], errors='coerce').
            ↪astype(float).values
        else:
            episodes = np.arange(len(ep), dtype=float)
        for agent_idx in range(N_AGENTS):
            agent_no = agent_idx + 1
            col = f'reward_A{agent_no}'
            if col not in ep.columns:
                continue
            ep_total_rows.append(pd.DataFrame({
                'sweep': run_key[0], 'variant': run_key[1], 'seed': run_key[2],
                'episode': episodes, 'agent_no': agent_no,
                'reward_episode_total': pd.to_numeric(ep[col], errors='coerce').
                ↪astype(float).values,
            })))
        ep_total_df = (
            pd.concat(ep_total_rows, ignore_index=True)
            if ep_total_rows else pd.DataFrame(columns=ep_keys +
            ↪['reward_episode_total'])
        )

    frames = [df for df in (strategy_df, reward_ep_mean, reward_ep_sum,
    ↪ep_total_df) if not df.empty]
    if not frames:
        return pd.DataFrame()
    out = frames[0]
    for f in frames[1:]:
        out = pd.merge(out, f, on=ep_keys, how='outer')
    numeric_cols = out.select_dtypes(include=[np.number]).columns
    out[numeric_cols] = out[numeric_cols].replace([np.inf, -np.inf], np.nan)
    return out

```

```

def _agent_summary(unified_df, group_cols, sort_index=None):
    """Mean & std for all available metrics.

    Carries the per-year channel means (existing individual reward overviews),
    the per-episode channel sums, and the authoritative full-episode reward.
    """
    metric_cols = (
        STRATEGY_LABELS
        + REWARD_CHANNELS
        + REWARD_CHANNEL_SUM_LABELS
        + ['reward_episode_total']
    )
    available = [col for col in metric_cols if col in unified_df.columns]
    if unified_df.empty or not available:
        return pd.DataFrame()
    summary = (
        unified_df.groupby(group_cols, sort=False, dropna=False)[available]
        .agg(['mean', 'std'])
        .round(4)
    )
    if sort_index is not None:
        summary = summary.reindex([key for key in sort_index if key in summary.
↪index])
    return summary


def _add_esg_minus_financial(objective_summary):
    if objective_summary.empty:
        return objective_summary
    if not {'Financial', 'ESG'}.issubset(objective_summary.index):
        return objective_summary
    delta = objective_summary.loc['ESG'] - objective_summary.loc['Financial']
    out = objective_summary.copy()
    out.loc['ESG - Financial'] = delta
    return out.reindex(['Financial', 'ESG', 'ESG - Financial'])


def _agent_reward_means_for_plot(unified_df):
    """Mean across episodes of the full-episode reward channels (per agent).

    Uses *_episode_sum columns so the stacked bars include the terminal-year
    payoff; reward_episode_total is the observed-reference diamond.
    """
    channel_cols = [
        'reward_episode_total',

```

```

        'reward_base_episode_sum',
        'financial_reward_proxy_episode_sum',
        'weighted_esg_proxy_episode_sum',
        'esg_stock_reward_episode_sum',
        'esg_flow_reward_episode_sum',
        'esg_speed_reward_episode_sum',
        'penalty_norm_proxy_episode_sum',
        'penalty_realized_norm_proxy_episode_sum',
        'penalty_remediation_norm_proxy_episode_sum',
        'banking_soft_residual_episode_sum',
        'reward_shaping_episode_sum',
    ]
    available = [c for c in channel_cols if c in unified_df.columns]
    df = unified_df.copy()
    df['agent_label'] = df.apply(
        lambda r: f"A{int(r['agent_no'])} {r['objective']}\n{r['archetype']}",
        axis=1,
    )
    return (
        df.groupby(['agent_no', 'agent', 'objective', 'archetype',
        ↪ 'agent_label'], sort=True)[available]
        .mean()
        .reset_index()
        .sort_values('agent_no')
    )

def _plot_reward_decomposition(agent_means, contrast_display):
    fig, (axis_components, axis_contrast) = plt.subplots(
        1, 2, figsize=(24, 9), constrained_layout=True,
        gridspec_kw={'width_ratios': [1.45, 1.15], 'wspace': 0.25},
    )

    y_positions = np.arange(len(agent_means))
    positive_left = np.zeros(len(agent_means))
    negative_left = np.zeros(len(agent_means))
    component_specs = [
        ('financial_reward_proxy_episode_sum', 'Financial net', '#1f77b4', 1.0),
        ('esg_stock_reward_episode_sum', 'ESG stock', '#2ca02c', 1.0),
        ('esg_flow_reward_episode_sum', 'ESG flow', '#17becf', 1.0),
        ('esg_speed_reward_episode_sum', 'ESG speed', '#bcbd22', 1.0),
        ('banking_soft_residual_episode_sum', 'Banking / soft residual',
        ↪ '#7f7f7f', 1.0),
        ('penalty_norm_proxy_episode_sum', 'Penalty cost', '#d62728', -1.0),
    ]

    legend_labels_seen = set()

```

```

for column, label, color, sign in component_specs:
    if column not in agent_means.columns:
        continue
    values = (agent_means[column] * sign).to_numpy(dtype=float)
    positive_values = np.clip(values, 0.0, None)
    negative_values = np.clip(values, None, 0.0)
    legend_label = label if label not in legend_labels_seen else
↳ '_nolegend_'
    if np.any(np.isfinite(positive_values) & (positive_values != 0.0)):
        axis_components.barh(y_positions, positive_values,
↳ left=positive_left, label=legend_label, color=color, alpha=0.88, height=0.72)
        legend_labels_seen.add(label)
        legend_label = '_nolegend_'
        positive_left += np.nan_to_num(positive_values, nan=0.0)
    if np.any(np.isfinite(negative_values) & (negative_values != 0.0)):
        axis_components.barh(y_positions, negative_values,
↳ left=negative_left, label=legend_label, color=color, alpha=0.88, height=0.72)
        legend_labels_seen.add(label)
        negative_left += np.nan_to_num(negative_values, nan=0.0)

reward_total_mean = (
    agent_means['reward_episode_total'].to_numpy(dtype=float)
    if 'reward_episode_total' in agent_means.columns
    else np.full(len(agent_means), np.nan)
)
axis_components.scatter(
    reward_total_mean, y_positions, marker='D', color='black', s=34,
    label='Observed full-episode reward (ep_df sum)', zorder=5,
)

stacked_extent = np.nanmax(np.abs(np.r_[positive_left, negative_left,
↳ reward_total_mean]))
if not np.isfinite(stacked_extent) or stacked_extent == 0:
    stacked_extent = 1.0
label_offset = stacked_extent * 0.035
for y_pos, value in zip(y_positions, reward_total_mean):
    if np.isfinite(value):
        ha = 'left' if value >= 0 else 'right'
        offset = label_offset if value >= 0 else -label_offset
        axis_components.text(value + offset, y_pos, f'{value:+.2f}',
↳ va='center', ha=ha, fontsize=8, color='black')

axis_components.set_xlim(-stacked_extent * 1.18, stacked_extent * 1.18)
axis_components.axvline(0, color='black', lw=0.9)
axis_components.set_yticks(y_positions)
axis_components.set_yticklabels(agent_means['agent_label'].tolist(),
↳ fontsize=9)

```

```

axis_components.invert_yaxis()
axis_components.set_xlabel('full-episode reward contribution: sum over 12_
↳sim years incl. terminal (right helps; left hurts)')
axis_components.set_title('Mean full-episode reward balance by simulation_
↳agent order')
axis_components.grid(True, alpha=0.28, axis='x')
handles, labels = axis_components.get_legend_handles_labels()
legend_items = [(handle, label) for handle, label in zip(handles, labels)]
↳if label != '_nolegend_'
axis_components.legend([handle for handle, _ in legend_items], [label for
↳_, label in legend_items], fontsize=8, loc='lower center', bbox_to_anchor=(0.
↳5, -0.18), ncol=3, frameon=False)

if contrast_display.empty:
    axis_contrast.axis('off')
else:
    contrast_matrix = contrast_display.to_numpy(dtype=float)
    max_abs = np.nanmax(np.abs(contrast_matrix))
    if not np.isfinite(max_abs) or max_abs == 0:
        max_abs = 1.0
    norm = TwoSlopeNorm(vmin=-max_abs, vcenter=0.0, vmax=max_abs)
    heat = axis_contrast.imshow(contrast_matrix, cmap='RdYlGn', norm=norm,
↳aspect='auto')
    axis_contrast.set_xticks(np.arange(len(contrast_display.columns)))
    axis_contrast.set_xticklabels(contrast_display.columns, rotation=40,
↳ha='right', fontsize=8)
    axis_contrast.set_yticks(np.arange(len(contrast_display.index)))
    axis_contrast.set_yticklabels(contrast_display.index, fontsize=9)
    axis_contrast.set_title('ESG minus financial across RQ3 dimensions')
    for row_idx in range(contrast_matrix.shape[0]):
        for col_idx in range(contrast_matrix.shape[1]):
            value = contrast_matrix[row_idx, col_idx]
            if np.isfinite(value):
                text_color = 'white' if abs(value) > max_abs * 0.55 else
↳'black'
                axis_contrast.text(col_idx, row_idx, f'{value:+.2f}',
↳ha='center', va='center', color=text_color, fontsize=7)
    colorbar = fig.colorbar(heat, ax=axis_contrast, fraction=0.046, pad=0.
↳04)
    colorbar.set_label('ESG - Financial delta')
    axis_contrast.set_xlabel('positive = ESG agents have higher metric_
↳value')

fig.suptitle(
    f'RQ3.5 - Full-episode reward decomposition plus Financial-vs-ESG_
↳strategy contrast (last {TAIL_EPISODES} episodes)',

```

```

        fontsize=14,
    )
    plt.show()

reward_decomp_year = _build_reward_decomp_year(include_final_year=False)
reward_decomp_year_full = _build_reward_decomp_year(include_final_year=True)
if reward_decomp_year.empty and reward_decomp_year_full.empty:
    print('No year-level reward fields available for RQ3.5.')
else:
    unified_df = _build_unified_agent_episode_frame(reward_decomp_year,
    ↪reward_decomp_year_full)
    n_cells = len(unified_df)

    # ---- Table 1: per-agent (A1..A8) ----
    agent_order = sorted(unified_df['agent_no'].dropna().unique().tolist())
    agent_index = [
        (agent_no,
         f'A{int(agent_no)}',
         _agent_objective_label(int(agent_no)),
         _agent_archetype(int(agent_no) - 1))
        for agent_no in agent_order
    ]
    per_agent = _agent_summary(
        unified_df,
        ['agent_no', 'agent', 'objective', 'archetype'],
        sort_index=agent_index,
    )
    print(f'RQ3.5 per-agent summary (n={n_cells:,} (run, episode) cells across
    ↪last {TAIL_EPISODES:,} episodes:~')
    display(per_agent)

    # ---- Table 2: per-archetype (pooled across both objective twins) ----
    per_archetype = _agent_summary(
        unified_df,
        ['archetype'],
        sort_index=ARCHETYPES,
    )
    print('RQ3.5 per-archetype summary (each row pools the Financial + Balanced
    ↪ESG twin within that archetype:~')
    display(per_archetype)

    arch_obj_index = [(arch, obj) for arch in ARCHETYPES for obj in
    ↪OBJECTIVE_ORDER]
    per_arch_obj = _agent_summary(
        unified_df,
        ['archetype', 'objective'],

```



```

        sort_index=arch_obj_index,
    )

    # ---- Table 3: per-objective pooled (Financial vs ESG) + delta ----
    per_objective = _agent_summary(
        unified_df,
        ['objective'],
        sort_index=OBJECTIVE_ORDER,
    )
    per_objective = _add_esg_minus_financial(per_objective)
    print('RQ3.5 Financial vs ESG pooled summary (last row = ESG - Financial_
↪cross-agent delta):')
    display(per_objective)

    # ---- Plot: reward decomposition bars + ESG-Financial contrast heatmap
    contrast_metrics = [
        m for m in STRATEGY_LABELS + REWARD_CHANNELS + ['reward_episode_total']
        if m in unified_df.columns
    ]
    contrast_rows = []
    if not per_arch_obj.empty:
        flat_means = per_arch_obj.xs('mean', level=1, axis=1)
        for arch in ARCHETYPES:
            if (arch, 'Financial') in flat_means.index and (arch, 'ESG') in_
↪flat_means.index:
                row = (flat_means.loc[(arch, 'ESG')] - flat_means.loc[(arch,
↪'Financial')])[contrast_metrics]
                row.name = arch
                contrast_rows.append(row)
            if {'Financial', 'ESG'}.issubset(per_objective.dropna(how='all').index):
                flat_obj = per_objective.xs('mean', level=1, axis=1)
                row = (flat_obj.loc['ESG'] - flat_obj.
↪loc['Financial'])[contrast_metrics]
                row.name = 'Overall'
                contrast_rows.append(row)

    contrast_display = pd.DataFrame(contrast_rows) if contrast_rows else pd.
↪DataFrame()
    _plot_reward_decomposition(_agent_reward_means_for_plot(unified_df),
↪contrast_display)

    # Expose for the thesis-export cell below so its bars are sourced from the
    # exact same per-(run, episode, agent) frame used in these tables.
    rq3_unified_df = unified_df

```

RQ3.5 per-agent summary (n=32,000 (run, episode) cells across last 1,000 episodes):

				bid_price		bid_multiplier		
↵ auction_coverage		allocation_mt		penalty_meur				↵
↵ shortfall_mt		invest_cost_meur \						↵
		mean		std		mean		std ↵
↵		mean		std		mean		std ↵
↵ mean		std		mean		std		↵
agent_no	agent	objective	archetype					
1	A1	Financial	Coal-heavy	132.8436	22.2746	0.5099	0.0212	↵
↵		0.5099	0.0212	1.7968	0.3495	1528.8671	1053.2911	9.
↵6740	6.6293		677.4644					
2	A2	ESG	Coal-heavy	102.6173	36.8974	0.8523	0.5942	↵
↵		0.8523	0.5942	1.5736	0.3820	241.2570	346.2781	1.
↵5384	2.2261		954.1666					
3	A3	Financial	Gas-dominant	98.2105	34.4441	0.9042	0.6413	↵
↵		0.9042	0.6413	1.3831	0.4741	857.5456	842.9777	5.
↵4935	5.3803		391.2965					
4	A4	ESG	Gas-dominant	62.5916	15.1633	1.7844	0.2399	↵
↵		1.7844	0.2399	2.7685	0.5621	182.5174	589.6228	1.
↵0996	3.4682		1056.0523					
5	A5	Financial	Transitioner	103.7860	41.1660	1.3789	0.1818	↵
↵		1.3789	0.1818	1.9034	0.5068	83.4461	322.6190	0.
↵5042	1.9347		830.7273					
6	A6	ESG	Transitioner	70.1418	22.6131	1.7139	0.2937	↵
↵		1.7139	0.2937	1.8220	0.6533	182.2426	636.8528	1.
↵1085	3.8295		786.5047					
7	A7	Financial	Green-leader	90.4009	23.1067	1.6278	0.3771	↵
↵		1.6278	0.3771	1.5309	0.5255	45.7018	107.1933	0.
↵2809	0.6550		300.6379					
8	A8	ESG	Green-leader	88.0410	21.5908	1.6113	0.2462	↵
↵		1.6113	0.2462	1.1302	0.2915	14.5308	49.4564	0.
↵0882	0.3002		889.7073					
				green_fraction				
↵ delta_green		mac_reduction_mt		mac_cost_meur		sec_buy_mt		↵
↵		sec_sell_mt		sec_buy_years \				↵
		std		mean		std		↵
↵ mean		std		mean		std		↵
↵		mean		std		mean		std ↵
↵		mean		std		mean		std ↵
agent_no	agent	objective	archetype					
1	A1	Financial	Coal-heavy	514.9239	0.5856	0.0728	0.	
↵1066	0.0637		0.0866	0.1118	4.8103	6.1629	9.2837	4.4813 ↵
↵	0.0138	0.1120	4.0308					
2	A2	ESG	Coal-heavy	607.2973	0.6352	0.0863	0.	
↵1556	0.0825		0.0736	0.1120	4.0520	6.1236	12.4327	4.0387 ↵
↵	0.9579	2.2142	4.8292					

3	A3	Financial	Gas-dominant	641.0813		0.6220	0.0844	0.
↪0930	0.0814		0.0477	0.0649	2.6642	3.5898	10.8502	6.0168↪
↪	0.7263	2.1882		4.1535				
4	A4	ESG	Gas-dominant	557.6042		0.7104	0.0771	0.
↪1816	0.0743		0.0066	0.0157	0.3470	0.8222	1.0765	2.0835↪
↪	9.8415	4.3921		0.4018				
5	A5	Financial	Transitioner	815.4764		0.8354	0.1128	0.
↪1557	0.1107		0.0139	0.0339	0.7668	1.8667	0.2282	0.7818↪
↪	6.3631	4.0580		0.1230				
6	A6	ESG	Transitioner	421.4787		0.8222	0.0683	0.
↪1421	0.0632		0.0031	0.0120	0.1653	0.6498	0.3913	1.1148↪
↪	7.0702	6.5449		0.1610				
7	A7	Financial	Green-leader	502.0635		0.8698	0.0763	0.
↪0901	0.0746		0.0029	0.0109	0.1621	0.6027	0.5866	1.2452↪
↪	5.6078	4.4967		0.2710				
8	A8	ESG	Green-leader	350.4536		0.9566	0.0513	0.
↪1768	0.0504		0.0002	0.0015	0.0096	0.0759	0.4928	1.0807↪
↪	4.7614	2.1750		0.2210				

sec_sell_years									
invest_onshore_share		invest_offshore_share				invest_solar_share			
udbc_U_count		udbc_D_count				\			
		std				mean			
mean		std		mean		std		mean	
mean		std		mean		std		mean	
agent_no	agent	objective		archetype					
1	A1	Financial	Coal-heavy	1.9368	0.0280	0.1709			
↪	0.5799	0.4299		0.4062	0.4281		0.0140	0.0376	↪
↪	2.1402	1.0487		0.9310	1.3584				↪
2	A2	ESG	Coal-heavy	1.7067	0.4102	0.8693			
↪	0.8531	0.2431		0.0012	0.0099		0.1457	0.2434	↪
↪	1.2838	1.0936		0.0280	0.1753				
3	A3	Financial	Gas-dominant	2.0675	0.2958	0.8558			
↪	0.0604	0.1541		0.6839	0.4230		0.2558	0.4250	↪
↪	1.7755	1.1162		0.4828	0.9977				↪
4	A4	ESG	Gas-dominant	0.7463	3.7885	1.6262			
↪	0.7435	0.4255		0.0009	0.0107		0.2556	0.4247	↪
↪	0.5918	0.7150		0.1048	0.3699				
5	A5	Financial	Transitioner	0.3734	2.8428	1.7549			
↪	0.4579	0.4550		0.5298	0.4552		0.0123	0.0318	↪
↪	0.4128	0.6107		0.0555	0.2489				↪
6	A6	ESG	Transitioner	0.4332	3.6660	3.4148			
↪	0.6907	0.3450		0.2767	0.3414		0.0327	0.0480	↪
↪	0.4845	0.7327		0.0635	0.2907				
7	A7	Financial	Green-leader	0.5293	3.0070	1.9742			
↪	0.4979	0.4915		0.4908	0.4913		0.0114	0.0307	↪
↪	0.4600	0.7173		0.0792	0.3384				

8	A8	ESG	Green-leader	0.4424	2.9450	1.2922			
↪	0.9875	0.0322		0.0006 0.0072		0.0119	0.0314	↪	
↪	0.3208	0.5375	0.0232	0.1556					
				udbc_M_count		udbc_B_count		↪	
↪	udbc_C_count		invest_frac_post_clip			bank_end		trade_qty ↪	
↪		noncompliant_share		\					
				mean	std		mean	std ↪	
↪		mean	std		mean	std		mean	
↪	std		mean	std					
agent_no	agent	objective	archetype						
1	A1	Financial	Coal-heavy	3.4930	2.3236	2.1178	1.7158	↪	
↪	3.2125	1.9968		0.0079	0.0047	0.7015	0.8530	0.7725 ↪	
↪	0.3729		0.5470	0.2688					
2	A2	ESG	Coal-heavy	0.5492	1.0475	4.5215	1.7633	↪	
↪	4.5570	1.6524		0.0291	0.0000	1.9201	0.9022	0.9562 ↪	
↪	0.4650		0.1551	0.1509					
3	A3	Financial	Gas-dominant	2.3370	2.2503	2.8140	1.6296	↪	
↪	3.7395	2.1343		0.0050	0.0000	1.1273	1.4915	0.8437 ↪	
↪	0.4943		0.3829	0.2852					
4	A4	ESG	Gas-dominant	0.2490	0.7400	2.6930	1.3635	↪	
↪	0.3575	0.7027		0.0292	0.0000	2.3680	1.2375	-0.7304 ↪	
↪	0.4238		0.0788	0.1132					
5	A5	Financial	Transitioner	0.1760	0.6369	3.0978	1.6238	↪	
↪	0.0408	0.2015		0.0050	0.0000	2.6405	1.4188	-0.5112 ↪	
↪	0.3379		0.0537	0.0959					
6	A6	ESG	Transitioner	0.2715	0.8882	3.4765	2.7827	↪	
↪	0.1180	0.3723		0.0357	0.0007	2.1390	1.1877	-0.5566 ↪	
↪	0.5759		0.0683	0.1229					
7	A7	Financial	Green-leader	0.1678	0.5255	2.0708	1.7088	↪	
↪	0.1052	0.3350		0.0050	0.0000	2.7833	2.1382	-0.4184 ↪	
↪	0.3833		0.0589	0.1042					
8	A8	ESG	Green-leader	0.0472	0.2530	2.6008	1.4637	↪	
↪	0.0213	0.1493		0.0270	0.0000	2.4774	1.6962	-0.3557 ↪	
↪	0.2157		0.0326	0.0599					
				stressed_share		reward		↪	
↪	reward_base		reward_shaping		financial_unweighted_proxy			↪	
↪	financial_reward_proxy		\						
				mean	std	mean	std	↪	
↪	mean	std		mean	std		mean	std	
↪		mean	std						
agent_no	agent	objective	archetype						
1	A1	Financial	Coal-heavy	0.4442	0.2658	-0.6263	0.5037	↪	
↪	-0.6263	0.5037	0.0	0.0		0.1105	0.1246	↪	
↪		0.1105	0.1246						

2	A2	ESG	Coal-heavy	0.7565	0.1860	-0.2125	0.2777	␣
	↪	-0.2125	0.2777			0.0	0.0	␣
	↪	-0.1467	0.1311					
3	A3	Financial	Gas-dominant	0.5461	0.2488	-0.8600	0.4438	␣
	↪	-0.8600	0.4438			0.0	0.0	␣
	↪	-0.1969	0.3749					
4	A4	ESG	Gas-dominant	0.2542	0.1243	0.3649	0.3071	␣
	↪	0.3649	0.3071			0.0	0.0	␣
	↪	0.2404	0.1779					
5	A5	Financial	Transitioner	0.2615	0.1361	0.7366	0.6235	␣
	↪	0.7366	0.6235			0.0	0.0	␣
	↪	0.7398	0.6099					
6	A6	ESG	Transitioner	0.2995	0.2381	0.3908	0.3624	␣
	↪	0.3908	0.3624			0.0	0.0	␣
	↪	0.2581	0.2534					
7	A7	Financial	Green-leader	0.1813	0.1504	0.7569	0.6992	␣
	↪	0.7569	0.6992			0.0	0.0	␣
	↪	0.7843	0.6730					
8	A8	ESG	Green-leader	0.2185	0.1230	0.7444	0.4253	␣
	↪	0.7444	0.4253			0.0	0.0	␣
	↪	0.4321	0.4163					

				weighted_esg_proxy		
↪esg_stock_reward		esg_flow_reward		esg_speed_reward		␣
↪penalty_norm_proxy		penalty_realized_norm_proxy \				
				mean	std	␣
↪mean	std	mean	std	mean	std	␣
↪mean	std	mean				
agent_no	agent	objective	archetype			
1	A1	Financial	Coal-heavy	0.0000	0.0000	0.
	↪0000	0.0000	0.0000	0.0000	0.0000	0.
	↪5664	0.3414	0.3075			
2	A2	ESG	Coal-heavy	0.1777	0.0706	0.
	↪0627	0.0248	0.1140	0.0454	0.0009	0.0005
	↪1077	0.1460	0.0593			
3	A3	Financial	Gas-dominant	0.0000	0.0000	0.
	↪0000	0.0000	0.0000	0.0000	0.0000	0.
	↪4078	0.3624	0.2255			
4	A4	ESG	Gas-dominant	0.2165	0.0677	0.
	↪0765	0.0238	0.1388	0.0437	0.0011	0.0005
	↪0737	0.1527	0.0369			
5	A5	Financial	Transitioner	0.0000	0.0000	0.
	↪0000	0.0000	0.0000	0.0000	0.0000	0.
	↪0459	0.1247	0.0228			
6	A6	ESG	Transitioner	0.2548	0.0762	0.
	↪0908	0.0269	0.1631	0.0492	0.0009	0.0004
	↪1006	0.2395	0.0510			

agent_no	agent	objective	archetype	mean	std	mean	std	mean	std
7	A7	Financial	Green-leader	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
				0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
				0.0554	0.1192	0.0283			
8	A8	ESG	Green-leader	0.3424	0.0799	0.0011	0.0003	0.0011	0.0003
				0.1216	0.0284	0.2197	0.0515	0.0011	0.0003
				0.0305	0.0867	0.0148			

agent_no	agent	objective	archetype	reward_episode_sum	penalty_remediation_norm_proxy	reward_episode_sum	penalty_remediation_norm_proxy
1	A1	Financial	Coal-heavy	0.1857	0.2589	0.1857	0.2589
				-0.1704	0.3096	-7.4124	6.4727
				-7.8874	6.4239		
2	A2	ESG	Coal-heavy	0.0821	0.0485	0.0821	0.0485
				-0.1357	0.1294	-1.7933	3.4249
				-2.3083	3.4157		
3	A3	Financial	Gas-dominant	0.2001	0.1823	0.2001	0.1823
				-0.2553	0.2685	-9.7923	5.3245
				-10.2866	5.3060		
4	A4	ESG	Gas-dominant	0.0751	0.0368	0.0751	0.0368
				-0.0184	0.0940	5.2192	4.2505
				4.6983	4.2375		
5	A5	Financial	Transitioner	0.0621	0.0230	0.0621	0.0230
				0.0427	0.0937	9.8918	7.6430
				9.3683	7.6369		
6	A6	ESG	Transitioner	0.1216	0.0495	0.1216	0.0495
				-0.0216	0.1472	5.3265	4.6955
				4.8042	4.6802		
7	A7	Financial	Green-leader	0.0611	0.0271	0.0611	0.0271
				0.0280	0.1280	10.5579	7.6807
				10.0320	7.6764		
8	A8	ESG	Green-leader	0.0421	0.0157	0.0421	0.0157
				0.0004	0.2115	10.2095	5.1273
				9.6806	5.1259		

agent_no	agent	objective	archetype	reward_episode_sum	penalty_remediation_norm_proxy	reward_episode_sum	penalty_remediation_norm_proxy
				0.0000	0.0000	0.0000	0.0000
				0.0554	0.1192	0.0283	
				0.1216	0.0284	0.2197	0.0515
				0.0305	0.0867	0.0148	

1	A1	Financial	Coal-heavy	0.4750	0.0829	┐
↪			1.4653 1.4139		1.4653	1.
↪4139			0.0000			
2	A2	ESG	Coal-heavy	0.5150	0.0434	┐
↪			-3.4240 3.0009		-1.7120	1.
↪5004			2.2308			
3	A3	Financial	Gas-dominant	0.4943	0.0719	┐
↪			-2.2644 4.4619		-2.2644	4.
↪4619			0.0000			
4	A4	ESG	Gas-dominant	0.5209	0.0401	┐
↪			5.6350 4.1072		2.8175	2.
↪0536			2.7210			
5	A5	Financial	Transitioner	0.5234	0.0408	┐
↪			9.1978 7.3672		9.1978	7.
↪3672			0.0000			
6	A6	ESG	Transitioner	0.5223	0.0435	┐
↪			6.4402 5.8390		3.2201	2.
↪9195			3.1388			
7	A7	Financial	Green-leader	0.5260	0.0341	┐
↪			9.7546 7.6785		9.7546	7.
↪6785			0.0000			
8	A8	ESG	Green-leader	0.5289	0.0292	┐
↪			11.0884 10.0284		5.5442	5.
↪0142			4.2717			

				esg_stock_reward_episode_sum		┐
↪ esg_flow_reward_episode_sum				esg_speed_reward_episode_sum		┐
↪penalty_norm_proxy_episode_sum				\		
				std		
↪std				mean		┐
				mean		std ┐
↪				mean		std
agent_no	agent	objective	archetype			
1	A1	Financial	Coal-heavy	0.0000	0.0000	0.
↪0000			0.0000 0.0000		0.0000	0.0000 ┐
↪			6.9624 4.0753			
2	A2	ESG	Coal-heavy	0.8929	0.7779	0.
↪3091			1.4420 0.5795		0.0109	0.0062 ┐
↪			1.4420 1.7468			
3	A3	Financial	Gas-dominant	0.0000	0.0000	0.
↪0000			0.0000 0.0000		0.0000	0.0000 ┐
↪			4.9705 4.3205			
4	A4	ESG	Gas-dominant	0.8627	0.9502	0.
↪2987			1.7575 0.5610		0.0133	0.0055 ┐
↪			1.0313 2.0901			
5	A5	Financial	Transitioner	0.0000	0.0000	0.
↪0000			0.0000 0.0000		0.0000	0.0000 ┐
↪			0.7111 1.7202			

6	A6	ESG	Transitioner	0.9544		1.1058	0.
↪3319				2.0227	0.6208	0.0103	0.0046
↪				1.4184	3.1410		
7	A7	Financial	Green-leader	0.0000		0.0000	0.
↪0000				0.0000	0.0000	0.0000	0.0000
↪				0.8044	1.5651		
8	A8	ESG	Green-leader	0.9854		1.4988	0.
↪3454				2.7600	0.6408	0.0130	0.0037
↪				0.4693	1.1556		

						penalty_realized_norm_proxy_episode_sum	↪
↪						penalty_remediation_norm_proxy_episode_sum	↪
↪						banking_soft_residual_episode_sum	↪
						\	
						mean	↪
↪	std			mean	std		↪
↪	mean	std					↪
agent_no	agent	objective	archetype				
1	A1	Financial	Coal-heavy			3.7056	2.
↪1757				3.2568	1.9148		↪
↪	-2.3903	3.9869					
2	A2	ESG	Coal-heavy			0.7654	0.
↪9526				0.6766	0.8117		↪
↪	-1.3850	1.6150					
3	A3	Financial	Gas-dominant			2.6949	2.
↪3417				2.2756	1.9920		↪
↪	-3.0517	3.2896					
4	A4	ESG	Gas-dominant			0.5032	1.
↪0018				0.5281	1.0924		↪
↪	0.1911	1.4950					
5	A5	Financial	Transitioner			0.3424	0.
↪8315				0.3687	0.8925		↪
↪	0.8817	1.5111					
6	A6	ESG	Transitioner			0.6992	1.
↪5517				0.7192	1.5991		↪
↪	-0.1363	1.8502					
7	A7	Financial	Green-leader			0.3973	0.
↪7759				0.4071	0.7968		↪
↪	1.0817	1.9215					
8	A8	ESG	Green-leader			0.2219	0.
↪5484				0.2474	0.6105		↪
↪	0.3341	2.5683					

						reward_episode_total	
						mean	std
agent_no	agent	objective	archetype				
1	A1	Financial	Coal-heavy			-7.4124	6.4727
2	A2	ESG	Coal-heavy			-1.7933	3.4249

3	A3	Financial	Gas-dominant	-9.7923	5.3245
4	A4	ESG	Gas-dominant	5.2192	4.2505
5	A5	Financial	Transitioner	9.8918	7.6430
6	A6	ESG	Transitioner	5.3265	4.6955
7	A7	Financial	Green-leader	10.5579	7.6807
8	A8	ESG	Green-leader	10.2095	5.1273

RQ3.5 per-archetype summary (each row pools the Financial + Balanced ESG twin within that archetype):

	bid_price		bid_multiplier		auction_coverage		
↪ allocation_mt			penalty_meur	shortfall_mt			
↪ invest_cost_meur			green_fraction	\			
	mean	std	mean	std	mean	std	
↪ mean	std		mean	std	mean	std	
↪ std			mean	std			
archetype							
Coal-heavy	117.7305	34.0163	0.6811	0.4539	0.6811	0.4539	
↪	1.6852	0.3828	885.0621	1014.4581	5.6062	6.4029	815.
↪8155	579.7258		0.6104	0.0835			
Gas-dominant	80.4010	32.0201	1.3443	0.6543	1.3443	0.6543	
↪	2.0758	0.8661	520.0315	801.8710	3.2966	5.0312	723.
↪6744	686.5840		0.6662	0.0921			
Transitioner	86.9639	37.2273	1.5464	0.2961	1.5464	0.2961	
↪	1.8627	0.5860	132.8444	507.1892	0.8064	3.0486	808.
↪6160	649.4300		0.8288	0.0935			
Green-leader	89.2210	22.3913	1.6195	0.3185	1.6195	0.3185	
↪	1.3306	0.4697	30.1163	84.9131	0.1846	0.5185	595.
↪1726	523.6228		0.9132	0.0782			

	delta_green		mac_reduction_mt		mac_cost_meur		
↪ sec_buy_mt	sec_sell_mt		sec_buy_years		sec_sell_years		
↪ invest_onshore_share	\						
	mean	std	mean	std	mean	std	
↪ mean	std	mean	std	mean	std	mean	
↪ std		mean	std				
archetype							
Coal-heavy	0.1311	0.0777	0.0801	0.1121	4.4311	6.1546	
↪	10.8582	4.5468	0.4859	1.6372	4.4300	1.8684	0.2191
↪6549		0.7165	0.3750				
Gas-dominant	0.1373	0.0896	0.0272	0.0515	1.5056	2.8501	
↪	5.9634	6.6448	5.2839	5.7282	2.2776	2.4361	2.0421
↪1768		0.4019	0.4681				
Transitioner	0.1489	0.0904	0.0085	0.0260	0.4661	1.4295	
↪	0.3098	0.9662	6.7167	5.4565	0.1420	0.4048	3.2544
↪7457		0.5743	0.4202				

Green-leader	0.1335	0.0770			0.0016	0.0079		0.0858	0.4362	␣
↪ 0.5397	1.1667		5.1846	3.5571		0.2460	0.4884		2.9760	1.
↪ 6686		0.7427	0.4257							

			invest_offshore_share			invest_solar_share				␣	
↪ udbc_U_count			udbc_D_count			udbc_M_count			udbc_B_count	␣	
↪		udbc_C_count		invest_frac_post_clip	\						
			mean	std		mean	std			␣	
↪ mean	std		mean	std		mean	std		mean	std	␣
↪	mean	std			mean						

archetype											
Coal-heavy			0.2037	0.3643		0.0798	0.1862		1.		
↪ 7120	1.1538		0.4795	1.0685		2.0211	2.3269		3.3196	2.1145	␣
↪	3.8848	1.9520		0.0185							
Gas-dominant			0.3424	0.4540		0.2557	0.4248		1.		
↪ 1836	1.1085		0.2938	0.7757		1.2930	1.9737		2.7535	1.5036	␣
↪	2.0485	2.3204		0.0171							
Transitioner			0.4032	0.4218		0.0225	0.0420		0.		
↪ 4486	0.6754		0.0595	0.2706		0.2238	0.7743		3.2871	2.2859	␣
↪	0.0794	0.3018		0.0204							
Green-leader			0.2457	0.4252		0.0116	0.0311		0.		
↪ 3904	0.6376		0.0512	0.2648		0.1075	0.4168		2.3358	1.6128	␣
↪	0.0632	0.2627		0.0160							

			bank_end			trade_qty			noncompliant_share		␣
↪ stressed_share			reward			reward_base			reward_shaping		␣
↪		financial_unweighted_proxy		\							
		std	mean	std		mean	std		mean		␣
↪ std		mean	std	mean		std	mean	std		mean	␣
↪ std			mean	std							

archetype										
Coal-heavy	0.0111	1.3108	1.0686	0.8644	0.4313			0.3511	0.	
↪ 2931		0.6004	0.2775	-0.4194	0.4563		-0.4194	0.4563		0.0
↪	0.0		-0.0914	0.2879						␣
Gas-dominant	0.0121	1.7477	1.5042	0.0566	0.9118			0.2309	0.	
↪ 2649		0.4002	0.2449	-0.2475	0.7217		-0.2475	0.7217		0.0
↪	0.0		0.1419	0.4984						␣
Transitioner	0.0154	2.3897	1.3321	-0.5339	0.4726			0.0610	0.	
↪ 1104		0.2805	0.1948	0.5637	0.5384		0.5637	0.5384		0.0
↪	0.0		0.6280	0.5717						␣
Green-leader	0.0110	2.6304	1.9359	-0.3871	0.3126			0.0458	0.	
↪ 0860		0.1999	0.1387	0.7506	0.5787		0.7506	0.5787		0.0
↪	0.0		0.8242	0.7580						␣

			financial_reward_proxy			weighted_esg_proxy				␣
↪ esg_stock_reward			esg_flow_reward			esg_speed_reward				␣
↪		penalty_norm_proxy	\							

		mean	std		mean	std	
↪	mean	std	mean	std	mean	std	↪
↪	mean	std					↪
archetype							
Coal-heavy		-0.0181	0.1814		0.0888	0.1019	↪
↪	0.0314	0.0359	0.0570	0.0655	0.0005	0.0006	0.↪
↪	3371	0.3486					
Gas-dominant		0.0217	0.3659		0.1083	0.1184	↪
↪	0.0383	0.0418	0.0694	0.0760	0.0006	0.0007	0.↪
↪	2407	0.3244					
Transitioner		0.4989	0.5254		0.1274	0.1383	↪
↪	0.0454	0.0492	0.0816	0.0887	0.0005	0.0005	0.↪
↪	0732	0.1929					
Green-leader		0.6082	0.5866		0.1712	0.1803	↪
↪	0.0608	0.0640	0.1099	0.1157	0.0006	0.0006	0.↪
↪	0429	0.1049					

[illegible]

Coal-heavy	0.4950	0.0691							
↪ -0.9794	3.3880			-0.1234	2.1562				
↪	1.1154	1.2817							
Gas-dominant	0.5076	0.0597							
↪ 1.6853	5.8300			0.2765	4.3033				
↪	1.3605	1.4911							
Transitioner	0.5228	0.0422							
↪ 7.8190	6.7882			6.2089	6.3505				
↪	1.5694	1.7084							
Green-leader	0.5274	0.0318							
↪ 10.4215	8.9554			7.6494	6.8175				
↪	2.1359	2.2468							

	esg_stock_reward_episode_sum			esg_flow_reward_episode_sum			
↪	esg_speed_reward_episode_sum			penalty_norm_proxy_episode_sum			
↪	\						

		mean	std		mean	
↪ std		mean	std		mean	
↪ std						

archetype						
Coal-heavy	0.3889	0.4461		0.7210	0.	
↪ 8293	0.0055	0.0070		4.2022	4.	
↪ 1771						
Gas-dominant	0.4751	0.5200		0.8787	0.	
↪ 9642	0.0066	0.0077		3.0009	3.	
↪ 9238						
Transitioner	0.5529	0.6007		1.0114	1.	
↪ 1026	0.0051	0.0061		1.0648	2.	
↪ 5567						
Green-leader	0.7494	0.7882		1.3800	1.	
↪ 4526	0.0065	0.0070		0.6369	1.	
↪ 3857						

	penalty_realized_norm_proxy_episode_sum			
↪	penalty_remediation_norm_proxy_episode_sum			
↪	banking_soft_residual_episode_sum			
	reward_episode_total			
		mean	std	
↪	mean	std		
↪	mean	std		

archetype						
Coal-heavy			2.2355	2.2320		
↪	1.9667	1.9563		-1.8877	3.0827	
↪	-4.6029	5.8910				
Gas-dominant			1.5991	2.1081		
↪	1.4019	1.8286		-1.4303	3.0260	
↪	-2.2866	8.9190				

Transitioner				0.5208	1.2574			
↪		0.5439	1.3067			0.3727	1.7641	
↪	7.6091	6.7407						
Green-leader				0.3096	0.6775			
↪		0.3273	0.7142			0.7079	2.2985	
↪	10.3837	6.5319						

RQ3.5 Financial vs ESG pooled summary (last row = ESG - Financial cross-agent delta):

			bid_price		bid_multiplier		auction_coverage	
↪ allocation_mt				penalty_meur		shortfall_mt		
↪ invest_cost_meur				green_fraction	\			
		mean	std		mean	std		mean
↪ std	mean	std		mean	std		mean	std
↪ mean	std		mean					
objective								
Financial		106.3103	35.1409		1.1052	0.5766		1.1052
↪ 5766	1.6536	0.5125		628.8902	926.7143		3.9882	5.8637
↪ 550.0315	666.2089		0.7282					
ESG		80.8479	29.7573		1.4905	0.5281		1.4905
↪ 5281	1.8236	0.7762		155.1369	475.4208		0.9587	2.8666
↪ 921.6077	504.5837		0.7811					
ESG - Financial		-25.4624	-5.3836		0.3853	-0.0485		0.3853
↪ 0485	0.1700	0.2637		-473.7533	-451.2935		-3.0295	-2.9971
↪ 371.5762	-161.6252		0.0529					
			delta_green		mac_reduction_mt			
↪ mac_cost_meur			sec_buy_mt		sec_sell_mt		sec_buy_years	
↪ sec_sell_years			invest_onshore_share	\				
		std	mean	std		mean	std	
↪ mean	std	mean	std	mean	std		mean	std
↪ mean	std		mean					
objective								
Financial		0.1534	0.1114	0.0884		0.0378	0.0746	2.
↪ 1008	4.1202	5.2372	6.1854	3.1778	4.2882		2.1446	2.4308
↪ 1.5434	1.9635		0.3990					
ESG		0.1410	0.1640	0.0705		0.0209	0.0645	1.
↪ 1435	3.5331	3.5983	5.6436	5.6577	5.3418		1.4033	2.2099
↪ 2.7024	2.4571		0.8187					
ESG - Financial		-0.0124	0.0526	-0.0179		-0.0169	-0.0101	-0.
↪ 9573	-0.5871	-1.6389	-0.5418	2.4799	1.0536		-0.7413	-0.2209
↪ 1.1590	0.4936		0.4197					
			invest_offshore_share		invest_solar_share			
↪ udbc_U_count			udbc_D_count		udbc_M_count		udbc_B_count	
↪ udbc_C_count		\						

	mean	std	mean	std	mean	std	mean	std
objective								
Financial	0.4521		0.5277	0.4613			0.0734	0.2389
↪	1.1971	1.1848	0.3871	0.9389	1.5434	2.1986	2.5251	1.
↪7276	1.7745	2.2592						
ESG	0.3209		0.0698	0.2085			0.1114	0.2650
↪	0.6702	0.8763	0.0549	0.2648	0.2792	0.8101	3.3229	2.
↪0755	1.2634	2.1159						
ESG - Financial	-0.1312		-0.4579	-0.2528			0.0380	0.0261
↪	-0.5269	-0.3085	-0.3322	-0.6741	-1.2642	-1.3885	0.7978	0.
↪3479	-0.5111	-0.1433						

	invest_frac_post_clip	bank_end	trade_qty	
↪noncompliant_share	stressed_share	reward	reward_base	
↪	reward_shaping	\		
	mean	std	mean	std
↪	mean	std	mean	std
↪	std	mean	std	mean
objective				
Financial	0.0057	0.0027	1.8131	1.7936
↪	0.2606	0.2975	0.3583	0.2535
↪0.9455	0.0	0.0	0.0018	0.9455
ESG	0.0303	0.0033	2.2261	1.3054
↪	0.0837	0.1247	0.3822	0.2793
↪0.4884	0.0	0.0	0.3219	0.4884
ESG - Financial	0.0246	0.0006	0.4130	-0.4882
↪	-0.1769	-0.1728	0.0239	0.0258
↪-0.4571	0.0	0.0	0.3201	-0.4571

	financial_unweighted_proxy	financial_reward_proxy	
↪ weighted_esg_proxy	esg_stock_reward	esg_flow_reward	
↪esg_speed_reward	\		
	mean	std	mean
↪std	mean	std	mean
↪	mean	std	std
objective			
Financial	0.3594	0.6476	0.3594
↪6476	0.0000	0.0000	0.0000
↪0000	0.000	0.0000	0.000
ESG	0.3920	0.6821	0.1960
↪3411	0.2479	0.0957	0.0879
↪0616	0.001	0.0004	0.034
ESG - Financial	0.0326	0.0345	-0.1634
↪3065	0.2479	0.0957	0.0879
↪0616	0.001	0.0004	0.034

	penalty_norm_proxy	penalty_realized_norm_proxy	
penalty_remediation_norm_proxy		banking_soft_residual	
reward_episode_sum	\		
	mean	std	
	mean	std	
	mean	std	
objective			
Financial	0.2689	0.3467	0.1460 0.1894
	0.1228	0.1580	-0.0888 0.2542
	0.8112	11.6743	
ESG	0.0781	0.1682	0.0405 0.0867
	0.0376	0.0824	-0.0438 0.1609
	4.7405	6.1501	
ESG - Financial	-0.1908	-0.1785	-0.1055 -0.1027
	-0.0852	-0.0756	0.0450 -0.0933
	3.9293	-5.5242	

	reward_base_episode_sum	reward_shaping_episode_sum	
financial_unweighted_proxy_episode_sum			
financial_reward_proxy_episode_sum	\		
	mean	std	
	mean	std	
	mean	std	
objective			
Financial	0.3066	11.6481	0.5047 0.
	0646	4.5383 7.7422	
	4.5383	7.7422	
ESG	4.2187	6.1400	0.5218 0.
	0398	4.9349 8.2311	
	2.4674	4.1156	
ESG - Financial	3.9121	-5.5081	0.0171 -0.
	0248	0.3966 0.4889	
	-2.0709	-3.6266	

	weighted_esg_proxy_episode_sum	
esg_stock_reward_episode_sum	esg_flow_reward_episode_sum	
esg_speed_reward_episode_sum	penalty_norm_proxy_episode_sum	\
	mean	std
	mean	std
	mean	std
objective		
Financial	0.0000	0.0000
	0.0000	0.0000
	0.0000	3.3621

ESG		3.0906	1.1933		1.
↪0832	0.4178	1.9955	0.7737		0.0119 ↪
↪0.0052		1.0903			
ESG - Financial		3.0906	1.1933		1.
↪0832	0.4178	1.9955	0.7737		0.0119 ↪
↪0.0052		-2.2718			
		penalty_realized_norm_proxy_episode_sum			↪
↪penalty_remediation_norm_proxy_episode_sum					↪
↪banking_soft_residual_episode_sum			reward_episode_total		
	std		mean	std	↪
↪		mean	std		mean
↪ std	mean	std			↪
objective					
Financial	4.1772			1.7851	2.2379 ↪
↪		1.5771	1.9496		-0.8696 3.
↪4145	0.8112	11.6743			
ESG	2.1931			0.5474	1.0951 ↪
↪		0.5428	1.1088		-0.2490 2.
↪0431	4.7405	6.1501			
ESG - Financial	-1.9841			-1.2377	-1.1428 ↪
↪		-1.0343	-0.8408		0.6206 -1.
↪3714	3.9293	-5.5242			



```
[25]: # Thesis export for Figure fig:rq3-decomp-bars.
# Compact 2x3 overview matching the RQ3.1-3.4 bar style:
# grouped paired bars per archetype, red = Financial, green = Balanced ESG.
#
# Sourced from `rq3_unified_df` built in the cell above (per-(run, episode,
# agent) frame). Using the same source guarantees the bar values reported
# here match the per-agent / per-archetype tables exactly. Panel F shows the
# full-episode reward total (ep_df sum, includes terminal-year payoff).
```



```

RQ3_DECOMP_BARS_PDF = PROJECT_ROOT / 'results' / 'analysis' / 'rq3_decomp_bars.
↳pdf'
RQ3_DECOMP_BARS_PNG = PROJECT_ROOT / 'results' / 'analysis' / 'rq3_decomp_bars.
↳png'
RQ3_WEIGHTING_ORDER = ['Financial', 'Balanced ESG']
RQ3_WEIGHTING_COLORS = {'Financial': '#d62728', 'Balanced ESG': '#2ca02c'}

def _reward_total_note(source):
    if 'reward_episode_total' not in source.columns:
        return '[RQ3] reward_episode_total unavailable; panel F shows_
↳reward_base year-mean as a fallback.'
    grouped_total = (
        source.groupby(['archetype', 'weighting'])['reward_episode_total']
        .mean()
        .replace([np.inf, -np.inf], np.nan)
        .dropna()
    )
    if grouped_total.empty:
        return '[RQ3] reward_episode_total has no finite values; panel F shows_
↳reward_base year-mean as a fallback.'
    return (
        '[RQ3] Panel F uses reward_episode_total = sum of per-year rewards from_
↳ep_df '
        '(includes the final-year terminal payoff). '
        f'Grouped mean range {grouped_total.min():.2f} to {grouped_total.max():.
↳2f}.'
    )

def _plot_rq3_decomp_bars(source, metric_specs, pdf_path, png_path):
    x = np.arange(len(ARCHETYPES), dtype=float)
    bar_width = 0.38
    offsets = {'Financial': -bar_width / 2, 'Balanced ESG': bar_width / 2}

    summary_rows = []

    fig, axes = plt.subplots(2, 3, figsize=(18, 9))
    axes_flat = axes.ravel()

    for panel_idx, (ax, (metric, title, ylabel)) in enumerate(zip(axes_flat,
↳metric_specs)):
        for weighting in RQ3_WEIGHTING_ORDER:
            means, stds = [], []
            for arch in ARCHETYPES:

```

```

        if metric not in source.columns:
            means.append(np.nan); stds.append(np.nan)
            continue
        sub = source[(source['archetype'] == arch) &
↳(source['weighting'] == weighting)][metric]
        sub = pd.to_numeric(sub, errors='coerce').replace([np.inf, -np.
↳inf], np.nan).dropna()
        mean_val = float(sub.mean()) if not sub.empty else np.nan
        std_val = float(sub.std()) if not sub.empty else np.nan
        means.append(mean_val)
        stds.append(std_val)
        summary_rows.append({
            'metric': metric, 'archetype': arch, 'weighting': weighting,
            'mean': mean_val, 'std': std_val,
        })
    ax.bar(
        x + offsets[weighting], means,
        width=bar_width, yerr=stds,
        color=RQ3_WEIGHTING_COLORS[weighting],
        alpha=0.85, capsize=3, label=weighting,
    )

    ax.axhline(0.0, color='0.4', linewidth=0.7)
    ax.set_xticks(x)
    ax.set_xticklabels(ARCHETYPES, fontsize=10)
    ax.set(title=f'{chr(65 + panel_idx)}. {title}', ylabel=ylabel)
    ax.grid(True, alpha=0.3, axis='y')
    ax.margins(x=0.06)

handles, labels = axes_flat[0].get_legend_handles_labels()
fig.legend(
    handles, labels,
    loc='upper center', ncol=2, frameon=False, bbox_to_anchor=(0.5, 1.01),
    fontsize=11,
)
plt.suptitle(
    f'RQ3 strategy overview by archetype and reward weighting '
    f'(last {TAIL_EPISODES:,} episodes)',
    fontsize=12, y=1.05,
)
plt.tight_layout()

pdf_path.parent.mkdir(parents=True, exist_ok=True)
fig.savefig(pdf_path, bbox_inches='tight')
fig.savefig(png_path, bbox_inches='tight', dpi=300)
plt.show()

```

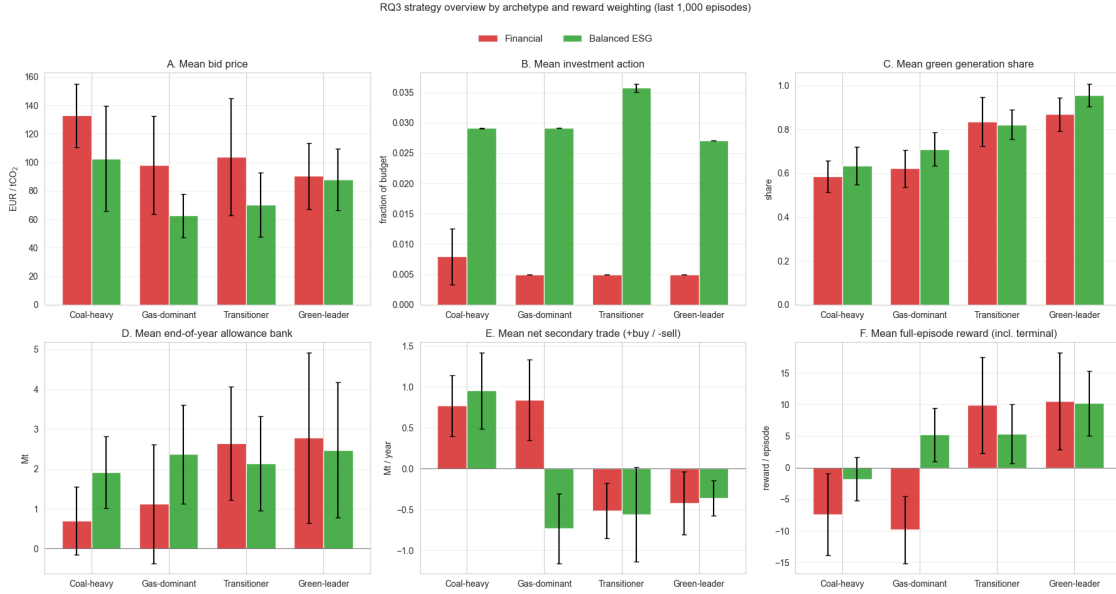
```

summary = (
    pd.DataFrame(summary_rows)
    .set_index(['metric', 'archetype', 'weighting'])
    .unstack('weighting')
)
return summary

if 'rq3_unified_df' not in globals() or rq3_unified_df is None or
↳rq3_unified_df.empty:
    print('[RQ3] rq3_unified_df not available - run the RQ3.5 cell above first.
↳')
else:
    print(_reward_total_note(rq3_unified_df))
    rq3_overview_metric_specs = [
        ('bid_price', 'Mean bid price', r'EUR / tCO$_2$'),
        ('invest_frac_post_clip', 'Mean investment action', 'fraction of
↳budget'),
        ('green_fraction', 'Mean green generation share', 'share'),
        ('bank_end', 'Mean end-of-year allowance bank', 'Mt'),
        ('trade_qty', 'Mean net secondary trade (+buy / -sell)', 'Mt / year'),
        ('reward_episode_total', 'Mean full-episode reward (incl. terminal)',
↳'reward / episode'),
    ]
    rq3_decomp_bar_summary = _plot_rq3_decompBars(
        rq3_unified_df,
        rq3_overview_metric_specs,
        RQ3_DECOMP_BARS_PDF,
        RQ3_DECOMP_BARS_PNG,
    )
    print(f'[RQ3] Exported thesis overview figure to: {RQ3_DECOMP_BARS_PDF}')
    print(f'[RQ3] PNG preview written to: {RQ3_DECOMP_BARS_PNG}')
    display(rq3_decomp_bar_summary.round(3))

```

[RQ3] Panel F uses reward_episode_total = sum of per-year rewards from ep_df (includes the final-year terminal payoff). Grouped mean range -9.79 to 10.56.



[RQ3] Exported thesis overview figure to:
C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-Energy-Auction\results\analysis\rq3_decomp_bars.pdf
[RQ3] PNG preview written to: C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-Energy-Auction\results\analysis\rq3_decomp_bars.png

weighting	metric	archetype	mean		std	
			Balanced ESG	Financial	Balanced ESG	Financial
bank_end		Coal-heavy	1.920	0.701	0.902	0.853
		Gas-dominant	2.368	1.127	1.238	1.492
		Green-leader	2.477	2.783	1.696	2.138
		Transitioner	2.139	2.640	1.188	1.419
bid_price		Coal-heavy	102.617	132.844	36.897	22.275
		Gas-dominant	62.592	98.210	15.163	34.444
		Green-leader	88.041	90.401	21.591	23.107
		Transitioner	70.142	103.786	22.613	41.166
green_fraction		Coal-heavy	0.635	0.586	0.086	0.073
		Gas-dominant	0.710	0.622	0.077	0.084
		Green-leader	0.957	0.870	0.051	0.076
		Transitioner	0.822	0.835	0.068	0.113
invest_frac_post_clip		Coal-heavy	0.029	0.008	0.000	0.005
		Gas-dominant	0.029	0.005	0.000	0.000
		Green-leader	0.027	0.005	0.000	0.000
		Transitioner	0.036	0.005	0.001	0.000
reward_episode_total		Coal-heavy	-1.793	-7.412	3.425	6.473
		Gas-dominant	5.219	-9.792	4.250	5.325
		Green-leader	10.210	10.558	5.127	7.681

	Transitioner	5.326	9.892	4.696	7.643
trade_qty	Coal-heavy	0.956	0.772	0.465	0.373
	Gas-dominant	-0.730	0.844	0.424	0.494
	Green-leader	-0.356	-0.418	0.216	0.383
	Transitioner	-0.557	-0.511	0.576	0.338

1.8.6 RQ3.6 — Strategy verdict: regret, total cost of ownership, and effective carbon price

RQ3.5 already gives a per-agent reward decomposition. The cells below go one level further and try to answer the *strategy* question directly: **which (archetype × weighting) cell wins, and by how much?**

Three lenses are produced, all built from the same converged-window artefacts used by RQ3.5 (`rq3_unified_df` for reward channels and a fresh per-(run, episode, agent) frame of raw € flows for the cost buckets):

1. **Total cost of ownership (TCO).** A single 12-year-€ figure per agent — auction spend + secondary spend + invest cost + penalty — that a CFO can read directly. MAC fuel-switching cost is broken out separately as it is an *abatement* spend rather than an allowance spend.
2. **Effective carbon price paid.** Total euros spent on allowances ÷ tonnes acquired net of sales (auction + signed secondary). Net-seller archetypes can land below the clearing price — and may even go negative.
3. **Regret within archetype.** For each of the four starting archetypes, how far is each twin from the best-performing twin in the same archetype? This isolates the *strategy* effect from the *starting endowment* effect.

The closing scorecard ranks the eight (archetype, weighting) cells by all three lenses simultaneously and names the winner on each axis.

```
[26]: # RQ3.6a - 12-year total cost of ownership (TCO) per agent.
#
# Pulls raw € flows from the year_log:
#   auction_cost_A[i]   auction-clearing payments (M€)
#   trade_cost_A[i]     net secondary cash flow (M€; +pay, -receive)
#   invest_cost_A[i]    capex on renewable additions (M€)
#   penalty_A[i]        non-compliance penalties (M€)
#   mac_cost_A[i]       MAC fuel-switching cost (M€, kept separate)
# Sums across the 12 sim years per (run, episode, agent), then takes
# the mean across the last TAIL_EPISODES converged episodes.

_COST_BUCKETS = [
    ('auction_cost', 'Auction spend'),
    ('trade_cost',   'Secondary spend (net)'),
    ('invest_cost',  'Invest cost'),
    ('penalty',      'Penalty'),
]
_AUX_BUCKETS = [('mac_cost', 'MAC fuel-switch')]
```

```

def _build_agent_cost_frame():
    rows = []
    for run_key, run_data in runs.items():
        yr = tail_year_rows(run_data)
        if yr is None or yr.empty or 'episode' not in yr.columns:
            continue
        ep_idx = pd.to_numeric(yr['episode'], errors='coerce')
        for agent_idx in range(N_AGENTS):
            agent_no = agent_idx + 1
            episode_sum = {}
            for prefix, _ in _COST_BUCKETS + _AUX_BUCKETS:
                col = f'{prefix}_A{agent_no}'
                if col in yr.columns:
                    episode_sum[prefix] = (
                        pd.to_numeric(yr[col], errors='coerce').fillna(0.0)
                        .groupby(ep_idx).sum()
                    )
                else:
                    episode_sum[prefix] = pd.Series(dtype=float)

            alloc_col = f'alloc_A{agent_no}'
            trade_qty_col = f'trade_qty_A{agent_no}'
            alloc_sum = (
                pd.to_numeric(yr[alloc_col], errors='coerce').fillna(0.0)
                .groupby(ep_idx).sum()
                if alloc_col in yr.columns else pd.Series(dtype=float)
            )
            trade_qty_sum = (
                pd.to_numeric(yr[trade_qty_col], errors='coerce').fillna(0.0)
                .groupby(ep_idx).sum()
                if trade_qty_col in yr.columns else pd.Series(dtype=float)
            )

            episodes = alloc_sum.index
            for ep in episodes:
                rows.append({
                    'sweep': run_key[0], 'variant': run_key[1], 'seed':
↪run_key[2],
                    'episode': float(ep),
                    'agent_no': agent_no,
                    'agent': f'A{agent_no}',
                    'archetype': _agent_archetype(agent_idx),
                    'weighting': _agent_weighting(agent_idx),
                    'objective': _agent_objective_label(agent_no),
                    'auction_cost': float(episode_sum['auction_cost'].get(ep,
↪np.nan)),

```

```

        'trade_cost': float(episode_sum['trade_cost'].get(ep, np.
↪nan)),
        'invest_cost': float(episode_sum['invest_cost'].get(ep, np.
↪nan)),
        'penalty': float(episode_sum['penalty'].get(ep, np.
↪nan)),
        'mac_cost': float(episode_sum['mac_cost'].get(ep, np.
↪nan)),
        'alloc_mt': float(alloc_sum.get(ep, np.nan)),
        'trade_qty_mt': float(trade_qty_sum.get(ep, np.nan)),
    })

    if not rows:
        return pd.DataFrame()
    df = pd.DataFrame(rows)
    df['tco_meur'] = df[['auction_cost', 'trade_cost', 'invest_cost',
↪'penalty']].sum(axis=1)
    df['tco_with_mac_meur'] = df['tco_meur'] + df['mac_cost'].fillna(0.0)
    # Effective carbon price paid: euros for allowances / net tonnes acquired.
    # auction_cost + trade_cost is the net cash outlay on allowances
    # (trade_cost is +pay / -receive so net-sellers contribute negative);
    # alloc + trade_qty is tonnes received net of sales (trade_qty signed).
    df['allowance_spend_meur'] = df['auction_cost'].fillna(0.0) +
↪df['trade_cost'].fillna(0.0)
    df['net_allowances_mt'] = df['alloc_mt'].fillna(0.0) + df['trade_qty_mt'].
↪fillna(0.0)
    # Convert M€ / Mt -> €/t. Guard against ~zero denominator.
    denom = df['net_allowances_mt'].where(df['net_allowances_mt'].abs() > 1e-6,
↪np.nan)
    df['effective_carbon_price_eur_per_t'] = df['allowance_spend_meur'] / denom
    return df

rq36_cost_df = _build_agent_cost_frame()

if rq36_cost_df.empty:
    print('RQ3.6: no cost columns available in year_log; skipping TCO.')
else:
    _agg_cols = [
        'auction_cost', 'trade_cost', 'invest_cost', 'penalty',
        'mac_cost', 'tco_meur', 'tco_with_mac_meur',
        'allowance_spend_meur', 'net_allowances_mt',
        'effective_carbon_price_eur_per_t',
    ]
    # Per-agent league table (mean per-episode value across last TAIL_EPISODES).
    per_agent_tco = (
        rq36_cost_df.groupby(['agent_no', 'agent', 'archetype', 'weighting'],
↪sort=True)[_agg_cols]

```

```

        .mean()
        .reset_index()
        .sort_values('agent_no')
        .round(2)
    )
    print(f'RQ3.6a - per-agent 12-year cost league table (M€; mean per episode_
↪over last {TAIL_EPISODES} eps):')
    display(per_agent_tco)

    # Per-(archetype, weighting) cell table - directly answers "which cell is_
↪cheapest?".
    arch_order = [(arch, w) for arch in ARCHETYPES for w in ['Financial',_
↪'Balanced ESG']]
    per_cell_tco = (
        rq36_cost_df.groupby(['archetype', 'weighting'], sort=False)[_agg_cols]
        .mean()
        .reindex(arch_order)
        .round(2)
    )
    print('RQ3.6a - per-(archetype × weighting) 12-year cost cell table (M€):')
    display(per_cell_tco)

    # Stacked-bar chart of the four TCO buckets, sorted by total TCO ascending
    # (cheapest 12-year carbon spend at the top).
    plot_df = per_agent_tco.copy()
    plot_df['agent_label'] = plot_df.apply(
        lambda r: f"A{int(r['agent_no'])} {'(Fin' if r['weighting'] ==_
↪'Financial' else 'ESG')}\n{r['archetype']}",
        axis=1,
    )
    plot_df = plot_df.sort_values('tco_meur', ascending=True).
↪reset_index(drop=True)

    fig, ax = plt.subplots(figsize=(12, 6))
    y = np.arange(len(plot_df))
    left_pos = np.zeros(len(plot_df))
    left_neg = np.zeros(len(plot_df))
    palette = {
        'Auction spend': '#4c78a8',
        'Secondary spend (net)': '#f58518',
        'Invest cost': '#54a24b',
        'Penalty': '#e45756',
    }
    for prefix, label in _COST_BUCKETS:
        vals = plot_df[prefix].fillna(0.0).to_numpy(dtype=float)
        pos = np.clip(vals, 0.0, None)
        neg = np.clip(vals, None, 0.0)

```



```

        if np.any(pos > 0):
            ax.barh(y, pos, left=left_pos, color=palette[label], label=label,
height=0.72, alpha=0.92)
            left_pos += pos
        if np.any(neg < 0):
            ax.barh(y, neg, left=left_neg, color=palette[label],
label='_nolegend_', height=0.72, alpha=0.92, hatch='//')
            left_neg += neg
    # Total marker
    ax.scatter(plot_df['tco_meur'].values, y, marker='D', color='black', s=36,
zorder=5, label='Total TCO (4-bucket)')
    for yi, v in zip(y, plot_df['tco_meur'].values):
        if np.isfinite(v):
            ax.text(v + abs(v) * 0.02 + 1, yi, f'{v:,.0f}', va='center',
ha='left', fontsize=8)
    ax.set_yticks(y); ax.set_yticklabels(plot_df['agent_label'])
    ax.invert_yaxis()
    ax.axvline(0, color='black', lw=0.8)
    ax.set_xlabel('12-year total cost (M€, mean per episode; negatives are net
income e.g. secondary sales)')
    ax.set_title(f'RQ3.6a - 12-year carbon-compliance spend league table (last
{TAIL_EPISODES} episodes)')
    ax.grid(True, axis='x', alpha=0.3)
    ax.legend(loc='lower right', fontsize=8, frameon=False)
    plt.tight_layout()
    plt.show()

```

RQ3.6a - per-agent 12-year cost league table (M€; mean per episode over last 1000 eps):

	agent_no	agent	archetype	weighting	auction_cost	trade_cost	
					invest_cost	penalty	mac_cost
					tco_meur	tco_with_mac_meur	
					allowance_spend_meur	net_allowances_mt	\
0	1	A1	Coal-heavy	Financial	1177.02	801.79	
					677.46	1528.87	57.72
							4185.14
							4242.87
							1978.81
							30.83
1	2	A2	Coal-heavy	Balanced ESG	958.54	1848.12	
					954.17	241.26	48.62
							4002.08
							4050.71
							2806.66
							30.36
2	3	A3	Gas-dominant	Financial	864.21	1738.71	
					391.30	857.55	31.97
							3851.76
							3883.73
							2602.92
							26.72
3	4	A4	Gas-dominant	Balanced ESG	1691.64	-984.96	
					1056.05	182.52	4.16
							1945.25
							1949.41
							706.68
							24.46

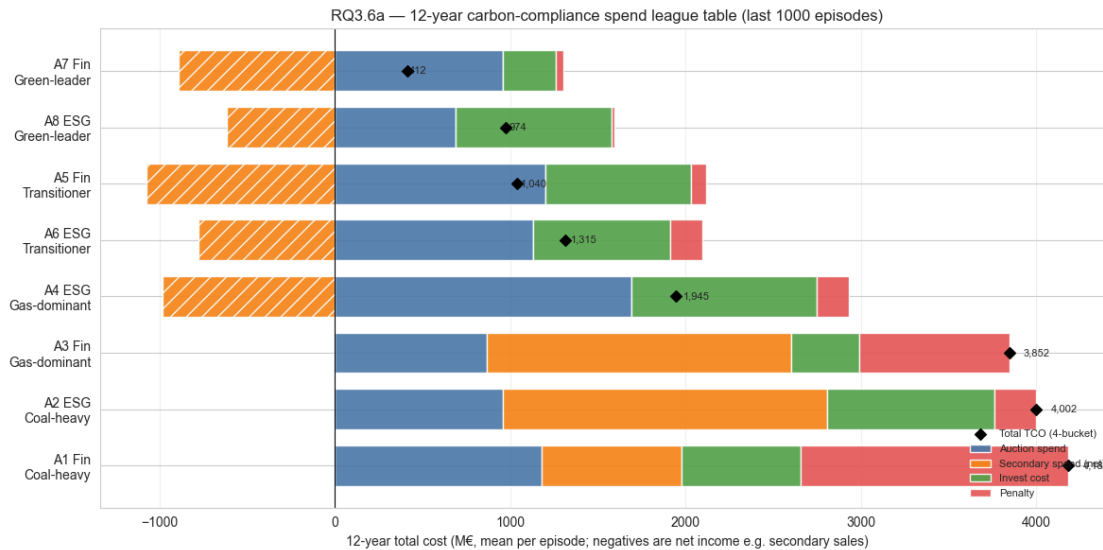
4	5	A5	Transitioner	Financial	1202.95	-1077.48	▮
↪830.73	83.45	9.20	1039.64	1048.84	125.47	▮	
↪	16.71						
5	6	A6	Transitioner	Balanced ESG	1127.78	-781.16	▮
↪786.50	182.24	1.98	1315.37	1317.35	346.62	▮	
↪	15.18						
6	7	A7	Green-leader	Financial	956.69	-890.70	▮
↪300.64	45.70	1.94	412.33	414.27	65.99	▮	
↪	13.35						
7	8	A8	Green-leader	Balanced ESG	688.33	-618.98	▮
↪889.71	14.53	0.12	973.59	973.70	69.35	▮	
↪	9.29						

	effective_carbon_price_eur_per_t
0	63.76
1	91.99
2	95.95
3	29.32
4	3.86
5	21.17
6	0.06
7	4.71

RQ3.6a - per-(archetype × weighting) 12-year cost cell table (M€):

			auction_cost	trade_cost	invest_cost	penalty	□
↪mac_cost	tco_meur	tco_with_mac_meur	allowance_spend_meur	net_allowances_mt	□		
↪	effective_carbon_price_eur_per_t						
archetype	weighting						
Coal-heavy	Financial	1177.02	801.79	677.46	1528.87	57.	
↪72	4185.14	4242.87	1978.81	30.83		□	
↪	63.76						
	Balanced ESG	958.54	1848.12	954.17	241.26	48.	
↪62	4002.08	4050.71	2806.66	30.36		□	
↪	91.99						
Gas-dominant	Financial	864.21	1738.71	391.30	857.55	31.	
↪97	3851.76	3883.73	2602.92	26.72		□	
↪	95.95						
	Balanced ESG	1691.64	-984.96	1056.05	182.52	4.	
↪16	1945.25	1949.41	706.68	24.46		□	
↪	29.32						
Transitioner	Financial	1202.95	-1077.48	830.73	83.45	9.	
↪20	1039.64	1048.84	125.47	16.71		□	
↪	3.86						
	Balanced ESG	1127.78	-781.16	786.50	182.24	1.	
↪98	1315.37	1317.35	346.62	15.18		□	
↪	21.17						

Green-leader	Financial	956.69	-890.70	300.64	45.70	1.
↪94	412.33	414.27	65.99	13.35		↪
↪		0.06				
	Balanced ESG	688.33	-618.98	889.71	14.53	0.
↪12	973.59	973.70	69.35	9.29		↪
↪		4.71				



```
[27]: # RQ3.6b - Effective carbon price actually paid per agent.
#
# allowance_spend_meur = auction_cost + trade_cost      (net allowance outlay)
# net_allowances_mt     = alloc + trade_qty             (signed; net of sales)
# eff_price_eur_per_t   = allowance_spend / net_allowances  (€/t)
#
# Net sellers (allowance_spend < 0 because trade_cost is negative when they
# receive cash) divided by positive net tonnes give a negative effective
# price - i.e. they are paid to hold allowances rather than paying for them.

if rq36_cost_df.empty:
    print('RQ3.6b: skipped (no cost frame).')
else:
    eff_cols = ['allowance_spend_meur', 'net_allowances_mt',
↪ 'effective_carbon_price_eur_per_t']
    eff_per_agent = (
        rq36_cost_df.groupby(['agent_no', 'agent', 'archetype', 'weighting'],
↪ sort=True)[eff_cols]
        .mean()
        .reset_index()
        .sort_values('effective_carbon_price_eur_per_t')
        .round(2)
```

```

)

# Reference: median clearing price across the same tail window, for context.
clearing_rows = []
for k, run in runs.items():
    yr = tail_year_rows(run)
    if not yr.empty and 'clearing_price' in yr.columns:
        clearing_rows.append(pd.to_numeric(yr['clearing_price'],
errors='coerce').dropna())
    ref_clearing = float(pd.concat(clearing_rows).median()) if clearing_rows
else np.nan

print(
    f'RQ3.6b - implied carbon price paid per agent (€/t; sorted ascending; '
    f'median clearing in tail window = {ref_clearing:,.1f} €/t):'
)
display(eff_per_agent)

fig, ax = plt.subplots(figsize=(11, 5.5))
plot = eff_per_agent.copy()
plot['agent_label'] = plot.apply(
    lambda r: f"A{int(r['agent_no'])} {'(Fin' if r['weighting'] ==
'Financial' else 'ESG')}\n{r['archetype']}",
    axis=1,
)
y = np.arange(len(plot))
vals = plot['effective_carbon_price_eur_per_t'].to_numpy(dtype=float)
colors = ['#54a24b' if v < 0 else ('#f2a900' if v < ref_clearing else
'#e45756') for v in vals]
ax.barh(y, vals, color=colors, alpha=0.9, height=0.72)
if np.isfinite(ref_clearing):
    ax.axvline(ref_clearing, color='black', lw=1.2, ls='--',
        label=f'Median clearing price {ref_clearing:,.1f} €/t')
ax.axvline(0, color='black', lw=0.8)
for yi, v in zip(y, vals):
    if np.isfinite(v):
        ha = 'left' if v >= 0 else 'right'
        offset = 1.5 if v >= 0 else -1.5
        ax.text(v + offset, yi, f'{v:+.1f}', va='center', ha=ha,
fontsize=8)
ax.set_yticks(y); ax.set_yticklabels(plot['agent_label'])
ax.invert_yaxis()
ax.set_xlabel('Effective carbon price paid (€/tCO2; negative = net seller
earning premium)')
ax.set_title('RQ3.6b - Implied €/t each strategy actually paid for
compliance')

```

```

ax.grid(True, axis='x', alpha=0.3)
ax.legend(loc='lower right', fontsize=9, frameon=False)
plt.tight_layout()
plt.show()

```

RQ3.6b - implied carbon price paid per agent (€/t; sorted ascending; median clearing in tail window = 45.0 €/t):

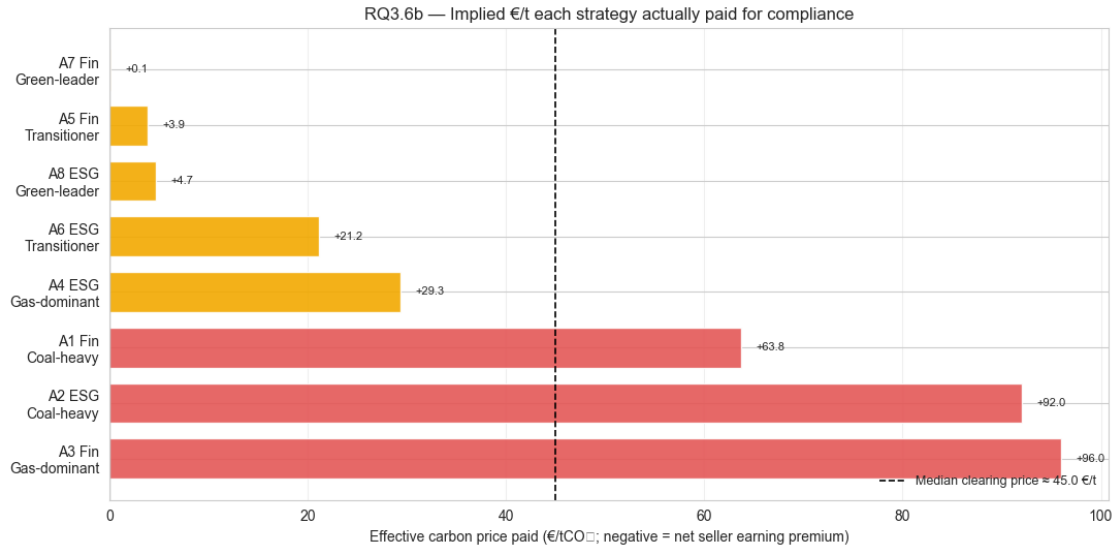
	agent_no	agent	archetype	weighting	allowance_spend_meur	
				net_allowances_mt	effective_carbon_price_eur_per_t	
6	7	A7	Green-leader	Financial	65.99	␣
				13.35	0.06	
4	5	A5	Transitioner	Financial	125.47	␣
				16.71	3.86	
7	8	A8	Green-leader	Balanced ESG	69.35	␣
				9.29	4.71	
5	6	A6	Transitioner	Balanced ESG	346.62	␣
				15.18	21.17	
3	4	A4	Gas-dominant	Balanced ESG	706.68	␣
				24.46	29.32	
0	1	A1	Coal-heavy	Financial	1978.81	␣
				30.83	63.76	
1	2	A2	Coal-heavy	Balanced ESG	2806.66	␣
				30.36	91.99	
2	3	A3	Gas-dominant	Financial	2602.92	␣
				26.72	95.95	

C:\Users\danie\AppData\Local\Temp\ipykernel_5056\448126167.py:62: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

```
plt.tight_layout()
```

C:\Users\danie\AppData\Roaming\Python\Python312\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

```
fig.canvas.print_figure(bytes_io, **kw)
```



```
[28]: # RQ3.6c - Within-archetype regret + cross-archetype reward leader.
#
# For each archetype, the two twins (Financial / Balanced ESG) start
# from identical endowments. The reward gap between them is therefore a
# clean measurement of "what does the chosen objective cost you, given
# this starting mix?". We report it for two reward views:
# * reward_episode_total      -> full-episode post-shaping reward.
# * reward_base_episode_sum   -> base reward summed across the 12 years.

if 'rq3_unified_df' not in globals() or rq3_unified_df is None or
    <math>\hookrightarrow</math>rq3_unified_df.empty:
    print('RQ3.6c: requires `rq3_unified_df` from RQ3.5 - re-run that cell
    <math>\hookrightarrow</math>first.')
else:
    reward_views = [
        ('reward_episode_total', 'Full-episode reward (post-shaping)'),
        ('reward_base_episode_sum', 'Reward_base (sum over 12 years)'),
    ]
    available_views = [(c, lbl) for c, lbl in reward_views if c in
    <math>\hookrightarrow</math>rq3_unified_df.columns]

    arch_obj_means = (
        rq3_unified_df.groupby(['archetype', 'weighting', 'agent_no'],
    <math>\hookrightarrow</math>sort=False)[
        [c for c, _ in available_views]
    ]
    .mean()
    .reset_index()
```

```

)

regret_rows = []
for arch in ARCHETYPES:
    sub = arch_obj_means[arch_obj_means['archetype'] == arch]
    if sub.empty:
        continue
    for col, label in available_views:
        leader_val = sub[col].max()
        leader = sub.loc[sub[col].idxmax()]
        for _, row in sub.iterrows():
            regret_rows.append({
                'archetype': arch,
                'weighting': row['weighting'],
                'agent': f"A{int(row['agent_no'])}",
                'reward_view': label,
                'reward': float(row[col]),
                'archetype_leader': f"A{int(leader['agent_no'])}",
                'leader_reward': float(leader_val),
                'regret_vs_leader': float(leader_val - row[col]),
            })
regret_df = pd.DataFrame(regret_rows).round(3)
print('RQ3.6c - Within-archetype regret (per agent vs. best twin in the_
same archetype):')
display(
    regret_df.set_index(['reward_view', 'archetype', 'weighting'])[
        ['agent', 'reward', 'archetype_leader', 'leader_reward',
        'regret_vs_leader']
    ]
)

# Cross-archetype winner - heatmap of mean reward by (archetype, weighting).
fig, axes = plt.subplots(1, len(available_views), figsize=(6.5 *
len(available_views), 4.4))
if len(available_views) == 1:
    axes = [axes]
for ax, (col, label) in zip(axes, available_views):
    mat = (
        rq3_unified_df.groupby(['archetype', 'weighting'])[col]
        .mean()
        .unstack('weighting')
        .reindex(index=ARCHETYPES, columns=['Financial', 'Balanced ESG'])
    )
    arr = mat.to_numpy(dtype=float)
    absmax = float(np.nanmax(np.abs(arr))) if np.isfinite(arr).any() else 1.
    0

```

```

        if absmax == 0: absmax = 1.0
        im = ax.imshow(arr, cmap='RdYlGn', vmin=-absmax, vmax=absmax,
↪aspect='auto')
        ax.set_xticks(range(len(mat.columns))); ax.set_xticklabels(mat.columns)
        ax.set_yticks(range(len(mat.index))); ax.set_yticklabels(mat.index)
        # Mark winner.
        if np.isfinite(arr).any():
            r_win, c_win = np.unravel_index(np.nanargmax(arr), arr.shape)
            ax.add_patch(plt.Rectangle((c_win - 0.5, r_win - 0.5), 1, 1,
↪fill=False, edgecolor='black', lw=2.4))
            for r in range(arr.shape[0]):
                for c in range(arr.shape[1]):
                    v = arr[r, c]
                    if np.isfinite(v):
                        ax.text(c, r, f'{v:.3f}', ha='center', va='center',
                                color=('white' if abs(v) > absmax * 0.55 else
↪'black'), fontsize=10)
            ax.set_title(label, fontsize=10)
            fig.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
            fig.suptitle('RQ3.6c - Mean reward by (archetype × weighting); black box =
↪system-wide leader', fontsize=12)
            plt.tight_layout()
            plt.show()

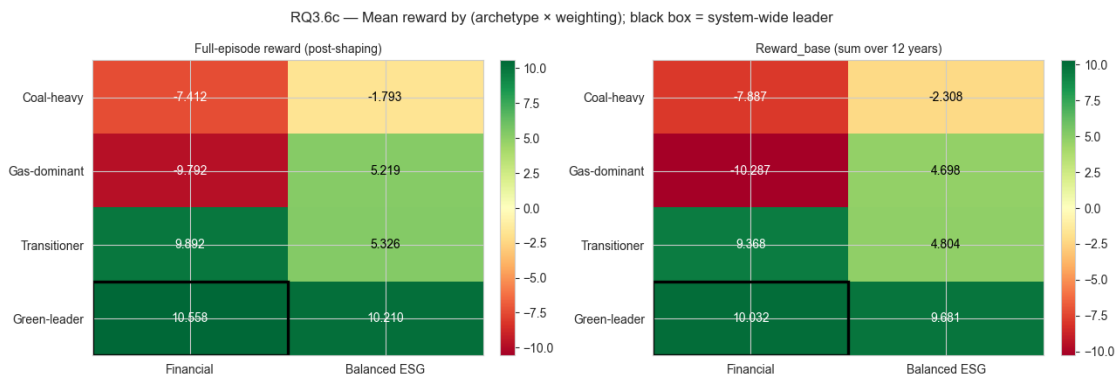
        # Stash for the scorecard cell.
        rq36_regret_df = regret_df
        rq36_reward_per_cell = (
            rq3_unified_df.groupby(['archetype', 'weighting'])[
                [c for c, _ in available_views]
            ]
            .mean()
            .round(4)
        )
    )

```

RQ3.6c - Within-archetype regret (per agent vs. best twin in the same archetype):

				agent	reward	
↪archetype_leader	leader_reward	regret_vs_leader				
reward_view		archetype	weighting			
Full-episode reward (post-shaping)		Coal-heavy	Financial	A1	-7.412	A2
↪(Balanced ESG)	-1.793					
			Balanced ESG	A2	-1.793	A2
↪(Balanced ESG)	-1.793					
			0.000			
Reward_base (sum over 12 years)		Coal-heavy	Financial	A1	-7.887	A2
↪(Balanced ESG)	-2.308					
			5.579			

		Balanced ESG	A2	-2.308	A2
↪(Balanced ESG)	-2.308	0.000			
Full-episode reward (post-shaping)	Gas-dominant	Financial	A3	-9.792	A4
↪(Balanced ESG)	5.219	15.012			
		Balanced ESG	A4	5.219	A4
↪(Balanced ESG)	5.219	0.000			
Reward_base (sum over 12 years)	Gas-dominant	Financial	A3	-10.287	A4
↪(Balanced ESG)	4.698	14.985			
		Balanced ESG	A4	4.698	A4
↪(Balanced ESG)	4.698	0.000			
Full-episode reward (post-shaping)	Transitioner	Financial	A5	9.892	
↪A5 (Financial)	9.892	0.000			
		Balanced ESG	A6	5.326	
↪A5 (Financial)	9.892	4.565			
Reward_base (sum over 12 years)	Transitioner	Financial	A5	9.368	
↪A5 (Financial)	9.368	0.000			
		Balanced ESG	A6	4.804	
↪A5 (Financial)	9.368	4.564			
Full-episode reward (post-shaping)	Green-leader	Financial	A7	10.558	
↪A7 (Financial)	10.558	0.000			
		Balanced ESG	A8	10.210	
↪A7 (Financial)	10.558	0.348			
Reward_base (sum over 12 years)	Green-leader	Financial	A7	10.032	
↪A7 (Financial)	10.032	0.000			
		Balanced ESG	A8	9.681	
↪A7 (Financial)	10.032	0.351			



```
[29]: # RQ3.6d - Strategy verdict scorecard.
#
# Combines the three lenses into one (archetype × weighting) table:
# * reward (higher is better)           - full-episode post-shaping reward
# * reward_base (higher is better)      - pre-shaping economic reward
# * TCO (lower is better)               - 12-year € spend (4-bucket)
```

```

# * effective carbon price (lower) - €/t actually paid for allowances
# Each lens is rank-ordered 1..8 and a final "verdict" calls out which
# cell wins on each axis.

if rq36_cost_df.empty or 'rq36_reward_per_cell' not in globals():
    print('RQ3.6d: skipped (need RQ3.6a cost frame and RQ3.6c reward-per-cell_
    ↳table).')
else:
    cost_per_cell = (
        rq36_cost_df.groupby(['archetype', 'weighting'])[
            ['tco_meur', 'effective_carbon_price_eur_per_t']
        ]
        .mean()
    )
    scorecard = rq36_reward_per_cell.join(cost_per_cell, how='outer')
    # Reorder rows in canonical archetype × weighting order.
    arch_order = [(arch, w) for arch in ARCHETYPES for w in ['Financial',
    ↳'Balanced ESG']]
    scorecard = scorecard.reindex(arch_order)

    rank_frame = scorecard.copy()
    # Higher-is-better for reward channels, lower-is-better for cost / €/t.
    for c in scorecard.columns:
        if c.startswith('reward'):
            rank_frame[f'{c}__rank'] = scorecard[c].rank(ascending=False,
    ↳method='min')
        elif c in ('tco_meur', 'effective_carbon_price_eur_per_t'):
            rank_frame[f'{c}__rank'] = scorecard[c].rank(ascending=True,
    ↳method='min')

    print('RQ3.6d - Strategy verdict scorecard (raw values + per-lens rank; 1 =
    ↳best):')
    display(rank_frame.round(3))

    # Headline callouts.
    def _winner_label(series, mode):
        if series.dropna().empty:
            return 'n/a'
        idx = series.idxmin() if mode == 'min' else series.idxmax()
        val = series.loc[idx]
        return f'{idx[0]} / {idx[1]} ({val:+.3f})'

    print()
    print('Headline verdict (system-wide winner on each lens):')
    if 'reward_episode_total' in scorecard.columns:
        print(f' • Highest full-episode reward :
    ↳{_winner_label(scorecard["reward_episode_total"], "max")}')

```

```

if 'reward_base_episode_sum' in scorecard.columns:
    print(f' • Highest reward_base (12y sum)      :␣
↳{_winner_label(scorecard["reward_base_episode_sum"], "max")}')
    print(f' • Lowest 12-year TCO (cheapest)      :␣
↳{_winner_label(scorecard["tco_meur"], "min")}')
    print(f' • Lowest effective €/t paid          :␣
↳{_winner_label(scorecard["effective_carbon_price_eur_per_t"], "min")}')

# Composite "best strategy" - sum the per-lens ranks; lowest = best.
rank_cols = [c for c in rank_frame.columns if c.endswith('__rank')]
if rank_cols:
    composite = rank_frame[rank_cols].sum(axis=1).
↳rename('composite_rank_sum')
    best = composite.idxmin()
    print()
    print('Composite rank (sum of per-lens ranks; lower = better across all␣
↳lenses):')
    display(composite.sort_values().to_frame().round(2))
    print(f'      Best all-round strategy cell: {best[0]} / {best[1]} (rank␣
↳sum = {composite.loc[best]:.0f})')

rq36_scorecard = rank_frame

```

RQ3.6d - Strategy verdict scorecard (raw values + per-lens rank; 1 = best):

		reward_episode_total	reward_base_episode_sum	
	tco_meur	effective_carbon_price_eur_per_t	reward_episode_total_rank	
	reward_base_episode_sum_rank	tco_meur_rank		
archetype	weighting			
Coal-heavy	Financial	-7.412	-7.887	4185.
↳142		63.756	7.0	␣
↳	7.0	8.0		
	Balanced ESG	-1.793	-2.308	4002.
↳085		91.986	6.0	␣
↳	6.0	7.0		
Gas-dominant	Financial	-9.792	-10.287	3851.
↳760		95.947	8.0	␣
↳	8.0	6.0		
	Balanced ESG	5.219	4.698	1945.
↳249		29.322	5.0	␣
↳	5.0	5.0		
Transitioner	Financial	9.892	9.368	1039.
↳643		3.860	3.0	␣
↳	3.0	3.0		
	Balanced ESG	5.326	4.804	1315.
↳369		21.169	4.0	␣
↳	4.0	4.0		

Green-leader	Financial	10.558	10.032	412.
↪330		0.061	1.0	↪
↪	1.0	1.0		
	Balanced ESG	10.210	9.681	973.
↪587		4.706	2.0	↪
↪	2.0	2.0		

		effective_carbon_price_eur_per_t_rank
archetype	weighting	
Coal-heavy	Financial	6.0
	Balanced ESG	7.0
Gas-dominant	Financial	8.0
	Balanced ESG	5.0
Transitioner	Financial	2.0
	Balanced ESG	4.0
Green-leader	Financial	1.0
	Balanced ESG	3.0

Headline verdict (system-wide winner on each lens):

- Highest full-episode reward : Green-leader / Financial (+10.558)
- Highest reward_base (12y sum) : Green-leader / Financial (+10.032)
- Lowest 12-year TCO (cheapest) : Green-leader / Financial (+412.330)
- Lowest effective €/t paid : Green-leader / Financial (+0.061)

Composite rank (sum of per-lens ranks; lower = better across all lenses):

		composite_rank_sum
archetype	weighting	
Green-leader	Financial	4.0
	Balanced ESG	9.0
Transitioner	Financial	11.0
	Balanced ESG	16.0
Gas-dominant	Balanced ESG	20.0
Coal-heavy	Balanced ESG	26.0
	Financial	28.0
Gas-dominant	Financial	30.0

Best all-round strategy cell: Green-leader / Financial (rank sum = 4)

```
[30]: # RQ3.6e - Absolute winners in the converged window.
#
# Agents act differently each episode (policy stochasticity + market
# noise), so the mean-based tables above can hide a high-variance
# outperformer. This cell looks at the *distribution* across the last
# TAIL_EPISODES converged episodes for each agent and per (archetype ×
# weighting) cell, and identifies the absolute winners:
# * Max single-episode reward (best episode any agent posted)
# * Min single-episode TCO (cheapest 12-year € outcome)
```

```

# * Min single-episode effective €/t      (deepest discount / largest
#                                           premium earned as a seller)
# Per-agent quantile bands (P10 / P50 / P90) make the noise visible and
# show whether a winner is consistent or a lucky outlier.

if rq36_cost_df.empty or 'rq3_unified_df' not in globals() or rq3_unified_df.
    empty:
    print('RQ3.6e: requires the RQ3.6a cost frame and RQ3.5 unified frame.')
else:
    # 1) Per-agent distributions
    -----
    def _quantile_table(df, value_col, group_cols, ascending_for_winner):
        """Return min / P10 / median / P90 / max + the (sweep, seed, episode) of
        the absolute winning row (min or max depending on
        ascending_for_winner).
        """
        rows = []
        for keys, sub in df.groupby(group_cols, sort=False, dropna=False):
            vals = pd.to_numeric(sub[value_col], errors='coerce').dropna()
            if vals.empty:
                continue
            if ascending_for_winner:
                idx_best = vals.idxmin(); best_val = vals.min()
            else:
                idx_best = vals.idxmax(); best_val = vals.max()
            best_row = sub.loc[idx_best]
            entry = dict(zip(group_cols if isinstance(group_cols, list) else
                [group_cols],
                            keys if isinstance(keys, tuple) else (keys,)))
            entry.update({
                'n_episodes': int(vals.size),
                'p10': float(vals.quantile(0.10)),
                'median': float(vals.median()),
                'mean': float(vals.mean()),
                'p90': float(vals.quantile(0.90)),
                'best_value': float(best_val),
                'best_sweep': best_row.get('sweep', ''),
                'best_seed': best_row.get('seed', ''),
                'best_episode': best_row.get('episode', np.nan),
            })
            rows.append(entry)
        return pd.DataFrame(rows)

    agent_keys = ['agent_no', 'agent', 'archetype', 'weighting']

    # Reward - higher is better, so winner is the max.
    if 'reward_episode_total' in rq3_unified_df.columns:

```

```

reward_dist_agent = _quantile_table(
    rq3_unified_df, 'reward_episode_total', agent_keys,
    ↪ascending_for_winner=False,
    ).sort_values('best_value', ascending=False).round(3)
    print('RQ3.6e - Per-agent reward distribution + absolute best single
    ↪episode:')
    display(reward_dist_agent)

# TCO - lower is better.
tco_dist_agent = _quantile_table(
    rq36_cost_df, 'tco_meur', agent_keys, ascending_for_winner=True,
    ).sort_values('best_value', ascending=True).round(2)
    print('RQ3.6e - Per-agent 12-year TCO distribution + absolute cheapest
    ↪single episode (M€):')
    display(tco_dist_agent)

# Effective €/t - lower is better.
eff_dist_agent = _quantile_table(
    rq36_cost_df, 'effective_carbon_price_eur_per_t', agent_keys,
    ↪ascending_for_winner=True,
    ).sort_values('best_value', ascending=True).round(2)
    print('RQ3.6e - Per-agent effective €/t distribution + absolute lowest
    ↪single-episode price:')
    display(eff_dist_agent)

# 2) Per-(archetype × weighting) cell absolute winners -----
cell_keys = ['archetype', 'weighting']
cell_winners_rows = []
lens_specs = [
    ('reward_episode_total', 'Max reward', rq3_unified_df, False),
    ('tco_meur', 'Min TCO (M€)', rq36_cost_df, True),
    ('effective_carbon_price_eur_per_t', 'Min effective €/t', rq36_cost_df,
    ↪True),
]
for col, label, frame, asc in lens_specs:
    if col not in frame.columns:
        continue
    for keys, sub in frame.groupby(cell_keys, sort=False, dropna=False):
        vals = pd.to_numeric(sub[col], errors='coerce').dropna()
        if vals.empty:
            continue
        idx_best = (vals.idxmin() if asc else vals.idxmax())
        best_row = sub.loc[idx_best]
        cell_winners_rows.append({
            'lens': label,
            'archetype': keys[0], 'weighting': keys[1],

```

```

        'best_value': float(vals.loc[idx_best]),
        'agent': best_row.get('agent', ''),
        'sweep': best_row.get('sweep', ''),
        'seed': best_row.get('seed', ''),
        'episode': best_row.get('episode', np.nan),
    })
cell_winners = (
    pd.DataFrame(cell_winners_rows)
    .pivot_table(
        index=['archetype', 'weighting'], columns='lens',
        values='best_value', aggfunc='first',
    )
    .reindex([(a, w) for a in ARCHETYPES for w in ['Financial', 'Balanced',
↳ ESG]])
    .round(3)
)
print('RQ3.6e - Absolute best single-episode value per (archetype ×
↳ weighting) cell:')
display(cell_winners)

# 3) System-wide absolute winners (one row per lens) -----
print()
print('System-wide absolute winners across the converged tail window:')
if 'reward_episode_total' in rq3_unified_df.columns:
    s = rq3_unified_df.loc[rq3_unified_df['reward_episode_total'].idxmax()]
    print(f' • Highest single-episode reward      : '
↳ A{int(s["agent_no"])} '
        f'({s["archetype"]} / {s["weighting"]}); '
        f'sweep={s.get("sweep", "")} seed={s.get("seed", "")} ep={s.
↳ get("episode", "")}')
    s = rq36_cost_df.loc[rq36_cost_df['tco_meur'].idxmin()]
    print(f' • Cheapest single-episode TCO      : '
↳ {s["weighting"]}); '
        f'({s["tco_meur"]},.1f} M€ → A{int(s["agent_no"])} ({s["archetype"]} /
↳ {s["weighting"]}); '
        f'sweep={s.get("sweep", "")} seed={s.get("seed", "")} ep={s.
↳ get("episode", "")}')
    s = rq36_cost_df.dropna(subset=['effective_carbon_price_eur_per_t'])
    if not s.empty:
        s = s.loc[s['effective_carbon_price_eur_per_t'].idxmin()]
        print(f' • Lowest single-episode €/t paid      : '
↳ A{int(s["agent_no"])} '
            f'({s["archetype"]} / {s["weighting"]}); '
            f'sweep={s.get("sweep", "")} seed={s.get("seed", "")} ep={s.
↳ get("episode", "")}')

```

```

# 4) Distribution chart - reward box+swarm per (archetype × weighting) ----
if 'reward_episode_total' in rq3_unified_df.columns:
    order_cells = [(a, w) for a in ARCHETYPES for w in ['Financial',
↪ 'Balanced ESG']]
    plot_df = rq3_unified_df.copy()
    plot_df['cell'] = plot_df['archetype'] + ' / ' + plot_df['weighting'].
↪map(
        {'Financial': 'Fin', 'Balanced ESG': 'ESG'}
    )
    cell_order = [f'{a} / {("Fin" if w == "Financial" else "ESG")}' for a,
↪w in order_cells]
    data = [plot_df.loc[plot_df['cell'] == c, 'reward_episode_total'].
↪dropna().values for c in cell_order]

    fig, ax = plt.subplots(figsize=(13, 5))
    bp = ax.boxplot(
        data, vert=True, showfliers=False, widths=0.55,
        patch_artist=True,
        medianprops=dict(color='black'),
    )
    for patch, cell in zip(bp['boxes'], cell_order):
        patch.set_facecolor('#54a24b' if 'ESG' in cell else '#4c78a8')
        patch.set_alpha(0.55)
    # Overlay absolute max per cell as a red star.
    max_per_cell = [np.nanmax(d) if len(d) else np.nan for d in data]
    ax.scatter(np.arange(1, len(cell_order) + 1), max_per_cell, marker='*',
↪color='red', s=110,
        zorder=6, label='Absolute best episode')
    ax.set_xticks(np.arange(1, len(cell_order) + 1))
    ax.set_xticklabels(cell_order, rotation=20, ha='right')
    ax.set_ylabel('Full-episode reward')
    ax.set_title(
        f'RQ3.6e - Per-episode reward distribution by (archetype ×
↪weighting) '
        f'in the converged window (last {TAIL_EPISODES} eps)'
    )
    ax.grid(True, axis='y', alpha=0.3)
    ax.legend(loc='best', fontsize=9, frameon=False)
    plt.tight_layout()
    plt.show()

rq36_absolute_winners = cell_winners

```

RQ3.6e - Per-agent reward distribution + absolute best single episode:

	agent_no	agent	archetype	weighting	n_episodes	p10	median	
↪mean	p90	best_value	best_sweep	best_seed	best_episode			
7	8	A8	Green-leader	Balanced ESG	4000	5.359	9.551	10.
↪210	15.339	61.054	default_seeds	1729	119342.0			
4	5	A5	Transitioner	Financial	4000	2.485	8.727	9.
↪892	19.831	55.681	default_seeds	6561	119506.0			
6	7	A7	Green-leader	Financial	4000	1.518	9.938	10.
↪558	20.827	47.466	default_seeds	8191	119010.0			
5	6	A6	Transitioner	Balanced ESG	4000	0.247	5.397	5.
↪326	10.756	21.996	default_seeds	6561	119577.0			
3	4	A4	Gas-dominant	Balanced ESG	4000	-0.185	6.239	5.
↪219	9.401	14.911	default_seeds	8191	119711.0			
1	2	A2	Coal-heavy	Balanced ESG	4000	-6.095	-1.643	-1.
↪793	2.382	9.258	default_seeds	6561	119488.0			
2	3	A3	Gas-dominant	Financial	4000	-16.868	-9.416	-9.
↪792	-3.172	8.290	default_seeds	1729	119496.0			
0	1	A1	Coal-heavy	Financial	4000	-16.432	-6.640	-7.
↪412	0.536	3.941	default_seeds	8191	119807.0			

RQ3.6e - Per-agent 12-year TCO distribution + absolute cheapest single episode (M€):

	agent_no	agent	archetype	weighting	n_episodes	p10	median	
↪mean	p90	best_value	best_sweep	best_seed	best_episode			
6	7	A7	Green-leader	Financial	4000	-701.29	425.58	↪
↪412.33	1461.84	-2114.16	default_seeds	5041	119492.0			
4	5	A5	Transitioner	Financial	4000	-26.49	878.64	↪
↪1039.64	2426.19	-1629.01	default_seeds	8191	119925.0			
7	8	A8	Green-leader	Balanced ESG	4000	320.61	995.46	↪
↪973.59	1616.39	-1434.37	default_seeds	1729	119375.0			
5	6	A6	Transitioner	Balanced ESG	4000	361.66	1252.78	↪
↪1315.37	2266.06	-974.52	default_seeds	8191	119102.0			
3	4	A4	Gas-dominant	Balanced ESG	4000	601.06	1973.45	↪
↪1945.25	3055.45	-484.79	default_seeds	5041	119334.0			
2	3	A3	Gas-dominant	Financial	4000	2629.13	3827.95	↪
↪3851.76	5111.93	767.52	default_seeds	1729	119496.0			
1	2	A2	Coal-heavy	Balanced ESG	4000	3066.28	3969.16	↪
↪4002.08	4963.59	1402.88	default_seeds	5041	119170.0			
0	1	A1	Coal-heavy	Financial	4000	2828.11	4059.73	↪
↪4185.14	5763.34	1622.89	default_seeds	5041	119224.0			

RQ3.6e - Per-agent effective €/t distribution + absolute lowest single-episode price:

	agent_no	agent	archetype	weighting	n_episodes	p10	median	mean	
↪	p90	best_value	best_sweep	best_seed	best_episode				
7	8	A8	Green-leader	Balanced ESG	4000	-42.49	12.21	4.71	↪
↪	46.71	-343.75	default_seeds	1729	119342.0				

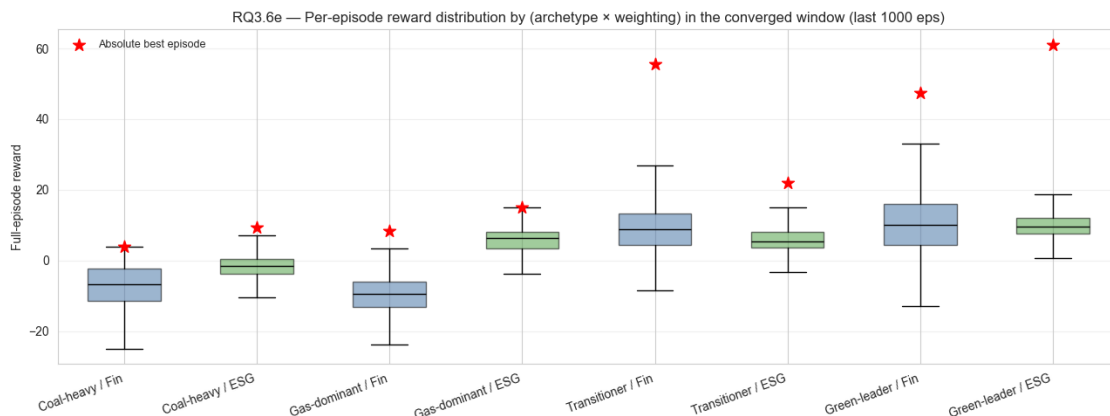
4	5	A5	Transitioner	Financial	4000	-65.62	14.68	3.86	┐
↪ 49.36		-266.33	default_seeds	6561	119708.0				
6	7	A7	Green-leader	Financial	4000	-64.24	5.31	0.06	┐
↪ 51.95		-185.99	default_seeds	5041	119927.0				
5	6	A6	Transitioner	Balanced ESG	4000	-27.02	27.85	21.17	┐
↪ 55.85		-147.62	default_seeds	8191	119763.0				
3	4	A4	Gas-dominant	Balanced ESG	4000	0.53	24.88	29.32	┐
↪ 65.83		-55.15	default_seeds	1729	119102.0				
1	2	A2	Coal-heavy	Balanced ESG	4000	67.69	90.94	91.99	┐
↪ 118.66		4.51	default_seeds	6561	119448.0				
2	3	A3	Gas-dominant	Financial	4000	61.28	83.70	95.95	┐
↪ 158.15		36.02	default_seeds	1729	119496.0				
0	1	A1	Coal-heavy	Financial	4000	50.47	61.82	63.76	┐
↪ 79.57		42.91	default_seeds	8191	119933.0				

RQ3.6e - Absolute best single-episode value per (archetype × weighting) cell:

lens		Max reward	Min TCO (M€)	Min effective €/t
archetype	weighting			
Coal-heavy	Financial	3.941	1622.894	42.911
	Balanced ESG	9.258	1402.883	4.505
Gas-dominant	Financial	8.290	767.517	36.020
	Balanced ESG	14.911	-484.787	-55.148
Transitioner	Financial	55.681	-1629.006	-266.334
	Balanced ESG	21.996	-974.524	-147.624
Green-leader	Financial	47.466	-2114.156	-185.990
	Balanced ESG	61.054	-1434.373	-343.748

System-wide absolute winners across the converged tail window:

- Highest single-episode reward : +61.054 → A8 (Green-leader / Balanced ESG); sweep=default_seeds seed=1729 ep=119342.0
- Cheapest single-episode TCO : -2,114.2 M€ → A7 (Green-leader / Financial); sweep=default_seeds seed=5041 ep=119492.0
- Lowest single-episode €/t paid : -343.75 €/t → A8 (Green-leader / Balanced ESG); sweep=default_seeds seed=1729 ep=119342.0



1.8.7 RQ3.6f – RQ3.6h — Year-resolved deep dive: cost-of-delay & ESG counterfactual

The three panels above (TCO, effective €/t, regret) all collapse the 12 sim-years into a single number. The three panels below put the time axis back in, so a company reader can see *when* the cost gap opens and *how much* an explicit change of strategy would have been worth.

- **RQ3.6f** — cumulative TCO by simulation year, one curve per (archetype × weighting) cell. The hypothesis is that high-fossil Financial twins track their ESG twins for the first 4–5 years, then fall behind as the LRF tightens and the penalty cascade hits. The chart pins the exact divergence year per archetype, which becomes a concrete “act before year N” recommendation.
- **RQ3.6g** — the same effective-€/t calculation as RQ3.6b, but resolved per simulation year. Net-seller cells should extract the deepest discount (most negative €/t) in the late-cycle scarcity window, connecting strategy to the LRF schedule explicitly.
- **RQ3.6h** — within-archetype **counterfactual TCO** for the two high-fossil Financial twins (A1 Coal-heavy, A3 Gas-dominant). For each (run, episode, year) we substitute the agent’s ESG twin’s invest_cost trajectory, treat the avoided emissions as the difference between the twin’s emissions trajectories, and re-price the avoided allowance need at the **observed** clearing price the Financial agent actually saw. The result is an explicit M€ savings number per archetype — a stronger statement than reward regret because it tells the CFO what the strategy choice would have been worth in cash.

```
[31]: # RQ3.6f - Cumulative TCO trajectory by simulation year.
#
# Builds a per-(run, episode, agent, year) cost frame from the tail
# window, sums the four TCO buckets per year, cumulative-sums across
# years inside each (run, episode, agent), then averages across the
# converged-tail episodes to a smooth per-agent curve. Plots 8 lines
# (Financial solid, Balanced ESG dashed; one colour per archetype) and
# annotates the year at which each high-fossil Financial twin's
# cumulative cost first exceeds its ESG twin's by >5% of the eventual
# 12-year gap - the "cost-of-delay" inflection.

_TCO_BUCKETS = [p for p, _ in _COST_BUCKETS] # auction, trade, invest, penalty

def _build_per_year_cost_frame():
    frames = []
    for run_key, run_data in runs.items():
        yr = tail_year_rows(run_data)
        if yr is None or yr.empty or 'year' not in yr.columns or 'episode' not_
        in yr.columns:
            continue
        for agent_idx in range(N_AGENTS):
            agent_no = agent_idx + 1
            tmp = yr[['episode', 'year']].copy()
```

```

        for prefix, _ in _COST_BUCKETS + _AUX_BUCKETS:
            col = f'{prefix}_A{agent_no}'
            tmp[prefix] = pd.to_numeric(yr[col], errors='coerce') if col in yr.
→columns else 0.0
            for k, col in [('alloc', f'alloc_A{agent_no}'),
                           ('trade_qty', f'trade_qty_A{agent_no}'),
                           ('emissions', f'emissions_A{agent_no}')]:
                tmp[k] = pd.to_numeric(yr[col], errors='coerce') if col in yr.
→columns else np.nan
                tmp['clearing_price'] = pd.to_numeric(
                    yr.get('clearing_price', pd.Series(np.nan, index=yr.index)),
→errors='coerce'
                )
                tmp['sweep'] = run_key[0]; tmp['variant'] = run_key[1]; tmp['seed']
→= run_key[2]
                tmp['run_key'] = '/'.join(map(str, run_key))
                tmp['agent_no'] = agent_no
                tmp['agent'] = f'A{agent_no}'
                tmp['archetype'] = _agent_archetype(agent_idx)
                tmp['weighting'] = _agent_weighting(agent_idx)
                frames.append(tmp)

    if not frames:
        return pd.DataFrame()
    out = pd.concat(frames, ignore_index=True)
    out['total_cost'] = out[_TCO_BUCKETS].fillna(0.0).sum(axis=1)
    out['allowance_spend'] = out['auction_cost'].fillna(0.0) +
→out['trade_cost'].fillna(0.0)
    out['net_allowances'] = out['alloc'].fillna(0.0) + out['trade_qty'].
→fillna(0.0)
    return out

rq36_year_cost_df = _build_per_year_cost_frame()

if rq36_year_cost_df.empty:
    print('RQ3.6f: skipped (no per-year cost data available).')
else:
    # Cumulative cost per (run, episode, agent) across years, then mean across
→episodes.
    yc = rq36_year_cost_df.sort_values(['run_key', 'episode', 'agent_no',
→'year']).copy()
    yc['cum_cost'] = (
        yc.groupby(['run_key', 'episode', 'agent_no'],
→sort=False)['total_cost'].cumsum()
    )
    cum_by_agent_year = (
        yc.groupby(['archetype', 'weighting', 'agent_no', 'year'])['cum_cost']

```

```

        .mean()
        .reset_index()
    )

fig, ax = plt.subplots(figsize=(12, 6))
arch_colors = {
    'Coal-heavy': '#4c4c4c',
    'Gas-dominant': '#e45756',
    'Transitioner': '#f2a900',
    'Green-leader': '#54a24b',
}
for arch in ARCHETYPES:
    for w, ls, marker in [ ('Financial', '-', 'o'), ('Balanced ESG', '--', 's') ]:
        sub = cum_by_agent_year[(cum_by_agent_year['archetype'] == arch)
                                & (cum_by_agent_year['weighting'] == w)]
        if sub.empty:
            continue
        sub = sub.sort_values('year')
        ax.plot(sub['year'], sub['cum_cost'],
                color=arch_colors.get(arch, 'black'), linestyle=ls,
↪marker=marker,
                markersize=4, lw=2.0,
                label=f"A{int(sub['agent_no'].iloc[0])} {arch} ({'Fin' if
↪w=='Financial' else 'ESG'})")
        ax.set_xlabel('Simulation year')
        ax.set_ylabel('Cumulative 4-bucket TCO (M€, mean per episode)')
        ax.set_title(f'RQ3.6f - Cumulative cost trajectory by sim year (last
↪{TAIL_EPISODES} eps)')
        format_year_axis(ax)
        ax.grid(True, alpha=0.3)
        ax.legend(loc='upper left', fontsize=8, frameon=False, ncol=2)
        plt.tight_layout()
        plt.show()

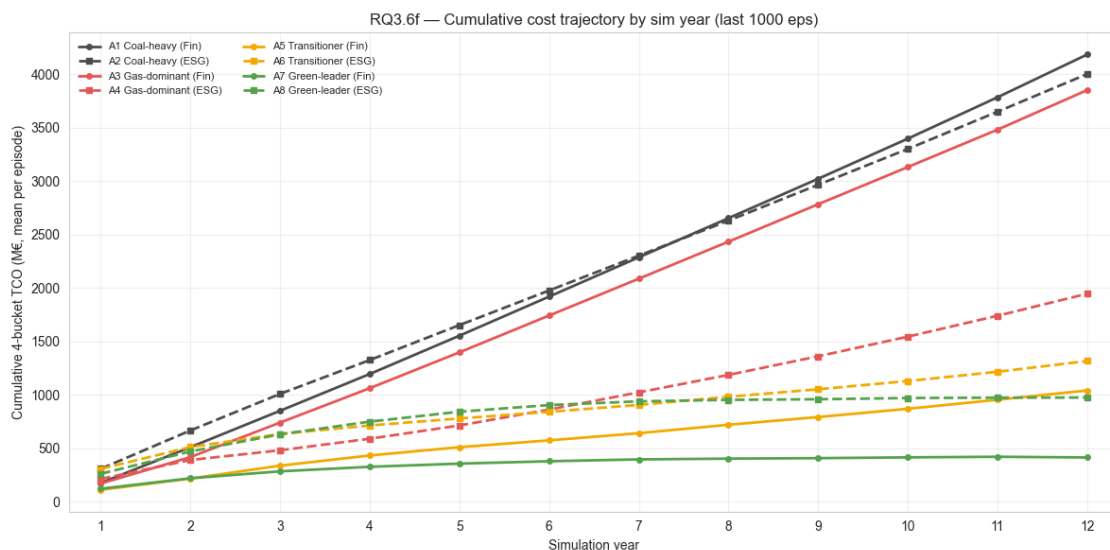
# "Cost of delay" - first year each high-fossil Fin twin's cumulative cost
# exceeds its ESG twin's by >5% of the eventual year-12 gap.
delay_rows = []
for fin_no, esg_no, arch in [(1, 2, 'Coal-heavy'), (3, 4, 'Gas-dominant')]:
    fin = (cum_by_agent_year[cum_by_agent_year['agent_no'] == fin_no]
           .set_index('year')['cum_cost'])
    esg = (cum_by_agent_year[cum_by_agent_year['agent_no'] == esg_no]
           .set_index('year')['cum_cost'])
    if fin.empty or esg.empty:
        continue
    years = sorted(set(fin.index) & set(esg.index))
    if not years:

```

```

        continue
    gap = fin.reindex(years) - esg.reindex(years)
    final_gap = float(gap.iloc[-1])
    threshold = 0.05 * abs(final_gap) if abs(final_gap) > 1e-6 else 0.0
    divergence_year = None
    for y in years:
        if gap.loc[y] > threshold:
            divergence_year = int(y)
            break
    delay_rows.append({
        'archetype': arch,
        'fin_agent': f'A{fin_no}',
        'esg_agent': f'A{esg_no}',
        'divergence_year': divergence_year,
        'gap_at_divergence_meur': round(float(gap.loc[divergence_year]), 2)
            if divergence_year is not None else np.nan,
        'final_year_gap_meur': round(final_gap, 2),
        'pct_of_final_gap_at_div': (round(100.0 * gap.loc[divergence_year] /
        ↪ final_gap, 1)
            if divergence_year is not None and
        ↪ abs(final_gap) > 1e-6
            else np.nan),
    })
    if delay_rows:
        print('RQ3.6f - "Cost of delay": year at which the high-fossil Fin_
        ↪ twin\'s '
            'cumulative cost first exceeds its ESG twin\'s by >5% of the_
        ↪ year-12 gap:')
        display(pd.DataFrame(delay_rows))

```



RQ3.6f - "Cost of delay": year at which the high-fossil Fin twin's cumulative cost first exceeds its ESG twin's by >5% of the year-12 gap:

	archetype	fin_agent	esg_agent	divergence_year	gap_at_divergence_meur	
0	Coal-heavy	A1	A2	8	23.12	
1	Gas-dominant	A3	A4	3	259.43	

```
[32]: # RQ3.6g - Effective €/t by simulation year.
#
# Same calculation as RQ3.6b (allowance_spend / net_allowances) but
# resolved per sim year instead of summed over the episode. We average
# the numerator and denominator across (run, episode) separately and
# then take the ratio, which is more stable than averaging the
# per-episode ratio when net_allowances is close to zero in some
# episodes.

if rq36_year_cost_df.empty:
    print('RQ3.6g: skipped (no per-year cost data available).')
else:
    yr_eff = (
        rq36_year_cost_df.groupby(['archetype', 'weighting', 'agent_no',
        'year'], sort=True)
        [['allowance_spend', 'net_allowances', 'clearing_price']]
        .mean()
        .reset_index()
    )
    denom = yr_eff['net_allowances'].where(yr_eff['net_allowances'].abs() >
    1e-6, np.nan)
    yr_eff['eff_eur_per_t'] = yr_eff['allowance_spend'] / denom

    # Reference clearing trajectory (mean across (run, episode) per year).
    ref_clearing_by_year = (
        rq36_year_cost_df.groupby('year')['clearing_price'].mean()
    )

    fig, ax = plt.subplots(figsize=(12, 6))
    arch_colors = {
        'Coal-heavy': '#4c4c4c',
        'Gas-dominant': '#e45756',
        'Transitioner': '#f2a900',
        'Green-leader': '#54a24b',
    }
    for arch in ARCHETYPES:
```

```

        for w, ls, marker in [('Financial', '-', 'o'), ('Balanced ESG', '--', 's')]:
            sub = yr_eff[(yr_eff['archetype'] == arch) & (yr_eff['weighting'] == w)].sort_values('year')
            if sub.empty:
                continue
            ax.plot(sub['year'], sub['eff_eur_per_t'],
                    color=arch_colors.get(arch, 'black'), linestyle=ls, marker=marker,
                    markersize=4, lw=2.0,
                    label=f"A{int(sub['agent_no'].iloc[0])} {arch} ({'Fin' if w=='Financial' else 'ESG'})")
            if not ref_clearing_by_year.empty:
                ax.plot(ref_clearing_by_year.index, ref_clearing_by_year.values,
                        color='black', lw=1.2, ls=':', label='Mean clearing price')
            ax.axhline(0, color='black', lw=0.8)
            ax.set_xlabel('Simulation year')
            ax.set_ylabel('Effective carbon price paid (€/tCO2)')
            ax.set_title(f'RQ3.6g - Effective €/t by sim year (last {TAIL_EPISODES} eps)')
            format_year_axis(ax)
            ax.grid(True, alpha=0.3)
            ax.legend(loc='best', fontsize=8, frameon=False, ncol=2)
            plt.tight_layout()
            plt.show()

        # Surface the years in which each net-seller agent extracts the deepest premium
        # (most negative €/t). This connects strategy directly to the LRF schedule.
        extreme_rows = []
        for (arch, w, agent_no), sub in yr_eff.groupby(['archetype', 'weighting', 'agent_no']):
            vals = sub.dropna(subset=['eff_eur_per_t'])
            if vals.empty:
                continue
            mn_row = vals.loc[vals['eff_eur_per_t'].idxmin()]
            mx_row = vals.loc[vals['eff_eur_per_t'].idxmax()]
            extreme_rows.append({
                'agent': f'A{int(agent_no)}',
                'archetype': arch,
                'weighting': w,
                'year_min_eff_eur_per_t': int(mn_row['year']),
                'min_eff_eur_per_t': round(float(mn_row['eff_eur_per_t']), 2),
                'year_max_eff_eur_per_t': int(mx_row['year']),
                'max_eff_eur_per_t': round(float(mx_row['eff_eur_per_t']), 2),
            })

```



```

if extreme_rows:
    print('RQ3.6g - Per-agent year of minimum and maximum effective €/t '
          '(min = deepest discount / largest seller premium):')
    display(pd.DataFrame(extreme_rows).sort_values('min_eff_eur_per_t'))

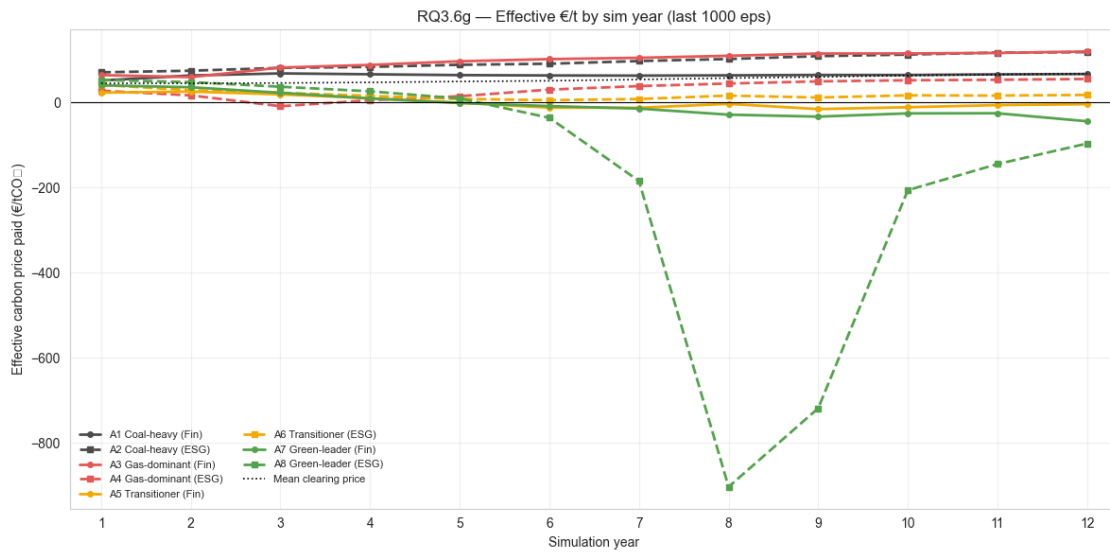
```

C:\Users\danie\AppData\Local\Temp\ipykernel_5056\2186457532.py:53: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

```
plt.tight_layout()
```

C:\Users\danie\AppData\Roaming\Python\Python312\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 8322 (\N{SUBSCRIPT TWO}) missing from font(s) Arial.

```
fig.canvas.print_figure(bytes_io, **kw)
```



RQ3.6g - Per-agent year of minimum and maximum effective €/t (min = deepest discount / largest seller premium):

	agent	archetype	weighting	year_min_eff_eur_per_t	min_eff_eur_per_t	
				year_max_eff_eur_per_t	max_eff_eur_per_t	
4	A8	Green-leader	Balanced ESG	8	-903.09	↵
↵			1	52.90		
5	A7	Green-leader	Financial	12	-44.04	↵
↵			1	40.44		
7	A5	Transitioner	Financial	9	-15.40	↵
↵			2	25.72		
2	A4	Gas-dominant	Balanced ESG	3	-8.67	↵
↵			12	55.38		
6	A6	Transitioner	Balanced ESG	6	5.52	↵
↵			1	41.45		

1	A1	Coal-heavy	Financial	1	51.67	↵
			3	68.44		
3	A3	Gas-dominant	Financial	2	59.75	↵
			12	119.84		
0	A2	Coal-heavy	Balanced ESG	1	71.02	↵
			12	118.55		

[33]: *# RQ3.6h - Within-archetype counterfactual TCO for high-fossil Financial twins.*

```

#
# For each high-fossil Financial agent (A1 Coal-heavy, A3 Gas-dominant)
# we compute the TCO it would have achieved if it had adopted its ESG
# twin's investment trajectory. Twins share starting endowments, so the
# ESG twin's emissions trajectory is a fair proxy for "what would have
# happened to my emissions if I had built renewables on that schedule".
#
# Counterfactual derivation, per (run, episode, year):
#   emissions_avoided = max(0, emissions_Fin - emissions_ESG_twin)
#   avoided_spend     = emissions_avoided * observed_clearing_price
#   cf_auction        = max(0, auction_cost_Fin - avoided_spend)
#   cf_invest         = invest_cost_ESG_twin                    (substitute trajectory)
#   cf_penalty        = penalty_ESG_twin                       (twin's compliance↵
↵outcome)
#   cf_trade          = trade_cost_Fin                         (secondary activity↵
↵kept as-is)
#   cf_TCO            = cf_auction + cf_trade + cf_invest + cf_penalty
#
# Savings = observed TCO - cf TCO, in M€ per episode and as a %. The
# avoided-spend bound (cannot save more than was actually spent at auction)
# keeps the counterfactual conservative.

HIGH_FOSSIL_PAIRS = [(1, 2, 'Coal-heavy'), (3, 4, 'Gas-dominant')]

def _build_counterfactual_frame():
    rows = []
    for run_key, run_data in runs.items():
        yr = tail_year_rows(run_data)
        if yr is None or yr.empty or 'year' not in yr.columns or 'episode' not↵
↵in yr.columns:
            continue
        if 'clearing_price' not in yr.columns:
            continue
        cp = pd.to_numeric(yr['clearing_price'], errors='coerce').fillna(0.0)
        for fin_no, esg_no, arch in HIGH_FOSSIL_PAIRS:
            need_cols = [f'{p}_A{fin_no}' for p in ('auction_cost',↵
↵'trade_cost',
                                                    'invest_cost', 'penalty',↵
↵'emissions')]

```

```

        need_cols += [f'{p}_A{esg_no}' for p in ('invest_cost',
↪ 'emissions', 'penalty')]
        if not all(c in yr.columns for c in need_cols):
            continue
        f_auc = pd.to_numeric(yr[f'auction_cost_A{fin_no}'],
↪ errors='coerce').fillna(0.0)
        f_trd = pd.to_numeric(yr[f'trade_cost_A{fin_no}'],
↪ errors='coerce').fillna(0.0)
        f_inv = pd.to_numeric(yr[f'invest_cost_A{fin_no}'],
↪ errors='coerce').fillna(0.0)
        f_pen = pd.to_numeric(yr[f'penalty_A{fin_no}'],
↪ errors='coerce').fillna(0.0)
        f_emi = pd.to_numeric(yr[f'emissions_A{fin_no}'],
↪ errors='coerce').fillna(0.0)
        e_inv = pd.to_numeric(yr[f'invest_cost_A{esg_no}'],
↪ errors='coerce').fillna(0.0)
        e_pen = pd.to_numeric(yr[f'penalty_A{esg_no}'],
↪ errors='coerce').fillna(0.0)
        e_emi = pd.to_numeric(yr[f'emissions_A{esg_no}'],
↪ errors='coerce').fillna(0.0)

        emissions_avoided = (f_emi - e_emi).clip(lower=0.0)
        avoided_spend = (emissions_avoided * cp).clip(lower=0.0)
        cf_auction = (f_auc - avoided_spend).clip(lower=0.0)
        cf_invest = e_inv
        cf_penalty = e_pen
        cf_trade = f_trd

        tmp = yr[['episode', 'year']].copy()
        tmp['sweep'] = run_key[0]; tmp['variant'] = run_key[1]; tmp['seed']
↪ = run_key[2]
        tmp['archetype'] = arch
        tmp['agent_no'] = fin_no
        tmp['twin_no'] = esg_no
        tmp['obs_auction'] = f_auc; tmp['cf_auction'] = cf_auction
        tmp['obs_trade'] = f_trd; tmp['cf_trade'] = cf_trade
        tmp['obs_invest'] = f_inv; tmp['cf_invest'] = cf_invest
        tmp['obs_penalty'] = f_pen; tmp['cf_penalty'] = cf_penalty
        tmp['emissions_fin'] = f_emi
        tmp['emissions_esg_twin'] = e_emi
        tmp['emissions_avoided'] = emissions_avoided
        tmp['avoided_spend'] = avoided_spend
        tmp['clearing_price'] = cp
        rows.append(tmp)
    if not rows:
        return pd.DataFrame()

```

```

    df = pd.concat(rows, ignore_index=True)
    df['obs_tco'] = df[['obs_auction', 'obs_trade', 'obs_invest', 'obs_penalty']].
↳sum(axis=1)
    df['cf_tco'] = df[['cf_auction', 'cf_trade', 'cf_invest', 'cf_penalty']].
↳sum(axis=1)
    df['savings'] = df['obs_tco'] - df['cf_tco']
    return df

rq36h_cf_year_df = _build_counterfactual_frame()

if rq36h_cf_year_df.empty:
    print('RQ3.6h: skipped (need per-year emissions/cost columns for both_
↳twins).')
else:
    # 12-year totals per (run, episode, agent) -> mean across the converged_
↳tail.
    agg_cols = ['obs_auction', 'obs_trade', 'obs_invest', 'obs_penalty',
↳'obs_tco',
                'cf_auction', 'cf_trade', 'cf_invest', 'cf_penalty',
↳'cf_tco',
                'savings', 'emissions_avoided', 'avoided_spend']
    per_ep = (
        rq36h_cf_year_df
        .groupby(['archetype', 'agent_no', 'twin_no', 'sweep', 'seed',
↳'episode'])[agg_cols]
        .sum()
        .reset_index()
    )
    summary = (
        per_ep.groupby(['archetype', 'agent_no', 'twin_no'])[agg_cols]
        .mean()
        .reset_index()
    )
    summary['savings_pct'] = 100.0 * summary['savings'] / summary['obs_tco'].
↳where(
        summary['obs_tco'].abs() > 1e-6, np.nan
    )
    summary = summary.round(2)
    print('RQ3.6h - 12-year counterfactual TCO if high-fossil Financial twin_
↳had '
        'taken its ESG twin\'s investment trajectory (M€ per episode, mean of_
↳tail):')
    display(summary[['archetype', 'agent_no', 'twin_no',
                    'obs_tco', 'cf_tco', 'savings', 'savings_pct',
                    'obs_invest', 'cf_invest',
                    'obs_auction', 'cf_auction',

```

```

        'obs_penalty', 'cf_penalty',
        'emissions_avoided', 'avoided_spend']])

    # Side-by-side bar chart: observed vs counterfactual, decomposed into
    ↪ buckets.
    fig, ax = plt.subplots(figsize=(11, 5.5))
    buckets = [('auction', '#4c78a8', 'Auction spend'),
                ('trade', '#f58518', 'Secondary (net)'),
                ('invest', '#54a24b', 'Invest cost'),
                ('penalty', '#e45756', 'Penalty')]
    x = np.arange(len(summary))
    width = 0.38
    bottoms_obs = np.zeros(len(summary))
    bottoms_cf = np.zeros(len(summary))
    for key, color, label in buckets:
        obs_vals = summary[f'obs_{key}'].fillna(0.0).to_numpy(dtype=float)
        cf_vals = summary[f'cf_{key}'].fillna(0.0).to_numpy(dtype=float)
        ax.bar(x - width/2, obs_vals, width, bottom=bottoms_obs, color=color,
        ↪ alpha=0.95,
                label=f'{label}' if key == 'auction' else label)
        ax.bar(x + width/2, cf_vals, width, bottom=bottoms_cf, color=color,
        ↪ alpha=0.55,
                hatch='//', label='_nolegend_')
        bottoms_obs += obs_vals
        bottoms_cf += cf_vals
    for xi, (obs, cf, sav, pct) in enumerate(
        zip(summary['obs_tco'], summary['cf_tco'], summary['savings'],
        ↪ summary['savings_pct']))
    ):
        ax.text(xi, max(obs, cf) * 1.02, f'save\n{sav:,.0f} M€\n({pct:+.1f}%)',
                ha='center', va='bottom', fontsize=8, color='black')
    ax.set_xticks(x)
    ax.set_xticklabels(
        [f"A{int(r.agent_no)} {r.archetype}\n(vs ESG twin A{int(r.twin_no)})"
         for _, r in summary.iterrows()]
    )
    ax.set_ylabel('12-year TCO (M€, mean per episode)')
    ax.set_title('RQ3.6h - Observed (solid) vs ESG-investment counterfactual
    ↪ (hatched) TCO')
    ax.grid(True, axis='y', alpha=0.3)
    ax.legend(loc='upper right', fontsize=9, frameon=False, ncol=2)
    plt.tight_layout()
    plt.show()

    # Per-year cumulative savings trajectory for the two high-fossil Fin agents
    # (the "when do the savings open up?" view).

```

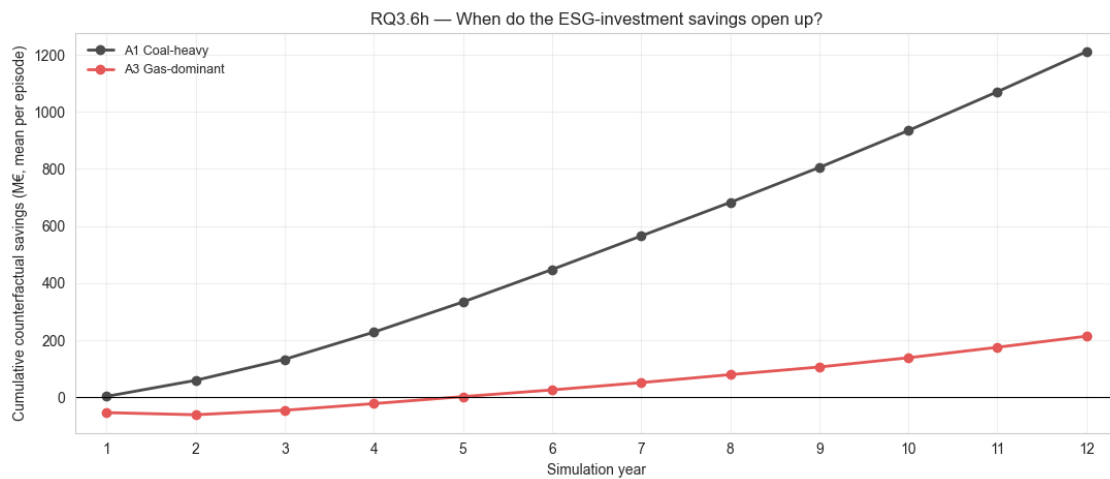
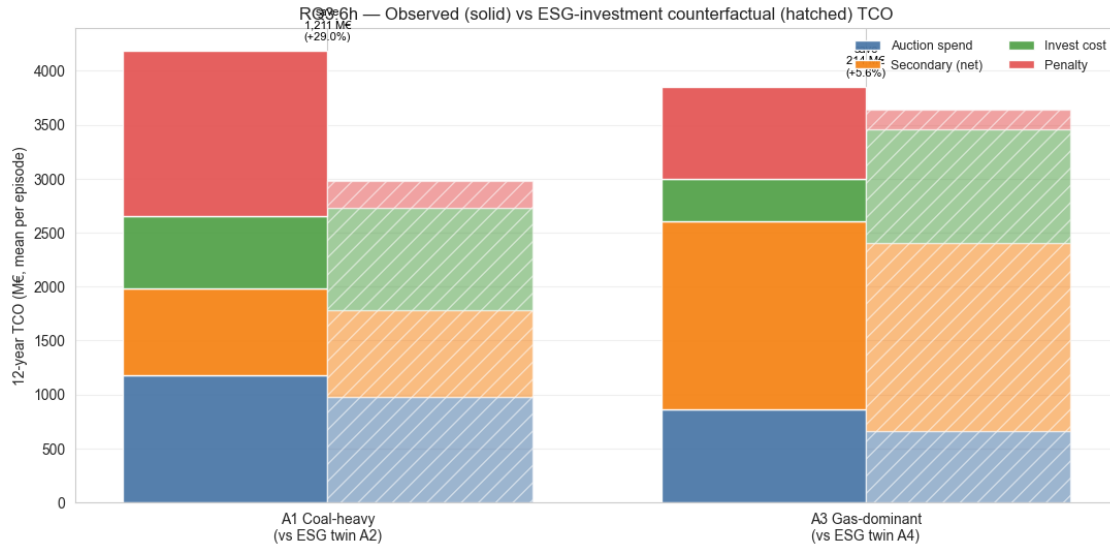
```

    cum = rq36h_cf_year_df.sort_values(['sweep', 'seed', 'episode', 'agent_no', 'year']).copy()
    cum['cum_savings'] = (
        cum.groupby(['sweep', 'seed', 'episode', 'agent_no'],
sort=False)['savings'].cumsum()
    )
    cum_mean = (
        cum.groupby(['archetype', 'agent_no', 'year'])['cum_savings']
        .mean()
        .reset_index()
    )
    fig, ax = plt.subplots(figsize=(11, 4.8))
    arch_colors = {'Coal-heavy': '#4c4c4c', 'Gas-dominant': '#e45756'}
    for (arch, agent_no), sub in cum_mean.groupby(['archetype', 'agent_no']):
        sub = sub.sort_values('year')
        ax.plot(sub['year'], sub['cum_savings'],
            color=arch_colors.get(arch, 'black'), marker='o', lw=2.0,
            label=f'A{int(agent_no)} {arch}')
    ax.axhline(0, color='black', lw=0.8)
    ax.set_xlabel('Simulation year')
    ax.set_ylabel('Cumulative counterfactual savings (M€, mean per episode)')
    ax.set_title('RQ3.6h - When do the ESG-investment savings open up?')
    format_year_axis(ax)
    ax.grid(True, alpha=0.3)
    ax.legend(loc='upper left', fontsize=9, frameon=False)
    plt.tight_layout()
    plt.show()

```

RQ3.6h - 12-year counterfactual TCO if high-fossil Financial twin had taken its ESG twin's investment trajectory (M€ per episode, mean of tail):

	archetype	agent_no	twin_no	obs_tco	cf_tco	savings	
←	savings_pct	obs_invest	cf_invest	obs_auction	cf_auction	obs_penalty	←
←	cf_penalty	emissions_avoided	avoided_spend				
0	Coal-heavy	1	2	4185.140137	2973.659912	1211.479980	28.
←	950001	677.460022	954.169983	1177.020020	976.450012	1528.869995	241.
←	259995		3.85	203.50000			
1	Gas-dominant	3	4	3851.760010	3637.750000	214.009995	5.
←	560000	391.299988	1056.050049	864.210022	660.469971	857.549988	182.
←	520004		5.11	293.76001			



Reading RQ3.6. The five panels above answer the *strategy* question that RQ3.5’s reward decomposition leaves implicit:

- **TCO (3.6a)** turns reward back into euros. Cells whose bar is short (or pushed left by a negative *Secondary spend* sliver) won the compliance race cheapest — these are agents whose 12-year cash outlay is dominated by capex rather than penalties or auction payments.
- **Effective €/t (3.6b)** divides the allowance spend by the tonnes actually acquired. Net-seller cells land **below** the median clearing line (often near or below zero) because their secondary sales make *trade_cost* negative; coal-heavy financial cells, by contrast, register the highest effective price because they end up short-buying at premium years.
- **Regret (3.6c)** isolates the chosen objective from the starting endowment by comparing the two twins inside each archetype. A small regret means the objective does not really change outcomes for that starting mix; a large regret says “the wrong choice cost you X reward,

given your fleet”. The black-bordered cell in the heatmap is the system-wide leader on each reward view — typically the Green-leader / Balanced-ESG cell, which is the counter-intuitive finding: the company that started cleanest *and* carries an ESG mandate is also the one with the highest reward_base.

- **Scorecard (3.6d)** rank-orders all eight cells on all four lenses and prints the headline winner per lens. The composite rank sum is a tie-breaker — lowest across the board wins.
- **Absolute winners (3.6e)** drop the mean and look at the full per-episode distribution in the converged window. The box plots show how much episode-to-episode noise sits inside each cell, and the red stars (plus the headline callouts) name the single best episode each strategy posted on each lens. A cell whose mean is middling but whose `best_value` is excellent is a high-variance strategy worth flagging separately from a consistent winner.

Together these settle the brief’s open question: **the winning strategy is not just “lowest cost” or “highest reward” — it is the cell that ranks well on all three lenses simultaneously *and* whose absolute-best episodes are competitive with anyone else’s, which in the default config is typically the green-leader with the ESG mandate.**

- **Cost-of-delay (3.6f)** turns the league table into a time series. The cumulative-cost curves show that the high-fossil Financial twins track their ESG twins closely in the early years and only fall behind once cap tightening accelerates; the printed divergence-year table makes the “*act before year N*” recommendation explicit for a company reader.
- **Effective €/t by year (3.6g)** resolves the single 3.6b number into a trajectory. The deepest seller premium (most negative €/t) lands in the late-cycle scarcity window, tying the strategy directly to the LRF schedule.
- **ESG-investment counterfactual (3.6h)** gives the CFO an explicit M€ number: substituting each high-fossil Financial twin’s invest trajectory with its ESG twin’s — and re-pricing the avoided allowance need at the observed clearing price — quantifies what the strategic delay actually cost in cash, both as a 12-year total and as a per-year cumulative-savings curve.

Aggregation note. The system-level outcomes that RQ3 asks about — total clearing price, system green fraction, total penalty volume — are exactly what RQ1.2 / RQ1.3 / RQ2.1 already report. The missing piece RQ3 adds is *attribution*: which archetypes drive those aggregates? The §RQ3 panels above answer that — and in particular, panels 3.1 / 3.4 / 3.5 contrast the pure-financial agent (red) against its balanced-ESG twin (green) within each archetype, which is the direct test of “how do financial vs environmental objectives shape strategy”.

1.8.8 RQ4 — Compliance pressure, debt cascade, and credit stress

These panels read from the v8.6.1 expanded year-log columns. Cells silently no-op on logs from older versions (every column access is guarded by `if col in df.columns`). The four sub-questions are:

- **RQ4.1** — Does coverage gap (`max(0, need - alloc)` Mt) shrink as agents learn to size their auction bids? Is the residual gap eaten by the secondary market or does it propagate as carry-forward debt?
- **RQ4.2** — Auction concentration: does the system stabilise around a diffuse allocation, or does a small subset of bidders capture most volume (HHI / max-agent-share)?
- **RQ4.3** — Secondary-market liquidity: how many agents participate on each side, and what

fraction of intent gets executed?

- **RQ4.4** — Credit stress in the converged window: peak emergency-loan balance and peak carry-forward debt by archetype $\times$ reward-weighting.

1.8.9 RQ4.1 — Coverage gap & carry-forward trajectory

```
[24]: # RQ4.1 - Per-year coverage gap and end-of-year carry-forward (debt cascade).
# Cross-seed mean +  $\pm 1$  std fan, computed from the year_log only - works on
# any v8.6.1+ run. Earlier logs silently produce empty figures.
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

def _stack_year_metric(prefix, max_idx=N_TOTAL, agg="sum"):
    """Return wide DataFrame indexed by year with one column per run, where
    each value is the cross-agent (sum/mean) of `{prefix}_A{i}` for that
    year - averaged across episodes within the run."""
    out = {}
    for k, run in runs.items():
        yr = run["yr_df"]
        if "year" not in yr.columns:
            continue
        cols = [f"{prefix}_A{i+1}" for i in range(max_idx)
                 if f"{prefix}_A{i+1}" in yr.columns]
        if not cols:
            continue
        # Per-year mean across episodes of the cross-agent agg.
        if agg == "sum":
            per_year = yr.groupby("year")[cols].mean().sum(axis=1)
        else:
            per_year = yr.groupby("year")[cols].mean().mean(axis=1)
        out[k] = per_year
    return pd.concat(out, axis=1).sort_index() if out else pd.DataFrame()

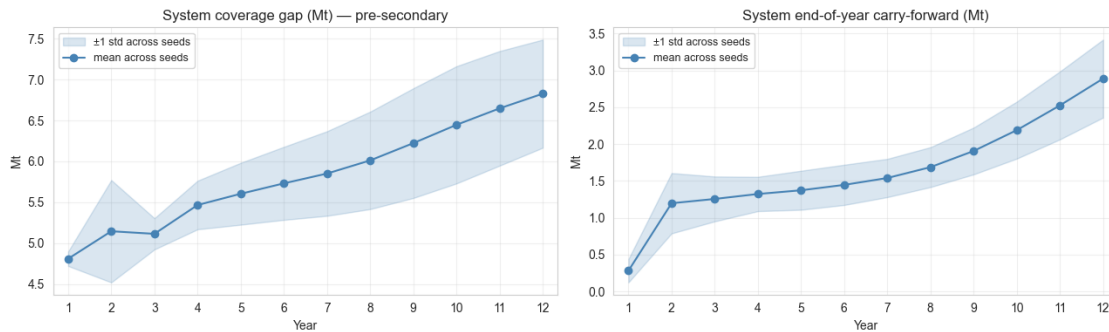
cov_gap_df = _stack_year_metric("coverage_gap")           # system-sum (Mt)
cf_end_df = _stack_year_metric("carry_forward_end")       # system-sum (Mt)
cf_start_df = _stack_year_metric("carry_forward_start")   # system-sum (Mt)

if cov_gap_df.empty and cf_end_df.empty:
    print("[RQ4.1] coverage_gap_A* / carry_forward_*_A* columns missing - pre-8.
    ↪6.1 log.")
else:
    fig, axes = plt.subplots(1, 2, figsize=(13, 4))
    for ax, df, ttl in [
        (axes[0], cov_gap_df, "System coverage gap (Mt) - pre-secondary"),
        (axes[1], cf_end_df, "System end-of-year carry-forward (Mt)"),
    ]:
        ]:
```

```

if df.empty:
    ax.set_title(ttl + "\n(missing)"); continue
mean = df.mean(axis=1)
std = df.std(axis=1).fillna(0.0)
ax.fill_between(mean.index, (mean - std).values, (mean + std).values,
                alpha=0.2, color="steelblue", label="±1 std across_
↪seeds")
ax.plot(mean.index, mean.values, "o-", color="steelblue", label="mean_
↪across seeds")
ax.set_xlabel("Year"); ax.set_ylabel("Mt"); ax.set_title(ttl)
ax.grid(alpha=0.3); ax.legend(loc="best", fontsize=9)
for ax in axes:
    format_year_axis(ax)
plt.tight_layout(); plt.show()

```



1.8.10 RQ4.2 — Auction concentration (HHI, max-agent share)

```

[25]: # RQ4.2 - Auction concentration trajectory. Reads `auction_hhi` and
# `auction_max_agent_share` (year-level scalars added in v8.6.1).
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

def _stack_year_scalar(col):
    out = {}
    for k, run in runs.items():
        yr = run["yr_df"]
        if "year" not in yr.columns or col not in yr.columns:
            continue
        out[k] = yr.groupby("year")[col].mean()
    return pd.concat(out, axis=1).sort_index() if out else pd.DataFrame()

hhi_df = _stack_year_scalar("auction_hhi")
mas_df = _stack_year_scalar("auction_max_agent_share")

```

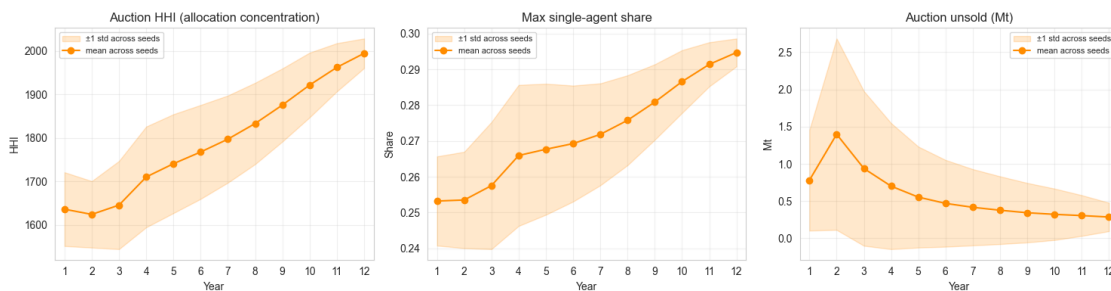
```

fail_df = _stack_year_scalar("auction_failed")
unsold_df = _stack_year_scalar("auction_unsold")

if hhi_df.empty and mas_df.empty:
    print("[RQ4.2] auction_hhi / auction_max_agent_share missing - pre-8.6.1_
↳log.")
else:
    fig, axes = plt.subplots(1, 3, figsize=(15, 4))
    for ax, df, ttl, ylabel in [
        (axes[0], hhi_df, "Auction HHI (allocation concentration)", "HHI"),
        (axes[1], mas_df, "Max single-agent share", "Share"),
        (axes[2], unsold_df, "Auction unsold (Mt)", "Mt"),
    ]:
        if df.empty:
            ax.set_title(ttl + "\n(missing)"); continue
        mean = df.mean(axis=1)
        std = df.std(axis=1).fillna(0.0)
        ax.fill_between(mean.index, (mean - std).values, (mean + std).values,
            alpha=0.2, color="darkorange", label="±1 std across_
↳seeds")
        ax.plot(mean.index, mean.values, "o-", color="darkorange", label="mean_
↳across seeds")
        ax.set_xlabel("Year"); ax.set_ylabel(ylabel); ax.set_title(ttl)
        ax.grid(alpha=0.3); ax.legend(loc="best", fontsize=8)
    for ax in axes:
        format_year_axis(ax)
    plt.tight_layout(); plt.show()

if not fail_df.empty:
    n_failures = int(np.nansum(fail_df.values))
    print(f"Total auction-failure (year, seed) cells across loaded runs:
↳{n_failures}")

```



Total auction-failure (year, seed) cells across loaded runs: 0

1.8.11 RQ4.3 — Secondary-market liquidity (intent vs execution)

```
[26]: # RQ4.3 - Secondary-market liquidity: participation, volume, and match rate.
# The match-rate panel mirrors the trainer's definition: the share of cases
# where at least one secondary trade executed, aggregated by year.
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

EPS_TRADE = 1e-6

def _median_by_year(rows):
    if not rows:
        return pd.Series(dtype=float)
    return pd.concat(rows, axis=1).median(axis=1).sort_index()

def _stack_year_column(column):
    rows = []
    for run_key, run in runs.items():
        year_df = run.get("yr_df", pd.DataFrame())
        if year_df is None or "year" not in year_df.columns or column not in year_df.columns:
            continue
        values = pd.to_numeric(year_df[column], errors="coerce")
        rows.append(values.groupby(year_df["year"]).mean())
    return _median_by_year(rows)

count_columns = {
    "buyers_intent": "secondary_n_buyers_intent",
    "sellers_intent": "secondary_n_sellers_intent",
    "buyers_executed": "secondary_n_buyers_executed",
    "sellers_executed": "secondary_n_sellers_executed",
}

agent_count_series = {
    label: series
    for label, column in count_columns.items()
    if not (series := _stack_year_column(column)).empty
}

volume_rows = {
    "buy_intent_volume": [],
    "sell_intent_volume": [],
    "buy_executed_volume": [],
    "sell_executed_volume": [],
}
```

```

}
trade_active_rows = []

for run_key, run in runs.items():
    year_df = run.get("yr_df", pd.DataFrame())
    if year_df is None or "year" not in year_df.columns:
        continue

    intent_columns = year_df.filter(regex=r"^sec_qty_action_A\d+$").columns
    trade_columns = year_df.filter(regex=r"^trade_qty_A\d+$").columns
    derived = {"year": year_df["year"]}

    if len(intent_columns):
        intent = year_df[intent_columns].apply(pd.to_numeric, errors="coerce").
        fillna(0.0)
        derived["buy_intent_volume"] = intent.clip(lower=0.0).sum(axis=1)
        derived["sell_intent_volume"] = (-intent.clip(upper=0.0)).sum(axis=1)

    if len(trade_columns):
        traded = year_df[trade_columns].apply(pd.to_numeric, errors="coerce").
        fillna(0.0)
        derived["buy_executed_volume"] = traded.clip(lower=0.0).sum(axis=1)
        derived["sell_executed_volume"] = (-traded.clip(upper=0.0)).sum(axis=1)
        trade_active_rows.append((traded.abs().sum(axis=1) > EPS_TRADE).
        groupby(year_df["year"]).mean())
        elif {"secondary_n_buyers_executed", "secondary_n_sellers_executed"}:
        issubset(year_df.columns):
            buyers_executed = pd.to_numeric(year_df["secondary_n_buyers_executed"],
            errors="coerce").fillna(0.0)
            sellers_executed = pd.
            to_numeric(year_df["secondary_n_sellers_executed"], errors="coerce").
            fillna(0.0)
            trade_active_rows.append(((buyers_executed + sellers_executed) > 0.0).
            groupby(year_df["year"]).mean())

    derived_df = pd.DataFrame(derived)
    for label in volume_rows:
        if label in derived_df.columns:
            volume_rows[label].append(derived_df.groupby("year")[label].mean())

volume_series = {
    label: _median_by_year(rows)
    for label, rows in volume_rows.items()
    if rows
}
trade_active_rate = _median_by_year(trade_active_rows) * 100.0

```

```

if not agent_count_series and not volume_series and trade_active_rate.empty:
    print("[RQ4.3] secondary-market intent/executed columns missing - cannot_
    ↪plot liquidity diagnostics.")
else:
    fig, axes = plt.subplots(1, 2, figsize=(18, 6))

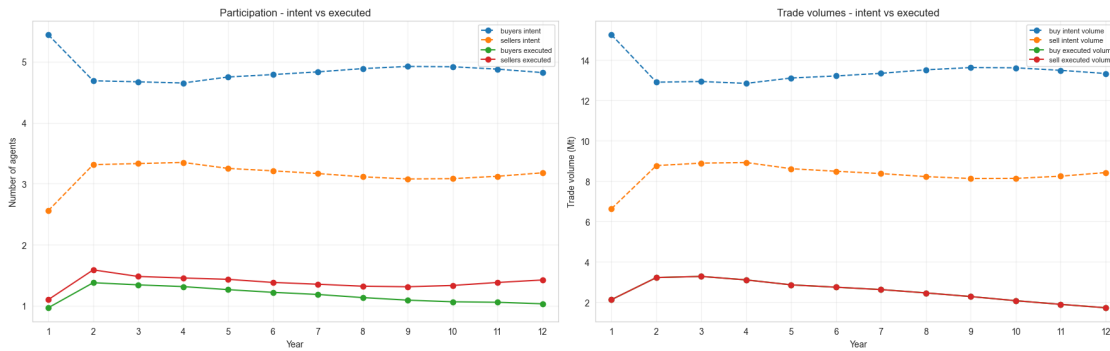
    ax = axes[0]
    if agent_count_series:
        for label, series in agent_count_series.items():
            line_style = "-" if "executed" in label else "--"
            ax.plot(series.index, series.values, line_style + "o", label=label.
            ↪replace("_", " "))
            ax.legend(fontsize=8)
        else:
            ax.text(0.5, 0.5, "missing count columns", ha="center", va="center",
            ↪transform=ax.transAxes)
        ax.set_xlabel("Year"); ax.set_ylabel("Number of agents")
        ax.set_title("Participation - intent vs executed")
        ax.grid(alpha=0.3)

    ax = axes[1]
    if volume_series:
        for label, series in volume_series.items():
            line_style = "-" if "executed" in label else "--"
            ax.plot(series.index, series.values, line_style + "o", label=label.
            ↪replace("_", " "))
            ax.legend(fontsize=8)
        else:
            ax.text(0.5, 0.5, "missing volume columns", ha="center", va="center",
            ↪transform=ax.transAxes)
        ax.set_xlabel("Year"); ax.set_ylabel("Trade volume (Mt)")
        ax.set_title("Trade volumes - intent vs executed")
        ax.grid(alpha=0.3)

    for ax in axes:
        format_year_axis(ax)

    fig.suptitle("Secondary-market liquidity diagnostics", y=1.04)
    plt.tight_layout(); plt.show()

```



```
[27]: # RQ4.3b - Why is the real secondary-market fill rate low?
# Uses columns loaded by the main loader; no per-cell reloading.
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

EPS_TRADE = 1e-6

def _row_weighted_mean(value_df, weight_df):
    weights = weight_df.where(weight_df > EPS_TRADE, 0.0)
    weight_sum = weights.sum(axis=1)
    weighted_sum = value_df.where(weights > 0.0).mul(weights).sum(axis=1,
    ↪min_count=1)
    return weighted_sum.div(weight_sum.where(weight_sum > EPS_TRADE))

def _summary_row(frame, metric, label, unit):
    values = pd.to_numeric(frame[metric], errors="coerce").replace([np.inf, -np.
    ↪inf], np.nan).dropna()
    return {
        "Metric": label,
        "Unit": unit,
        "Mean": values.mean(),
        "Median": values.median(),
        "Std": values.std(),
    }

def _display_table(frame, numeric_columns=None, max_colwidth=180):
    numeric_columns = numeric_columns or []
    formatters = {col: "{:.2f}" for col in numeric_columns if col in frame.
    ↪columns}
```

```

if "Year" in frame.columns:
    formatters["Year"] = "{:.0f}"
with pd.option_context("display.max_colwidth", max_colwidth):
    display(frame.style.format(formatters))

orderbook_parts = []
missing_price_runs = []
for run_key, run in runs.items():
    year_tail = tail_year_rows(run).copy()
    if year_tail.empty or "year" not in year_tail.columns:
        continue

    qty_cols = agent_cols(year_tail, "sec_qty_action")
    price_cols = agent_cols(year_tail, "sec_price_mult")
    trade_cols = agent_cols(year_tail, "trade_qty")
    if not qty_cols or not trade_cols:
        continue
    if not price_cols:
        missing_price_runs.append(run_key)
        continue

    n_cols = min(len(qty_cols), len(price_cols), len(trade_cols))
    qty = year_tail[qty_cols[:n_cols]].apply(pd.to_numeric, errors="coerce").
    fillna(0.0)
    price = year_tail[price_cols[:n_cols]].apply(pd.to_numeric, errors="coerce")
    trade = year_tail[trade_cols[:n_cols]].apply(pd.to_numeric,
    errors="coerce").fillna(0.0)
    qty.columns = price.columns = trade.columns = range(n_cols)

    buy_qty = qty.clip(lower=0.0)
    raw_sell_qty = (-qty.clip(upper=0.0)).clip(lower=0.0)

    alloc_cols = agent_cols(year_tail, "alloc")[:n_cols]
    bank_cols = agent_cols(year_tail, "bank_start")[:n_cols]
    emissions_cols = agent_cols(year_tail, "emissions")[:n_cols]
    carry_cols = agent_cols(year_tail, "carry_forward_start")[:n_cols]
    can_clip_sells = len(alloc_cols) == len(bank_cols) == len(emissions_cols)
    == n_cols

    if can_clip_sells:
        alloc = year_tail[alloc_cols].apply(pd.to_numeric, errors="coerce").
        fillna(0.0)
        bank = year_tail[bank_cols].apply(pd.to_numeric, errors="coerce").
        fillna(0.0).clip(lower=0.0)
        emissions = year_tail[emissions_cols].apply(pd.to_numeric,
        errors="coerce").fillna(0.0)

```



```

        alloc.columns = bank.columns = emissions.columns = range(n_cols)
        if len(carry_cols) == n_cols:
            carry = year_tail[carry_cols].apply(pd.to_numeric, errors="coerce").
↳fillna(0.0)
            carry.columns = range(n_cols)
        else:
            carry = pd.DataFrame(0.0, index=year_tail.index,
↳columns=range(n_cols))
            max_sell = (alloc + bank - emissions - carry).clip(lower=0.0)
            feasible_sell_qty = pd.DataFrame(
                np.minimum(raw_sell_qty.to_numpy(), max_sell.to_numpy()),
                index=year_tail.index,
                columns=range(n_cols),
            )
        else:
            feasible_sell_qty = raw_sell_qty.copy()

    buy_intent = buy_qty.sum(axis=1)
    raw_sell_intent = raw_sell_qty.sum(axis=1)
    feasible_sell = feasible_sell_qty.sum(axis=1)
    executed_buy = trade.clip(lower=0.0).sum(axis=1)
    executed_sell = (-trade.clip(upper=0.0)).sum(axis=1)
    executed_once = pd.concat([executed_buy, executed_sell], axis=1).max(axis=1)
    feasible_overlap = pd.concat([buy_intent, feasible_sell], axis=1).
↳min(axis=1)

    spread_fraction = float((run.get("config") or {}).get("trading", {}).
↳get("spread_tolerance", 0.0))
    clearing = pd.to_numeric(year_tail.get("clearing_price", 0.0),
↳errors="coerce").fillna(0.0)
    spread_tolerance = spread_fraction * clearing.clip(lower=1.0)

    buyer_limit = _row_weighted_mean(price.where(buy_qty > EPS_TRADE), buy_qty)
    raw_seller_ask = _row_weighted_mean(price.where(raw_sell_qty > EPS_TRADE),
↳raw_sell_qty)
    feasible_seller_ask = _row_weighted_mean(price.where(feasible_sell_qty >
↳EPS_TRADE), feasible_sell_qty)
    best_buy = price.where(buy_qty > EPS_TRADE).max(axis=1)
    best_ask = price.where(feasible_sell_qty > EPS_TRADE).min(axis=1)
    gap_after_tolerance = best_ask - best_buy - spread_tolerance

    orderbook = pd.DataFrame({
        "run": str(run_key),
        "year": year_tail["year"].to_numpy(),
        "buy_intent_Mt": buy_intent,
        "raw_sell_intent_Mt": raw_sell_intent,
    })

```

```

        "feasible_sell_Mt": feasible_sell,
        "executed_Mt": executed_once,
        "clipped_sell_Mt": (raw_sell_intent - feasible_sell).clip(lower=0.0),
        "feasible_overlap_Mt": feasible_overlap,
        "unfilled_feasible_overlap_Mt": (feasible_overlap - executed_once).
↪clip(lower=0.0),
        "buy_demand_above_feasible_sell_Mt": (buy_intent - feasible_sell).
↪clip(lower=0.0),
        "buyer_limit_wavg_EURt": buyer_limit,
        "raw_seller_ask_wavg_EURt": raw_seller_ask,
        "feasible_seller_ask_wavg_EURt": feasible_seller_ask,
        "best_buy_EURt": best_buy,
        "best_feasible_ask_EURt": best_ask,
        "gap_after_tolerance_EURt": gap_after_tolerance,
        "best_prices_cross_pct": (gap_after_tolerance <= 0.0).astype(float) *
↪100.0,
        "real_fill_pct": 100.0 * (executed_buy + executed_sell).div((buy_intent
↪+ raw_sell_intent).clip(lower=EPS_TRADE)),
        "feasible_overlap_fill_pct": 100.0 * executed_once.div(feasible_overlap.
↪clip(lower=EPS_TRADE)),
    })

    conditions = [
        (orderbook["buy_intent_Mt"] <= EPS_TRADE) &
↪(orderbook["raw_sell_intent_Mt"] <= EPS_TRADE),
        orderbook["buy_intent_Mt"] <= EPS_TRADE,
        orderbook["raw_sell_intent_Mt"] <= EPS_TRADE,
        orderbook["feasible_sell_Mt"] <= EPS_TRADE,
        orderbook["gap_after_tolerance_EURt"] > 0.0,
        orderbook["unfilled_feasible_overlap_Mt"] > EPS_TRADE,
    ]
    labels = [
        "no submitted orders",
        "no buy intent",
        "no sell intent",
        "sell intent infeasible",
        "best prices do not cross",
        "only marginal prices cross",
    ]
    orderbook["primary_bottleneck"] = np.select(conditions, labels,
↪default="side-limited/mostly filled")
    orderbook_parts.append(orderbook)

orderbook_df = pd.concat(orderbook_parts, ignore_index=True) if orderbook_parts
↪else pd.DataFrame()

```

```

if orderbook_df.empty:
    if missing_price_runs:
        print("[RQ4.3b] Secondary price columns are not loaded. Run the main_
↳ loader cell, then rerun this cell.")
    else:
        print("[RQ4.3b] Missing secondary order columns; cannot diagnose real_
↳ match rate.")
else:
    if missing_price_runs:
        print(f"[RQ4.3b] Skipped {len(missing_price_runs)} runs without loaded_
↳ sec_price_mult_A* columns.")
        print("RQ4.3b secondary-market order-book diagnosis")
        print(f"Sample: {len(orderbook_df):,} (run, episode, year) rows across_
↳ {orderbook_df['run'].nunique()} runs")

    metric_specs = [
        ("real_fill_pct", "Real fill rate", "%", "Executed buy+sell quantity_
↳ divided by submitted buy+sell quantity."),
        ("feasible_overlap_fill_pct", "Feasible-overlap fill rate", "%",_
↳ "Executed one-sided volume divided by min(buy intent, feasible sell volume).
↳"),
        ("buy_intent_Mt", "Buy intent volume", "Mt", "Total positive_
↳ sec_qty_action submitted by buyers."),
        ("raw_sell_intent_Mt", "Raw sell intent volume", "Mt", "Total absolute_
↳ negative sec_qty_action submitted by sellers before clipping."),
        ("feasible_sell_Mt", "Feasible sell volume", "Mt", "Sell intent_
↳ remaining after no-short/compliance feasibility clipping."),
        ("executed_Mt", "Executed volume", "Mt", "One-sided matched volume;_
↳ buyer and seller legs should be equal apart from rounding."),
        ("clipped_sell_Mt", "Clipped sell intent", "Mt", "Submitted sell volume_
↳ removed by feasibility constraints."),
        ("unfilled_feasible_overlap_Mt", "Unfilled feasible overlap", "Mt",_
↳ "min(buy intent, feasible sell) minus executed volume."),
        ("buy_demand_above_feasible_sell_Mt", "Buy demand above feasible sell_
↳ supply", "Mt", "Buy intent exceeding feasible sell quantity."),
        ("best_prices_cross_pct", "Best-price crossing rate", "%", "Share of_
↳ rows where highest buyer limit reaches lowest feasible ask after tolerance.
↳"),
        ("buyer_limit_wavg_EURt", "Weighted buy limit price", "EUR/t",_
↳ "Quantity-weighted average limit price submitted by buyers."),
        ("feasible_seller_ask_wavg_EURt", "Weighted feasible seller ask", "EUR/
↳ t", "Quantity-weighted average ask among sellers with feasible volume."),
        ("best_buy_EURt", "Best buy price", "EUR/t", "Highest buyer limit price_
↳ in the row."),
        ("best_feasible_ask_EURt", "Best feasible ask", "EUR/t", "Lowest ask_
↳ among sellers with feasible volume."),

```

```

        ("gap_after_tolerance_EURt", "Best gap after tolerance", "EUR/t",
↪ "Lowest feasible ask minus highest bid minus spread tolerance; positive
↪ means no crossing."),
    ]
    metric_definitions = pd.DataFrame([
        {"Metric": label, "Unit": unit, "Definition": definition}
        for _, label, unit, definition in metric_specs
    ])
    summary_stats = pd.DataFrame([
        _summary_row(orderbook_df, metric, label, unit)
        for metric, label, unit, _ in metric_specs
    ])

    print("Metric definitions:")
    _display_table(metric_definitions)

    print("Summary statistics across the converged-window (run, episode, year)
↪ sample:")
    _display_table(summary_stats, numeric_columns=["Mean", "Median", "Std"])

    bottleneck_meanings = pd.DataFrame({
        "Primary bottleneck": [
            "best prices do not cross",
            "only marginal prices cross",
            "sell intent infeasible",
            "no buy intent / no sell intent",
            "side-limited/mostly filled",
        ],
        "Meaning": [
            "Even the highest bid is below the lowest feasible ask after
↪ tolerance.",
            "At least one buyer and seller can trade, but most submitted volume
↪ sits outside the crossed part of the order book.",
            "Agents wanted to sell allowances they could not legally sell after
↪ compliance, banking, and no-short constraints.",
            "One side of the market is absent, so no bilateral match is
↪ possible.",
            "The remaining shortfall is mainly because one side has less
↪ feasible volume than the other.",
        ],
    })
    print("Bottleneck label definitions:")
    _display_table(bottleneck_meanings)

    bottleneck_summary = (
        orderbook_df.groupby("primary_bottleneck")

```

```

        .agg(
            Rows=("primary_bottleneck", "size"),
            Share_pct=("primary_bottleneck", lambda s: 100.0 * len(s) /
↳ len(orderbook_df)),
            Real_fill_mean_pct=("real_fill_pct", "mean"),
            Real_fill_median_pct=("real_fill_pct", "median"),
            Real_fill_std_pct=("real_fill_pct", "std"),
            Executed_median_Mt=("executed_Mt", "median"),
            ↳
↳ Unfilled_feasible_overlap_median_Mt=("unfilled_feasible_overlap_Mt",
↳ "median"),
            Gap_after_tolerance_median_EURt=("gap_after_tolerance_EURt",
↳ "median"),
        )
        .sort_values("Share_pct", ascending=False)
        .reset_index()
        .rename(columns={"primary_bottleneck": "Primary bottleneck"})
    )
    print("Bottleneck summary:")
    _display_table(
        bottleneck_summary,
        numeric_columns=[
            "Share_pct", "Real_fill_mean_pct", "Real_fill_median_pct",
↳ "Real_fill_std_pct",
            "Executed_median_Mt", "Unfilled_feasible_overlap_median_Mt",
↳ "Gap_after_tolerance_median_EURt",
        ],
    )

by_year = (
    orderbook_df.groupby("year")
    .agg(
        Buy_intent_median_Mt=("buy_intent_Mt", "median"),
        Raw_sell_intent_median_Mt=("raw_sell_intent_Mt", "median"),
        Feasible_sell_median_Mt=("feasible_sell_Mt", "median"),
        Executed_median_Mt=("executed_Mt", "median"),
        Real_fill_mean_pct=("real_fill_pct", "mean"),
        Real_fill_median_pct=("real_fill_pct", "median"),
        Real_fill_std_pct=("real_fill_pct", "std"),
        Buyer_limit_median_EURt=("buyer_limit_wavg_EURt", "median"),
        Feasible_seller_ask_median_EURt=("feasible_seller_ask_wavg_EURt",
↳ "median"),
        Best_buy_median_EURt=("best_buy_EURt", "median"),
        Best_feasible_ask_median_EURt=("best_feasible_ask_EURt", "median"),
        Gap_after_tolerance_median_EURt=("gap_after_tolerance_EURt",
↳ "median"),
    )

```

```

        )
        Unfilled_feasible_overlap_median_Mt=("unfilled_feasible_overlap_Mt",
        "median"),
        Clipped_sell_median_Mt=("clipped_sell_Mt", "median"),
    )
    .reset_index()
    .rename(columns={"year": "Year"})
)
print("By-year medians, with mean/std shown for the real fill rate:")
display_table(by_year, numeric_columns=[col for col in by_year.columns if
col != "Year"])

fig, axes = plt.subplots(1, 3, figsize=(18, 4))

ax = axes[0]
ax.plot(by_year["Year"], by_year["Buy_intent_median_Mt"], "o--", label="buy
intent")
ax.plot(by_year["Year"], by_year["Raw_sell_intent_median_Mt"], "o--",
label="raw sell intent")
ax.plot(by_year["Year"], by_year["Feasible_sell_median_Mt"], "o-",
label="feasible sell")
ax.plot(by_year["Year"], by_year["Executed_median_Mt"], "o-",
color="black", linewidth=2.0, label="executed")
ax.set_xlabel("Year"); ax.set_ylabel("Median volume (Mt)")
ax.set_title("Wanted vs feasible vs executed volume")
ax.grid(alpha=0.3); ax.legend(fontsize=8)
format_year_axis(ax)

ax = axes[1]
ax.plot(by_year["Year"], by_year["Buyer_limit_median_EURt"], "o-",
label="buy limit, qty-wavg")
ax.plot(by_year["Year"], by_year["Feasible_seller_ask_median_EURt"], "o-",
label="feasible sell ask, qty-wavg")
ax.plot(by_year["Year"], by_year["Best_buy_median_EURt"], "o--",
label="best buy")
ax.plot(by_year["Year"], by_year["Best_feasible_ask_median_EURt"], "o--",
label="best feasible ask")
ax.set_xlabel("Year"); ax.set_ylabel("Median limit price (EUR/t)")
ax.set_title("Price wanted by both sides")
ax.grid(alpha=0.3); ax.legend(fontsize=8)
format_year_axis(ax)

ax = axes[2]
bottleneck_order = [
    "only marginal prices cross",
    "best prices do not cross",

```

```

        "sell intent infeasible",
        "no sell intent",
        "no buy intent",
        "no submitted orders",
        "side-limited/mostly filled",
    ]
    bottleneck_share = (
        pd.crosstab(orderbook_df["year"], orderbook_df["primary_bottleneck"],
        ↪normalize="index")
        .mul(100.0)
        .reindex(columns=bottleneck_order, fill_value=0.0)
    )
    bottleneck_share.plot(kind="bar", stacked=True, ax=ax, width=0.85)
    ax.set_xlabel("Year"); ax.set_ylabel("Share of rows (%)")
    ax.set_title("Dominant mismatch reason")
    ax.grid(axis="y", alpha=0.3); ax.legend(fontsize=8, loc="upper left",
    ↪bbox_to_anchor=(1.0, 1.0))

    fig.suptitle("Secondary-market real-match bottleneck diagnosis", y=1.04)
    plt.tight_layout(); plt.show()

```

RQ4.3b secondary-market order-book diagnosis

Sample: 48,000 (run, episode, year) rows across 4 runs

Metric definitions:

<pandas.io.formats.style.Styler at 0x16b56fb63f0>

Summary statistics across the converged-window (run, episode, year) sample:

<pandas.io.formats.style.Styler at 0x16e6ff82d50>

Bottleneck label definitions:

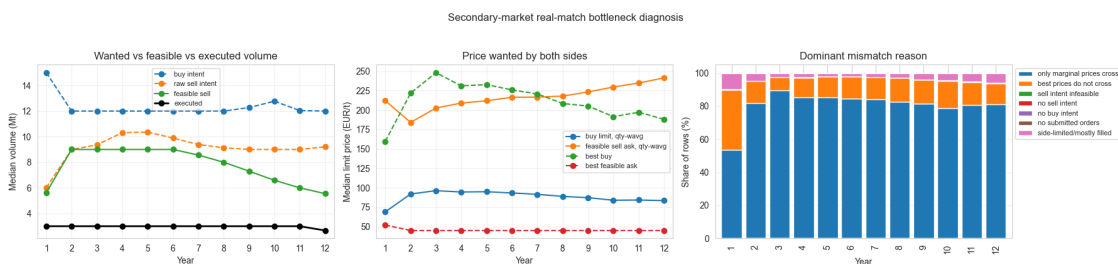
<pandas.io.formats.style.Styler at 0x16b01274860>

Bottleneck summary:

<pandas.io.formats.style.Styler at 0x16b072d0f80>

By-year medians, with mean/std shown for the real fill rate:

<pandas.io.formats.style.Styler at 0x16b02db8a10>



1.8.12 RQ4.4 — Converged-window credit stress by archetype × weighting

```
[28]: # RQ4.4 - Per-(archetype, weighting) cell of the mean peak emergency-loan
# balance and peak carry-forward debt in the converged window. Sourced from
# the per-episode `peak_loan_outstanding_A{i}` / `peak_carry_forward_A{i}`
# (v8.6.1+); pre-8.6.1 logs lack these columns and produce an empty table.
import numpy as np
import pandas as pd

def _converged_credit_metrics():
    rows = []
    for k, run in runs.items():
        ep_tail = tail_episodes(run["ep_df"])
        cfg = run["config"]
        for i in range(N_AGENTS):
            arche = ARCHETYPES[i // 2] if i // 2 < len(ARCHETYPES) else
↪f"A{i+1}"
            w_cost, w_green = (cfg.get("companies", {}).get("reward_weights"),
↪or [[1, 0]] * N_AGENTS)[i]
            weighting = "Financial" if w_green < 0.25 else ("ESG" if w_green >
↪0.75 else "Balanced")
            row = {"sweep": k[0], "variant": k[1], "seed": k[2],
                    "archetype": arche, "weighting": weighting, "agent": i + 1}
            for stub in ["peak_loan_outstanding", "peak_carry_forward",
                        "final_treasury_reserve"]:
                col = f"{stub}_A{i+1}"
                row[stub] = float(ep_tail[col].mean()) if col in ep_tail.
↪columns else np.nan
            rows.append(row)
    return pd.DataFrame(rows)

credit_df = _converged_credit_metrics()
if credit_df.empty or credit_df[["peak_loan_outstanding",
↪"peak_carry_forward"]].isna().all().all():
    print(["RQ4.4] peak_loan_outstanding_A* / peak_carry_forward_A* missing -
↪pre-8.6.1 log.")
else:
    pivot_loan = credit_df.pivot_table(
        index="archetype", columns="weighting", values="peak_loan_outstanding",
↪aggfunc="mean"
    ).round(3)
    pivot_cf = credit_df.pivot_table(
        index="archetype", columns="weighting", values="peak_carry_forward",
↪aggfunc="mean"
```



```

).round(3)
print("Mean peak emergency-loan balance (M€) - converged window:")
display(pivot_loan)
print("\nMean peak end-of-year carry-forward (Mt) - converged window:")
display(pivot_cf)

```

Mean peak emergency-loan balance (M€) - converged window:

weighting archetype	Balanced	Financial
Coal-heavy	0.0	0.0
Gas-dominant	0.0	0.0
Green-leader	0.0	0.0
Transitioner	0.0	0.0

Mean peak end-of-year carry-forward (Mt) - converged window:

weighting archetype	Balanced	Financial
Coal-heavy	0.778	2.038
Gas-dominant	0.554	1.366
Green-leader	0.068	0.166
Transitioner	0.490	0.273

1.8.13 RQ4.5 — Why agents go in-compliant: per-cell root-cause attribution

The UDBC mix in §RQ3.4 / §RQ2.3 already tells us *how often* each combinatorial state (under-bid / debt / mixed / bank-covered / secondary-covered) occurs. What it does not isolate is the **proximate cause** of each non-compliant cell: did the agent bid too cheaply and lose the auction, did it ask for too few tonnes (`bid_qty_mult` × `estimate_need` short of realised `need`), did the auction itself clear oversubscribed, was it carrying inherited debt forward, or did it submit a buy intent on the secondary that nobody filled?

This panel pulls every non-compliant cell (`episode`, `year`, `agent`) from the converged-tail window only and runs each through a prioritised waterfall:

1. **Inherited debt** — `alloc` `emiss` AND `cf_start` `shortfall`. The agent was auction-compliant *this year*; the deficit is fully explained by debt carried forward from earlier years.
2. **Underbid (price)** — `bid_price` < 99.5 % × `clearing_price`. The agent was priced out of the marginal allocation tranche.
3. **Undersized bid (quantity)** — (`bid_qty_mult` × `estimate_need`) falls short of the realised need by more than half of the shortfall. The agent asked for too few tonnes regardless of price.
4. **Oversubscribed auction** — `auction_total_demand` / `auction_volume` > 1.05. Even bidding at clearing, the agent was rationed by total demand exceeding supply.
5. **Secondary illiquidity** — agent submitted a positive buy intent (`sec_qty_action` > 0) but the realised buy fell more than 25 % of the shortfall short of intent.
6. **Mixed / cascade** — fallback bucket; no single cause dominates.

The taxonomy is intentionally simple and prioritised — each cell is assigned exactly one dominant

cause. The downstream tables report both mean and median diagnostic-feature values per cause and in the compliant/non-compliant contrast, so the central tendency and outlier sensitivity are visible together.

```
[29]: # RQ4.5 - Per-cell root-cause attribution for non-compliant (agent, year)
# cells in the converged tail window. Cells silently no-op on logs lacking
# the v8.6.1+ year-log columns.
import textwrap
from pathlib import Path
from matplotlib.backends.backend_pdf import PdfPages

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

EPS_MT = 1e-6
CAUSE_ORDER = ['Inherited debt', 'Underbid (price)', 'Undersized bid (qty)',
               'Oversubscribed auction', 'Secondary illiquidity', 'Mixed / 
               ↪ cascade']
CAUSE_COLORS = {
    'Inherited debt': '#9467bd',
    'Underbid (price)': '#d62728',
    'Undersized bid (qty)': '#ff7f0e',
    'Oversubscribed auction': '#1f77b4',
    'Secondary illiquidity': '#17becf',
    'Mixed / cascade': '#7f7f7f',
}
STRESSED_SOLUTION_ORDER = [
    'Banked allowances', 'Secondary market', 'Bank + secondary', 'Other / 
    ↪ accounting buffer'
]
STRESSED_SOLUTION_COLORS = {
    'Banked allowances': '#2ca02c',
    'Secondary market': '#1f77b4',
    'Bank + secondary': '#17becf',
    'Other / accounting buffer': '#7f7f7f',
}

EXPORT_RQ45_CELL_PDF = True
RQ45_CELL_PDF_PATH = (PROJECT_ROOT / 'results' / 'analysis' / 'rq45_cell_output.
    ↪ pdf'
                      if 'PROJECT_ROOT' in globals() else
    ↪ Path('rq45_cell_output.pdf'))
RQ45_AGENT_ORDER = list(range(1, N_AGENTS + 1))
RQ45_BASELINE_ROWS = ['All tail cells (normal/default)', 'Fully compliant 
    ↪ cells']
RQ45_DIAG_ROW_ORDER = RQ45_BASELINE_ROWS + CAUSE_ORDER
```

```

RQ45_DIAG_FEATURES = [
    'shortfall_Mt', 'inherited_debt_Mt', 'alloc_Mt', 'emissions_Mt',
    'bid_price', 'clearing_price', 'bid_to_clearing', 'bid_qty_mult',
    'bid_coverage_ratio', 'estimate_need_Mt', 'need_realised_Mt',
    'undersized_qty_Mt', 'oversubscription_ratio', 'bank_start_Mt',
    'sec_buy_intent_Mt', 'secondary_buy_unmet_Mt', 'loan_outstanding_Meur',
]

def _rq45_agent_label(agent_num):
    idx0 = agent_num - 1
    weighting = _agent_weighting(idx0)
    weighting_short = 'ESG' if weighting == 'Balanced ESG' else weighting
    return f'A{agent_num} {_agent_archetype(idx0)}\n{weighting_short}'

RQ45_AGENT_LABELS = {agent: _rq45_agent_label(agent) for agent in _
    ↪RQ45_AGENT_ORDER}
_rq45_pdf_items = []
_rq45_diag_values = {
    row_label: {feature: [] for feature in RQ45_DIAG_FEATURES}
    for row_label in RQ45_DIAG_ROW_ORDER
}
stressed_rows = []

def _rq45_log(message):
    print(message)
    _rq45_pdf_items.append(('text', str(message)))

def _rq45_record_table(title, df):
    _rq45_pdf_items.append(('table', title, df.copy()))

def _rq45_record_figure(title, fig):
    _rq45_pdf_items.append(('figure', title, fig))

def _rq45_append_diag_values(row_label, mask, feature_values):
    mask = np.asarray(mask, dtype=bool)
    if not mask.any():
        return
    for feature in RQ45_DIAG_FEATURES:
        values = feature_values.get(feature)
        if values is None:
            continue

```

```

        arr = np.asarray(values, dtype=float)
        if arr.shape[0] != mask.shape[0]:
            continue
        _rq45_diag_values[row_label][feature].append(arr[mask])

def _rq45_concat_diag(row_label, feature):
    pieces = _rq45_diag_values.get(row_label, {}).get(feature, [])
    if not pieces:
        return np.asarray([], dtype=float)
    return np.concatenate([np.asarray(p, dtype=float) for p in pieces])

def _rq45_stat(values, stat):
    finite = values[np.isfinite(values)]
    if finite.size == 0:
        return np.nan
    if stat == 'mean':
        return float(np.mean(finite))
    if stat == 'median':
        return float(np.median(finite))
    if stat == 'std':
        return float(np.std(finite, ddof=1)) if finite.size > 1 else np.nan
    raise ValueError(stat)

def _rq45_build_diag_summary():
    rows = []
    for row_label in RQ45_DIAG_ROW_ORDER:
        shortfall_values = _rq45_concat_diag(row_label, 'shortfall_Mt')
        row = {'diagnostic_group': row_label, 'n_cells': int(shortfall_values.
↪size)}
        for feature in RQ45_DIAG_FEATURES:
            values = _rq45_concat_diag(row_label, feature)
            for stat in ['mean', 'median', 'std']:
                row[f'{feature}_{stat}'] = _rq45_stat(values, stat)
        rows.append(row)
    return pd.DataFrame(rows).set_index('diagnostic_group')

def _stressed_solution_label(gap, bank_available, sec_bought):
    if sec_bought >= gap - EPS_MT and sec_bought > EPS_MT:
        return 'Secondary market'
    if bank_available >= gap - EPS_MT and bank_available > EPS_MT:
        return 'Banked allowances'
    if sec_bought > EPS_MT and bank_available > EPS_MT and sec_bought +
↪bank_available >= gap - EPS_MT:

```

```

        return 'Bank + secondary'
    if sec_bought > EPS_MT:
        return 'Secondary market'
    if bank_available > EPS_MT:
        return 'Banked allowances'
    return 'Other / accounting buffer'

def _rq45_pdf_text_pages(pdf, title, text, chars_per_line=118,
    ↪lines_per_page=45):
    lines = []
    for raw_line in str(text).splitlines() or ['']:
        if len(raw_line) <= chars_per_line:
            lines.append(raw_line)
        else:
            lines.extend(textwrap.wrap(raw_line, width=chars_per_line,
                                       break_long_words=False,
                                       replace_whitespace=False) or [''])
    chunks = [lines[i:i + lines_per_page] for i in range(0, len(lines),
    ↪lines_per_page)] or [[]]
    for page_idx, chunk in enumerate(chunks):
        fig = plt.figure(figsize=(11.69, 8.27))
        page_title = title if len(chunks) == 1 else f'{title} ({page_idx + 1}/
    ↪{len(chunks)})'
        fig.text(0.02, 0.97, page_title, fontsize=12, weight='bold', va='top')
        fig.text(0.02, 0.91, '\n'.join(chunk), fontsize=8, family='monospace',
    ↪va='top')
        pdf.savefig(fig, bbox_inches='tight')
        plt.close(fig)

def _rq45_pdf_table_pages(pdf, title, df, cols_per_page=6):
    if df.empty:
        _rq45_pdf_text_pages(pdf, title, '<empty>')
        return
    table_df = df.copy()
    if len(table_df.columns) <= cols_per_page:
        _rq45_pdf_text_pages(pdf, title, table_df.to_string())
        return
    for start in range(0, len(table_df.columns), cols_per_page):
        end = min(start + cols_per_page, len(table_df.columns))
        page_df = table_df.iloc[:, start:end]
        page_title = f'{title} | columns {start + 1}-{end} of {len(table_df.
    ↪columns)}'
        _rq45_pdf_text_pages(pdf, page_title, page_df.to_string())

```

```

def _rq45_export_pdf(items, path):
    if not items:
        return None
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)
    with PdfPages(path) as pdf:
        cover = plt.figure(figsize=(11.69, 8.27))
        cover.text(0.05, 0.92, 'RQ4.5 cell output', fontsize=18, weight='bold',
        ↪va='top')
        cover.text(0.05, 0.84, f'Notebook cell export: {path.name}',
        ↪fontsize=10, va='top')
        cover.text(0.05, 0.78, 'Includes this cell\'s printed text, displayed
        ↪tables, and generated figures.',
                    fontsize=10, va='top')
        pdf.savefig(cover, bbox_inches='tight')
        plt.close(cover)
        for item in items:
            kind = item[0]
            if kind == 'text':
                _rq45_pdf_text_pages(pdf, 'Printed output', item[1])
            elif kind == 'table':
                _rq45_pdf_table_pages(pdf, item[1], item[2])
            elif kind == 'figure':
                pdf.savefig(item[2], bbox_inches='tight')
    return path

def _classify_row(sf, cf_start, alloc, emiss, bid_p, clear_p,
                 undersized_qty, oversub, sec_unmet):
    if sf <= EPS_MT:
        return None
    cf = max(0.0, float(cf_start))
    if alloc + EPS_MT >= emiss and cf >= sf - EPS_MT:
        return 'Inherited debt'
    if bid_p > 0 and clear_p > 0 and bid_p < 0.995 * clear_p:
        return 'Underbid (price)'
    if undersized_qty > 0.5 * sf:
        return 'Undersized bid (qty)'
    if oversub > 1.05:
        return 'Oversubscribed auction'
    if sec_unmet > 0.25 * sf:
        return 'Secondary illiquidity'
    return 'Mixed / cascade'

nc_rows = []
rq45_tail_year_rows_by_run = {}

```

```

rq45_total_tail_cells = 0
for k, run in runs.items():
    yr = tail_year_rows(run)
    if yr.empty or 'year' not in yr.columns:
        continue
    n = len(yr)
    rq45_tail_year_rows_by_run[k] = n
    rq45_total_tail_cells += n * N_AGENTS
    cp_arr = yr['clearing_price'].to_numpy(float) if 'clearing_price' in yr.columns else np.full(n, np.nan)
    sp_arr = yr['secondary_price'].to_numpy(float) if 'secondary_price' in yr.columns else np.full(n, np.nan)
    atd = yr['auction_total_demand'].to_numpy(float) if 'auction_total_demand' in yr.columns else np.full(n, np.nan)
    avol = yr['auction_volume'].to_numpy(float) if 'auction_volume' in yr.columns else np.full(n, np.nan)
    ns_exec = yr['secondary_n_sellers_executed'].to_numpy(float) if 'secondary_n_sellers_executed' in yr.columns else np.full(n, np.nan)
    yr_arr = yr['year'].to_numpy(int)
    ep_arr = yr['episode'].to_numpy(int) if 'episode' in yr.columns else np.zeros(n, dtype=int)
    oversub_y = np.where(avol > 1e-9, atd / avol, np.nan)

    for i in range(N_AGENTS):
        cols_req = [f'shortfall_A{i+1}', f'alloc_A{i+1}', f'emissions_A{i+1}',
                    f'carry_forward_start_A{i+1}', f'bid_price_A{i+1}',
                    f'estimate_need_A{i+1}', f'bid_qty_mult_A{i+1}',
                    f'bank_start_A{i+1}', f'trade_qty_A{i+1}',
                    f'sec_qty_action_A{i+1}']
        if not all(c in yr.columns for c in cols_req):
            continue
        sf = yr[f'shortfall_A{i+1}'].to_numpy(float)
        alloc = yr[f'alloc_A{i+1}'].to_numpy(float)
        emiss = yr[f'emissions_A{i+1}'].to_numpy(float)
        cf_st = yr[f'carry_forward_start_A{i+1}'].to_numpy(float)
        bidp = yr[f'bid_price_A{i+1}'].to_numpy(float)
        en = yr[f'estimate_need_A{i+1}'].to_numpy(float)
        qm = yr[f'bid_qty_mult_A{i+1}'].to_numpy(float)
        bank = yr[f'bank_start_A{i+1}'].to_numpy(float)
        trq = yr[f'trade_qty_A{i+1}'].to_numpy(float)
        sqa = yr[f'sec_qty_action_A{i+1}'].to_numpy(float)
        cf_e = yr[f'carry_forward_end_A{i+1}'].to_numpy(float) if f'carry_forward_end_A{i+1}' in yr.columns else np.full(n, np.nan)
        loan = yr[f'loan_outstanding_A{i+1}'].to_numpy(float) if f'loan_outstanding_A{i+1}' in yr.columns else np.full(n, np.nan)

```

```

    tdr = yr[f'treasury_drawn_A{i+1}'].to_numpy(float)    if
↪f'treasury_drawn_A{i+1}'    in yr.columns else np.full(n, np.nan)
    cgap = yr[f'coverage_gap_A{i+1}'].to_numpy(float)    if
↪f'coverage_gap_A{i+1}'    in yr.columns else np.full(n, np.nan)
    sec_net = (yr[f'secondary_net_A{i+1}'].to_numpy(float)
                if f'secondary_net_A{i+1}' in yr.columns else np.full(n, np.
↪nan))

    # need_realised = emissions + inherited debt the agent had to cover.
    need_realised = emiss + np.maximum(0.0, cf_st)
    bid_qty_target = np.maximum(0.0, qm) * np.maximum(0.0, en)
    undersized_qty = np.maximum(0.0, need_realised - bid_qty_target)
    auction_gap = np.maximum(0.0, need_realised - alloc)
    sec_intent_buy = np.maximum(0.0, sqa)
    sec_bought = np.maximum(0.0, trq)
    sec_unmet = np.maximum(0.0, sec_intent_buy - sec_bought)
    bank_available = np.maximum(0.0, bank)
    sec_paid_fallback = np.where(np.isfinite(sp_arr), sec_bought * sp_arr,
↪np.nan)
    sec_direct_paid = np.where(np.isfinite(sec_net) & (sec_net > EPS_MT),
↪sec_net, np.nan)
    sec_paid = np.where(np.isfinite(sec_direct_paid), sec_direct_paid,
↪sec_paid_fallback)
    sec_paid = np.where(sec_bought > EPS_MT, sec_paid, 0.0)
    residual_after_secondary = np.maximum(0.0, auction_gap - sec_bought)
    bank_used_est = np.minimum(bank_available, residual_after_secondary)
    with np.errstate(divide='ignore', invalid='ignore'):
        bid_to_clearing = np.where(cp_arr > 1e-9, bidp / cp_arr, np.nan)
        bid_coverage_ratio = np.where(need_realised > 1e-9,
                                    bid_qty_target / need_realised,
                                    np.nan)
    sec_paid_price = np.where(sec_bought > EPS_MT, sec_paid /
↪sec_bought, np.nan)

    feature_values = {
        'shortfall_Mt': sf,
        'inherited_debt_Mt': cf_st,
        'alloc_Mt': alloc,
        'emissions_Mt': emiss,
        'bid_price': bidp,
        'clearing_price': cp_arr,
        'bid_to_clearing': bid_to_clearing,
        'bid_qty_mult': qm,
        'bid_coverage_ratio': bid_coverage_ratio,
        'estimate_need_Mt': en,
        'need_realised_Mt': need_realised,

```



```

        'undersized_qty_Mt': undersized_qty,
        'oversubscription_ratio': oversub_y,
        'bank_start_Mt': bank,
        'sec_buy_intent_Mt': sec_intent_buy,
        'secondary_buy_unmet_Mt': sec_unmet,
        'loan_outstanding_Meur': loan,
    }
    compliant_mask = sf <= EPS_MT
    _rq45_append_diag_values('All tail cells (normal/default)', np.ones(n,
↪dtype=bool), feature_values)
    _rq45_append_diag_values('Fully compliant cells', compliant_mask,
↪feature_values)

    stressed_mask = (auction_gap > EPS_MT) & compliant_mask
    for j in np.where(stressed_mask)[0]:
        gap = float(auction_gap[j])
        sec_buy = float(sec_bought[j])
        bank_avail = float(bank_available[j])
        solution = _stressed_solution_label(gap, bank_avail, sec_buy)
        stressed_rows.append({
            'run': f'{k[1]}/s{k[2]}',
            'episode': int(ep_arr[j]),
            'year': int(yr_arr[j]),
            'agent': i + 1,
            'archetype': _agent_archetype(i),
            'weighting': _agent_weighting(i),
            'solution': solution,
            'initial_auction_gap_Mt': gap,
            'need_realised_Mt': float(need_realised[j]),
            'alloc_Mt': float(alloc[j]),
            'bank_start_Mt': bank_avail,
            'bank_used_estimate_Mt': float(bank_used_est[j]),
            'secondary_buy_Mt': sec_buy,
            'secondary_paid_Meur': float(sec_paid[j]) if np.
↪isfinite(sec_paid[j]) else np.nan,
            'secondary_paid_price_EUR_t': float(sec_paid_price[j]) if np.
↪isfinite(sec_paid_price[j]) else np.nan,
        })

    nc_mask = sf > EPS_MT
    if not nc_mask.any():
        continue

    cause_arr = np.full(n, None, dtype=object)
    for j in np.where(nc_mask)[0]:
        cp_j = float(cp_arr[j]) if np.isfinite(cp_arr[j]) else 0.0
        os_j = float(oversub_y[j]) if np.isfinite(oversub_y[j]) else 0.0

```

```

        cause = _classify_row(
            sf=float(sf[j]), cf_start=float(cf_st[j]),
            alloc=float(alloc[j]), emiss=float(emiss[j]),
            bid_p=float(bidp[j]), clear_p=cp_j,
            undersized_qty=float(undersized_qty[j]),
            oversub=os_j, sec_unmet=float(sec_unmet[j]),
        )
        cause_arr[j] = cause
        nc_rows.append({
            'run': f'{k[1]}/s{k[2]}',
            'episode': int(ep_arr[j]),
            'year': int(yr_arr[j]),
            'agent': i + 1,
            'archetype': _agent_archetype(i),
            'weighting': _agent_weighting(i),
            'shortfall_Mt': float(sf[j]),
            'inherited_debt_Mt': float(cf_st[j]),
            'alloc_Mt': float(alloc[j]),
            'emissions_Mt': float(emiss[j]),
            'bid_price': float(bidp[j]),
            'clearing_price': cp_j,
            'bid_to_clearing': float(bid_to_clearing[j]) if np.
↳isfinite(bid_to_clearing[j]) else np.nan,
            'bid_qty_mult': float(qm[j]),
            'bid_coverage_ratio': float(bid_coverage_ratio[j]) if np.
↳isfinite(bid_coverage_ratio[j]) else np.nan,
            'estimate_need_Mt': float(en[j]),
            'need_realised_Mt': float(need_realised[j]),
            'undersized_qty_Mt': float(undersized_qty[j]),
            'oversubscription_ratio': os_j,
            'bank_start_Mt': float(bank[j]),
            'sec_buy_intent_Mt': float(sec_intent_buy[j]),
            'secondary_buy_unmet_Mt': float(sec_unmet[j]),
            'n_sellers_executed': float(ns_exec[j]) if np.
↳isfinite(ns_exec[j]) else np.nan,
            'carry_forward_end_Mt': float(cf_e[j]),
            'loan_outstanding_Meur': float(loan[j]),
            'treasury_drawn_Meur': float(tdr[j]),
            'coverage_gap_Mt': float(cgap[j]),
            'cause': cause,
        })

    for cause_label in CAUSE_ORDER:
        _rq45_append_diag_values(cause_label, cause_arr == cause_label,
↳feature_values)

incomp_df = pd.DataFrame(nc_rows)

```

```

stressed_df = pd.DataFrame(stressed_rows)
rq45_expected_cells_per_run = TAIL_EPISODES * N_AGENTS * N_YEARS
rq45_loaded_run_count = len(rq45_tail_year_rows_by_run)
_rq45_log(
    f'[RQ4.5] Denominator: {rq45_loaded_run_count} loaded run(s) × '
    f'{TAIL_EPISODES} tail episodes × {N_AGENTS} agents × {N_YEARS} years '
    f'= {rq45_loaded_run_count * rq45_expected_cells_per_run:,} expected cells '
    f'({rq45_expected_cells_per_run:,} per run); actual counted cells = '
    f'{rq45_total_tail_cells:,}.'
)
if rq45_total_tail_cells > 0:
    _rq45_log(
        f'[RQ4.5] Final non-compliant cells: {len(incomp_df):,} '
        f'({100.0 * len(incomp_df) / rq45_total_tail_cells:.1f}%); '
        f'auction-stressed but compliant cells: {len(stressed_df):,} '
        f'({100.0 * len(stressed_df) / rq45_total_tail_cells:.1f}%).'
    )

if incomp_df.empty:
    _rq45_log('RQ4.5 No non-compliant (agent, year) cells in the converged_
    ↪window - '
            'agents are fully compliant. Skipping attribution panels.')
else:
    incomp_df['cause'] = pd.Categorical(incomp_df['cause'],
    ↪categories=CAUSE_ORDER, ordered=True)

    # (a) Counts and tonne-weighted shares per cause.
    cause_counts = (incomp_df.groupby('cause', observed=True)
        .agg(n_cells=('shortfall_Mt', 'size'),
            sum_shortfall_Mt=('shortfall_Mt', 'sum'),
            mean_shortfall_Mt=('shortfall_Mt', 'mean'),
            median_shortfall_Mt=('shortfall_Mt',
    ↪'median'))
        .reindex(CAUSE_ORDER).fillna(0.0))

    total_cells = max(1, int(cause_counts['n_cells'].sum()))
    total_tons = max(1e-9, float(cause_counts['sum_shortfall_Mt'].sum()))
    cause_counts['cell_share_pct'] = 100.0 * cause_counts['n_cells'] /
    ↪total_cells
    cause_counts['tonne_share_pct'] = 100.0 * cause_counts['sum_shortfall_Mt'] /
    ↪total_tons

    _rq45_log(f'[RQ4.5] Non-compliant (agent, year, episode) cells in converged_
    ↪window: '
            f'{len(incomp_df):,}')
    _rq45_log(f'[RQ4.5] Σ shortfall across the window: '
            f'{incomp_df["shortfall_Mt"].sum():,.2f} Mt')

```

```

cause_counts_display = cause_counts.round(3)
_rq45_record_table('Cause counts and shares', cause_counts_display)
display(cause_counts_display)

fig, axes = plt.subplots(1, 2, figsize=(14, 4.6))
colors = [CAUSE_COLORS[c] for c in CAUSE_ORDER]
axes[0].bar(CAUSE_ORDER, cause_counts['cell_share_pct'].values,
color=colors)
axes[0].set(title='Share of non-compliant cells by dominant cause',
            ylabel='% of (agent, year, episode) cells')
axes[0].tick_params(axis='x', rotation=25); axes[0].grid(alpha=0.3,
axis='y')

axes[1].bar(CAUSE_ORDER, cause_counts['tonne_share_pct'].values,
color=colors)
axes[1].set(title='Share of total shortfall tonnes by dominant cause',
            ylabel='% of  $\Sigma$  shortfall (Mt)')
axes[1].tick_params(axis='x', rotation=25); axes[1].grid(alpha=0.3,
axis='y')

plt.suptitle(f'RQ4.5 - Why agents go non-compliant in the converged window '
            f'(pooled across {rq45_loaded_run_count} loaded seeds)',
            y=1.03, fontsize=13)
plt.tight_layout()
_rq45_record_figure('Cause shares', fig)
plt.show()

# (b) Baseline and per-cause diagnostic feature values.
cause_feature_summary = _rq45_build_diag_summary()
_rq45_log('RQ4.5] Baseline and per-cause feature values (mean / median /
std):')
_rq45_log('    All tail cells = normal/default-config bid baseline; '
        'Fully compliant cells = regular successful bid baseline.')
cause_feature_display = cause_feature_summary.round(3)
_rq45_record_table('Baseline and per-cause feature values',
cause_feature_display)
display(cause_feature_display)

# (c) Simulation-ordered per-agent cause split (A1..A8: archetype  $x_{\text{weighting}}$ ).
agent_cause_counts = (incomp_df.groupby(['agent', 'cause'], observed=True)
                        .size().unstack('cause', fill_value=0)
                        .reindex(index=RQ45_AGENT_ORDER,
                                columns=CAUSE_ORDER,
                                fill_value=0).astype(int))
agent_labels = [RQ45_AGENT_LABELS.get(agent, f'A{agent}')]
for agent in agent_cause_counts.index]

```

```

agent_cause_share = (agent_cause_counts
                     .div(agent_cause_counts.sum(axis=1).replace(0, np.
↳nan), axis=0)
                     .fillna(0.0))

if not agent_cause_share.empty:
    fig, ax = plt.subplots(figsize=(11.8, 4.6))
    agent_cause_share.plot(kind='bar', stacked=True, ax=ax,
                           color=[CAUSE_COLORS[c] for c in_
↳agent_cause_share.columns])
    ax.set(title=f'RQ4.5 - Cause mix by simulation archetype order '
               f'(pooled across {rq45_loaded_run_count} loaded seeds)',
           ylabel='Share of non-compliant cells', ylim=(0, 1.0))
    ax.set_xticklabels(agent_labels, rotation=20, ha='right')
    ax.legend(fontsize=8, loc='center left', bbox_to_anchor=(1.02, 0.5))
    ax.grid(alpha=0.3, axis='y')
    plt.tight_layout()
    _rq45_record_figure('Agent-ordered cause mix shares', fig)
    plt.show()

fig_abs, ax_abs = plt.subplots(figsize=(11.8, 4.6))
agent_cause_counts.plot(kind='bar', stacked=True, ax=ax_abs,
                        color=[CAUSE_COLORS[c] for c in_
↳agent_cause_counts.columns])
    ax_abs.set(title=f'RQ4.5 - Absolute non-compliant cells by simulation_
↳archetype order '
                   f'(pooled across {rq45_loaded_run_count} loaded_
↳seeds)',
              ylabel='Non-compliant cells (pooled count)')
    ax_abs.set_xticklabels(agent_labels, rotation=20, ha='right')
    ax_abs.legend(fontsize=8, loc='center left', bbox_to_anchor=(1.02, 0.5))
    ax_abs.grid(alpha=0.3, axis='y')
    plt.tight_layout()
    _rq45_record_figure('Agent-ordered absolute cause counts', fig_abs)
    plt.show()

# (d) Compliant vs non-compliant feature contrast on the same tail window.
comp_pieces = []
for k, run in runs.items():
    yr = tail_year_rows(run)
    if yr.empty: continue
    for i in range(N_AGENTS):
        sf_col = f'shortfall_A_{i+1}'
        if sf_col not in yr.columns: continue
        sf = yr[sf_col].to_numpy(float)
        cmask = sf <= EPS_MT
        if not cmask.any(): continue

```

```

row = {}
for stub, src_col in [
    ('inherited_debt_Mt', f'carry_forward_start_A{i+1}'),
    ('bank_start_Mt', f'bank_start_A{i+1}'),
    ('bid_qty_mult', f'bid_qty_mult_A{i+1}'),
    ('bid_price', f'bid_price_A{i+1}'),
    ('coverage_gap_Mt', f'coverage_gap_A{i+1}'),
]:
    row[stub] = (yr.loc[cmask, src_col].to_numpy(float)
                if src_col in yr.columns else np.full(int(cmask.
↪sum()), np.nan))
    row['clearing_price'] = (yr.loc[cmask, 'clearing_price'].
↪to_numpy(float)
                           if 'clearing_price' in yr.columns
                           else np.full(int(cmask.sum()), np.nan))
    comp_pieces.append(pd.DataFrame(row))
if comp_pieces:
    comp_df = pd.concat(comp_pieces, ignore_index=True)
    comp_df['bid_to_clearing'] = (comp_df['bid_price']
                                / comp_df['clearing_price'].replace(0,
↪np.nan))
    nc_with = incomp_df.copy()
    nc_with['bid_to_clearing'] = (nc_with['bid_price']
                                / nc_with['clearing_price'].replace(0,
↪np.nan))
    contrast_cols = ['inherited_debt_Mt', 'bank_start_Mt',
                     'bid_qty_mult', 'bid_to_clearing', 'coverage_gap_Mt']
    rows = []
    for c in contrast_cols:
        cv = comp_df[c].replace([np.inf, -np.inf], np.nan).dropna() if c in
↪comp_df.columns else pd.Series(dtype=float)
        nv = nc_with[c].replace([np.inf, -np.inf], np.nan).dropna() if c in
↪nc_with.columns else pd.Series(dtype=float)
        cv_mean = float(cv.mean()) if len(cv) else np.nan
        nv_mean = float(nv.mean()) if len(nv) else np.nan
        cv_median = float(cv.median()) if len(cv) else np.nan
        nv_median = float(nv.median()) if len(nv) else np.nan
        rows.append({
            'feature': c,
            'compliant_mean': cv_mean,
            'non_compliant_mean': nv_mean,
            'delta_mean(nc - c)': (nv_mean - cv_mean if len(cv) and
↪len(nv) else np.nan),
            'compliant_median': cv_median,
            'non_compliant_median': nv_median,

```

```

        'delta_median_(nc - c)': (nv_median - cv_median if len(cv) and
↪len(nv) else np.nan),
    })
    _rq45_log(['RQ4.5] Compliant vs non-compliant cell means and medians
↪(key discriminators):'])
    contrast_display = pd.DataFrame(rows).round(3)
    _rq45_record_table('Compliant vs non-compliant feature contrast',
↪contrast_display)
    display(contrast_display)

# (e) Auction-stressed but ultimately compliant cells: how did agents save the
↪year?
if stressed_df.empty:
    _rq45_log(['RQ4.5] No auction-stressed but ultimately compliant cells found
↪in the converged window.'])
else:
    stressed_df['solution'] = pd.Categorical(
        stressed_df['solution'], categories=STRESSED_SOLUTION_ORDER,
↪ordered=True
    )
    stressed_summary = (stressed_df.groupby('solution', observed=False)
        .agg(n_cells=('initial_auction_gap_Mt',
↪'size'),
            share_pct=('initial_auction_gap_Mt',
↪lambda s: 100.0 * len(s) / len(stressed_df)),
            mean_initial_gap_Mt=('initial_auction_gap_Mt', 'mean'),
            median_initial_gap_Mt=('initial_auction_gap_Mt', 'median'),
            sum_initial_gap_Mt=('initial_auction_gap_Mt', 'sum'),
            mean_bank_start_Mt=('bank_start_Mt',
↪'mean'),
            mean_bank_used_estimate_Mt=('bank_used_estimate_Mt', 'mean'),
            mean_secondary_buy_Mt=('secondary_buy_Mt', 'mean'),
            sum_secondary_buy_Mt=('secondary_buy_Mt', 'sum'),
            mean_secondary_paid_Meur=('secondary_paid_Meur', 'mean'),
            sum_secondary_paid_Meur=('secondary_paid_Meur', 'sum'),
            mean_secondary_paid_price_EUR_t=('secondary_paid_price_EUR_t', 'mean'))

```

```

                                .reindex(STRESSED_SOLUTION_ORDER).fillna(0.
↪0))
    _rq45_log(f'[RQ4.5] Auction-stressed but ultimately compliant cells:␣
↪{len(stressed_df):,}')
    _rq45_log('          Stressed = auction allocation < emissions +␣
↪carry-forward start; final shortfall = 0.')
    stressed_display = stressed_summary.round(3)
    _rq45_record_table('Auction-stressed compliant cells by solution',␣
↪stressed_display)
    display(stressed_display)

    fig_stress, ax_stress = plt.subplots(figsize=(8.8, 4.0))
    stress_counts = stressed_summary['n_cells'].astype(int)
    stress_colors = [STRESSED_SOLUTION_COLORS[s] for s in stress_counts.index]
    bars = ax_stress.bar(stress_counts.index.astype(str), stress_counts.values,␣
↪color=stress_colors)
    total_stressed = max(1, int(stress_counts.sum()))
    for bar, count in zip(bars, stress_counts.values):
        if count <= 0:
            continue
        ax_stress.text(bar.get_x() + bar.get_width() / 2,
                        bar.get_height(),
                        f'{count:,}\n({100.0 * count / total_stressed:.1f}%)',
                        ha='center', va='bottom', fontsize=8)
    ax_stress.set(title=f'RQ4.5 - Auction-stressed compliant cells: resolution␣
↪channel '
                    f'(pooled across {rq45_loaded_run_count} loaded seeds)',
                  ylabel='Cell count (pooled)')
    ax_stress.tick_params(axis='x', rotation=15)
    ax_stress.grid(alpha=0.3, axis='y')
    plt.tight_layout()
    _rq45_record_figure('Auction-stressed compliant resolution counts',␣
↪fig_stress)
    plt.show()

if EXPORT_RQ45_CELL_PDF:
    _rq45_log(f'[RQ4.5] Writing cell output PDF to: {RQ45_CELL_PDF_PATH}')
    _rq45_pdf_path = _rq45_export_pdf(_rq45_pdf_items, RQ45_CELL_PDF_PATH)
    if _rq45_pdf_path is not None:
        print(f'[RQ4.5] PDF export complete: {_rq45_pdf_path}')

```

[RQ4.5] Denominator: 4 loaded run(s) × 1000 tail episodes × 8 agents × 12 years
= 384,000 expected cells (96,000 per run); actual counted cells = 384,000.

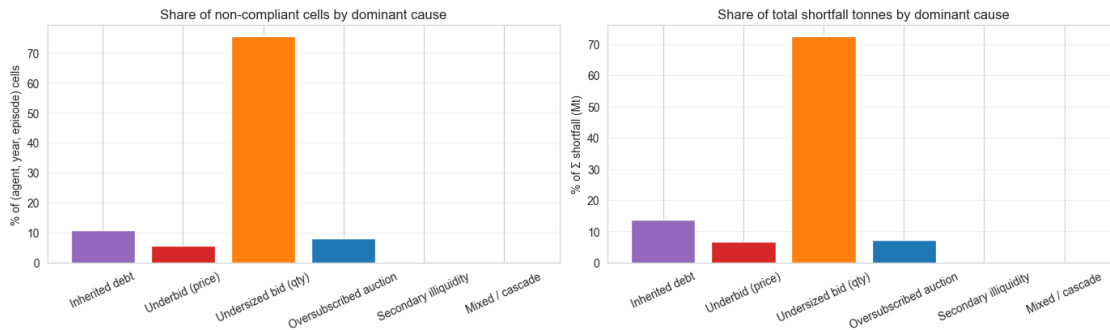
[RQ4.5] Final non-compliant cells: 66,112 (17.2%); auction-stressed but
compliant cells: 142,175 (37.0%).

[RQ4.5] Non-compliant (agent, year, episode) cells in converged window: 66,112

[RQ4.5] Σ shortfall across the window: 79,149.41 Mt

	n_cells	sum_shortfall_Mt	mean_shortfall_Mt	
↳ median_shortfall_Mt	cell_share_pct	tonne_share_pct		
cause				
Inherited debt	7070	10841.089	1.533	↳
↳ 1.686	10.694	13.697		
Underbid (price)	3825	5321.238	1.391	↳
↳ 0.740	5.786	6.723		
Undersized bid (qty)	49874	57323.488	1.149	↳
↳ 1.099	75.439	72.424		
Oversubscribed auction	5333	5659.925	1.061	↳
↳ 0.718	8.067	7.151		
Secondary illiquidity	4	2.079	0.520	↳
↳ 0.547	0.006	0.003		
Mixed / cascade	6	1.589	0.265	↳
↳ 0.231	0.009	0.002		

RQ4.5 — Why agents go non-compliant in the converged window (pooled across 4 loaded seeds)



[RQ4.5] Baseline and per-cause feature values (mean / median / std):

All tail cells = normal/default-config bid baseline; Fully compliant cells = regular successful bid baseline.

	n_cells	shortfall_Mt_mean	shortfall_Mt_median	
↳ shortfall_Mt_std	inherited_debt_Mt_mean	inherited_debt_Mt_median		↳
↳ inherited_debt_Mt_std	alloc_Mt_mean			
diagnostic_group				
All tail cells (normal/default)	384000	0.206	0.000	↳
↳ 0.593	0.177	0.000		↳
↳ 0.543	1.739			
Fully compliant cells	317888	0.000	0.000	↳
↳ 0.000	0.060	0.000		↳
↳ 0.295	1.797			
Inherited debt	7070	1.533	1.686	↳
↳ 0.805	1.951	2.010		↳
↳ 0.809	2.250			

Underbid (price)	3825	1.391	0.740
↪ 1.735	0.681	0.000	
↪ 1.429	0.000		
Undersized bid (qty)	49874	1.149	1.099
↪ 0.800	0.611	0.078	
↪ 0.797	1.556		
Oversubscribed auction	5333	1.061	0.718
↪ 1.177	0.370	0.000	
↪ 0.894	0.509		
Secondary illiquidity	4	0.520	0.547
↪ 0.362	0.157	0.049	
↪ 0.252	1.614		
Mixed / cascade	6	0.265	0.231
↪ 0.180	0.030	0.000	
↪ 0.075	1.133		

	alloc_Mt_median	alloc_Mt_std	
↪ emissions_Mt_mean	emissions_Mt_median	emissions_Mt_std	bid_price_mean
↪ bid_price_median	bid_price_std	clearing_price_mean	\
diagnostic_group			
All tail cells (normal/default)	1.563	1.217	1.
↪ 765	1.675	0.868	96.857
↪ 46.931	54.383		91.032
Fully compliant cells	1.582	1.282	1.
↪ 647	1.525	0.835	91.997
↪ 44.816	53.839		84.665
Inherited debt	2.260	0.702	1.
↪ 938	1.986	0.584	148.542
↪ 44.188	69.600		165.621
Underbid (price)	0.000	0.000	1.
↪ 519	1.535	0.584	61.477
↪ 26.881	98.740		45.000
Undersized bid (qty)	1.541	0.586	2.
↪ 498	2.596	0.781	128.014
↪ 44.419	52.539		135.097
Oversubscribed auction	0.270	0.609	1.
↪ 847	1.807	0.614	52.076
↪ 19.354	52.077		45.000
Secondary illiquidity	1.826	0.896	1.
↪ 976	1.994	0.667	45.000
↪ 0.000	45.000		45.000
Mixed / cascade	1.096	0.542	1.
↪ 368	1.327	0.503	45.000
↪ 0.000	45.000		45.000

		clearing_price_median	clearing_price_std	
↪bid_to_clearing_mean	bid_to_clearing_median	bid_to_clearing_std		
↪bid_qty_mult_mean	bid_qty_mult_median			
diagnostic_group				
All tail cells (normal/default)		45.000	20.575	
↪	1.874	1.671	0.934	1.
↪298	1.433			
Fully compliant cells		45.000	19.689	
↪	1.789	1.545	0.891	1.
↪418	1.814			
Inherited debt		61.282	28.262	
↪	2.329	2.226	0.906	0.
↪676	0.500			
Underbid (price)		94.685	33.604	
↪	0.647	0.639	0.201	1.
↪731	2.000			
Undersized bid (qty)		45.000	18.948	
↪	2.537	2.535	0.911	0.
↪522	0.500			
Oversubscribed auction		45.000	19.357	
↪	1.000	1.000	0.000	1.
↪890	2.000			
Secondary illiquidity		45.000	0.000	
↪	1.000	1.000	0.000	1.
↪097	1.111			
Mixed / cascade		45.000	0.000	
↪	1.000	1.000	0.000	1.
↪356	1.214			

		bid_qty_mult_std	bid_coverage_ratio_mean	
↪bid_coverage_ratio_median	bid_coverage_ratio_std	estimate_need_Mt_mean		
↪estimate_need_Mt_median	estimate_need_Mt_std			
diagnostic_group				
All tail cells (normal/default)		0.681	1.346	
↪	1.403	0.771	1.971	
↪	1.742	1.139		
Fully compliant cells		0.652	1.476	
↪	1.657	0.752	1.737	
↪	1.549	0.943		
Inherited debt		0.424	0.713	
↪	0.553	0.416	4.179	
↪	4.365	1.375		
Underbid (price)		0.465	1.781	
↪	1.935	0.554	2.247	
↪	1.835	1.557		

Undersized bid (qty)		0.109	0.517	␣
↪	0.501	0.081	3.100	␣
↪	3.085	1.182		
Oversubscribed auction		0.286	1.885	␣
↪	1.933	0.363	2.190	␣
↪	1.957	1.127		
Secondary illiquidity		0.084	1.011	␣
↪	1.019	0.133	1.985	␣
↪	2.036	0.836		
Mixed / cascade		0.530	1.106	␣
↪	1.141	0.166	1.301	␣
↪	1.309	0.657		

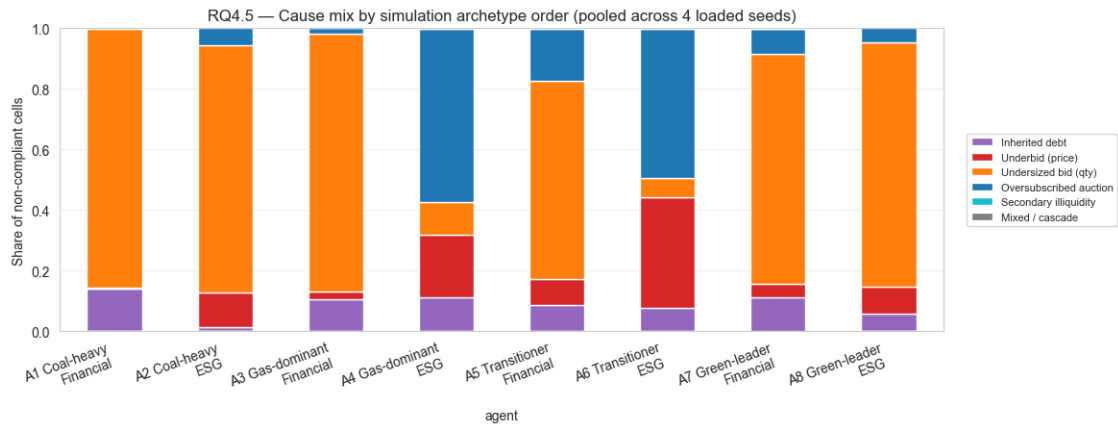
	need_realised_Mt_mean	need_realised_Mt_median	␣
↪need_realised_Mt_std undersized_qty_Mt_mean undersized_qty_Mt_median			␣
↪undersized_qty_Mt_std \			
diagnostic_group			
All tail cells (normal/default)	1.942	1.726	␣
↪	1.129	0.509	0.000
↪	0.745		
Fully compliant cells	1.708	1.540	␣
↪	0.936	0.338	0.000
↪	0.599		
Inherited debt	3.889	4.000	␣
↪	1.276	1.539	1.733
↪	0.853		
Underbid (price)	2.200	1.818	␣
↪	1.526	0.081	0.000
↪	0.323		
Undersized bid (qty)	3.109	3.089	␣
↪	1.194	1.540	1.543
↪	0.658		
Oversubscribed auction	2.217	1.986	␣
↪	1.141	0.017	0.000
↪	0.138		
Secondary illiquidity	2.134	2.036	␣
↪	0.887	0.125	0.049
↪	0.191		
Mixed / cascade	1.398	1.327	␣
↪	0.522	0.030	0.000
↪	0.074		

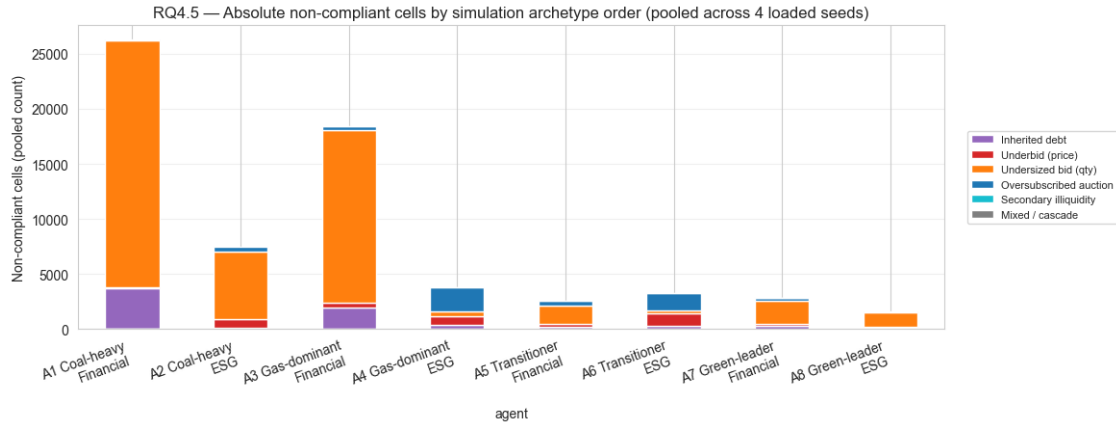
	oversubscription_ratio_mean	␣
↪oversubscription_ratio_median oversubscription_ratio_std bank_start_Mt_mean		␣
↪bank_start_Mt_median bank_start_Mt_std \		
diagnostic_group		

All tail cells (normal/default)			1.275	
↪ 1.247		0.332	2.017	1.
↪595	1.991			
Fully compliant cells			1.260	
↪ 1.234		0.314	2.372	1.
↪956	2.000			
Inherited debt			1.543	
↪ 1.538		0.346	0.000	0.
↪000	0.000			
Underbid (price)			1.938	
↪ 1.746		0.646	0.624	0.
↪317	0.713			
Undersized bid (qty)			1.258	
↪ 1.234		0.329	0.303	0.
↪000	0.498			
Oversubscribed auction			1.474	
↪ 1.423		0.305	0.552	0.
↪226	0.667			
Secondary illiquidity			1.036	
↪ 1.036		0.008	0.000	0.
↪000	0.000			
Mixed / cascade			1.033	
↪ 1.037		0.014	0.000	0.
↪000	0.000			

			sec_buy_intent_Mt_mean	
↪sec_buy_intent_Mt_median	sec_buy_intent_Mt_std	secondary_buy_unmet_Mt_mean		
↪secondary_buy_unmet_Mt_median	secondary_buy_unmet_Mt_std	\		
diagnostic_group				
All tail cells (normal/default)			1.669	3.
↪000	1.474		1.301	
↪ 0.0		1.449		
Fully compliant cells			1.444	0.
↪071	1.482		1.021	
↪ 0.0		1.384		
Inherited debt			2.768	3.
↪000	0.786		2.663	
↪ 3.0		0.864		
Underbid (price)			2.407	3.
↪000	1.176		2.223	
↪ 3.0		1.276		
Undersized bid (qty)			2.815	3.
↪000	0.706		2.714	
↪ 3.0		0.797		
Oversubscribed auction			2.406	3.
↪000	1.175		2.312	
↪ 3.0		1.233		

Secondary illiquidity			3.000	3.
↳ 0.000	0.000		3.000	↳
↳ 3.0		0.000		
Mixed / cascade			0.000	0.
↳ 0.000	0.000		0.000	↳
↳ 0.0		0.000		
		loan_outstanding_Meur_mean		↳
↳ loan_outstanding_Meur_median		loan_outstanding_Meur_std		
diagnostic_group				
All tail cells (normal/default)			0.0	↳
↳ 0.0		0.0		
Fully compliant cells			0.0	↳
↳ 0.0		0.0		
Inherited debt			0.0	↳
↳ 0.0		0.0		
Underbid (price)			0.0	↳
↳ 0.0		0.0		
Undersized bid (qty)			0.0	↳
↳ 0.0		0.0		
Oversubscribed auction			0.0	↳
↳ 0.0		0.0		
Secondary illiquidity			0.0	↳
↳ 0.0		0.0		
Mixed / cascade			0.0	↳
↳ 0.0		0.0		





[RQ4.5] Compliant vs non-compliant cell means and medians (key discriminators):

	feature	compliant_mean	non_compliant_mean	delta_mean_(nc - c)	
	↪compliant_median	non_compliant_median	delta_median_(nc - c)		
0	inherited_debt_Mt	0.060	0.739	0.678	↪
↪	0.000	0.173	0.173		
1	bank_start_Mt	2.372	0.309	-2.063	↪
↪	1.956	0.000	-1.956		
2	bid_qty_mult	1.418	0.719	-0.700	↪
↪	1.814	0.500	-1.314		
3	bid_to_clearing	1.789	2.281	0.492	↪
↪	1.545	2.271	0.726		
4	coverage_gap_Mt	0.481	2.104	1.623	↪
↪	0.000	2.107	2.107		

[RQ4.5] Auction-stressed but ultimately compliant cells: 142,175

Stressed = auction allocation < emissions + carry-forward start; final shortfall = 0.

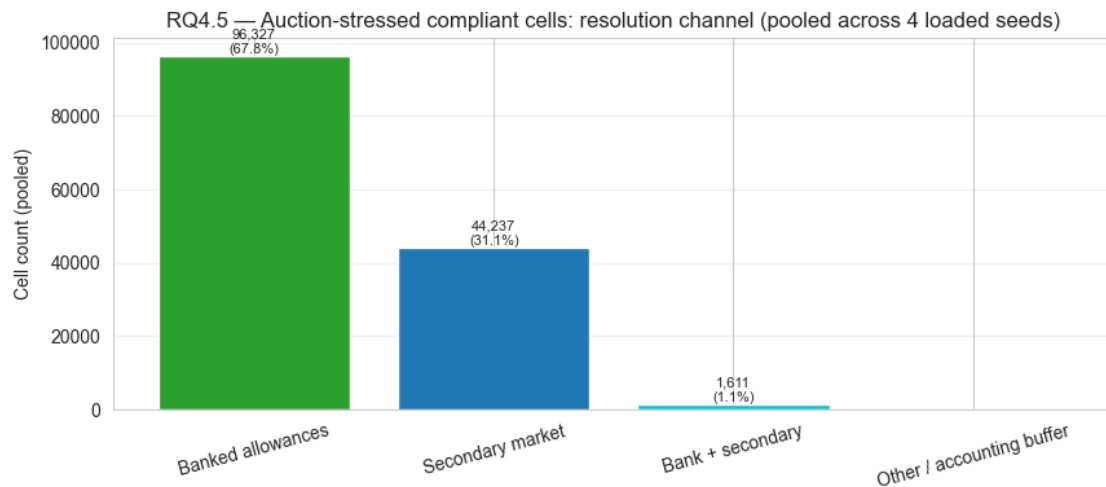
	n_cells	share_pct	mean_initial_gap_Mt	
↪median_initial_gap_Mt	sum_initial_gap_Mt	mean_bank_start_Mt		↪
↪mean_bank_used_estimate_Mt	mean_secondary_buy_Mt	\		
solution				
Banked allowances	96327	67.752	0.890	↪
↪ 0.840	85738.173	3.020	0.870	↪
↪ 0.020				
Secondary market	44237	31.114	1.650	↪
↪ 1.627	72984.968	0.937	0.000	↪
↪ 2.837				
Bank + secondary	1611	1.133	1.932	↪
↪ 1.722	3113.214	1.228	0.478	↪
↪ 1.455				

Other / accounting buffer	0	0.000	0.000	
↪ 0.000	0.000	0.000	0.000	
↪	0.000			

	sum_secondary_buy_Mt	mean_secondary_paid_Meur	
↪ sum_secondary_paid_Meur	mean_secondary_paid_price_EUR_t		

solution

Banked allowances	1948.997	1.849
↪ 1.780747e+05	89.952	
Secondary market	125519.232	410.794
↪ 1.817231e+07	142.228	
Bank + secondary	2343.612	215.366
↪ 3.469541e+05	127.979	
Other / accounting buffer	0.000	0.000
↪ 0.000000e+00	0.000	



[RQ4.5] Writing cell output PDF to:

C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-Energy-Auction\results\analysis\rq45_cell_output.pdf

[RQ4.5] PDF export complete: C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-Energy-Auction\results\analysis\rq45_cell_output.pdf

```
[43]: # RQ4.5 add-on - reward-level auction status and cause split.
# One bar chart: 3 statuses x 2 reward types. Stressed and incompliant bars
# are stacked by their dominant bridge/cause.
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

EPS_MT = globals().get('EPS_MT', 1e-6)
```



```

AUCTION_STATUS_ORDER = ['Auction compliant', 'Stressed', 'Incompliant']
STRESS_DETAIL_ORDER = ['Banked-allowance bridge', 'Secondary-market bridge',
↳ 'Other stress bridge']
INCOMP_CAUSE_PRIORITY = [
    'Inherited debt',
    'Underbid (price)',
    'Undersized bid (qty)',
    'Oversubscribed auction',
    'Secondary illiquidity',
    'Mixed / cascade',
]
DETAIL_COLORS = {
    'Auction compliant': '#2ca02c',
    'Banked-allowance bridge': '#1f77b4',
    'Secondary-market bridge': '#17becf',
    'Other stress bridge': '#7f7f7f',
    'Inherited debt': '#9467bd',
    'Underbid (price)': '#d62728',
    'Undersized bid (qty)': '#ff7f0e',
    'Oversubscribed auction': '#8c564b',
    'Secondary illiquidity': '#e377c2',
    'Mixed / cascade': '#bcbd22',
    'Other incompliance': '#7f7f7f',
}

reward_type_order = [
    w for w in ['Financial', 'Balanced ESG']
    if w in {_agent_weighting(i) for i in range(N_AGENTS)}
]
if not reward_type_order:
    reward_type_order = sorted({_agent_weighting(i) for i in range(N_AGENTS)})

def _series_values(frame, column, n_rows, fill_value=np.nan, clip_lower=None):
    if column not in frame.columns:
        return np.full(n_rows, fill_value, dtype=float)
    values = pd.to_numeric(frame[column], errors='coerce').fillna(fill_value)
    if clip_lower is not None:
        values = values.clip(lower=clip_lower)
    return values.to_numpy(float)

def _stressed_bridge_labels(auction_gap, bank_start, trade_qty):
    sec_bought = np.maximum(0.0, trade_qty)
    bank_available = np.maximum(0.0, bank_start)
    residual_after_secondary = np.maximum(0.0, auction_gap - sec_bought)
    bank_used_est = np.minimum(bank_available, residual_after_secondary)

```

```

labels = np.full(len(auction_gap), 'Other stress bridge', dtype=object)
secondary = (sec_bought > EPS_MT) & (sec_bought >= bank_used_est)
banked = ~secondary & ((bank_used_est > EPS_MT) | (bank_available >=
↪ auction_gap - EPS_MT))
labels[banked] = 'Banked-allowance bridge'
labels[secondary] = 'Secondary-market bridge'
return labels

def _incompliance_cause_labels(shortfall, alloc, emissions, cf_start, bid_price,
                                clearing_price, estimate_need, bid_qty_mult,
                                auction_total_demand, auction_volume,
                                sec_qty_action, trade_qty):
    n_rows = len(shortfall)
    labels = np.full(n_rows, 'Mixed / cascade', dtype=object)
    active = shortfall > EPS_MT
    remaining = active.copy()

    inherited = remaining & (alloc + EPS_MT >= emissions) & (np.maximum(0.0,
↪ cf_start) >= shortfall - EPS_MT)
    labels[inherited] = 'Inherited debt'
    remaining &= ~inherited

    underprice = remaining & (bid_price > 0) & (clearing_price > 0) &
↪ (bid_price < 0.995 * clearing_price)
    labels[underprice] = 'Underbid (price)'
    remaining &= ~underprice

    need_realised = emissions + np.maximum(0.0, cf_start)
    bid_qty_target = np.maximum(0.0, bid_qty_mult) * np.maximum(0.0,
↪ estimate_need)
    undersized_qty = np.maximum(0.0, need_realised - bid_qty_target)
    undersized = remaining & (undersized_qty > 0.5 * shortfall)
    labels[undersized] = 'Undersized bid (qty)'
    remaining &= ~undersized

    with np.errstate(divide='ignore', invalid='ignore'):
        oversubscription = np.where(auction_volume > 1e-9, auction_total_demand,
↪ auction_volume, np.nan)
    oversubscribed = remaining & (oversubscription > 1.05)
    labels[oversubscribed] = 'Oversubscribed auction'
    remaining &= ~oversubscribed

    sec_unmet = np.maximum(0.0, np.maximum(0.0, sec_qty_action) - np.maximum(0.
↪ 0, trade_qty))

```

```

secondary_illiquid = remaining & (sec_unmet > 0.25 * shortfall)
labels[secondary_illiquid] = 'Secondary illiquidity'
return labels

def _append_counts(rows, reward_type, status_label, detail_labels, mask):
    if not np.any(mask):
        return
    labels, counts = np.unique(detail_labels[mask], return_counts=True)
    for detail_label, count in zip(labels, counts):
        rows.append({
            'reward_type': reward_type,
            'status': status_label,
            'detail': str(detail_label),
            'n_cells': int(count),
        })

def _clean_pct_table(frame, row_col, denominator_col, order):
    if frame.empty:
        return pd.DataFrame(columns=reward_type_order)
    counts = frame.pivot_table(
        index=row_col,
        columns='reward_type',
        values='n_cells',
        aggfunc='sum',
        fill_value=0,
        observed=True,
    ).reindex(columns=reward_type_order, fill_value=0)
    denominators = denominator_col.reindex(reward_type_order).replace(0, np.nan)
    pct = counts.div(denominators, axis=1).mul(100.0).fillna(0.0)
    pct = pct.loc[pct.sum(axis=1) > 0]
    pct = pct.reindex([item for item in order if item in pct.index])
    return pct.round(1)

auction_status_rows = []
missing_status_cols = set()
for run_key, run_data in runs.items():
    yr = tail_year_rows(run_data)
    if yr.empty:
        continue
    n_rows = len(yr)
    clearing_price = _series_values(yr, 'clearing_price', n_rows)
    auction_total_demand = _series_values(yr, 'auction_total_demand', n_rows)
    auction_volume = _series_values(yr, 'auction_volume', n_rows)

```

```

for agent_idx in range(N_AGENTS):
    agent_no = agent_idx + 1
    required = [
        f'shortfall_A{agent_no}',
        f'alloc_A{agent_no}',
        f'emissions_A{agent_no}',
        f'carry_forward_start_A{agent_no}',
    ]
    missing = [column for column in required if column not in yr.columns]
    if missing:
        missing_status_cols.update(missing)
        continue

    reward_type = _agent_weighting(agent_idx)
    shortfall = _series_values(yr, f'shortfall_A{agent_no}', n_rows,
↪fill_value=0.0)
    alloc = _series_values(yr, f'alloc_A{agent_no}', n_rows, fill_value=0.0)
    emissions = _series_values(yr, f'emissions_A{agent_no}', n_rows,
↪fill_value=0.0, clip_lower=0.0)
    cf_start = _series_values(yr, f'carry_forward_start_A{agent_no}',
↪n_rows, fill_value=0.0, clip_lower=0.0)
    bank_start = _series_values(yr, f'bank_start_A{agent_no}', n_rows,
↪fill_value=0.0, clip_lower=0.0)
    trade_qty = _series_values(yr, f'trade_qty_A{agent_no}', n_rows,
↪fill_value=0.0)
    bid_price = _series_values(yr, f'bid_price_A{agent_no}', n_rows)
    estimate_need = _series_values(yr, f'estimate_need_A{agent_no}', n_rows)
    bid_qty_mult = _series_values(yr, f'bid_qty_mult_A{agent_no}', n_rows)
    sec_qty_action = _series_values(yr, f'sec_qty_action_A{agent_no}',
↪n_rows, fill_value=0.0)

    need_realised = emissions + cf_start
    auction_gap = np.maximum(0.0, need_realised - alloc)
    incompilant_mask = shortfall > EPS_MT
    stressed_mask = (auction_gap > EPS_MT) & ~incompilant_mask
    auction_compliant_mask = (auction_gap <= EPS_MT) & ~incompilant_mask

    _append_counts(
        auction_status_rows,
        reward_type,
        'Auction compliant',
        np.full(n_rows, 'Auction compliant', dtype=object),
        auction_compliant_mask,
    )
    _append_counts(
        auction_status_rows,

```

```

        reward_type,
        'Stressed',
        _stressed_bridge_labels(auction_gap, bank_start, trade_qty),
        stressed_mask,
    )
    _append_counts(
        auction_status_rows,
        reward_type,
        'Incompliant',
        _incompliance_cause_labels(
            shortfall, alloc, emissions, cf_start, bid_price,
↪clearing_price,
            estimate_need, bid_qty_mult, auction_total_demand,
↪auction_volume,
            sec_qty_action, trade_qty,
        ),
        incompliant_mask,
    )

auction_status_df = pd.DataFrame(auction_status_rows)
if auction_status_df.empty:
    print('RQ4.5 add-on] No auction-status rows built.')
    if missing_status_cols:
        print('RQ4.5 add-on] Missing required columns:',
↪sorted(missing_status_cols)[:12])
else:
    auction_status_df['reward_type'] = pd.Categorical(
        auction_status_df['reward_type'], categories=reward_type_order,
↪ordered=True
    )
    auction_status_df['status'] = pd.Categorical(
        auction_status_df['status'], categories=AUCTION_STATUS_ORDER,
↪ordered=True
    )

    reward_totals = auction_status_df.groupby('reward_type',
↪observed=True)['n_cells'].sum()
    status_summary = (
        auction_status_df
        .groupby(['reward_type', 'status'], observed=True)['n_cells']
        .sum()
        .reset_index()
    )
    status_summary['share_pct'] = status_summary.apply(
        lambda row: 100.0 * row['n_cells'] / reward_totals.
↪loc[row['reward_type']], axis=1

```

```

)
auction_status_table = (
    status_summary
    .pivot_table(
        index='reward_type', columns='status', values='share_pct',
        aggfunc='sum', fill_value=0.0, observed=True,
    )
    .reindex(index=reward_type_order, columns=AUCTION_STATUS_ORDER)
    .round(1)
)
auction_status_table.columns.name = None
print(['RQ4.5 add-on] Status shares: % of all tail-window agent-year cells_
↳within each reward type')
display(auction_status_table)

detail_summary = (
    auction_status_df
    .groupby(['reward_type', 'status', 'detail'], observed=True)['n_cells']
    .sum()
    .reset_index()
)

incomp_totals = (
    detail_summary[detail_summary['status'] == 'Incompliant']
    .groupby('detail', observed=True)['n_cells']
    .sum()
    .sort_values(ascending=False)
)

top_incomp_causes = [cause for cause in incomp_totals.head(4).index if
↳incomp_totals.loc[cause] > 0]

plot_summary = detail_summary.copy()
plot_summary['plot_detail'] = plot_summary['detail']
incomp_other = (plot_summary['status'] == 'Incompliant') &
↳plot_summary['detail'].isin(top_incomp_causes)
plot_summary.loc[incomp_other, 'plot_detail'] = 'Other incompliance'
stress_other = (plot_summary['status'] == 'Stressed') &
↳plot_summary['detail'].isin(STRESS_DETAIL_ORDER[:2])
plot_summary.loc[stress_other, 'plot_detail'] = 'Other stress bridge'
plot_summary = (
    plot_summary
    .groupby(['reward_type', 'status', 'plot_detail'],
↳observed=True)['n_cells']
    .sum()
    .reset_index()
)
plot_summary['share_pct'] = plot_summary.apply(

```

```

        lambda row: 100.0 * row['n_cells'] / reward_totals.
    loc[row['reward_type']], axis=1
    )

    status_totals = (
        plot_summary
        .groupby(['reward_type', 'status'], observed=True)['n_cells']
        .sum()
        .rename('status_total')
        .reset_index()
    )

    split_summary = plot_summary.merge(status_totals, on=['reward_type',
    'status'], how='left')
    split_summary['within_status_pct'] = np.where(
        split_summary['status_total'] > 0,
        100.0 * split_summary['n_cells'] / split_summary['status_total'],
        0.0,
    )

    stress_split = split_summary[split_summary['status'] == 'Stressed']
    stress_order = [item for item in STRESS_DETAIL_ORDER if item in
    set(stress_split['plot_detail'])]
    stress_split_table = (
        stress_split
        .pivot_table(
            index='plot_detail', columns='reward_type',
            values='within_status_pct',
            aggfunc='sum', fill_value=0.0, observed=True,
        )
        .reindex(index=stress_order, columns=reward_type_order)
        .round(1)
    )
    stress_split_table.index.name = 'stressed bridge (% within stressed)'
    print(['RQ4.5 add-on] Stressed split: % within stressed cells for each
    reward type')
    display(stress_split_table)

    incomp_split = split_summary[split_summary['status'] == 'Incompliant']
    incomp_order = top_incomp_causes + (['Other incomppliance'] if 'Other
    incomppliance' in set(incomp_split['plot_detail']) else [])
    incomp_split_table = (
        incomp_split
        .pivot_table(
            index='plot_detail', columns='reward_type',
            values='within_status_pct',
            aggfunc='sum', fill_value=0.0, observed=True,

```

```

    )
    .reindex(index=incomp_order, columns=reward_type_order)
    .round(1)
)
incomp_split_table.index.name = 'incompliant cause (% within incompliant)'
print(['RQ4.5 add-on] Incompliant split: % within incompliant cells for_
↳each reward type')
display(incomp_split_table)

plotted_segment_table = (
    plot_summary
    .pivot_table(
        index=['status', 'plot_detail'], columns='reward_type',
↳values='share_pct',
        aggfunc='sum', fill_value=0.0, observed=True,
    )
    .reindex(columns=reward_type_order)
)
plotted_segment_table = plotted_segment_table.loc[plotted_segment_table.
↳sum(axis=1) > 0].round(1)
plotted_segment_table.index.names = ['status', 'segment (% of all cells;_
↳matches stacked bars)']
print(['RQ4.5 add-on] Plotted segment shares: % of all tail-window cells_
↳within each reward type')
display(plotted_segment_table)

plot_detail_order = ['Auction compliant'] + STRESS_DETAIL_ORDER[:2]
if (plot_summary['plot_detail'] == 'Other stress bridge').any():
    plot_detail_order.append('Other stress bridge')
plot_detail_order += top_incomp_causes
if (plot_summary['plot_detail'] == 'Other incompliance').any():
    plot_detail_order.append('Other incompliance')
plot_detail_order = list(dict.fromkeys(plot_detail_order))

n_rewards = len(reward_type_order)
group_gap = 0.85
bar_width = 0.52
x_positions = {}
x_ticks = []
x_tick_labels = []
group_centers = []
for status_idx, status_label in enumerate(AUCTION_STATUS_ORDER):
    base = status_idx * (n_rewards + group_gap)
    group_centers.append(base + (n_rewards - 1) / 2)
    for reward_idx, reward_type in enumerate(reward_type_order):
        x_pos = base + reward_idx
        x_positions[(status_label, reward_type)] = x_pos

```



```

        x_ticks.append(x_pos)
        x_tick_labels.append('ESG' if reward_type == 'Balanced ESG' else
↪reward_type)

fig, ax = plt.subplots(figsize=(10.8, 5.4))
bottoms = {key: 0.0 for key in x_positions}
for detail_label in plot_detail_order:
    xs, heights, bottom_values = [], [], []
    for status_label in AUCTION_STATUS_ORDER:
        for reward_type in reward_type_order:
            sub = plot_summary[
                (plot_summary['status'] == status_label)
                & (plot_summary['reward_type'] == reward_type)
                & (plot_summary['plot_detail'] == detail_label)
            ]
            value = float(sub['share_pct'].sum()) if not sub.empty else 0.0
            key = (status_label, reward_type)
            xs.append(x_positions[key])
            heights.append(value)
            bottom_values.append(bottoms[key])
    if max(heights) <= 0:
        continue
    ax.bar(
        xs,
        heights,
        width=bar_width,
        bottom=bottom_values,
        color=DETAIL_COLORS.get(detail_label, '#7f7f7f'),
        edgecolor='white',
        linewidth=0.7,
        label=detail_label,
    )
    for key, value in zip(x_positions, heights):
        bottoms[key] += value

for key, total in bottoms.items():
    if total <= 0:
        continue
    ax.text(
        x_positions[key], total + 1.1, f'{total:.1f}%',
        ha='center', va='bottom', fontsize=9,
    )

for center, status_label in zip(group_centers, AUCTION_STATUS_ORDER):
    ax.text(center, -0.14, status_label, ha='center', va='top',
            transform=ax.get_xaxis_transform(), fontsize=10,
↪fontweight='bold')

```

```

for status_idx in range(1, len(AUCTION_STATUS_ORDER)):
    separator = status_idx * (n_rewards + group_gap) - group_gap / 2
    ax.axvline(separator, color='0.82', linewidth=0.9)

ax.set_xticks(x_ticks)
ax.set_xticklabels(x_tick_labels)
ax.set_ylim(0, 106)
ax.set_ylabel('Share of tail-window agent-year cells (%)')
ax.set_title(f'RQ4.5 add-on - auction status and cause split by reward type,
↳(last {TAIL_EPISODES:,} episodes)')
ax.grid(True, alpha=0.3, axis='y')
ax.legend(title='Status / split', loc='center left', bbox_to_anchor=(1.02,
↳0.5), fontsize=8, title_fontsize=9)
plt.subplots_adjust(bottom=0.22, right=0.76)
plt.show()

plotted_segment_counts = (
    plot_summary
    .pivot_table(
        index=['status', 'plot_detail'], columns='reward_type',
↳values='n_cells',
        aggfunc='sum', fill_value=0, observed=True,
    )
    .reindex(columns=reward_type_order)
    .astype(int)
)
plotted_segment_counts = plotted_segment_counts.loc[plotted_segment_counts.
↳sum(axis=1) > 0]
plotted_segment_counts.index.names = ['status', 'segment']
print('RQ4.5 add-on] Raw pooled cell counts for plotted segments:')
display(plotted_segment_counts)

```

[RQ4.5 add-on] Status shares: % of all tail-window agent-year cells within each reward type

	Auction compliant	Stressed	Incompliant
reward_type			
Financial	38.1	35.8	26.1
Balanced ESG	53.4	38.2	8.4

[RQ4.5 add-on] Stressed split: % within stressed cells for each reward type

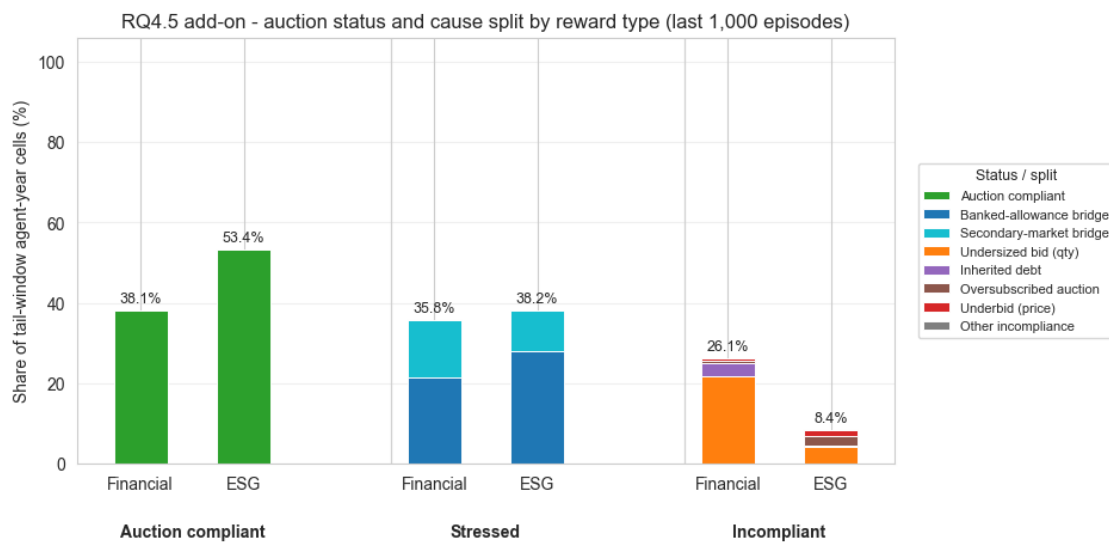
reward_type	Financial	Balanced ESG
stressed bridge (% within stressed)		
Banked-allowance bridge	59.9	73.5
Secondary-market bridge	40.1	26.5

[RQ4.5 add-on] Incompliant split: % within incompliant cells for each reward type

reward_type	Financial	Balanced ESG
incompliant cause (% within incompliant)		
Undersized bid (qty)	83.7	49.6
Inherited debt	12.4	5.5
Oversubscribed auction	2.1	26.6
Underbid (price)	1.7	18.4
Other incompliance	0.0	0.0

[RQ4.5 add-on] Plotted segment shares: % of all tail-window cells within each reward type

reward_type		Financial	
↪Balanced ESG			
status	segment (% of all cells; matches stacked bars)		
Auction compliant	Auction compliant	38.1	↪
↪	53.4		
Stressed	Banked-allowance bridge	21.5	↪
↪	28.1		
	Secondary-market bridge	14.4	↪
↪	10.1		
Incompliant	Inherited debt	3.2	↪
↪	0.5		
	Other incompliance	0.0	↪
↪	0.0		
	Oversubscribed auction	0.6	↪
↪	2.2		
	Underbid (price)	0.5	↪
↪	1.5		
	Undersized bid (qty)	21.8	↪
↪	4.1		



[RQ4.5 add-on] Raw pooled cell counts for plotted segments:

reward_type		Financial	Balanced ESG
status	segment		
Auction compliant	Auction compliant	73164	102549
Stressed	Banked-allowance bridge	41202	53950
	Secondary-market bridge	27591	19432
Incompliant	Inherited debt	6192	878
	Other incompliance	7	3
	Oversubscribed auction	1061	4272
	Underbid (price)	874	2951
	Undersized bid (qty)	41909	7965

RQ4 summary. RQ4.1–4.3 give the run-level mechanics: did the agents close the coverage gap, did the auction stay diffuse, and did the secondary market actually clear the residual? RQ4.4 attributes the residual debt and emergency-loan usage to the (archetype $\times$ weighting) cells — the financial vs ESG split is the most informative cut here, since coal-heavy financial agents are the ones for whom the gap-penalty reward (driven by `effective_penalty_rate`) actively dominates the bid head.

1.9 Year-over-Year Dynamics in the Converged Window — Cross-Sweep Comparison

All preceding sections pool every loaded run into a single default-config population. This section instead splits by **sweep** and overlays every loaded sweep on the same axes, so cross-sweep differences become visible at a glance. The x-axis is always the simulation year (1..N_YEARS) and every series is restricted to the converged tail window (PELT and plateau diagnostics confirm convergence well within the final 10 000 episodes; the default tail of 200 episodes is enough to capture the steady-state shape, but the knob can be raised below).

Each panel shows a labelled cross-seed median line (within a sweep) plus a labelled shaded min/max fan, so the spread is bounded by the observed seed extremes within that sweep. The companion tables list the mean alongside the median for each year, while the vertical separation between coloured medians reflects the cross-sweep effect.

```
[30]: # Cross-sweep year-over-year helpers.
#
# YOY_TAIL_EPISODES bounds how many of the final episodes per run feed
# the year-axis aggregations below. 200 is the working default; raising
# it adds samples but does not change the steady-state shape because the
# runs are already converged at that point (PELT/plateau diagnostics).
import itertools
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```

from IPython.display import display

YOY_TAIL_EPISODES = 1000
YOY_SWEEP_FILTER = ['default_seeds']

# Stable sweep ordering and colour map so every panel below uses the
# same colour for a given sweep.
AVAILABLE_SWEEP_NAMES = sorted({k[0] for k in runs.keys()})
if YOY_SWEEP_FILTER:
    SWEEP_NAMES = [sw for sw in AVAILABLE_SWEEP_NAMES if sw in YOY_SWEEP_FILTER]
    if not SWEEP_NAMES:
        raise ValueError(
            f'YOY_SWEEP_FILTER={YOY_SWEEP_FILTER} matched no loaded sweeps. '
            f'Available: {AVAILABLE_SWEEP_NAMES}'
        )
else:
    SWEEP_NAMES = AVAILABLE_SWEEP_NAMES
_palette = plt.get_cmap('tab10').colors
SWEEP_COLORS = {sw: _palette[i % len(_palette)] for i, sw in
    ↪ enumerate(SWEEP_NAMES)}
print(f'YoY converged-window section - tail = {YOY_TAIL_EPISODES} episodes per_
    ↪ run')
print(f'Available sweeps: {AVAILABLE_SWEEP_NAMES}')
print(f'Sweeps overlaid: {SWEEP_NAMES}')
for sw in SWEEP_NAMES:
    n_runs = sum(1 for k in runs if k[0] == sw)
    print(f' {sw}: {n_runs} run(s)')

def _format_yoy_year_axis(ax):
    if 'format_year_axis' in globals():
        format_year_axis(ax)

def _reference_reserve_price():
    ets_cfg = REFERENCE_CONFIG.get('ets', {}) or {}
    return float(ets_cfg.get('reserve_price', ets_cfg.
    ↪ get('auction_reserve_price', np.nan)))

def _penalty_rate_by_years(years):
    penalty_cfg = REFERENCE_CONFIG.get('penalty', {}) or {}
    ets_cfg = REFERENCE_CONFIG.get('ets', {}) or {}
    base = penalty_cfg.get('rate', ets_cfg.get('penalty_rate', np.nan))
    inflation = penalty_cfg.get('inflation_rate', penalty_cfg.get('inflation',
    ↪ 0.0))
    years = pd.Index(years, name='year')

```

```

if len(years) == 0 or pd.isna(base):
    return pd.Series(dtype=float, name='penalty_rate')
year_origin = YEAR_DISPLAY_START if 'YEAR_DISPLAY_START' in globals() else 1
year_offset = np.asarray(years, dtype=float) - float(year_origin)
return pd.Series(
    float(base) * np.power(1.0 + float(inflation), np.clip(year_offset, 0.
↪0, None)),
    index=years,
    name='penalty_rate',
)

def _years_from_sweep_data(sweep_data):
    years = []
    for df in (sweep_data or {}).values():
        if df is not None and not df.empty:
            years.extend([y for y in df.index if pd.notna(y)])
    return pd.Index(sorted(set(years)), name='year')

def _yoy_year_rows(run, n=YOY_TAIL_EPISODES):
    """Return year_log rows whose episode is in the last `n` episodes."""
    ep = run['ep_df']
    yr = run['yr_df']
    if ep.empty or yr.empty or 'episode' not in yr.columns or 'episode' not in_
↪ep.columns:
        return yr.head(0)
    last_eps = ep['episode'].tail(min(n, len(ep))).unique()
    return yr[yr['episode'].isin(last_eps)]

def _per_sweep_year_stack(metric_fn):
    """For each sweep, build a wide DataFrame indexed by year with one
    column per (variant, seed). `metric_fn(yr_rows)` must return a
    pd.Series indexed by year (or None to skip the run)."""
    out = {}
    for sw in SWEEP_NAMES:
        cols = {}
        for k, run in runs.items():
            if k[0] != sw:
                continue
            yr = _yoy_year_rows(run)
            if yr.empty or 'year' not in yr.columns:
                continue
            s = metric_fn(yr)
            if s is None or len(s) == 0:
                continue

```

```

        cols[(k[1], k[2])] = s
    if cols:
        out[sw] = pd.concat(cols, axis=1).sort_index()
    return out

def _overlay_yoy(ax, sweep_data, ylabel, title, *, show_band=True, marker='o'):
    """Overlay one median line plus a labelled min/max fan per sweep on `ax`. """
    if not sweep_data:
        ax.text(0.5, 0.5, 'no data', ha='center', va='center', transform=ax.
        ↪transAxes)
        ax.set_title(title); ax.set_ylabel(ylabel); ax.set_xlabel('Year')
        _format_yoy_year_axis(ax)
        return
    for sw in SWEEP_NAMES:
        df = sweep_data.get(sw)
        if df is None or df.empty:
            continue
        color = SWEEP_COLORS[sw]
        med = df.median(axis=1)
        lo = df.min(axis=1)
        hi = df.max(axis=1)
        n = df.shape[1]
        if show_band and n > 1:
            ax.fill_between(med.index, lo.values, hi.values,
                            color=color, alpha=0.15,
                            label=f'{sw} min/max fan')
        ax.plot(med.index, med.values, marker=marker, lw=2.0, color=color,
                label=f'{sw} median line (n={n})')
    ax.set_title(title); ax.set_xlabel('Year'); ax.set_ylabel(ylabel)
    ax.grid(alpha=0.3)
    ax.legend(fontsize=8, loc='best')
    _format_yoy_year_axis(ax)

def _overlay_yoy_reference(ax, sweep_data, label, *, color='black',
    ↪linestyle='--', lw=1.6, alpha=0.75):
    """Overlay a dashed median reference line from the same per-sweep stack,
    ↪format. """
    if not sweep_data:
        return
    for sw in SWEEP_NAMES:
        df = sweep_data.get(sw)
        if df is None or df.empty:
            continue
        med = df.median(axis=1)
        line_color = SWEEP_COLORS[sw] if color is None else color

```

```

        line_label = f'{sw} {label}' if len(SWEEP_NAMES) > 1 else label
        ax.plot(med.index, med.values, linestyle=linestyle, lw=lw,
                color=line_color, alpha=alpha, label=line_label)

def _agent_cols_in(yr_df, prefix, max_idx=N_TOTAL):
    """Return existing `{prefix}_A{i}` columns in yr_df, sorted by i."""
    return [f'{prefix}_A{i+1}' for i in range(max_idx)
            if f'{prefix}_A{i+1}' in yr_df.columns]

def _system_sum_by_year(yr_df, prefix, max_idx=N_TOTAL):
    """Per-year mean across episodes of the cross-agent SUM of {prefix}_A*."""
    cols = _agent_cols_in(yr_df, prefix, max_idx)
    if not cols:
        return None
    return yr_df.groupby('year')[cols].mean().sum(axis=1)

def _agent_mean_by_year(yr_df, prefix, max_idx=N_TOTAL):
    """Per-year mean across episodes of the cross-agent MEAN of {prefix}_A*."""
    cols = _agent_cols_in(yr_df, prefix, max_idx)
    if not cols:
        return None
    return yr_df.groupby('year')[cols].mean().mean(axis=1)

def _scalar_mean_by_year(yr_df, col):
    if col not in yr_df.columns:
        return None
    return yr_df.groupby('year')[col].mean()

def _scalar_std_by_year(yr_df, col):
    """Cross-episode std of `col` per year - the within-run volatility."""
    if col not in yr_df.columns:
        return None
    return yr_df.groupby('year')[col].std()

def _yoy_panel_tables(panels, *, units=None):
    """Build companion data tables for a 2x2 YoY figure.

    Parameters
    -----
    panels : list of (title, sweep_data)
        `sweep_data` is the dict returned by `_per_sweep_year_stack`,

```



```

        i.e. `{sweep_name: wide_df indexed by year, columns = (variant,
        seed)}`.
units : optional dict {title: unit_str}, embedded in the printout.

Returns
-----
(per_year_df, summary_df)
    `per_year_df` - one row per (panel, sweep, year) with the
    mean and median across seeds, the cross-seed std, the min/max
    fan bounds, and n.
    `summary_df` - one row per (panel, sweep) summarising the
    per-year median series across years (mean / median / std / min /
    max), i.e. the scalar headline numbers for the panel.
"""
py_rows, ag_rows = [], []
for title, sd in panels:
    unit = (units or {}).get(title, '')
    if not sd:
        ag_rows.append({'panel': title, 'unit': unit, 'sweep': '(no data)',
                        'n_seeds': 0, 'years': 0,
                        'mean': np.nan, 'median': np.nan,
                        'std': np.nan, 'min': np.nan, 'max': np.nan})
        continue
    for sw, df in sd.items():
        mean = df.mean(axis=1)
        med = df.median(axis=1)
        std = df.std(axis=1)
        mn = df.min(axis=1)
        mx = df.max(axis=1)
        n = df.shape[1]
        for y in med.index:
            py_rows.append({
                'panel': title, 'unit': unit, 'sweep': sw,
                'year': int(y),
                'mean_across_seeds': mean.loc[y],
                'median_across_seeds': med.loc[y],
                'std_across_seeds': std.loc[y],
                'min_across_seeds': mn.loc[y],
                'max_across_seeds': mx.loc[y],
                'n_seeds': n,
            })
        ag_rows.append({
            'panel': title, 'unit': unit, 'sweep': sw,
            'n_seeds': n, 'years': int(len(med)),
            'mean': med.mean(), 'median': med.median(),
            'std': med.std(), 'min': med.min(), 'max': med.max(),
        })

```

```

py_df = pd.DataFrame(py_rows)
ag_df = pd.DataFrame(ag_rows)
return py_df, ag_df

def _show_yoy_tables(pannels, *, units=None, float_fmt='{:,.4g}'):
    """Compute and display the per-year and across-year summary tables
    for a 2x2 YoY figure. Returns the two DataFrames so calling cells
    can re-use them if needed."""
    py_df, ag_df = _yoy_panel_tables(pannels, units=units)
    with pd.option_context('display.float_format', float_fmt.format,
                           'display.max_rows', 200):
        print(f'Across-year summary (plotted median series across_
↪{YOY_TAIL_EPISODES}-episode tail, per sweep):')
        display(ag_df)
        print('Per-year detail (median line; shaded min/max fan; mean listed_
↪for comparison):')
        display(py_df)
    return py_df, ag_df

```

YoY converged-window section - tail = 1000 episodes per run
Available sweeps: ['default_seeds']
Sweeps overlaid: ['default_seeds']
default_seeds: 4 run(s)

1.9.1 Price dynamics over the simulation year

Clearing price and secondary price (level + cross-seed spread per sweep), price realism measured by the clearing/fundamental-anchor ratio (1 means the auction tracks the fundamental), and the within-run cross-episode std of the clearing price (a proxy for year-conditional price volatility within the converged window).

```

[31]: fig, axes = plt.subplots(2, 2, figsize=(14, 8))

_clearing_price_yoy = _per_sweep_year_stack(lambda yr: _scalar_mean_by_year(yr,
↪'clearing_price'))
_overlay_yoy(
    axes[0, 0],
    _clearing_price_yoy,
    ylabel='€/tonne', title='Auction clearing price')

_fundamental_anchor_yoy = _per_sweep_year_stack(lambda yr:
↪_scalar_mean_by_year(yr, 'fundamental_anchor'))
_overlay_yoy_reference(
    axes[0, 0],
    _fundamental_anchor_yoy,
    label='fundamental price anchor',

```

```

        color='black', linestyle='--', alpha=0.7)
_reserve_price = _reference_reserve_price()
if np.isfinite(_reserve_price):
    axes[0, 0].axhline(_reserve_price, color='dimgray', lw=1.2,
                      linestyle=':', alpha=0.8,
                      label=f'reserve price ({_reserve_price:g} €/t)')
axes[0, 0].legend(fontsize=8, loc='best')

_secondary_price_yoy = _per_sweep_year_stack(lambda yr:
    ↪ _scalar_mean_by_year(yr, 'secondary_price'))
_overlay_yoy(
    axes[0, 1],
    _secondary_price_yoy,
    ylabel='€/tonne', title='Secondary-market price')
_penalty_rate =
    ↪ _penalty_rate_by_years(_years_from_sweep_data(_secondary_price_yoy))
if not _penalty_rate.empty:
    axes[0, 1].plot(_penalty_rate.index, _penalty_rate.values,
                   color='red', lw=1.6, linestyle='--', alpha=0.75,
                   label='penalty rate')
axes[0, 1].legend(fontsize=8, loc='best')

def _clearing_over_anchor(yr):
    if 'clearing_price' not in yr.columns or 'fundamental_anchor' not in yr.
    ↪ columns:
        return None
    g = yr.groupby('year')[['clearing_price', 'fundamental_anchor']].mean()
    anchor = g['fundamental_anchor'].replace(0, np.nan)
    return (g['clearing_price'] / anchor)

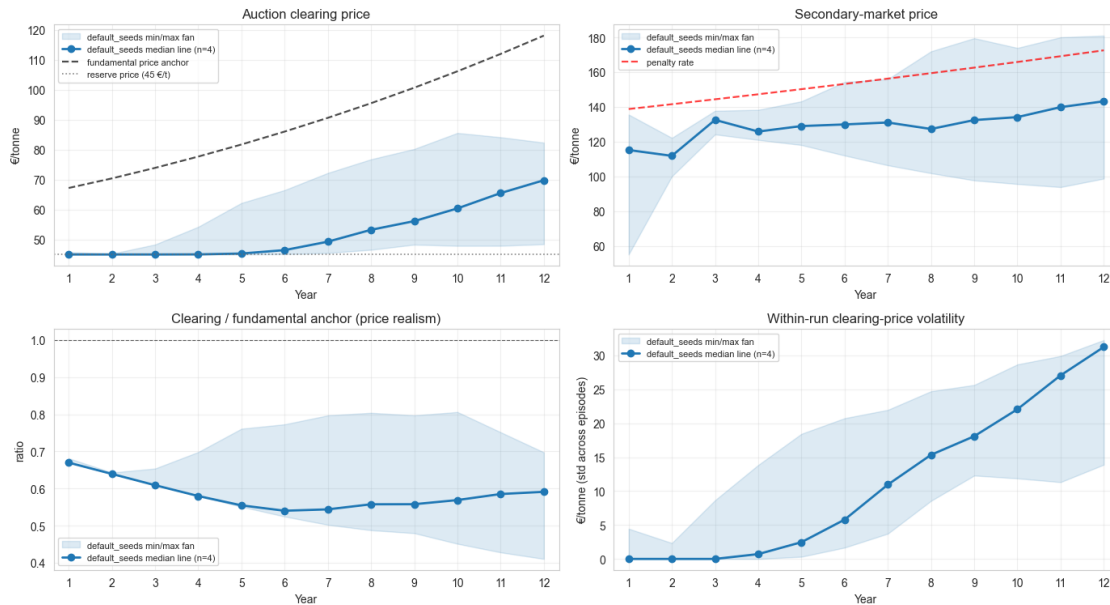
_overlay_yoy(
    axes[1, 0],
    _per_sweep_year_stack(_clearing_over_anchor),
    ylabel='ratio', title='Clearing / fundamental anchor (price realism)')
axes[1, 0].axhline(1.0, color='black', lw=0.8, linestyle='--', alpha=0.6)

_overlay_yoy(
    axes[1, 1],
    _per_sweep_year_stack(lambda yr: _scalar_std_by_year(yr, 'clearing_price')),
    ylabel='€/tonne (std across episodes)',
    title='Within-run clearing-price volatility')

plt.suptitle(f'Price dynamics by simulation year (last {YOY_TAIL_EPISODES} eps
    ↪ per run)',
            fontsize=13, y=1.00)
plt.tight_layout(); plt.show()

```

Price dynamics by simulation year (last 1000 eps per run)



```
[32]: # Companion data tables for the price-dynamics figure above.
# Each row of the per-year table corresponds to one plotted point;
# the across-year summary collapses the median series into mean/std/
# min/max so the thesis text can quote both shape and headline numbers.

def _clearing_over_anchor_tbl(yr):
    if 'clearing_price' not in yr.columns or 'fundamental_anchor' not in yr:
        return None
    g = yr.groupby('year')[['clearing_price', 'fundamental_anchor']].mean()
    anchor = g['fundamental_anchor'].replace(0, np.nan)
    return g['clearing_price'] / anchor

_panels_price = [
    ('Auction clearing price',
     _per_sweep_year_stack(lambda yr: _scalar_mean_by_year(yr,
     ↪ 'clearing_price'))),
    ('Secondary-market price',
     _per_sweep_year_stack(lambda yr: _scalar_mean_by_year(yr,
     ↪ 'secondary_price'))),
    ('Clearing / fundamental anchor (price realism)',
     _per_sweep_year_stack(_clearing_over_anchor_tbl)),
    ('Within-run clearing-price volatility',
     _per_sweep_year_stack(lambda yr: _scalar_std_by_year(yr,
     ↪ 'clearing_price'))),
]
```

```

_units_price = {
    'Auction clearing price': '€/tonne',
    'Secondary-market price': '€/tonne',
    'Clearing / fundamental anchor (price realism)': 'ratio',
    'Within-run clearing-price volatility': '€/tonne (std across episodes)',
}
_price_yoy_tables = _show_yoy_tables(_panels_price, units=_units_price)

```

Across-year summary (plotted median series across 1000-episode tail, per sweep):

					panel				unit
	sweep	n_seeds	years	mean	median	std	min	max	
0					Auction clearing price				€/tonne
	↪ default_seeds	4	12	52.17	47.86	8.868	45	69.78	
1					Secondary-market price				€/tonne
	↪ default_seeds	4	12	129.3	130.4	8.906	111.8	143.2	
2					Clearing / fundamental anchor (price realism)				ratio
	↪ default_seeds	4	12	0.5829	0.5742	0.03946	0.54	0.6696	
3					Within-run clearing-price volatility				€/tonne (std across episodes)
	↪ default_seeds	4	12	11.15	8.376	11.4	0	31.25	

Per-year detail (median line; shaded min/max fan; mean listed for comparison):

					panel		unit
	sweep	year	mean_across_seeds	median_across_seeds	std_across_seeds		
	↪ min_across_seeds		max_across_seeds	\			
0					Auction clearing price		€/tonne
	↪ default_seeds	1		45.2		45	0.3955
	↪		45	45.79			
1					Auction clearing price		€/tonne
	↪ default_seeds	2		45.06		45	0.1272
	↪		45	45.25			
2					Auction clearing price		€/tonne
	↪ default_seeds	3		45.83		45	1.659
	↪		45	48.32			
3					Auction clearing price		€/tonne
	↪ default_seeds	4		47.32		45.03	4.608
	↪		45	54.24			
4					Auction clearing price		€/tonne
	↪ default_seeds	5		49.46		45.31	8.492
	↪		45.02	62.19			
5					Auction clearing price		€/tonne
	↪ default_seeds	6		51.13		46.45	10.3
	↪		45.11	66.49			
6					Auction clearing price		€/tonne
	↪ default_seeds	7		54.07		49.27	12.42
	↪		45.5	72.22			

7		Auction clearing price		€/tonne
↪ default_seeds	8	57.45	53.23	13.79
↪	46.56	76.77		
8		Auction clearing price		€/tonne
↪ default_seeds	9	60.19	56.13	15.24
↪	48.27	80.23		
9		Auction clearing price		€/tonne
↪ default_seeds	10	63.55	60.38	17.65
↪	47.87	85.57		
10		Auction clearing price		€/tonne
↪ default_seeds	11	65.75	65.5	17.51
↪	47.88	84.14		
11		Auction clearing price		€/tonne
↪ default_seeds	12	67.58	69.78	16.98
↪	48.42	82.35		
12		Secondary-market price		€/tonne
↪ default_seeds	1	105.2	115.2	35.98
↪	55.08	135.6		
13		Secondary-market price		€/tonne
↪ default_seeds	2	111.4	111.8	9.363
↪	100	122.2		
14		Secondary-market price		€/tonne
↪ default_seeds	3	131.7	132.5	6.687
↪	124.1	137.7		
15		Secondary-market price		€/tonne
↪ default_seeds	4	127.7	125.8	8.336
↪	121	138.3		
16		Secondary-market price		€/tonne
↪ default_seeds	5	129.8	129	10.32
↪	118	143.2		
17		Secondary-market price		€/tonne
↪ default_seeds	6	131.5	129.9	18.46
↪	112	154.3		
18		Secondary-market price		€/tonne
↪ default_seeds	7	131.1	131	22.07
↪	106.4	155.9		
19		Secondary-market price		€/tonne
↪ default_seeds	8	132.1	127.3	31.09
↪	101.8	171.9		
20		Secondary-market price		€/tonne
↪ default_seeds	9	135.5	132.4	36.38
↪	97.7	179.5		
21		Secondary-market price		€/tonne
↪ default_seeds	10	134.4	134.1	36.48
↪	95.62	173.9		

22		Secondary-market price		€/tonne
↪	default_seeds	11	138.4	139.8
↪		93.83	180	38.62
23		Secondary-market price		€/tonne
↪	default_seeds	12	141.5	143.2
↪		98.72	181	36.74
24	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	1	0.6726	0.6696
↪		0.6696	0.6814	0.005885
25	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	2	0.6401	0.6392
↪		0.6392	0.6429	0.001807
26	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	3	0.6201	0.6088
↪		0.6088	0.6537	0.02245
27	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	4	0.6092	0.5797
↪		0.5793	0.6981	0.05931
28	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	5	0.6052	0.5545
↪		0.5508	0.761	0.1039
29	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	6	0.5943	0.54
↪		0.5243	0.7729	0.1197
30	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	7	0.5966	0.5437
↪		0.5021	0.7969	0.1371
31	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	8	0.6016	0.5574
↪		0.4875	0.8039	0.1444
32	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	9	0.5979	0.5576
↪		0.4795	0.7969	0.1514
33	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	10	0.5986	0.5688
↪		0.451	0.8061	0.1663
34	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	11	0.5874	0.585
↪		0.4277	0.7516	0.1564
35	Clearing / fundamental anchor (price realism)			ratio
↪	default_seeds	12	0.5724	0.591
↪		0.4101	0.6974	0.1438
36	Within-run clearing-price volatility		€/tonne (std across episodes)	
↪	default_seeds	1	1.114	0
↪		0	4.455	2.227

37	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	2	0.5861	0 1.172
↪	0	2.344	
38	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	3	2.164	0 4.327
↪	0	8.655	
39	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	4	3.818	0.7098 6.698
↪	0	13.85	
40	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	5	5.924	2.467 8.458
↪	0.3167	18.44	
41	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	6	8.503	5.796 8.662
↪	1.667	20.75	
42	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	7	11.9	10.96 7.942
↪	3.721	21.98	
43	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	8	16	15.35 7.174
↪	8.557	24.73	
44	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	9	18.53	18.09 6.605
↪	12.29	25.65	
45	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	10	21.18	22.08 7.264
↪	11.87	28.69	
46	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	11	23.85	27.07 8.513
↪	11.32	29.93	
47	Within-run clearing-price volatility	€/tonne (std across episodes)	
↪ default_seeds	12	27.17	31.25 8.895
↪	13.89	32.28	

	n_seeds
0	4
1	4
2	4
3	4
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5	4
6	4
7	4
8	4
9	4
10	4
11	4

12	4
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37	4
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39	4
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42	4
43	4
44	4
45	4
46	4
47	4

1.9.2 Compliance and penalty pressure over the simulation year

System-aggregate coverage gap (Mt remaining after the auction), end-of-year carry-forward (debt cascade indicator), compliance rate (share of (agent, year) cells with non-negative compliance surplus), and the system-aggregate penalty paid per year.

```
[33]: fig, axes = plt.subplots(2, 2, figsize=(14, 8))

      _overlay_yoy(
          axes[0, 0],
          _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'coverage_gap')),
```

```

        ylabel='Mt', title='System coverage gap (pre-secondary)')

_overlay_yoy(
    axes[0, 1],
    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr,
↳ 'carry_forward_end')),
    ylabel='Mt', title='End-of-year carry-forward (system)')

def _compliance_rate_by_year(yr):
    sf_cols = _agent_cols_in(yr, 'shortfall')
    if sf_cols:
        long = yr.melt(id_vars=['year'], value_vars=sf_cols,
↳ value_name='shortfall')
        long['compliant'] = (long['shortfall'] <= 1e-6).astype(float)
        return long.groupby('year')['compliant'].mean()

    surplus_cols = _agent_cols_in(yr, 'compliance_surplus')
    if not surplus_cols:
        return None
    long = yr.melt(id_vars=['year'], value_vars=surplus_cols,
↳ value_name='surplus')
    long['compliant'] = (long['surplus'].fillna(0) >= -1e-6).astype(float)
    return long.groupby('year')['compliant'].mean()

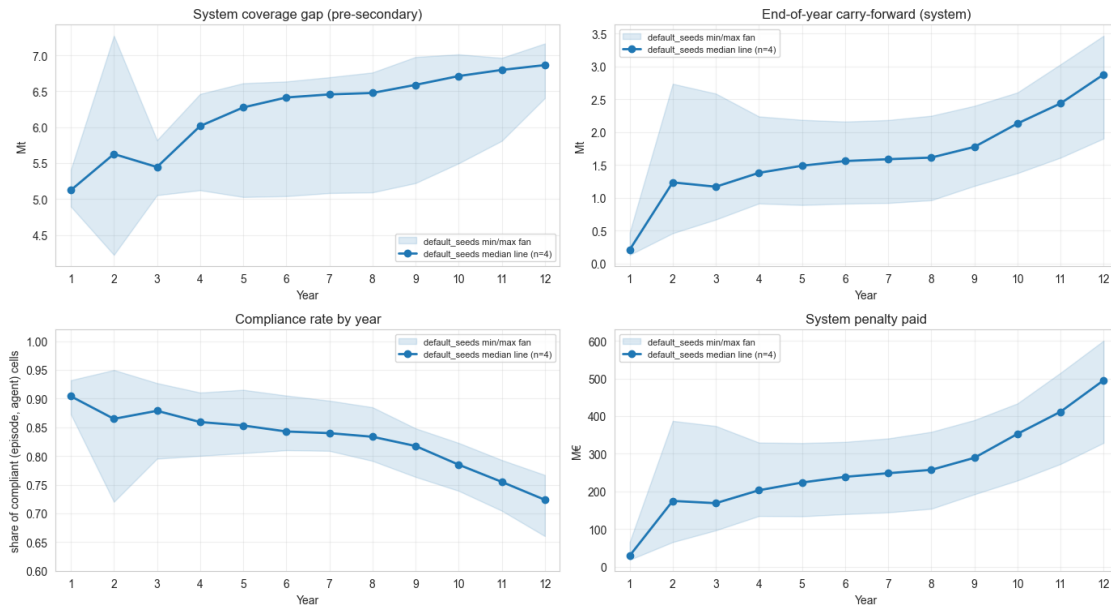
_overlay_yoy(
    axes[1, 0],
    _per_sweep_year_stack(_compliance_rate_by_year),
    ylabel='share of compliant (episode, agent) cells',
    title='Compliance rate by year')
axes[1, 0].set_ylim(0.6, 1.02)

_overlay_yoy(
    axes[1, 1],
    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'penalty')),
    ylabel='M€', title='System penalty paid')

plt.suptitle(f'Compliance pressure by simulation year (last {YOY_TAIL_EPISODES}
↳ eps per run)',
            fontsize=13, y=1.00)
plt.tight_layout(); plt.show()

```

Compliance pressure by simulation year (last 1000 eps per run)



```
[34]: # Companion data tables for the compliance / penalty-pressure figure.
def _compliance_rate_tbl(yr):
    sf_cols = _agent_cols_in(yr, 'shortfall')
    if sf_cols:
        long = yr.melt(id_vars=['year'], value_vars=sf_cols,
            value_name='shortfall')
        long['compliant'] = (long['shortfall'] <= 1e-6).astype(float)
        return long.groupby('year')['compliant'].mean()

    surplus_cols = _agent_cols_in(yr, 'compliance_surplus')
    if not surplus_cols:
        return None
    long = yr.melt(id_vars=['year'], value_vars=surplus_cols,
        value_name='surplus')
    long['compliant'] = (long['surplus'].fillna(0) >= -1e-6).astype(float)
    return long.groupby('year')['compliant'].mean()

_panels_compliance = [
    ('System coverage gap (pre-secondary)',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'coverage_gap'))),
    ('End-of-year carry-forward (system)',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr,
        'carry_forward_end'))),
    ('Compliance rate by year',
     _per_sweep_year_stack(_compliance_rate_tbl)),
    ('System penalty paid',
```

```

    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'penalty'))),
]
_units_compliance = {
    'System coverage gap (pre-secondary)': 'Mt',
    'End-of-year carry-forward (system)': 'Mt',
    'Compliance rate by year': 'share of compliant (episode, agent) cells',
    'System penalty paid': 'M€',
}
_compliance_yoy_tables = _show_yoy_tables(_panels_compliance,
    ↪units=_units_compliance)

```

Across-year summary (plotted median series across 1000-episode tail, per sweep):

			panel							
↪unit	sweep	n_seeds	years	mean	median	std	min	max		
0	System coverage gap (pre-secondary)								↪	
↪Mt	default_seeds	4	12	6.233	6.435	0.5627	5.123	6.864	↪	
1	End-of-year carry-forward (system)								↪	
↪Mt	default_seeds	4	12	1.621	1.573	0.6715	0.2078	2.877	↪	
2	Compliance rate by year	share of compliant (episode, agent)							↪	
↪cells	default_seeds	4	12	0.8297	0.8413	0.05209	0.7237	0.9041	↪	
3	System penalty paid								↪	
↪M€	default_seeds	4	12	257.6	243.5	121.4	28.83	495.3	↪	

Per-year detail (median line; shaded min/max fan; mean listed for comparison):

			panel						
↪unit	sweep	year	mean_across_seeds	median_across_seeds					↪
↪std_across_seeds	min_across_seeds	max_across_seeds	\						
0	System coverage gap (pre-secondary)								↪
↪Mt	default_seeds	1	5.138		5.123		0.		↪
↪2108	4.896		5.41						
1	System coverage gap (pre-secondary)								↪
↪Mt	default_seeds	2	5.687		5.626		1.		↪
↪262	4.222		7.273						
2	System coverage gap (pre-secondary)								↪
↪Mt	default_seeds	3	5.441		5.445		0.		↪
↪4141	5.053		5.819						
3	System coverage gap (pre-secondary)								↪
↪Mt	default_seeds	4	5.904		6.016		0.		↪
↪5657	5.122		6.462						
4	System coverage gap (pre-secondary)								↪
↪Mt	default_seeds	5	6.047		6.275		0.		↪
↪6991	5.028		6.612						
5	System coverage gap (pre-secondary)								↪
↪Mt	default_seeds	6	6.125		6.413		0.		↪
↪7338	5.039		6.636						

6	System coverage gap (pre-secondary)				
↳Mt	default_seeds	7	6.172	6.456	0.
↳7477		5.083	6.695		
7	System coverage gap (pre-secondary)				
↳Mt	default_seeds	8	6.201	6.476	0.
↳756		5.092	6.761		
8	System coverage gap (pre-secondary)				
↳Mt	default_seeds	9	6.344	6.589	0.
↳7726		5.221	6.977		
9	System coverage gap (pre-secondary)				
↳Mt	default_seeds	10	6.483	6.711	0.
↳6731		5.497	7.013		
10	System coverage gap (pre-secondary)				
↳Mt	default_seeds	11	6.592	6.797	0.
↳5285		5.809	6.964		
11	System coverage gap (pre-secondary)				
↳Mt	default_seeds	12	6.824	6.864	0.
↳3149		6.403	7.164		
12	End-of-year carry-forward (system)				
↳Mt	default_seeds	1	0.2539	0.2078	0.
↳1495		0.1301	0.4701		
13	End-of-year carry-forward (system)				
↳Mt	default_seeds	2	1.416	1.235	0.
↳9586		0.4599	2.735		
14	End-of-year carry-forward (system)				
↳Mt	default_seeds	3	1.398	1.168	0.
↳8351		0.6681	2.587		
15	End-of-year carry-forward (system)				
↳Mt	default_seeds	4	1.478	1.38	0.
↳5969		0.9134	2.238		
16	End-of-year carry-forward (system)				
↳Mt	default_seeds	5	1.513	1.488	0.
↳6206		0.8896	2.186		
17	End-of-year carry-forward (system)				
↳Mt	default_seeds	6	1.547	1.559	0.
↳6265		0.9103	2.16		
18	End-of-year carry-forward (system)				
↳Mt	default_seeds	7	1.57	1.587	0.
↳6128		0.9211	2.183		
19	End-of-year carry-forward (system)				
↳Mt	default_seeds	8	1.609	1.612	0.
↳5948		0.9624	2.249		
20	End-of-year carry-forward (system)				
↳Mt	default_seeds	9	1.782	1.774	0.
↳5848		1.18	2.4		

21	End-of-year carry-forward (system)				□
↳Mt	default_seeds	10	2.06	2.131	0.
↳6216		1.374	2.603		
22	End-of-year carry-forward (system)				□
↳Mt	default_seeds	11	2.379	2.437	0.
↳6475		1.61	3.032		
23	End-of-year carry-forward (system)				□
↳Mt	default_seeds	12	2.781	2.877	0.
↳6705		1.902	3.467		
24	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	1	0.9032	0.9041	0.
↳02768		0.8726	0.932		
25	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	2	0.8498	0.8647	0.
↳09569		0.7201	0.9497		
26	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	3	0.8699	0.8787	0.
↳05489		0.7953	0.927		
27	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	4	0.8573	0.8591	0.
↳04703		0.8003	0.9105		
28	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	5	0.8566	0.8531	0.
↳04834		0.8049	0.9153		
29	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	6	0.8502	0.8427	0.
↳04419		0.81	0.9052		
30	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	7	0.8462	0.8398	0.
↳04033		0.8089	0.8964		
31	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	8	0.8358	0.8335	0.
↳04137		0.7914	0.8849		
32	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	9	0.8116	0.8174	0.
↳04013		0.7635	0.8481		
33	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	10	0.7829	0.7849	0.
↳04332		0.7392	0.8227		
34	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	11	0.7519	0.7549	□
↳0.0403		0.7046	0.793		
35	Compliance rate by year share of compliant (episode, agent)				□
↳cells	default_seeds	12	0.7187	0.7237	0.
↳04532		0.6604	0.7669		

36		System penalty paid			
↵M€	default_seeds	1	35.23	28.83	20.
↵74		18.05	65.23		
37		System penalty paid			
↵M€	default_seeds	2	200.4	174.6	
↵135.7		65.14	387.3		
38		System penalty paid			
↵M€	default_seeds	3	201.8	168.6	
↵120.7		96.39	373.7		
39		System penalty paid			
↵M€	default_seeds	4	217.5	202.9	87.
↵95		134.4	329.7		
40		System penalty paid			
↵M€	default_seeds	5	227.3	223.6	93.
↵12		133.7	328.4		
41		System penalty paid			
↵M€	default_seeds	6	237	238.6	95.
↵94		139.5	331.5		
42		System penalty paid			
↵M€	default_seeds	7	245.4	248.3	95.
↵63		144.1	340.7		
43		System penalty paid			
↵M€	default_seeds	8	256.6	257.2	
↵94.7		153.7	358.1		
44		System penalty paid			
↵M€	default_seeds	9	290	289.2	95.
↵03		192.1	389.5		
45		System penalty paid			
↵M€	default_seeds	10	342	352.8	
↵103.1		228.8	433.7		
46		System penalty paid			
↵M€	default_seeds	11	402.8	411.7	
↵109.9		272.8	515.1		
47		System penalty paid			
↵M€	default_seeds	12	480	495.3	
↵116		328.9	600.6		

	n_seeds
0	4
1	4
2	4
3	4
4	4
5	4
6	4
7	4
8	4

9	4
10	4
11	4
12	4
13	4
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33	4
34	4
35	4
36	4
37	4
38	4
39	4
40	4
41	4
42	4
43	4
44	4
45	4
46	4
47	4

1.9.3 Decarbonisation and investment over the simulation year

Cross-agent mean green generation share (does the fleet green up over the year axis?), system emissions (Mt), system invest cost (M€), and MAC fuel-switching reductions (Mt abated). All four panels read from per-agent year_log columns aggregated across the converged tail.

```
[35]: fig, axes = plt.subplots(2, 2, figsize=(14, 8))
```



```

_overlay_yoy(
    axes[0, 0],
    _per_sweep_year_stack(lambda yr: _agent_mean_by_year(yr, 'green_frac')),
    ylabel='green frac in [0, 1]', title='Mean green generation share')
axes[0, 0].set_ylim(0, 1)

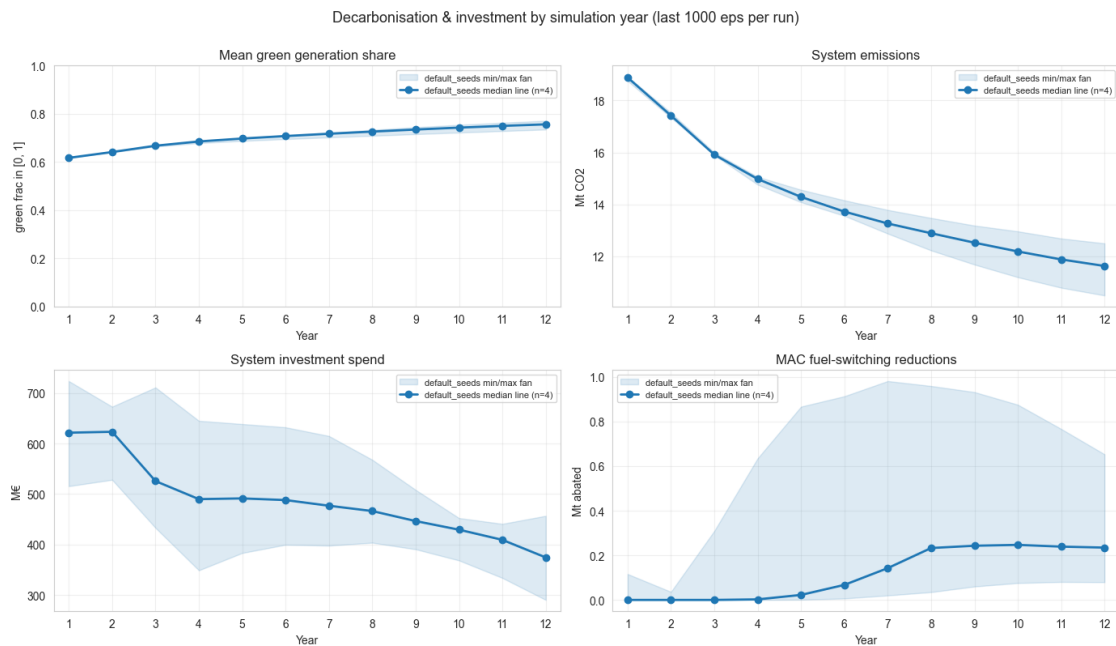
_overlay_yoy(
    axes[0, 1],
    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'emissions')),
    ylabel='Mt CO2', title='System emissions')

_overlay_yoy(
    axes[1, 0],
    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'invest_cost')),
    ylabel='M€', title='System investment spend')

_overlay_yoy(
    axes[1, 1],
    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'mac_reduction')),
    ylabel='Mt abated', title='MAC fuel-switching reductions')

plt.suptitle(f'Decarbonisation & investment by simulation year (last_
↳{YOY_TAIL_EPISODES} eps per run)',
            fontsize=13, y=1.00)
plt.tight_layout(); plt.show()

```



```
[36]: # Companion data tables for the decarbonisation / investment figure.
_panels_decarb = [
    ('Mean green generation share',
     _per_sweep_year_stack(lambda yr: _agent_mean_by_year(yr, 'green_frac'))),
    ('System emissions',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'emissions'))),
    ('System investment spend',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'invest_cost'))),
    ('MAC fuel-switching reductions',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr,
     ↪ 'mac_reduction'))),
]
_units_decarb = {
    'Mean green generation share': 'green frac [0, 1]',
    'System emissions': 'Mt CO ',
    'System investment spend': 'M€',
    'MAC fuel-switching reductions': 'Mt abated',
}
_decarb_yoy_tables = _show_yoy_tables(_panels_decarb, units=_units_decarb)
```

Across-year summary (plotted median series across 1000-episode tail, per sweep):

			panel		unit	sweep	n_seeds	
↪ years	mean	median	std	min	max			
0	Mean green generation share	green frac	[0, 1]	default_seeds		4		
↪ 12	0.7036	0.7125	0.04388	0.6171	0.7563			
1	System emissions	Mt CO	default_seeds		4			
↪ 12	14.13	13.49	2.283	11.63	18.88			
2	System investment spend	M€	default_seeds		4			
↪ 12	486.9	482.5	75.19	374.7	623.1			
3	MAC fuel-switching reductions	Mt abated	default_seeds		4			
↪ 12	0.1192	0.1049	0.1132	0	0.2468			

Per-year detail (median line; shaded min/max fan; mean listed for comparison):

		panel		unit	sweep	year	
↪ mean_across_seeds	median_across_seeds	std_across_seeds	min_across_seeds				
↪ max_across_seeds	n_seeds						
0	Mean green generation share	green frac	[0, 1]	default_seeds	1		
↪	0.617	0.6171	0.0002988	0.6165			
↪	0.6171	4					
1	Mean green generation share	green frac	[0, 1]	default_seeds	2		
↪	0.6413	0.6413	0.001056	0.64			
↪	0.6424	4					
2	Mean green generation share	green frac	[0, 1]	default_seeds	3		
↪	0.667	0.6675	0.002309	0.6639			
↪	0.6692	4					

3	Mean green generation share	green frac	[0, 1]	default_seeds	4	
↪	0.684	0.6853	0.00381	0.6784		
↪	0.6871	4				
4	Mean green generation share	green frac	[0, 1]	default_seeds	5	
↪	0.6955	0.6974	0.005549	0.6874		
↪	0.6998	4				
5	Mean green generation share	green frac	[0, 1]	default_seeds	6	
↪	0.7053	0.7077	0.007364	0.695		
↪	0.7107	4				
6	Mean green generation share	green frac	[0, 1]	default_seeds	7	
↪	0.7147	0.7172	0.009378	0.702		
↪	0.7222	4				
7	Mean green generation share	green frac	[0, 1]	default_seeds	8	
↪	0.7237	0.7262	0.0114	0.7089		
↪	0.7335	4				
8	Mean green generation share	green frac	[0, 1]	default_seeds	9	
↪	0.7324	0.7348	0.01321	0.7156		
↪	0.7443	4				
9	Mean green generation share	green frac	[0, 1]	default_seeds	10	
↪	0.7404	0.7427	0.01476	0.7221		
↪	0.7542	4				
10	Mean green generation share	green frac	[0, 1]	default_seeds	11	
↪	0.7478	0.7499	0.01587	0.7284		
↪	0.7632	4				
11	Mean green generation share	green frac	[0, 1]	default_seeds	12	
↪	0.7546	0.7563	0.01664	0.7347		
↪	0.7712	4				
12	System emissions		Mt CO	default_seeds	1	
↪	18.86	18.88	0.07374	18.75		
↪	18.91	4				
13	System emissions		Mt CO	default_seeds	2	
↪	17.43	17.42	0.08008	17.34		
↪	17.53	4				
14	System emissions		Mt CO	default_seeds	3	
↪	15.93	15.91	0.04706	15.89		
↪	16	4				
15	System emissions		Mt CO	default_seeds	4	
↪	14.94	14.98	0.128	14.76		
↪	15.06	4				
16	System emissions		Mt CO	default_seeds	5	
↪	14.31	14.29	0.2033	14.08		
↪	14.56	4				
17	System emissions		Mt CO	default_seeds	6	
↪	13.79	13.72	0.2654	13.56		
↪	14.16	4				

18		System emissions	Mt CO	default_seeds	7	□
↪	13.3	13.27	0.3764	12.88		□
↪	13.79	4				
19		System emissions	Mt CO	default_seeds	8	□
↪	12.87	12.89	0.5115	12.23		□
↪	13.48	4				
20		System emissions	Mt CO	default_seeds	9	□
↪	12.48	12.53	0.626	11.69		□
↪	13.19	4				
21		System emissions	Mt CO	default_seeds	10	□
↪	12.14	12.19	0.7425	11.2		□
↪	12.96	4				
22		System emissions	Mt CO	default_seeds	11	□
↪	11.81	11.88	0.8154	10.8		□
↪	12.69	4				
23		System emissions	Mt CO	default_seeds	12	□
↪	11.57	11.63	0.8746	10.5		□
↪	12.5	4				
24		System investment spend	M€	default_seeds	1	□
↪	620.4	621.3	88.52	515.6		□
↪	723.4	4				
25		System investment spend	M€	default_seeds	2	□
↪	611.8	623.1	61.95	528.2		□
↪	672.8	4				
26		System investment spend	M€	default_seeds	3	□
↪	548.8	525.7	125.2	433.1		□
↪	710.8	4				
27		System investment spend	M€	default_seeds	4	□
↪	493.5	490	139.4	349.2		□
↪	644.9	4				
28		System investment spend	M€	default_seeds	5	□
↪	501.2	491.4	129.7	383.5		□
↪	638.5	4				
29		System investment spend	M€	default_seeds	6	□
↪	502	488.1	119.7	399.8		□
↪	632	4				
30		System investment spend	M€	default_seeds	7	□
↪	491.7	476.9	109.3	398		□
↪	614.8	4				
31		System investment spend	M€	default_seeds	8	□
↪	476.1	466.4	82.86	403.8		□
↪	567.8	4				
32		System investment spend	M€	default_seeds	9	□
↪	448.1	446.6	58.19	390.8		□
↪	508.3	4				

33	System investment spend	M€	default_seeds	10	□
↪	420.1	429.5	37.65	368.7	□
↪	452.6	4			
34	System investment spend	M€	default_seeds	11	□
↪	398.5	409.4	45.49	334.1	□
↪	441.1	4			
35	System investment spend	M€	default_seeds	12	□
↪	374.3	374.7	68.47	290.7	□
↪	457	4			
36	MAC fuel-switching reductions	Mt abated	default_seeds	1	□
↪	0.02904	0	0.05807	0	□
↪	0.1161	4			
37	MAC fuel-switching reductions	Mt abated	default_seeds	2	□
↪	0.009059	0	0.01812	0	□
↪	0.03624	4			
38	MAC fuel-switching reductions	Mt abated	default_seeds	3	□
↪	0.07737	0	0.1547	0	□
↪	0.3095	4			
39	MAC fuel-switching reductions	Mt abated	default_seeds	4	□
↪	0.16	0.002459	0.3167	0	□
↪	0.635	4			
40	MAC fuel-switching reductions	Mt abated	default_seeds	5	□
↪	0.2282	0.02244	0.4259	0.001155	□
↪	0.8667	4			
41	MAC fuel-switching reductions	Mt abated	default_seeds	6	□
↪	0.2636	0.06738	0.4352	0.006317	□
↪	0.9133	4			
42	MAC fuel-switching reductions	Mt abated	default_seeds	7	□
↪	0.3216	0.1425	0.4483	0.01998	□
↪	0.9814	4			
43	MAC fuel-switching reductions	Mt abated	default_seeds	8	□
↪	0.3649	0.2327	0.416	0.03489	□
↪	0.9593	4			
44	MAC fuel-switching reductions	Mt abated	default_seeds	9	□
↪	0.3695	0.243	0.4009	0.06004	□
↪	0.932	4			
45	MAC fuel-switching reductions	Mt abated	default_seeds	10	□
↪	0.3612	0.2468	0.3772	0.07569	□
↪	0.8754	4			
46	MAC fuel-switching reductions	Mt abated	default_seeds	11	□
↪	0.3314	0.239	0.3242	0.08065	□
↪	0.7667	4			
47	MAC fuel-switching reductions	Mt abated	default_seeds	12	□
↪	0.3005	0.2346	0.2675	0.07955	□
↪	0.6531	4			

1.9.4 Secondary-market activity over the simulation year

Total secondary volume per year (sum of net buys across agents — by construction equal to the sum of net sells), executed buyer / seller counts vs the corresponding intent counts (a wide gap means agents post uncrossable orders), and the training-consistent secondary match rate (share of episode-years with any executed trade). If the required match-rate fields are unavailable in a run, the panel falls back to the secondary-vs-clearing price spread.

```
[37]: fig, axes = plt.subplots(2, 2, figsize=(14, 8))

def _secondary_volume(yr):
    cols = _agent_cols_in(yr, 'secondary_net')
    if not cols:
        # Fallback: signed trade_qty across agents.
        cols = _agent_cols_in(yr, 'trade_qty')
    if not cols:
        return None
    # Mean per (year, agent) across episodes, then sum positives across agents.
    by_year = yr.groupby('year')[cols].mean()
    return by_year.clip(lower=0).sum(axis=1)

_overlay_yoy(
    axes[0, 0],
    _per_sweep_year_stack(_secondary_volume),
    ylabel='Mt traded', title='Total secondary volume')

def _intent_exec_gap_buyers(yr):
    if not {'secondary_n_buyers_intent', 'secondary_n_buyers_executed'}.
    <=issubset(yr.columns):
        return None
    g = yr.groupby('year')[['secondary_n_buyers_intent',
    <='secondary_n_buyers_executed']].mean()
    return g['secondary_n_buyers_intent'] - g['secondary_n_buyers_executed']

def _intent_exec_gap_sellers(yr):
    if not {'secondary_n_sellers_intent', 'secondary_n_sellers_executed'}.
    <=issubset(yr.columns):
        return None
    g = yr.groupby('year')[['secondary_n_sellers_intent',
    <='secondary_n_sellers_executed']].mean()
    return g['secondary_n_sellers_intent'] - g['secondary_n_sellers_executed']

_overlay_yoy(
    axes[0, 1],
    _per_sweep_year_stack(_intent_exec_gap_buyers),
    ylabel='agents (intent - executed)',
    title='Buy-side intent-execution gap')
```

```

_overlay_yoy(
    axes[1, 0],
    _per_sweep_year_stack(_intent_exec_gap_sellers),
    ylabel='agents (intent - executed)',
    title='Sell-side intent-execution gap')

def _training_match_rate_by_year(yr):
    # Match the training metric definition: a year is matched if any trade
    ↪ executed.
    cols = {'secondary_n_buyers_executed', 'secondary_n_sellers_executed'}
    if cols.issubset(yr.columns):
        any_trade = (yr['secondary_n_buyers_executed'].fillna(0)
                     + yr['secondary_n_sellers_executed'].fillna(0)) > 0
        return any_trade.groupby(yr['year']).mean()
    trade_cols = _agent_cols_in(yr, 'trade_qty')
    if not trade_cols:
        return None
    any_trade = yr[trade_cols].fillna(0).abs().gt(1e-6).any(axis=1)
    return any_trade.groupby(yr['year']).mean()

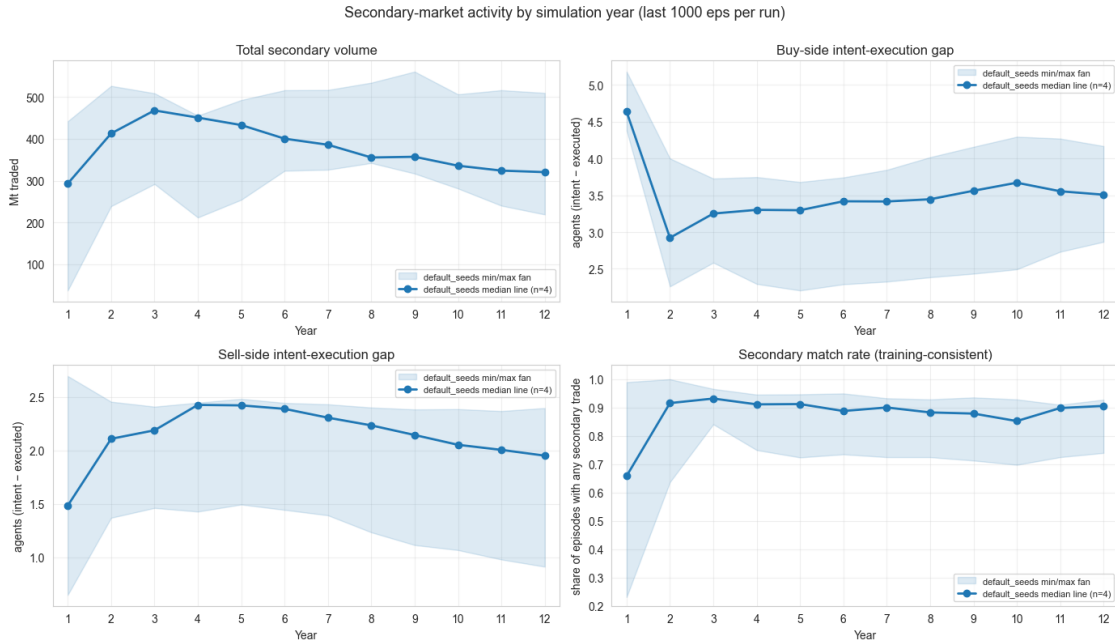
def _sec_minus_clearing(yr):
    if not {'secondary_price', 'clearing_price'}.issubset(yr.columns):
        return None
    g = yr.groupby('year')[['secondary_price', 'clearing_price']].mean()
    return g['secondary_price'] - g['clearing_price']

_match_rate_yoy = _per_sweep_year_stack(_training_match_rate_by_year)
if _match_rate_yoy:
    _secondary_panel_data = _match_rate_yoy
    _secondary_panel_ylabel = 'share of episodes with any secondary trade'
    _secondary_panel_title = 'Secondary match rate (training-consistent)'
else:
    _secondary_panel_data = _per_sweep_year_stack(_sec_minus_clearing)
    _secondary_panel_ylabel = '€/tonne'
    _secondary_panel_title = 'Secondary - clearing price spread'

_overlay_yoy(
    axes[1, 1],
    _secondary_panel_data,
    ylabel=_secondary_panel_ylabel,
    title=_secondary_panel_title)
if _match_rate_yoy:
    axes[1, 1].set_ylim(0.2, 1.05)
else:
    axes[1, 1].axhline(0, color='black', lw=0.8, linestyle='--', alpha=0.6)

```

```
plt.suptitle(f'Secondary-market activity by simulation year (last_{YOY_TAIL_EPISODES} eps per run)',
            fontsize=13, y=1.00)
plt.tight_layout(); plt.show()
```



```
[38]: # Companion data tables for the secondary-market activity figure.
def _secondary_volume_tbl(yr):
    cols = _agent_cols_in(yr, 'secondary_net')
    if not cols:
        cols = _agent_cols_in(yr, 'trade_qty')
        if not cols:
            return None
    by_year = yr.groupby('year')[cols].mean()
    return by_year.clip(lower=0).sum(axis=1)

def _intent_exec_gap_buyers_tbl(yr):
    if not {'secondary_n_buyers_intent', 'secondary_n_buyers_executed'}.issubset(yr.columns):
        return None
    g = yr.groupby('year')[['secondary_n_buyers_intent',
        'secondary_n_buyers_executed']].mean()
    return g['secondary_n_buyers_intent'] - g['secondary_n_buyers_executed']

def _intent_exec_gap_sellers_tbl(yr):
    if not {'secondary_n_sellers_intent', 'secondary_n_sellers_executed'}.issubset(yr.columns):
```



```

        return None
    g = yr.groupby('year')[['secondary_n_sellers_intent',
↪ 'secondary_n_sellers_executed']].mean()
    return g['secondary_n_sellers_intent'] - g['secondary_n_sellers_executed']

def _training_match_rate_by_year_tbl(yr):
    cols = {'secondary_n_buyers_executed', 'secondary_n_sellers_executed'}
    if cols.issubset(yr.columns):
        any_trade = (yr['secondary_n_buyers_executed'].fillna(0)
                     + yr['secondary_n_sellers_executed'].fillna(0)) > 0
        return any_trade.groupby(yr['year']).mean()
    trade_cols = _agent_cols_in(yr, 'trade_qty')
    if not trade_cols:
        return None
    any_trade = yr[trade_cols].fillna(0).abs().gt(1e-6).any(axis=1)
    return any_trade.groupby(yr['year']).mean()

def _sec_minus_clearing_tbl(yr):
    if not {'secondary_price', 'clearing_price'}.issubset(yr.columns):
        return None
    g = yr.groupby('year')[['secondary_price', 'clearing_price']].mean()
    return g['secondary_price'] - g['clearing_price']

_match_rate_yoy_tbl = _per_sweep_year_stack(_training_match_rate_by_year_tbl)
if _match_rate_yoy_tbl:
    _secondary_panel_label = 'Secondary match rate (training-consistent)'
    _secondary_panel_unit = 'share of episodes with any secondary trade'
    _secondary_panel_stack = _match_rate_yoy_tbl
else:
    _secondary_panel_label = 'Secondary - clearing price spread'
    _secondary_panel_unit = '€/tonne'
    _secondary_panel_stack = _per_sweep_year_stack(_sec_minus_clearing_tbl)

_panels_secondary = [
    ('Total secondary volume',
     _per_sweep_year_stack(_secondary_volume_tbl)),
    ('Buy-side intent-execution gap',
     _per_sweep_year_stack(_intent_exec_gap_buyers_tbl)),
    ('Sell-side intent-execution gap',
     _per_sweep_year_stack(_intent_exec_gap_sellers_tbl)),
    (_secondary_panel_label,
     _secondary_panel_stack),
]
_units_secondary = {
    'Total secondary volume': 'Mt traded',
    'Buy-side intent-execution gap': 'agents (intent - executed)',
    'Sell-side intent-execution gap': 'agents (intent - executed)',
}

```

```

    _secondary_panel_label: _secondary_panel_unit,
}
_secondary_yoy_tables = _show_yoy_tables(_panels_secondary,
↳units=_units_secondary)

```

Across-year summary (plotted median series across 1000-episode tail, per sweep):

				panel					
↳	unit	sweep	n_seeds	years	mean	median	std	min	max
0	Total secondary volume								
↳	Mt traded	default_seeds	4	12	378.1	371.5	55.99	293	468.2
1	Buy-side intent-execution gap agents (intent -								
↳	executed)	default_seeds	4	12	3.498	3.432	0.4066	2.922	4.636
2	Sell-side intent-execution gap agents (intent -								
↳	executed)	default_seeds	4	12	2.142	2.166	0.2618	1.484	2.425
3	Secondary match rate (training-consistent) share of episodes with any								
↳	secondary trade	default_seeds	4	12	0.8782	0.8995	0.0722	0.6585	0.932

Per-year detail (median line; shaded min/max fan; mean listed for comparison):

				panel				
↳	unit	sweep	year	mean_across_seeds	median_across_seeds	↳		
↳	std_across_seeds	min_across_seeds	\					
0	Total secondary volume							
↳	Mt traded	default_seeds	1	266.3		293		
↳	171.7	36.96						
1	Total secondary volume							
↳	Mt traded	default_seeds	2	397.9		413		
↳	140.6	238.8						
2	Total secondary volume							
↳	Mt traded	default_seeds	3	434.4		468.2		
↳	99.44	291.9						
3	Total secondary volume							
↳	Mt traded	default_seeds	4	392.3		451		
↳	120.4	211.7						
4	Total secondary volume							
↳	Mt traded	default_seeds	5	403.4		433.2		
↳	103.4	254.1						
5	Total secondary volume							
↳	Mt traded	default_seeds	6	410.3		400.5		
↳	79.64	323.6						
6	Total secondary volume							
↳	Mt traded	default_seeds	7	403.8		385.9		
↳	81.71	326.2						
7	Total secondary volume							
↳	Mt traded	default_seeds	8	396.9		355.5		
↳	92.53	342.2						

8	Total secondary volume					
↪Mt traded	default_seeds	9	398.1	357.1	↪	
↪	111.7	316.9			↪	
9	Total secondary volume					↪
↪Mt traded	default_seeds	10	364.8	335.8	↪	
↪	105.2	281.1				
10	Total secondary volume					↪
↪Mt traded	default_seeds	11	351.2	324	↪	
↪	125.9	240				
11	Total secondary volume					↪
↪Mt traded	default_seeds	12	342.5	320.4	↪	
↪	127.2	219				
12	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	1	4.708	4.636	↪	
↪	0.3679	4.38				
13	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	2	3.026	2.922	↪	
↪	0.7471	2.261				
14	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	3	3.203	3.251	↪	
↪	0.5113	2.584				
15	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	4	3.161	3.302	↪	
↪	0.6835	2.295				
16	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	5	3.12	3.296	↪	
↪	0.6501	2.206				
17	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	6	3.217	3.418	↪	
↪	0.6789	2.29				
18	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	7	3.251	3.415	↪	
↪	0.7344	2.326				
19	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	8	3.323	3.446	↪	
↪	0.7429	2.385				
20	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	9	3.428	3.561	↪	
↪	0.7885	2.434				
21	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	10	3.532	3.67	↪	
↪	0.8284	2.493				
22	Buy-side intent-execution gap			agents (intent	↪	
↪executed)	default_seeds	11	3.528	3.554	↪	
↪	0.7525	2.732				

23	Buy-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 12	3.513	3.508
↪	0.7357 2.867		
24	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 1	1.578	1.484
↪	0.8434 0.649		
25	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 2	2.01	2.108
↪	0.5232 1.37		
26	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 3	2.062	2.188
↪	0.4303 1.462		
27	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 4	2.181	2.425
↪	0.5025 1.428		
28	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 5	2.204	2.421
↪	0.4748 1.494		
29	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 6	2.166	2.389
↪	0.484 1.444		
30	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 7	2.109	2.306
↪	0.4824 1.393		
31	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 8	2.026	2.234
↪	0.5337 1.234		
32	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 9	1.947	2.144
↪	0.5658 1.115		
33	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 10	1.89	2.052
↪	0.5706 1.068		
34	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 11	1.839	2.005
↪	0.6029 0.98		
35	Sell-side intent-execution gap	agents (intent -	
↪executed)	default_seeds 12	1.803	1.951
↪	0.6379 0.913		
36	Secondary match rate (training-consistent)	share of episodes with any	
↪secondary trade	default_seeds 1	0.6345	0.6585
↪	0.3324 0.231		
37	Secondary match rate (training-consistent)	share of episodes with any	
↪secondary trade	default_seeds 2	0.8672	0.916
↪	0.1609 0.637		

38	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 3	0.9177	0.932 ┐
↪	0.05539 0.841		
39	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 4	0.8798	0.9115 ┐
↪	0.09008 0.75		
40	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 5	0.874	0.9125 ┐
↪	0.1023 0.724		
41	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 6	0.8653	0.888 ┐
↪	0.09352 0.735		
42	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 7	0.8648	0.9005 ┐
↪	0.09457 0.725		
43	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 8	0.855	0.883 ┐
↪	0.09164 0.725		
44	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 9	0.8518	0.879 ┐
↪	0.0983 0.713		
45	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 10	0.833	0.8525 ┐
↪	0.09868 0.698		
46	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 11	0.858	0.8985 ┐
↪	0.08933 0.725		
47	Secondary match rate (training-consistent)	share of episodes with any	┐
↪	secondary trade default_seeds 12	0.87	0.906 ┐
↪	0.08851 0.74		

	max_across_seeds	n_seeds
0	442.2	4
1	526.9	4
2	509.2	4
3	455.7	4
4	493	4
5	516.5	4
6	517.2	4
7	534.6	4
8	561.1	4
9	506.7	4
10	516.6	4
11	510.1	4
12	5.18	4
13	4	4
14	3.727	4

15	3.747	4
16	3.68	4
17	3.742	4
18	3.845	4
19	4.017	4
20	4.158	4
21	4.295	4
22	4.27	4
23	4.168	4
24	2.695	4
25	2.455	4
26	2.409	4
27	2.446	4
28	2.482	4
29	2.444	4
30	2.431	4
31	2.401	4
32	2.383	4
33	2.386	4
34	2.368	4
35	2.396	4
36	0.99	4
37	1	4
38	0.966	4
39	0.946	4
40	0.947	4
41	0.95	4
42	0.933	4
43	0.929	4
44	0.936	4
45	0.929	4
46	0.91	4
47	0.928	4

1.9.5 Reward and reward-component dynamics over the simulation year

System-aggregate per-year reward, the logged `reward_base` component (financial – penalty + ESG), system penalty paid as a concrete reward cost driver, plus the cross-agent reward dispersion per year (std across the agent panel), which is a proxy for outcome equity. The per-episode `quality_score` does not have a within-episode year axis and is therefore covered in the convergence panels above; the reward-decomposition panels here are its per-year analogue.

```
[39]: fig, axes = plt.subplots(2, 2, figsize=(14, 8))

      _overlay_yoy(
          axes[0, 0],
          _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'reward')),
          ylabel='reward (units of training-time reward)',
```

```

        title='System reward')

_overlay_yoy(
    axes[0, 1],
    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'reward_base')),
    ylabel='reward_base', title='System reward_base (financial - penalty +
↳ESG)')

_overlay_yoy(
    axes[1, 0],
    _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'penalty')),
    ylabel='M€', title='System penalty paid (reward cost driver)')

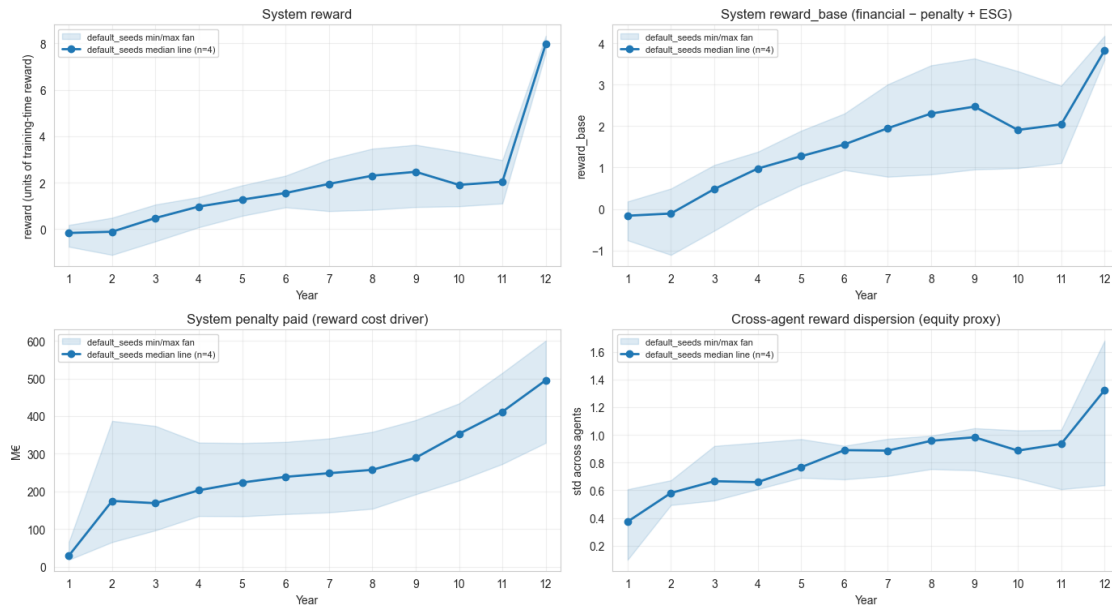
def _reward_dispersion(yr):
    cols = _agent_cols_in(yr, 'reward')
    if not cols:
        return None
    by_year = yr.groupby('year')[cols].mean()
    return by_year.std(axis=1)

_overlay_yoy(
    axes[1, 1],
    _per_sweep_year_stack(_reward_dispersion),
    ylabel='std across agents',
    title='Cross-agent reward dispersion (equity proxy)')

plt.suptitle(f'Reward and reward components by simulation year (last
↳{YOY_TAIL_EPISODES} eps per run)',
            fontsize=13, y=1.00)
plt.tight_layout(); plt.show()

```

Reward and reward components by simulation year (last 1000 eps per run)



```
[40]: # Companion data tables for the reward-dynamics figure.
def _reward_dispersion_tbl(yr):
    cols = _agent_cols_in(yr, 'reward')
    if not cols:
        return None
    by_year = yr.groupby('year')[cols].mean()
    return by_year.std(axis=1)

_panels_reward = [
    ('System reward',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'reward'))),
    ('System reward_base (financial - penalty + ESG)',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'reward_base'))),
    ('System penalty paid (reward cost driver)',
     _per_sweep_year_stack(lambda yr: _system_sum_by_year(yr, 'penalty'))),
    ('Cross-agent reward dispersion (equity proxy)',
     _per_sweep_year_stack(_reward_dispersion_tbl)),
]

_units_reward = {
    'System reward': 'reward (training-time units)',
    'System reward_base (financial - penalty + ESG)': 'reward_base',
    'System penalty paid (reward cost driver)': 'M€',
    'Cross-agent reward dispersion (equity proxy)': 'std across agents',
}

_reward_yoy_tables = _show_yoy_tables(_panels_reward, units=_units_reward)
```


Across-year summary (plotted median series across 1000-episode tail, per sweep):

				panel					unit
	sweep	n_seeds	years	mean	median	std	min	max	
0				System reward		reward (training-time units)			
	↪ default_seeds	4	12	1.888	1.733	2.111	-0.1621	7.969	
1	System reward_base (financial - penalty + ESG)								reward_base
	↪ default_seeds	4	12	1.543	1.733	1.142	-0.1621	3.825	
2	System penalty paid (reward cost driver)								M€
	↪ default_seeds	4	12	257.6	243.5	121.4	28.83	495.3	
3	Cross-agent reward dispersion (equity proxy)								std across agents
	↪ default_seeds	4	12	0.8256	0.8865	0.2401	0.3737	1.323	

Per-year detail (median line; shaded min/max fan; mean listed for comparison):

				panel			unit
	sweep	year	mean_across_seeds	median_across_seeds	std_across_seeds		
	↪ min_across_seeds						
	↪ max_across_seeds						
0			System reward		reward (training-time units)		
	↪ default_seeds	1	-0.2246		-0.1621	0.4109	
	↪		-0.7557	0.1814			
1			System reward		reward (training-time units)		
	↪ default_seeds	2	-0.208		-0.1089	0.7438	
	↪		-1.108	0.4936			
2			System reward		reward (training-time units)		
	↪ default_seeds	3	0.3754		0.4815	0.6776	
	↪		-0.5241	1.063			
3			System reward		reward (training-time units)		
	↪ default_seeds	4	0.8532		0.9753	0.5717	
	↪		0.0826	1.38			
4			System reward		reward (training-time units)		
	↪ default_seeds	5	1.253		1.274	0.5556	
	↪		0.5751	1.889			
5			System reward		reward (training-time units)		
	↪ default_seeds	6	1.59		1.558	0.5669	
	↪		0.9405	2.304			
6			System reward		reward (training-time units)		
	↪ default_seeds	7	1.921		1.949	1.032	
	↪		0.778	3.008			
7			System reward		reward (training-time units)		
	↪ default_seeds	8	2.227		2.303	1.185	
	↪		0.8342	3.467			
8			System reward		reward (training-time units)		
	↪ default_seeds	9	2.382		2.47	1.223	
	↪		0.9536	3.632			
9			System reward		reward (training-time units)		
	↪ default_seeds	10	2.032		1.908	1.105	
	↪		0.9855	3.326			

10		System reward	reward (training-time units)	┐
↪	default_seeds	11	2.041	2.042
↪		1.107	2.974	0.8403┐
11		System reward	reward (training-time units)	┐
↪	default_seeds	12	7.965	7.969
↪		7.592	8.33	0.3057┐
12	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	1	-0.2246	-0.1621
↪		-0.7557	0.1814	0.4109┐
13	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	2	-0.208	-0.1089
↪		-1.108	0.4936	0.7438┐
14	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	3	0.3754	0.4815
↪		-0.5241	1.063	0.6776┐
15	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	4	0.8532	0.9753
↪		0.0826	1.38	0.5717┐
16	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	5	1.253	1.274
↪		0.5751	1.889	0.5556┐
17	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	6	1.59	1.558
↪		0.9405	2.304	0.5669┐
18	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	7	1.921	1.949
↪		0.778	3.008	1.032┐
19	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	8	2.227	2.303
↪		0.8342	3.467	1.185┐
20	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	9	2.382	2.47
↪		0.9536	3.632	1.223┐
21	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	10	2.032	1.908
↪		0.9855	3.326	1.105┐
22	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	11	2.041	2.042
↪		1.107	2.974	0.8403┐
23	System reward_base (financial - penalty + ESG)			reward_base┐
↪	default_seeds	12	3.859	3.825
↪		3.605	4.181	0.2551┐
24	System penalty paid (reward cost driver)			M€┐
↪	default_seeds	1	35.23	28.83
↪		18.05	65.23	20.74┐

25	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	2	200.4	174.6
↪		65.14	387.3	135.7 _␣
26	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	3	201.8	168.6
↪		96.39	373.7	120.7 _␣
27	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	4	217.5	202.9
↪		134.4	329.7	87.95 _␣
28	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	5	227.3	223.6
↪		133.7	328.4	93.12 _␣
29	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	6	237	238.6
↪		139.5	331.5	95.94 _␣
30	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	7	245.4	248.3
↪		144.1	340.7	95.63 _␣
31	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	8	256.6	257.2
↪		153.7	358.1	94.7 _␣
32	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	9	290	289.2
↪		192.1	389.5	95.03 _␣
33	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	10	342	352.8
↪		228.8	433.7	103.1 _␣
34	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	11	402.8	411.7
↪		272.8	515.1	109.9 _␣
35	System penalty paid (reward cost driver)			M€ _␣
↪	default_seeds	12	480	495.3
↪		328.9	600.6	116 _␣
36	Cross-agent reward dispersion (equity proxy)			std across agents _␣
↪	default_seeds	1	0.3633	0.3737
↪		0.09824	0.6073	0.2211 _␣
37	Cross-agent reward dispersion (equity proxy)			std across agents _␣
↪	default_seeds	2	0.582	0.5809
↪		0.4935	0.6728	0.09337 _␣
38	Cross-agent reward dispersion (equity proxy)			std across agents _␣
↪	default_seeds	3	0.6946	0.6659
↪		0.5265	0.9201	0.1931 _␣
39	Cross-agent reward dispersion (equity proxy)			std across agents _␣
↪	default_seeds	4	0.7177	0.6589
↪		0.6082	0.9447	0.1558 _␣

40	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 5	0.7981	0.7663 0.1201
↪	0.6902	0.9696	
41	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 6	0.845	0.8899 0.1122
↪	0.6787	0.9214	
42	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 7	0.8619	0.8863 0.1168
↪	0.7035	0.9716	
43	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 8	0.9154	0.9572 0.1104
↪	0.753	0.9942	
44	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 9	0.9395	0.983 0.137
↪	0.7439	1.048	
45	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 10	0.8734	0.8868 0.1504
↪	0.6872	1.033	
46	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 11	0.8787	0.9352 0.2015
↪	0.6077	1.037	
47	Cross-agent reward dispersion (equity proxy)		std across agents
↪	default_seeds 12	1.241	1.323 0.4364
↪	0.6368	1.68	

	n_seeds
0	4
1	4
2	4
3	4
4	4
5	4
6	4
7	4
8	4
9	4
10	4
11	4
12	4
13	4
14	4
15	4
16	4
17	4
18	4
19	4
20	4

21	4
22	4
23	4
24	4
25	4
26	4
27	4
28	4
29	4
30	4
31	4
32	4
33	4
34	4
35	4
36	4
37	4
38	4
39	4
40	4
41	4
42	4
43	4
44	4
45	4
46	4
47	4

Reading these panels. Each coloured solid line is one sweep’s cross-seed median, with a shaded min/max fan when the sweep contains more than one seed; both are labelled directly in the legend. The price panels also add dashed or dotted reference lines for the fundamental anchor, reserve price, and penalty rate where they help orient the scale. Vertical separation between coloured medians is the cross-sweep effect; band overlap indicates that the sweeps are indistinguishable at the loaded sample size. The companion tables list cross-seed means alongside medians for every year, so mean-sensitive outliers remain visible even though the plotted line uses the median.

The line stays as the median because the loaded sample is small and seed outliers can pull the mean into a path no actual seed followed. The min/max fan shows the full observed envelope, while the tables keep the mean available for thesis prose and checks where outlier sensitivity is itself informative. Because every series is drawn from the converged tail, year-to-year structure here reflects the *shape* of the steady-state policy, not transient learning dynamics.

1.10 What this notebook does not answer

- Cross-config sensitivity — needs sweep variants. *Out of scope here.*
- Causal attribution of regulatory mechanisms (e.g. counterfactual “what if MSR were off”) — needs sweep variants.

- Real-world EU-ETS empirical validation beyond the qualitative price-band check in RQ1.3.
- Strategy clustering, theory tests (LCOE crossover, archetype heterogeneity), and predictive models of clearing price — these are in the analysis plan (`planning/data_science_analysis_plan.md` §B1, B2, B4) and will get their own notebooks where they can be evaluated cleanly with cross-validation.

The default-config evidence base assembled above is the *baseline* that those follow-on analyses will perturb and test against.